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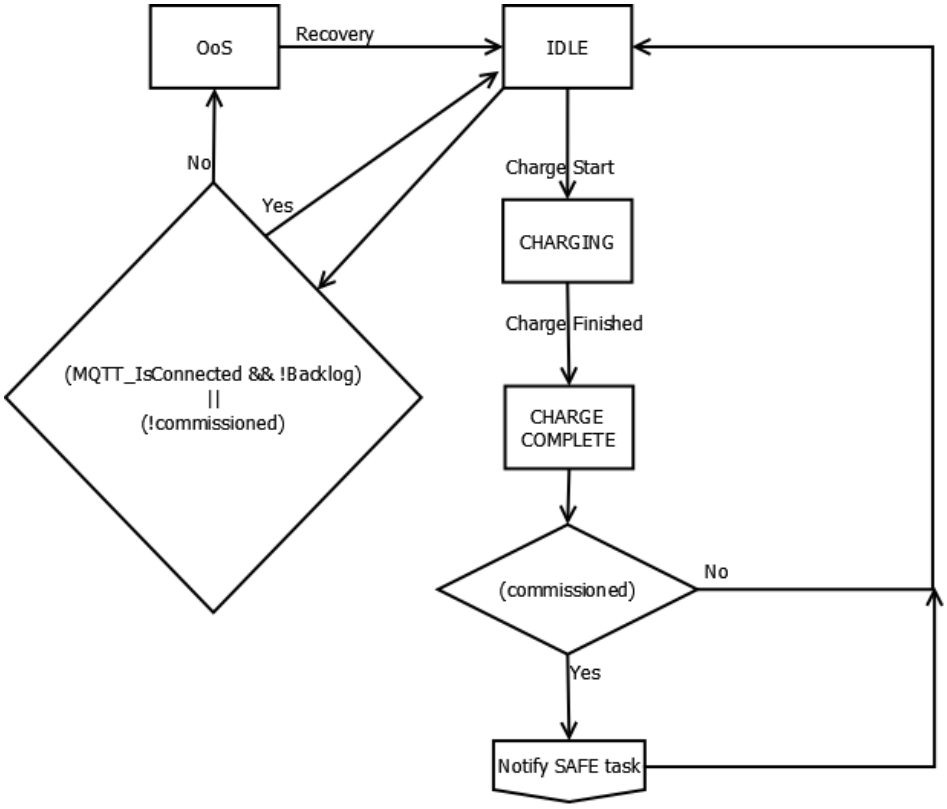
Charge Session State Machine

Overview

The Charge Session State Machine (CSSM) is executed periodically (approximately every 50ms). All states communicate to the AFE measurement module to extract measurement data.

It moves between several major states:

1. Idle State
    a. Monitors voltage/current for the start of a charge session
    b. When start of charging has been detected transitions to Charging State
2. Charging State
    a. Calculates the accumulating charge session energy in kWh
    b. Calculates timing for energy IR pulses
    c. Updates the charge session receipt
    d. Tests for the end of a charge session
    e. When end of charging has been detected transitions to Charge Completion State
3. Charge Completion State
    a. Completes the charge session receipt
    b. Notifies other relevant modules of completion of charge session
    c. Transitions to Idle State
4. Out Of Service (OoS) State
    a. Any state can transition to OoS on occurrence of an unexpected error, either from the CSSM or prompted by supervisor
    b. The CSSM will enter OoS if a previous SAFE packet wasn't successfully sent out over MQTT (Backlog flag shown in diagram)
    c. When allowed by supervisor, monitors conditions for recovery back to Idle - to recover all errors must be cleared and the voltage and current must be below the start thresholds



Tariffs

The DC Meter supports a tariff name to be displayed locally on the Trusted Screen Overlay and logged with the charge session information (as the OCMF optional "UT" field in the first record in the signed transaction data). No fiscal calculations are supported.

The DC Meter is configured in production with a default tariff name that will be used on power up.

The CPO can choose to update the active tariff throughout the meter's life through OCPP. Alternatively the Tariffs can be left with the default name.

Tariff name changes are signed with the CPO's private key. The signature is verified with the CPO public key which is pre-written into the meter's memory also during production. The CPO public key will conform to SECP256R1 (Key Curve), ASN.1 DER (Format), RFC-5480 (Standard).

Once tariffs are sent to the meter they will be used until new tariffs are sent or a power cycle. The tariff for a given charge session will be set once power delivery has started and changing the active tariff after this time will only apply to the next charge session.

Session Energy Calculations

Measured Energy Values (Ws)

Measured energy values of a CSSM calculation interval are calculated using 16 digit fixed point decimal numbers with a range of 10^6 to 10^-9,

These are the values that are reported during each cycle of the CSSM. Energy values are handled to .000000001 Ws precision.

Cumulative Energy Values (kWh)

Cumulative Energy Values are calculated using 19 digit fixed point decimal numbers with a range of 10^6 to 10^-12. Cumulative Energy values are handled to .000000000001 kWh precision.

Error Handling

In the case of an error in energy calculations the CSSM is placed in OoS state.

Accuracy Compensation

To improve the metrological accuracy several methods are applied, see Metrological Accuracy Compensation.

Out Of Service Causes

The following is a list of errors which can cause the CSSM to go out of service

- INVALID_STATE
- START_END_THRESHOLD
- ABOVE_IDLE_ENTRY_THRESHOLD
- MONOTONIC_TIMESTAMP
- AFE_READ
- AFE_CABLE_SOURCE
- GET_MEASUREMENT
- DECNUM_READ_VOLTAGE
- DECNUM_READ_CURRENT
- DECNUM_COMP_VOLTAGE
- DECNUM_COMP_CURRENT
- DECNUM_COMP_CABLE_SOURCE
- DECNUM_SESSION_ENERGY
- DECNUM_PULSE_CALCULATE
- DECNUM_TME
- CABLE_SELECT_SOURCE
- RTC_READ
- RD_CSS_DETAILS
- WR_CSS_DETAILS
- FORCED_CHARGE_COMPLETE
- MQTT_DISCONNECTED
- MQTT_BACKLOG
- TSO_NO_SESSION
- SUPERVISOR
    - Supervisor can put CSSM into OoS for several reasons:
        - Self checks fail (see Self Tests)
        - Firmware update in progress
        - PCBA or Sub-assembly test running

Analogue Front End (AFE)

Overview

The AFE is a physical modular component dedicated for measuring DC voltage and current as accurately as possible. The measurements are then passed to the Charge Session State Machine to manage the high level tasks related to charging.

The AFE is responsible for:

1. performing the Voltage and Current measurements of a charging session so power delivery can be accurately calculated
2. performing regular drift calibration of the ADC
3. performing regular temperature measurements of the ADC
4. automatically handling of ADC gain adjustments for different measured current values
5. performing 'dithering' on the request for measured values to provide a level of immunisation against harmonics
6. converting each measured sample to an actual nV and nA, TempC with a monotonic timestamp
7. providing an API allowing retrieval of measured samples
8. closed analogue loop control of the heating element to maintain 70 degrees Celsius

The Charge Session State Machine is responsible for:

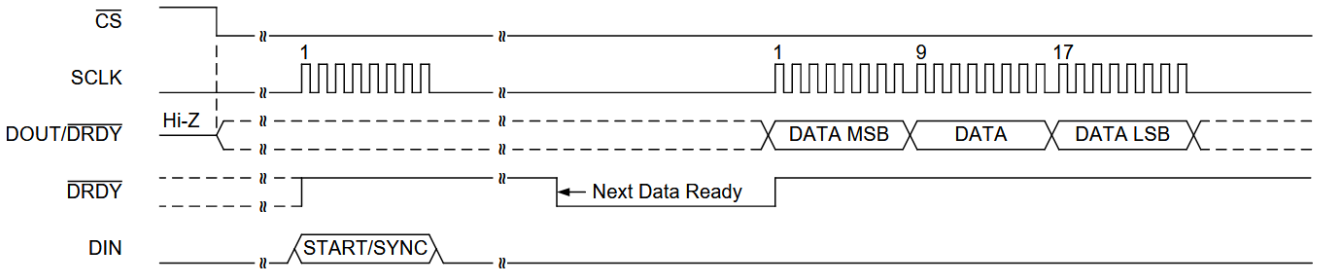
1. calculating kWh from the voltage and current measurements supplied by the AFE
2. accumulating the kWh value for a charge session
3. accumulating the kWh for the life of the meter
4. attribute kWh values to specific tariffs
5. pairing charge session id from the charger with the charge session
6. notify firmware module to sending completed charge session details to the CPO
7. notify firmware module to store the charge session details to persistent storage
8. determining when a charge session starts and ends

AFE Sequencing

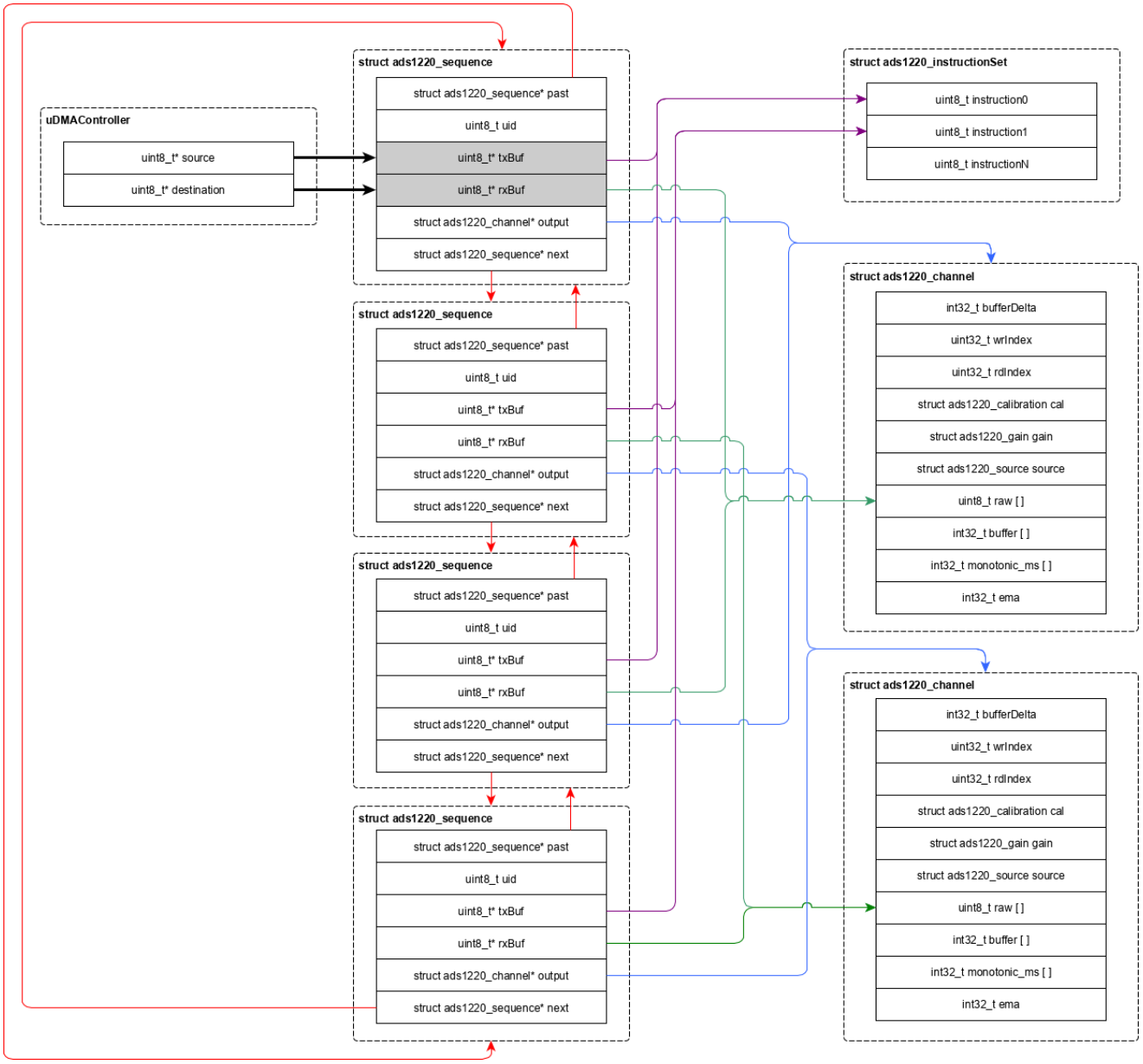
The DC Meter AFE utilises an external 4 channel 24-bit delta sigma ADC (ADS1220).

http://www.ti.com/product/ADS1220

The ADS1220 is configured to operate in continuous single-shot mode. As soon as the **Data-Ready** line of the ADS1220 is low, a ADS1220 instruction set is written to the bus. The data returned on the bus will not correspond to the present ADS1220 instruction. The data returned will be from the previous ADS1220 instruction.



To streamline communication to the ADS1220, the microcontroller's uDMA controller is used. At runtime a dynamically linked list of ADS1220 sequence is created. Every time data is available, the uDMA controller traverses and executes to the next ADS1220 instruction in the list.


uDMAController
uint8_t* source
uint8_t* destination
struct ads1220_sequence
struct ads1220_sequence* past
uint8_t uid
uint8_t* txBuf
uint8_t* rxBuf
struct ads1220_channel* output
struct ads1220_sequence* next
struct ads1220_sequence
struct ads1220_sequence* past
uint8_t uid
uint8_t* txBuf
uint8_t* rxBuf
struct ads1220_channel* output
struct ads1220_sequence* next
struct ads1220_sequence
struct ads1220_sequence* past
uint8_t uid
uint8_t* txBuf
uint8_t* rxBuf
struct ads1220_channel* output
struct ads1220_sequence* next
struct ads1220_sequence
struct ads1220_sequence* past
uint8_t uid
uint8_t* txBuf
uint8_t* rxBuf
struct ads1220_channel* output
struct ads1220_sequence* next
struct ads1220_instructionSet
uint8_t instruction0
uint8_t instruction1
uint8_t instructionN
struct ads1220_channel
int32_t bufferDelta
uint32_t wrIndex
uint32_t rdIndex
struct ads1220_calibration cal
struct ads1220_gain gain
struct ads1220_source source
uint8_t raw [ ]
int32_t buffer [ ]
int32_t monotonic_ms [ ]
int32_t ema
struct ads1220_channel
int32_t bufferDelta
uint32_t wrIndex
uint32_t rdIndex
struct ads1220_calibration cal
struct ads1220_gain gain
struct ads1220_source source
uint8_t raw [ ]
int32_t buffer [ ]
int32_t monotonic_ms [ ]
int32_t ema

LMT01 Pulse Count Temperature Sensor

Datasheet

http://www.ti.com/lit/ds/symlink/lmt01.pdf

Overview

The LMT01 device is a high-accuracy, 2-pin temperature sensor with an easy-to-use pulse count current loop interface, which makes it suitable for on-board and off-board applications in automotive, industrial, and consumer markets. The LMT01 digital pulse count output and high accuracy over a wide temperature range allow pairing with any MCU without concern for integrated ADC quality or availability, while minimising software overhead.

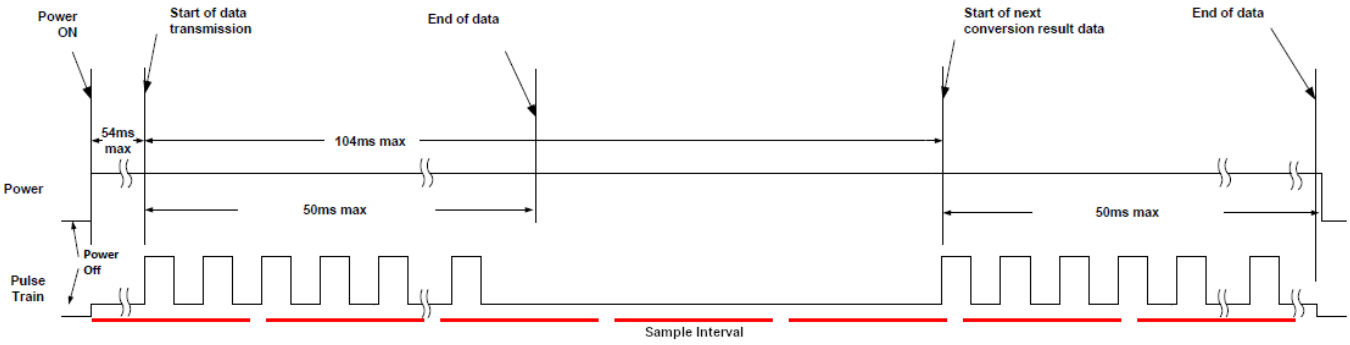
TI's LMT01 device achieves a maximum ±0.5°C accuracy (using non-linear transfer function) with very fine resolution (0.0625°C) over a temperature range of –20°C to 90°C without system calibration or hardware and software compensation. Overall valid measurement range is -50/150degC.

Firmware Considerations

The LMT01 is used for measuring the shunt temperature of the DC Meter. This is considered a safety critical measurement and therefore it's performance and reliability should be ensured.

The main difficulty arises from the fact that the LMT01 device operates asynchronous to the microcontroller.

Since the sensor operates asynchronously, pulse counts are sampled over fixed intervals. The interval time guarantees that one interval will not detect any pulses, signifying the end of a result transmission.



Hardware Considerations

- The LMT01 sensor pulse output to the microcontroller is inverted due to the transistor arrangement that the sensor drives.
  - Therefore the output is inverted relative to the figure shown above.
- The ISO7763 digital isolator holds its output high when the input side is not powered, or the input pin is floating.
- The LMT01 is continuously powered by the AFE supply and therefore its output cannot be easily synchronised to the microcontroller.

Metrological Accuracy Compensation

The DC Meter has a number of mechanisms to improve the accuracy of the measurements. This document describes the overall measurement method and major compensation elements. There is no long term compensation mechanism.

Hardware

The electronic hardware design has been designed to be robust and reliable. One major element implemented in hardware to improve accuracy is the use of an ovenised circuit section which maintains 70°C to minimise thermal drift.

Firmware

Note: Any value with a superscript X (e.g. A^X) is changed dependant on the detected charge cable. Any value with a superscript Y (e.g. A^Y) is changed dependant on the current gain selected.

Energy Calculation Overview

Energy is calculated during charging as a sum using the corrected voltage and current at a fixed interval.

$$E_{TOT} = \sum E_{INT}$$

$$E_{INT} = V_C^X * I_C * t_{INT}$$

Where:

E_{TOT} is total energy during a charge (kWh).

E_{INT} is the energy in the calculated interval (kWh).

V_C^X is the compensated voltage measurement (V).

I_C is the compensated current measurement (A).

t_{INT} is the fixed time interval between calculations (seconds).

Compensated Voltage

Voltage is compensated from the raw samples as follows.

$$V_C^X = [(V_F^X - O_F) * G_V - V_{OFF}^X] * C_V^X - I_C * R_{CABLE}^X$$

Where:

V_F^X is the filtered reading of voltage (ADC Counts).

O_F is the filtered reading of offset of the ADC with 1x gain (ADC Counts)

G_V is the fixed gain to convert to voltage (V/ADC Counts).

V_{OFF}^X is the calibrated offset voltage (V).

C_V^X is the calibrated scale for voltage (V/V).

R_{CABLE}^X is the loaded parameter for resistance of the detected cable (Ohms). Default value is 0.0.

The cable voltage drop based on current will be done within the firmware module doing the energy calculation.

Compensated Current

Current is compensated from the raw samples as follows.

$$I_C = [(I_F^Y - O_F^Y) * G_I^Y - I_{OFF}^Y] * C_I^Y$$

Where:

I_F^Y is the filtered reading of current (ADC Counts).

O_F^Y is the filtered reading of offset of the ADC (ADC Counts).

G_I^Y is the fixed overall gain to convert to current (A/ADC Counts).

I_{OFF}^Y is the calibrated offset current (A).

C_I^Y is the calibrated scale for current (A/A).

Current Sampling Gain

Current is sampled with a different gain depending on the last readings. When the current is rising above 150A the ADC gain is switched to low gain (x64). When the current is falling below 100A the ADC gain is switched to high gain (x128). This allows improved resolution at the lower currents.

Offset Sampling

The offset is obtained by changing the ADC multiplexer to short both positive and negative inputs pins and reading the result. The offset is read with the three different gain levels used, x1, x64, x128 and stored in different registers.

Optical Test Output

To act as an indicator for energy measured, an infrared LED is used as an optical output for certification testing purposes.

This output has two main functions:

- To indicate total measured energy through total number of pulses
- To indicate rate of energy delivery via time between pulses (to be compatible with existing industry standard test equipment)

As this output is used to validate the metrological accuracy, timing inconsistencies cannot be added as a result of execution delays.

Parameters

Constant: 50 Wh/impulse; 20 impulses/ kWh

Delay: 245 milliseconds

Pulse Width: Individual impulse width is 250 microseconds

Operating Points

Maximum Expected Power

At 720 kW, above the maximum power output 350kW of the charger, this will result in 4 pulses per second.

Minimum Expected Power

At 5A; 50V; 250W, the minimum accuracy operation conditions of the DC Meter, this will result in a pulse every 720 seconds.

Implementation

Energy calculations are carried out during a session by the CSSM. Nominally this is done every 50ms, with some variance. The energy calculations are performed based on the AFE samples acquired since the last execution period, the samples are taken nominally at 40ms, with some variance.

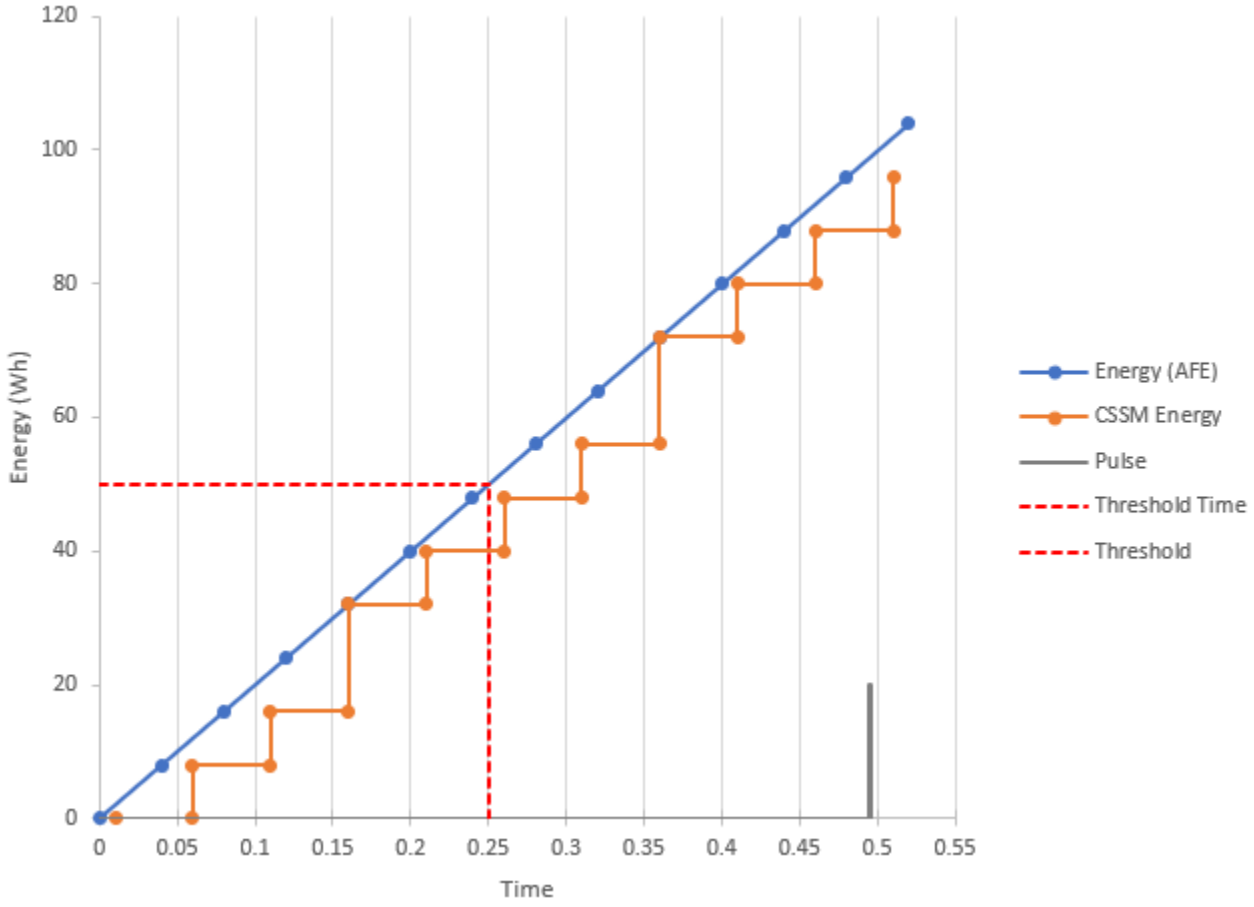
The implementation takes the timing variances into account and produces evenly spaced pulses when at a fixed power level.

Each AFE sample consists of a voltage and current measurement pair that are time-stamped when received by the processor. During the CSSM execution, if the accumulating energy since the last pulse goes above the constant threshold the next pulse is calculated. The time when that threshold was met is calculated by linearly interpolating the energy over the last set of sample measurements and timestamps. This calculation may occur up to ~85ms after the threshold was actually met. A pulse is then scheduled for exactly 245ms after the calculated time.

Peripherals

Three peripherals are used to implement this solution.

- 32 bit timer - used to give accurate timestamps to the measurement interval (with microsecond resolution)
- 32 bit timer - used to precisely schedule the pulse start time
- 16 bit PWM - used to provide a single shot pulse, to drive the hardware pin controlling the infrared LED output


120
100
80
60
40
20
0
Energy (Wh)
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
0.55
Time
Energy (AFE)
CSSM Energy
Pulse
Threshold Time
Threshold

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Supporting the SAFE Initiative

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- Figure – Diagram showing information flow between objects for a user receiving a session's bill.

SAFE Initiative Introduction

S.A.F.E stands for Software Alliance For E-mobility.

Transparenzsoftware has several supported formats for transmitting metering information:

1. Open Charge Metering Format (OCMF)
2. ASFEN
3. EDL_40_P
4. EDL_40_SIG

Tritium's DC Meter supports only OCMF and can be verified using the Transparenzsoftware.

References

ID	Title	Source	Notes
SSI1	Specification Fileformat Transparency-Software V2 KLMA 2018-07-10.pdf	Z:\Tritium\TRI110 - DC Meter\00 - Documents\TRI110.013 SAFE initiative\HTB file format info sent by HTB CEO 180725\	Format of information including charge session details to be sent from the DC Metering system to has.to.be transparency software.
SSI2	OCMF specification	http://hers.abl.de/SAFE/Datenformat_OCMF/Datenformat_OCMF_v1.0.pdf Z:\Tritium\TRI110 - DC Meter\00 - Documents\TRI110.013 SAFE initiative\supported formats\Datenformat_OCMF_v1.0.pdf	There is a google translate version at the location specified in "Source": "Datenformat_OCMF_v1.0-Google Translate.htm"

Fundamentals

1. Contents of the metering payload contains a cryptographic signature from the meter to ensure authenticity of the information, so data can be in easy to read plain text.
2. The user can search a CPO for their charge session to retrieve a previous metering payload.
3. The user can obtain the meter/charger public key from a trusted source (government server or TSO display).
4. The charge session details are retrieved from the portal (CPO, Tritium, has.to.be/Transparency, government agency, etc) that recorded it.
5. The user can verify the data with another application (such as a combination of grep and OpenSSL command line).

Solution

In order to be as interoperable as possible, the DC Meter will transmit measured data in the format specified in SSI2. This provides the most flexibility by supporting the SAFE initiative and supporting a mechanism that doesn't require the has.to.be software.

1. Uses the DC Meter as a storage device.
2. Uses the signing of information for authenticity.
3. Uses the retrieval of the meter's public key from a trusted authority to validate signed data.
4. The DC Meter will store the information for 3 months.
5. The Provider will supply the information when requested.
6. The "User" can validate using other independent 3rd party signature validation tools such as OpenSSL via command line arguments.

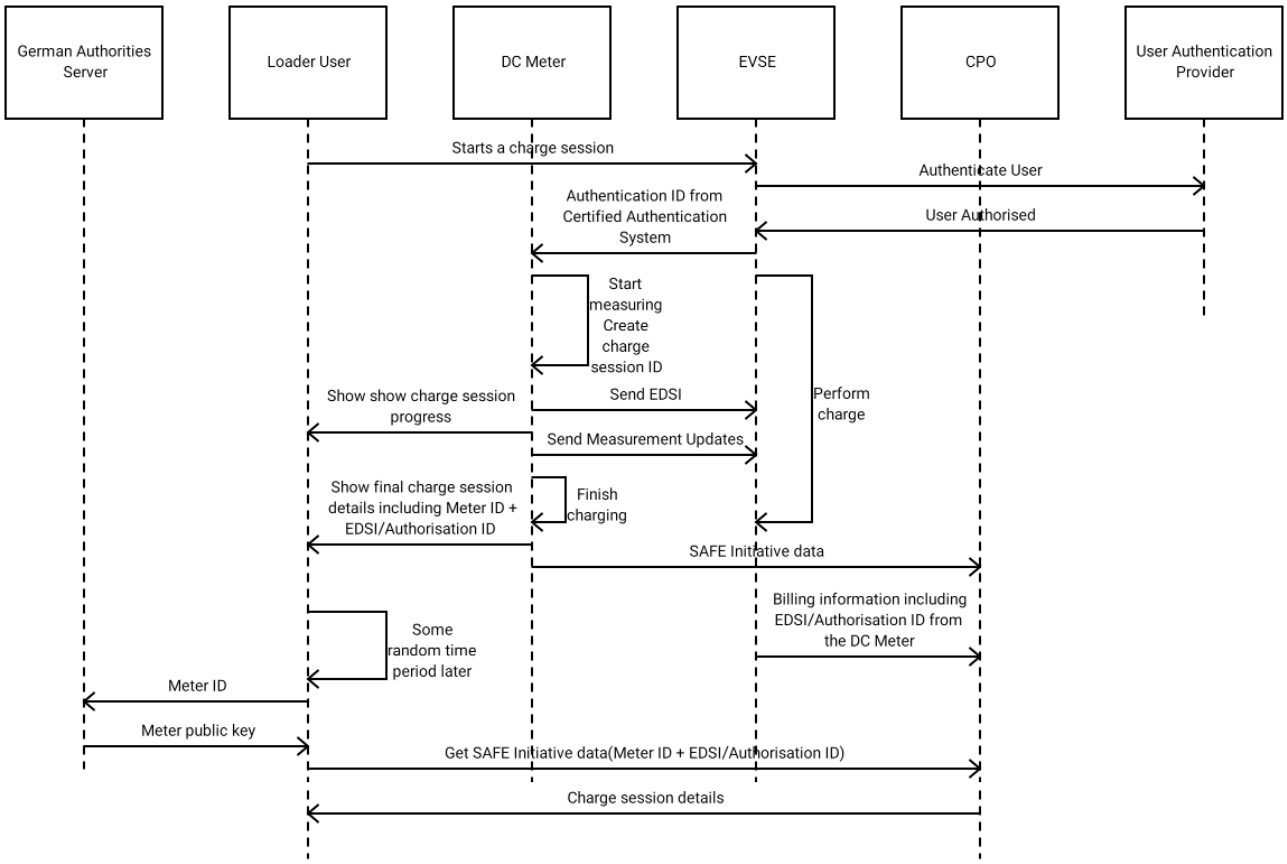


Figure – Sequence diagram of order of events to receive and verify a bill.

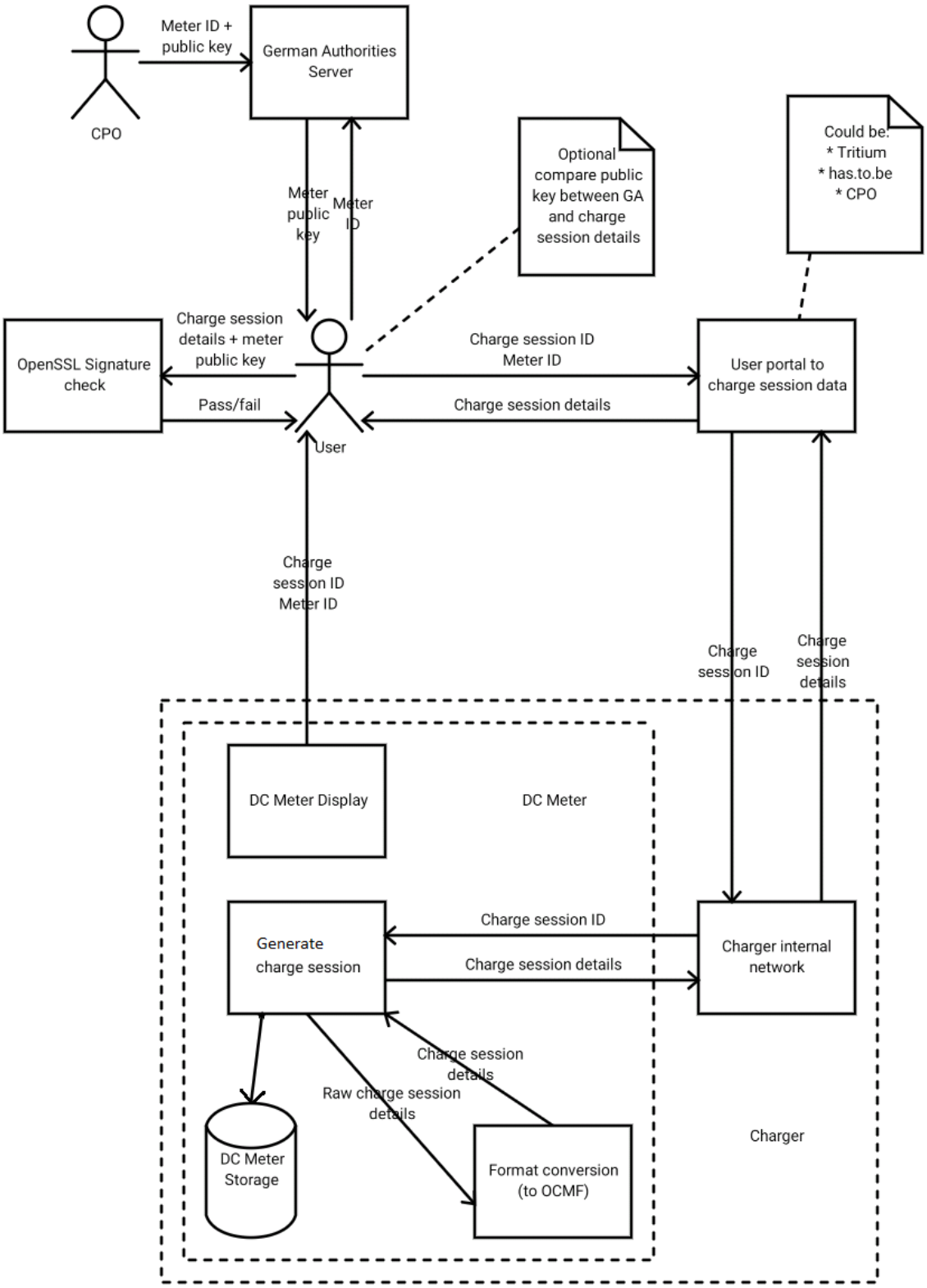


Figure – Diagram showing information flow between objects for a user receiving a session's bill.

Example OCMF message with added comments

The DC Meter complies with the OCMF v1.0 (Open Charge Metering Format) which can be obtained here:

http://hers.abl.de/SAFE/Datenformat_OCMF/Datenformat_OCMF_v1.0.pdf

To support the flexibility of external billing arrangements, reading entries are initally created every 15 minutes of charging with energy reading value. The end of transaction reading entry will contain the total session energy usage.

Due to memory constraints the 15 minute logging interval will continue for 6 hours, then reduce in rate to one reading entry per hour. After 48 hours of charging, the remainder of the charge session will be logged to the final reading entry, and no further reading entries will be created.

Tritium's DC Meter makes use of two optional manufacturer extension fields:

- VS - in user assignment portion of main data - contains the 'Session Signature', a globally unique identifier for the charge session, generated by the DC Meter.
- UT - contains the name of the tariff active in this charge (max 16 characters).

```
OCMF | {
        "FV": "1.0",                                                  // Protocol version number
        "GI": "Tritium Pty Ltd",                           // Gateway Identification: Manufacturer - same as MV
        "GS": "DCM202000001",                                  // Gateway serial - same as MS
        "PG": "T0000000010",                                  // Session ID (2^32) - Energy Delivery Session ID
        "MV": "Tritium Pty Ltd",                           // Meter vendor
        "MS": "DCM202000001",                                  // Meter serial ID (mandatory)
        "MF": "1.011",                                                 // Meter firmware
        "IS": "false",                                                 // Identification status of user
        "IL": "HEARSAY",                                            // Identification level
        "IF": ["RFID_NONE", "OCPP_NONE"],        // Identification flags
        "IT": "LOCAL",                                                 // Identification Type
        "ID": "1F2D3A4F5506C7",                                  // Identification Data - Charger Session ID
        "VS": "BAF9YAATLR68",                                  // Tritium special: Space for charge session
signature generated by DC Meter

        "RD" : [
                {"TM":"2021-09-15T00:50:57,000+0200 S","TX":"B","RV":"0.000","RU":"kWh","ST":"G","RI":"1-b:1.8.
e","UT":"Default Tariff"},
                {"TM":"2021-09-15T00:51:41,000+0200 S","TX":"C","RV":"3.959","RU":"kWh","ST":"G","RI":"1-b:1.8.
e","UT":"Default Tariff"},
                {"TM":"2021-09-15T00:51:41,000+0200 S","TX":"E","RV":"3.959","RU":"kWh","ST":"G","RI":"1-b:1.8.
e","UT":"Default Tariff"}
                ]
        } | {
                "SD":"
3044022079C63D04EA437654780D417063C6AFB4D6B61AEF50EEB3396EA5BA6E6DA792DE022008572037339682021 3B500ED637889F40220
5CE1C586B69AA95E8ED3F24537A5"
        }
```

Example SAFE packet

```
<?xml version="1.0" encoding="UTF-8"?>
<values>
        <value>
                <signedData format="OCMF" encoding="plain">OCMF|{"FV":"1.0","GI":"Tritium Pty Ltd","GS":"123","
PG":"T0000000083","MV":"Tritium Pty Ltd","MS":"123","MF":"1.009","IS":"false","IL":"HEARSAY","IF":["RFID_NONE",
"OCPP_NONE"],"IT":"LOCAL","ID":"","VS":"E9YJGGBHZ1ME","RD":[{"TM":"2021-09-15T07:24:34,000+0200 S","TX":"B","
RV":"0.000","RU":"kWh","ST":"G","RI":"1-b:1.8.e","UT":" "},{"TM":"2021-09-15T07:25:18,000+0200 S","TX":"C","
RV":"3.960","RU":"kWh","ST":"G","RI":"1-b:1.8.e","UT":" "},{"TM":"2021-09-15T07:25:18,000+0200 S","TX":"E","
RV":"3.960","RU":"kWh","ST":"G","RI":"1-b:1.8.e","UT":" "}]}|{"SD":"
304402201B15EE6420899D06A3C70880DDCB580BDB2E0F06EADF25719D7515C61D911785022004D42601796C2ADC9853D3AF82F3C052FCBE
B4F0BC20485D29FD3F2C73D3E632"}</signedData>
                <publicKey encoding="plain"
>3059301306072A8648CE3D020106082A8648CE3D030107034200044 7AFFAB654C32B8A802689456050E4FF36D5CE23960F7E20A0825D7DB
7F65B1EA09086ADB91047B891420B2D301E7441C141CFAEECE2A3E09DDBFF49F883AC37</publicKey>
        </value>
</values>
```

Energy Delivery Session Identifier (EDSI)

The EDSI is used to:

1. Identify a charge session to the customer.
2. Avoid 'authentication of the user' requirements on the DC Meter.

Note 1 of AUTHZ002 recommends "a randomly generated transaction ID". However, it doesn't recommend the length of the ID.

To comply with the OCMF's pagination field, the identifier should be non-random, incrementing after every session. 32 bit length ID.

Tritium will ensure that each delivery of energy carries an identifier for the legally relevant receipt of energy delivery through the creation of a "Energy Delivery Session Identifier" (EDSI).

1. The EDSI shall be created on the DC Meter in response to the start of an energy delivery session.
2. The EDSI shall be an input to a hashing function, the result of which will be shown on the legally relevant portion of the display.
3. The EDSI shall be stored with the legally relevant measured values on the DC Meter.
4. The EDSI shall be included in the legally relevant "Digital Receipt" that is transmitted to the Charge Point Operator via OCPP and forms part of the cryptographically verifiable legally relevant data receipt (SAFE Initiative).
5. When an authentication system is in use, which is covered by one or more standards and certifications suitable for use with EVSE, and a certified identity system can provide a verifiable ID for the authenticated Loader User, that ID can be included in the DC Meter's Legally relevant display, storage, and digital receipt.
6. The EDSI and the authentication ID (when available) shall be stored on the DC Meter together with the legally relevant measured energy delivered.

Session Signature

The session signature is a 12-character, 5-bit encoded string that allows a charge session to be uniquely identified.

The character set for the string is as follows:

```
0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F,G,H,J,K,L,M,N,P,R,S,T,U,V,X,Y,Z
```

The Session Signature is displayed on the legally relevant display, and also appears in the "VS" field of the SAFE message/OCMF receipt. The session signature can hence be used by an end user to search the CPO's receipt database with a unique string, and retrieve the OCMF receipt that matches their charging session.

Networking Interfaces

There are several interfaces with the DC Meter to a number of different actors/purposes. There are differing security requirements for the different interfaces. This document gives an overview of those different interfaces, their purpose and their security.

Interface	Purpose	Transport Layer	Security Mechanism	Trust Root/Private Key Management	Notes
Trusted Technician Interface	• Telemetry • Calibration • Commissioning • Configuration • Testing • Servicing	TLS 1.2	TLS connection. x.509 certificates with adjustable permissions	TTI Certificate Authority - Tritium controlled	See Trusted Technician Interface (TTI) Certificate for read-only can be made readily available. Server (DCM) authentication is not critical; only client (PC) authentication for inputs.
Trusted Screen Overlay*	Display of information to user	TCP echoed onto SPI	Information exchanged is signed by private keys. Keys are signed by Trust Anchor to authenticate.	TSO and DCM generate unique keys onboard	See Pairing TSO and DCM
Trust Anchor	Authentication of Tritium product	TCP	Physical location security, not networked (air gapped)	Trust Anchor physically secured - Tritium controlled	See Trust Anchor Communication
SAFE*	Signature of billing receipt	(via Charger Communications)	Information is signed by DCM private key.	DCM generates unique key onboard	See Supporting the SAFE Initiative
CPO Communication	• (Optional) Change tariff	(via Charger Communications)	Information is signed by CPO's private key	CPO public key written to DCM EEPROM during configuration	See Charger to DC Meter Communications
Charger Communications (a.k.a MQTT)	• Communication with charger • Forwards communication with CPO	MQTT via TLS 1.2	Security mechanism is not critical as legally-relevant information is signed	Charger has certificate generated by MQTT Certificate Authority - Tritium controlled	See Charger to DC Meter Communications
Debug print out	Debug messages out	UDP	N/A	N/A	Not legally-relevant data. No input
Time	Time synchronisation	UDP	N/A	N/A	NTP
Update Server	Hosting firmware image for bootloading	TLS 1.2	TLS connection.	Image Hosting Certificate Authority - Tritium controlled	See Application Package Manager
Update Image	Signing of firmware image	(via Update Server)	Binary image is signed by certificate authority's private key.	Image Signing Certificate Authority - Tritium controlled	See Application Package Manager

*DC Meter uses same private/public key for TSO and SAFE signing.

Certificate Authority

All Tritium controlled certificate authorities are kept on a secure GIT repository (separate from main source repository), only accessible by minimal personnel. The root keys are kept there and only leaf certificates are issued to the appropriate receiver e.g. TTI production access certificate generated and then given to secure room production personnel.

Charger to DC Meter Communications

The charger and DC meter communicate via a MQTT broker.

MQTT Topics

List of what topics the DC meter system should subscribe and publish to, over MQTT, as a tree structure is listed below:

Publish

- DCMS
  - Telemetry
    - Status
      - Heartbeat
      - State
      - Error Info
      - Secondary Info
      - OoS
      - Stop Charge
    - Voltage
    - Current
    - Power
    - Temperature
      - Board
      - Shunt
      - AFE
  - Info
    - TME (Total Metered Energy)
    - Firmware
    - Public Key
    - Serial ID
  - Session
    - EDSI*
    - Energy
    - Duration
  - CSMS
    - SAFE** ('JSON_base64': SessionID + EDSI + SAFE)
    - CPO Public Key
    - Active Tariff
    - Queued Tariff

Subscribe

- DCMS
  - Charger
    - Session*
      - ID (hexadecimal string, fixed-length 16 bytes, example: "74EF112C541A8741")
      - Signature
  - CSMS
    - Tarrifs**

*Session ID and EDSI are paired so Session ID should be pushed by MQTT broker when EDSI is received.

**File containing data and signature

Cryptographic Security

The mbedTLS library includes an entropy accumulator module that handles many requirements for random bit generation. Multiple inputs are SHA256 together. Sources of entropy can be seen in Entropy source.

All critical transmitted data is signed by the meter with a ECDSA with 256 bit key length.

All keys and signatures will be derived from secp256r1 curve (256-bit).

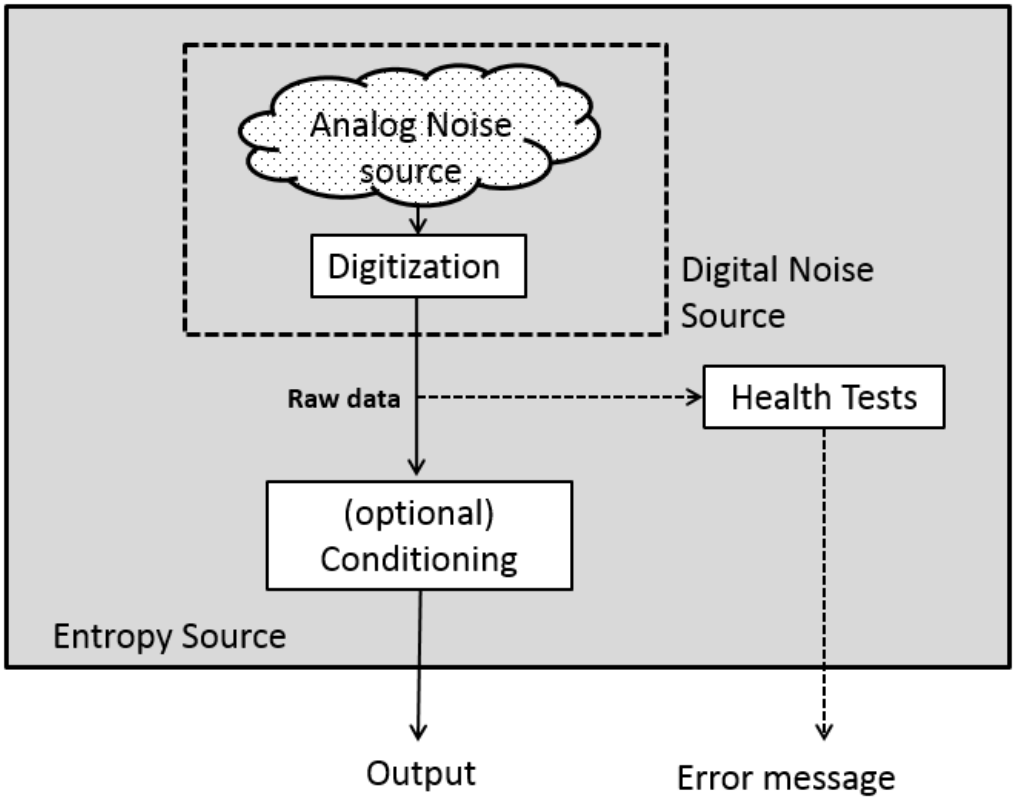
All hashes calculated on the meter are SHA256.

Private keys for TSO and SAFE signing are never displayed, exported, or read from the DC Meter at any time. A new private key may be generated, but never seen. A new private key is only generated in production prior to being signed by the Trust Anchor.

Entropy source

Model

Entropy source used in DCM project was created and evaluated according to recommendations in NIST SP 800-90B with below model:



Analogue noise source is composed of some set of the below:

- Internal processor temperature measurement (TSO)
- Voltage on a floating pin (TSO)
- Zener diode reverse voltage (DCM)
- System supply voltage (DCM)
- Board temperature measurement (DCM)

Analogue voltages from above sources are read by ADC. Only two least significant bits are used. Four independent inputs are sampled and ORed to create 8-bit sample of entropy which is transferred to entropy pool. No conditioning is performed.

Health Tests

Entropy source implements below health tests:

- Repetition Count Test
- Adaptive Proportion Test

These tests are run continuously during entropy source operation analysing every generated entropy sample. Startup self-test runs continuous tests on 1024 consecutive samples before allowing any reads from the entropy source (1024 entropy samples are discarded).

Cutoff values for health tests was calculated assuming the entropy would be at most 4 bits per 1 byte and false positive probability of $= 2^{20}$. For lower entropy estimates those values are more conservative and allow for faster detection of any problems with entropy source.

Entropy Estimate TSO

SP800-90B_EntropyAssessment test suit was used to perform estimation of the entropy source.

Initial entropy estimate results:

Tested data: TSO_ENTROPY 

Loaded 125000 samples of 256 distinct 8-bit-wide symbols
Number of Binary Symbols: 1000000

***** Warning: data contains less than 1000000 samples *****

Running non-IID tests...

Running Most Common Value Estimate...
Bitstring MCV Estimate: mode = 500588, p-hat = 0.50058800000000003, p_u = 0.501875914405154
Most Common Value Estimate (bit string) = 0.994597 / 1 bit(s)
Literal MCV Estimate: mode = 3008, p-hat = 0.024063999999999999, p_u = 0.025180499129802431
Most Common Value Estimate = 5.311549 / 8 bit(s)

Running Entropic Statistic Estimates (bit strings only)...
Bitstring Collision Estimate: X-bar = 2.4761705488932746, sigma-hat = 0.49943245278882376, p = 0.61369651531708058
Collision Test Estimate (bit string) = 0.704403 / 1 bit(s)
Bitstring Markov Estimate: P_0 = 0.49941200000000002, P_1 = 0.50058800000000003, P_0,0 = 0.51553930530164538, P_0,1 = 0.48446069469835462, P_1,0 = 0.48332161378219213, P_1,1 = 0.51667838621780793, p_max = 1.89870646284791 54e-37
Markov Test Estimate (bit string) = 0.953018 / 1 bit(s)
Bitstring Compression Estimate: X-bar = 5.095580078444673, sigma-hat = 1.0453171279805322, p = 0.095732601400006612
Compression Test Estimate (bit string) = 0.564141 / 1 bit(s)

Running Tuple Estimates...
Bitstring t-Tuple Estimate: t = 26, p-hat_max = 0.67867812579184883, p_u = 0.67988099766088372
Bitstring LRS Estimate: u = 27, v = 50, p-hat = 0.59412595716394034, p_u = 0.5953908455323621
T-Tuple Test Estimate (bit string) = 0.556646 / 1 bit(s)
Literal t-Tuple Estimate: t = 3, p-hat_max = 0.076970433136702238, p_u = 0.078912363185281567
Literal LRS Estimate: u = 4, v = 6, p-hat = 0.025198790585647265, p_u = 0.026340647506713166
T-Tuple Test Estimate = 3.663605 / 8 bit(s)
LRS Test Estimate (bit string) = 0.748091 / 1 bit(s)
LRS Test Estimate = 5.246565 / 8 bit(s)

Running Predictor Estimates...
Bitstring MultiMCW Prediction Estimate: N = 999937, Pglobal' = 0.5099518083247998 (C = 508632) Plocal can't affect result (r = 23)
Multi Most Common in Window (MultiMCW) Prediction Test Estimate (bit string) = 0.971567 / 1 bit(s)
Literal MultiMCW Prediction Estimate: N = 124937, Pglobal' = 0.026690541968189662 (C = 3191) Plocal = 0.066460295690674007 (r = 6)
Multi Most Common in Window (MultiMCW) Prediction Test Estimate = 3.911363 / 8 bit(s)
Bitstring Lag Prediction Estimate: N = 999999, Pglobal' = 0.62768868032127301 (C = 626442) Plocal can't affect result (r = 36)
Lag Prediction Test Estimate (bit string) = 0.671879 / 1 bit(s)
Literal Lag Prediction Estimate: N = 124999, Pglobal' = 0.024517925509921403 (C = 2927) Plocal = 0.038411357486915501 (r = 5)
Lag Prediction Test Estimate = 4.702323 / 8 bit(s)
Bitstring MultiMMC Prediction Estimate: N = 999998, Pglobal' = 0.63590459527860699 (C = 634663) Plocal = 0.67503520854902899 (r = 44)
Multi Markov Model with Counting (MultiMMC) Prediction Test Estimate (bit string) = 0.566965 / 1 bit(s)
Literal MultiMMC Prediction Estimate: N = 124998, Pglobal' = 0.046616736890156975 (C = 5638) Plocal = 0.066454823913203537 (r = 6)
Multi Markov Model with Counting (MultiMMC) Prediction Test Estimate = 3.911482 / 8 bit(s)
Bitstring LZ78Y Prediction Estimate: N = 999983, Pglobal' = 0.51735603414128084 (C = 516060) Plocal can't affect result (r = 22)
LZ78Y Prediction Test Estimate (bit string) = 0.950771 / 1 bit(s)
Literal LZ78Y Prediction Estimate: N = 124983, Pglobal' = 0.04666297151055105 (C = 5643) Plocal = 0.066456169129722908 (r = 6)
LZ78Y Prediction Test Estimate = 3.911453 / 8 bit(s)

H_original: 3.663605
H_bitstring: 0.556646

min(H_original, 8 X H_bitstring): 3.663605

Restart test result:



Tested data: TSO_ENTROPY_RESTART

```
Loaded 125000 samples of 256 distinct 8-bit-wide symbols
Number of Binary Symbols: 1000000

*** Warning: data contains less than 1000000 samples ***


Running non-IID tests...

Running Most Common Value Estimate...
Bitstring MCV Estimate: mode = 504230, p-hat = 0.50422999999999996, p_u = 0.50551786920582831
Most Common Value Estimate (bit string) = 0.984166 / 1 bit(s)
Literal MCV Estimate: mode = 2932, p-hat = 0.023456000000000001, p_u = 0.024558647493909624
Most Common Value Estimate = 5.347625 / 8 bit(s)

Running Entropic Statistic Estimates (bit strings only)...
Bitstring Collision Estimate: X-bar = 2.467260619727762, sigma-hat = 0.4989275971047431, p = 0.63182950613234823
Collision Test Estimate (bit string) = 0.662393 / 1 bit(s)
Bitstring Markov Estimate: P_0 = 0.49576999999999999, P_1 = 0.50422999999999996, P_0,0 = 0.51573212524381318, P_0,1 =
0.48426787475618682, P_1,0 = 0.47614184003331811, P_1,1 = 0.52385815996668184, p_max = 1.10
34947029349809e-36
Markov Test Estimate (bit string) = 0.933182 / 1 bit(s)
Bitstring Compression Estimate: X-bar = 5.0538494665350857, sigma-hat = 1.0781510509989141, p = 0.11185973493558654
Compression Test Estimate (bit string) = 0.526706 / 1 bit(s)

Running Tuple Estimates...
Bitstring t-Tuple Estimate: t = 30, p-hat_max = 0.7135138695884492, p_u = 0.71467845221570114
Bitstring LRS Estimate: u = 31, v = 51, p-hat = 0.59643604941784989, p_u = 0.59769978269478419
T-Tuple Test Estimate (bit string) = 0.484634 / 1 bit(s)
Literal t-Tuple Estimate: t = 3, p-hat_max = 0.070607343281027177, p_u = 0.072473672966833283
Literal LRS Estimate: u = 4, v = 6, p-hat = 0.022449570231439998, p_u = 0.023528858388111841
T-Tuple Test Estimate = 3.786399 / 8 bit(s)
LRS Test Estimate (bit string) = 0.742507 / 1 bit(s)
LRS Test Estimate = 5.409425 / 8 bit(s)

Running Predictor Estimates...
Bitstring MultiMCW Prediction Estimate: N = 999937, Pglobal' = 0.52001373198811962 (C = 518694) Plocal can't affect result (r = 18)
Multi Most Common in Window (MultiMCW) Prediction Test Estimate (bit string) = 0.943378 / 1 bit(s)
Literal MultiMCW Prediction Estimate: N = 124937, Pglobal' = 0.024415453299536687 (C = 2913) Plocal = 0.066460295709743725 (r = 6)
Multi Most Common in Window (MultiMCW) Prediction Test Estimate = 3.911363 / 8 bit(s)
Bitstring Lag Prediction Estimate: N = 999999, Pglobal' = 0.61445308847673119 (C = 613198) Plocal = 0.64771283626842591 (r = 40)
Lag Prediction Test Estimate (bit string) = 0.626574 / 1 bit(s)
Literal Lag Prediction Estimate: N = 124999, Pglobal' = 0.023781249039914971 (C = 2837) Plocal = 0.038411357486628259 (r = 5)
Lag Prediction Test Estimate = 4.702323 / 8 bit(s)
Bitstring MultiMMC Prediction Estimate: N = 999998, Pglobal' = 0.62500508953117651 (C = 623756) Plocal = 0.68724003211083529 (r = 46)
Multi Markov Model with Counting (MultiMMC) Prediction Test Estimate (bit string) = 0.541114 / 1 bit(s)
Literal MultiMMC Prediction Estimate: N = 124998, Pglobal' = 0.041086572026887987 (C = 4958) Plocal = 0.066454823919652975 (r = 6)
Multi Markov Model with Counting (MultiMMC) Prediction Test Estimate = 3.911482 / 8 bit(s)
Bitstring LZ78Y Prediction Estimate: N = 999983, Pglobal' = 0.52109775168740846 (C = 519802) Plocal can't affect result (r = 18)
LZ78Y Prediction Test Estimate (bit string) = 0.940374 / 1 bit(s)
Literal LZ78Y Prediction Estimate: N = 124983, Pglobal' = 0.041058945120268975 (C = 4954) Plocal = 0.066456169140905311 (r = 6)
LZ78Y Prediction Test Estimate = 3.911453 / 8 bit(s)

H_original: 3.786399
H_bitstring: 0.484634

min(H_original, 8 X H_bitstring): 3.786399
```

Conclusion

Lowest entropy estimate from performed tests was 3.66 bits per 1 byte. To avoid security risks entropy was assumed to be 2 bits per 1 byte. That required the seed for Random Number Generator to be 128-bytes long to meet the requirement of security_strength = 256 (according to NIST 800-90A).

Entropy Estimate DCMeter

SP800-90B_EntropyAssessment test suit was used to perform estimation of the entropy source.

Initial entropy estimate results:

Tested data: DCM_ENTROPY



Loaded 125000 samples of 256 distinct 8-bit-wide symbols
Number of Binary Symbols: 1000000

***** Warning: data contains less than 1000000 samples *****

Running non-IID tests...

Running Most Common Value Estimate...
Bitstring MCV Estimate: mode = 512150, p-hat = 0.51214999999999999, p_u = 0.51343753498903066
Most Common Value Estimate (bit string) = 0.961739 / 1 bit(s)
Literal MCV Estimate: mode = 697, p-hat = 0.0055760000000000002, p_u = 0.0061185140517987608
Most Common Value Estimate = 7.352603 / 8 bit(s)

Running Entropic Statistic Estimates (bit strings only)...
Bitstring Collision Estimate: X-bar = 2.5014445842474027, sigma-hat = 0.499998538531707, p = 0.51721002500903523
Collision Test Estimate (bit string) = 0.951178 / 1 bit(s)
Bitstring Markov Estimate: P_0 = 0.48785000000000001, P_1 = 0.51214999999999999, P_0,0 = 0.48643640463257148, P_0,1 = 0.51356359536742846, P_1,0 = 0.48919552708293873, P_1,1 = 0.51080447291706133, p_max = 4.54
71889905616428e-38
Markov Test Estimate (bit string) = 0.969127 / 1 bit(s)
Bitstring Compression Estimate: X-bar = 5.2170587287665198, sigma-hat = 1.0146958526204284, p = 0.029834688671164455
Compression Test Estimate (bit string) = 0.844478 / 1 bit(s)

Running Tuple Estimates...
Bitstring t-Tuple Estimate: t = 16, p-hat_max = 0.52662886552610955, p_u = 0.52791495301242475
Bitstring LRS Estimate: u = 17, v = 37, p-hat = 0.50098816908640509, p_u = 0.50227608186689265
T-Tuple Test Estimate (bit string) = 0.921623 / 1 bit(s)
Literal t-Tuple Estimate: t = 1, p-hat_max = 0.0055760000000000002, p_u = 0.0061185140517987608
Literal LRS Estimate: u = 2, v = 4, p-hat = 0.0047568950570268933, p_u = 0.0052581863764200347
T-Tuple Test Estimate = 7.352603 / 8 bit(s)
LRS Test Estimate (bit string) = 0.993448 / 1 bit(s)
LRS Test Estimate = 7.571219 / 8 bit(s)

Running Predictor Estimates...
Bitstring MultiMCW Prediction Estimate: N = 999937, Pglobal' = 0.50970480363292847 (C = 508385) Plocal can't affect result (r = 19)
Multi Most Common in Window (MultiMCW) Prediction Test Estimate (bit string) = 0.972266 / 1 bit(s)
Literal MultiMCW Prediction Estimate: N = 124937, Pglobal' = 0.0052219840364005248 (C = 590) Plocal can't affect result (r = 3)
Multi Most Common in Window (MultiMCW) Prediction Test Estimate = 7.581186 / 8 bit(s)
Bitstring Lag Prediction Estimate: N = 999999, Pglobal' = 0.51598887379460168 (C = 514701) Plocal can't affect result (r = 23)
Lag Prediction Test Estimate (bit string) = 0.954588 / 1 bit(s)
Literal Lag Prediction Estimate: N = 124999, Pglobal' = 0.0055895910672354868 (C = 634) Plocal can't affect result (r = 3)
Lag Prediction Test Estimate = 7.483042 / 8 bit(s)
Bitstring MultiMMC Prediction Estimate: N = 999998, Pglobal' = 0.51340756239112006 (C = 512119) Plocal can't affect result (r = 20)
Multi Markov Model with Counting (MultiMMC) Prediction Test Estimate (bit string) = 0.961824 / 1 bit(s)
Literal MultiMMC Prediction Estimate: N = 124998, Pglobal' = 0.0050425074976730745 (C = 569) Plocal can't affect result (r = 3)
Multi Markov Model with Counting (MultiMMC) Prediction Test Estimate = 7.631643 / 8 bit(s)
Bitstring LZ78Y Prediction Estimate: N = 999983, Pglobal' = 0.51341425354430159 (C = 512118) Plocal can't affect result (r = 20)
LZ78Y Prediction Test Estimate (bit string) = 0.961805 / 1 bit(s)
Literal LZ78Y Prediction Estimate: N = 124983, Pglobal' = 0.0050515424860314824 (C = 570) Plocal can't affect result (r = 3)
LZ78Y Prediction Test Estimate = 7.629060 / 8 bit(s)

H_original: 7.352603
H_bitstring: 0.844478

min(H_original, 8 X H_bitstring): 6.755821

Restart test result:



Tested data: DCM_ENTROPY_RESTART

Loaded 125000 samples of 256 distinct 8-bit-wide symbols
Number of Binary Symbols: 1000000

***** Warning: data contains less than 1000000 samples *****

Running non-IID tests...

Running Most Common Value Estimate...
Bitstring MCV Estimate: mode = 512090, p-hat = 0.51209000000000005, p_u = 0.51337753873641978
Most Common Value Estimate (bit string) = 0.961908 / 1 bit(s)
Literal MCV Estimate: mode = 692, p-hat = 0.0055360000000000001, p_u = 0.0060765755315594508
Most Common Value Estimate = 7.362526 / 8 bit(s)

Running Entropic Statistic Estimates (bit strings only)...
Bitstring Collision Estimate: X-bar = 2.5025876782469858, sigma-hat = 0.49999392951679228, p = 0.5
Collision Test Estimate (bit string) = 1.000000 / 1 bit(s)
Bitstring Markov Estimate: P_0 = 0.48791000000000001, P_1 = 0.51208999999999993, P_0,0 = 0.48456477629070938, P_0,1 = 0.51543522370929062, P_1,0 = 0.49109822706599832, P_1,1 = 0.50890177293400174, p_max = 2.8304775140527792e-38
Markov Test Estimate (bit string) = 0.974470 / 1 bit(s)
Bitstring Compression Estimate: X-bar = 5.2119763366628451, sigma-hat = 1.0181239430763833, p = 0.034900323605197014
Compression Test Estimate (bit string) = 0.806769 / 1 bit(s)

Running Tuple Estimates...
Bitstring t-Tuple Estimate: t = 16, p-hat_max = 0.52934264357354366, p_u = 0.53062833919023844
Bitstring LRS Estimate: u = 17, v = 41, p-hat = 0.51886258802425067, p_u = 0.52014958652029042
T-Tuple Test Estimate (bit string) = 0.914226 / 1 bit(s)
Literal t-Tuple Estimate: t = 1, p-hat_max = 0.0055360000000000001, p_u = 0.0060765755315594508
Literal LRS Estimate: u = 2, v = 4, p-hat = 0.0044267896543079858, p_u = 0.0049104548693724498
T-Tuple Test Estimate = 7.362526 / 8 bit(s)
LRS Test Estimate (bit string) = 0.943002 / 1 bit(s)
LRS Test Estimate = 7.669928 / 8 bit(s)

Running Predictor Estimates...
Bitstring MultiMCW Prediction Estimate: N = 999937, Pglobal' = 0.50962980214602827 (C = 508310) Plocal can't affect result (r = 23)
Multi Most Common in Window (MultiMCW) Prediction Test Estimate (bit string) = 0.972478 / 1 bit(s)
Literal MultiMCW Prediction Estimate: N = 124937, Pglobal' = 0.0054830023635195945 (C = 621) Plocal can't affect result (r = 3)
Multi Most Common in Window (MultiMCW) Prediction Test Estimate = 7.510818 / 8 bit(s)
Bitstring Lag Prediction Estimate: N = 999999, Pglobal' = 0.5051203819376 (C = 503832) Plocal can't affect result (r = 21)
Lag Prediction Test Estimate (bit string) = 0.985301 / 1 bit(s)
Literal Lag Prediction Estimate: N = 124999, Pglobal' = 0.0047218776996898939 (C = 531) Plocal can't affect result (r = 3)
Lag Prediction Test Estimate = 7.726424 / 8 bit(s)
Bitstring MultiMMC Prediction Estimate: N = 999998, Pglobal' = 0.51314257823112464 (C = 511854) Plocal can't affect result (r = 23)
Multi Markov Model with Counting (MultiMMC) Prediction Test Estimate (bit string) = 0.962568 / 1 bit(s)
Literal MultiMMC Prediction Estimate: N = 124998, Pglobal' = 0.0051436296964943876 (C = 581) Plocal can't affect result (r = 3)
Multi Markov Model with Counting (MultiMMC) Prediction Test Estimate = 7.602998 / 8 bit(s)
Bitstring LZ78Y Prediction Estimate: N = 999983, Pglobal' = 0.51328825923540855 (C = 511992) Plocal can't affect result (r = 23)
LZ78Y Prediction Test Estimate (bit string) = 0.962159 / 1 bit(s)
Literal LZ78Y Prediction Estimate: N = 124983, Pglobal' = 0.0051610973672557353 (C = 583) Plocal can't affect result (r = 3)
LZ78Y Prediction Test Estimate = 7.598106 / 8 bit(s)

H_original: 7.362526
H_bitstring: 0.806769

min(H_original, 8 X H_bitstring): 6.454154

Conclusion

Lowest entropy estimate from performed tests was 6.45 bits per 1 byte. To avoid security risks and having consistency in the project entropy was assumed to be 2 bits per 1 byte. That required the seed for Random Number Generator to be 128-bytes long to meet the requirement of security_strength = 256 (according to NIST 800-90A).

Random numbers

Random numbers generator was tested according to NIST SP 800-22 . (Bitstream count: 100, bitstream length 100000)

Random Numbers test TSO

Tested data: TSO_RNG_10M

All statistical tests in sts-2.1.2 test suite have passed. Full report: finalAnalysisReport.txt

Random Numbers test DCM

Tested data: DCM_RNG_10M

All statistical tests in sts-2.1.2 test suite have passed. Full report: finalAnalysisReport.txt

External Non Volatile Storage

Overview

The microcontroller on the DC Meter does not have enough internal non-volatile storage to satisfy all its purposes including a long-term logbook-based auditing system. The DC Meter has been designed to have service period of 10 years. During the service period, every event and configuration change needs to be logged.

The external non volatile storage is located on the controller board of the meter, protected by tamper evident seal.

The external non volatile storage is used for a number purposes:

- Audit Event Log - saves audit events over the entire life of the meter.
  - See Audit Event Log
  - More than 600 blocks or 150 MB are reserved for the Audit Event Log
- Charge Session Log - saves information to reproduce SAFE receipt
  - See Charge Session Log
  - More than 300 blocks or 75 MB are reserved for the Charge Session Log
- Application Package Manager - saves the downloaded firmware package of the meter to allow boot loading
  - See Application Package Manager
  - 4 blocks or 1 MB are reserved for the Application Package Manager
- Run Time Session Info - saves the auxiliary information needed to be kept in storage
  - 2 blocks or 512 kB are reserved for the Run Time Session Info

Run Time Session Info

The last EDSI used and Total Metered Energy are saved to the external flash after every charge session and read at startup. This allows these variables to be added upon throughout the whole life of the meter. These are saved with a hash to verify integrity.

Device Information and Write Methodology

Flash Memory Topology

Selecting the right flash-based storage device required a trade-off between cost, capacity and robustness. A planar Single-level Cell (SLC) topology NAND with Error Correcting Codes (ECC) was selected for its durability and longevity. NAND flash stores data in cells comprised of floating-gate transistors. The amount of charge stored in each cell determines the value of the data. If the charge in the cell become insufficient to determine the state, then the information will be lost. ECC technology can correct bits and restoring depleted erroneous cell charge.

Flash Memory Cell Endurance

Cell endurance is described as the number of program/erase (P/E) cycles a flash cell can undergo before data can no longer be sufficiently retained and the cell becomes unreliable. Every time a flash cell is written (one P/E cycle) the oxide layer of the cell is slightly degraded. As this layer weakens from the stress of P/E cycles it becomes easier for stored charge to leak, and the cell's ability to retain data is reduced.

Flash Data Retention

The level of charge in each cell must be kept within certain thresholds to maintain data integrity. Unfortunately, charge leaks from flash cells over time, and if too much charge is lost then the data stored will also be lost.

During normal operation, it is the flash driver's responsibility to routinely restore lost charge. However, when the flash is not powered the state of charge will naturally degrade with time. The rate of charge loss, and sensitivity of the flash to that loss, is impacted by the flash structure, amount of flash wear (number of P/E cycles performed on the cell), and the storage temperature.

Flash Memory Device Selection

Tritium selected the Toshiba 4Gb TC58CV SLC NAND with ECC device range for the DC Meter. Toshiba was contacted and provided internal reliability data.







Flash Memory Organisation

Feature	Size (in bytes)
Memory Capacity	(4096 + 128) x 64 pages x 2048 blocks
Memory Cell Buffer	4096 + 128
Page Size	4096 + 128
Block Size	256k + 8k

Flash Memory Performance

Symbol	Parameter	Min	Typ	Max	Unit
N_{VB}	Number of Valid Blocks	2008	-	2048	Block
t_{PROG}	Average Programming Time	-	450	600	us
t_{BERASE}	Block Erasing Time	-	2	7	ms
t_{R}	Cell Array to Buffer (with ECC)	-	115	280	us
N	Number of Partial Program Cycles in the Same Page	-	-	4	time

Flash Memory Limitations

1. NAND flash page cannot be individually erased.

- Tritium's strategy is to have a fixed length, page aligned, usage of blocks. Usage of memory is then easily detected between power cycles.
- File systems typically have some kind journaling system for meta-data to track file system changes. Changes to the file system, require changes in the journal. NAND Flash isn't well suited to journaling because the block sizes are large, and erasing is slow.
- Alternative: Third party options

Product	Comment
littlefs	https://github.com/ARMmbed/littlefs/blob/master/DESIGN.md "NAND flash already has many limitations that make it poorly suited for an embedded system: low erase cycles, very large blocks, errors that can develop even during reads, errors that can develop during writes of neighbouring blocks. Managing sensitive storage, such as NAND flash, is out of scope for the littlefs." Requires a Flash Translation Layer (FTL)
FreeRTOS+FAT	Requires a Flash Translation Layer (FTL)
yaffs	https://yaffs.net/ Yaffs supports a NAND flash. GPL-v2 Licence

2. Partial page programming.
   - ECC Parity Code generation restricts the number of writes (and where) to a page before invalidating the ECC system. Each page has a usable section of 4096 bytes. For each 512 bytes sector of a page, 16 bytes are reserved in an Out-of-band (OOB) area for ECC generation. The number of partial program cycles in the same page is 4, so the page is divided into 1024 bytes sectors.



Flash Memory Lifetime Concerns

The DC Meter could be running continuously at elevated temperature for its entire service life. The durability of the SLC NAND flash required internal quality testing.

Test	Iterations	Condition	Pass / Fail	Comment
Program/Erase each page (4096 bytes) in a block (64 pages) with alternating checker pattern data (0xAA/0x55)	100,000	25 °C	Pass	
Program/Erase each page (4096 bytes) in a block (64 pages) with alternating checker pattern data (0xAA/0x55)	100,000	80 °C	Fail	Some pages exhibited ECC uncorrectable after 45,000 iterations. Sufficient for our lifetime use.
Read disturb, continuous read of preprogrammed block, checking surrounding pages for corruption	100,000	25 °C	Pass	
Read disturb, continuous read of preprogrammed block, checking surrounding pages for corruption	100,000	80 °C	Pass	

Flash Memory Firmware: Page Organisation

- Records are fixed length (128 bytes) and include a SHA256 fingerprint for integrity checking.
- Records are to be sector aligned with eight records per sector.

Flash Memory Firmware: Power down

Several of the records kept are used in RAM and then flushed to disk. In the case of an unexpected power down (all power losses), the device has sufficient backup capacitance to detect the power drop, flush the records to flash before losing power.

Flash Memory Firmware: Bad Block Tracking

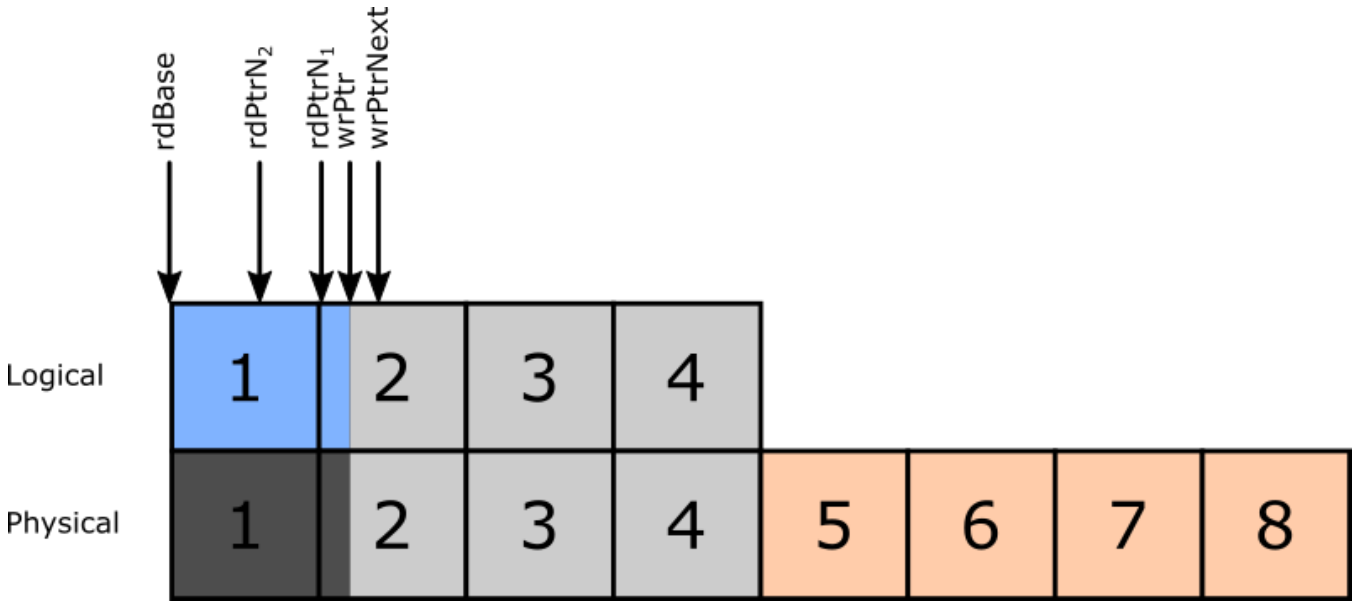
Block Allocation Table (BAT) section has been implemented to maintain a contiguous logical address space over a discontinuous physical address space.

- On first ever startup and when a new bad block is detected, the BAT is updated.
- There are 2048 blocks in a single plane of the TC58CV device.
    - 2048 bits (256 bytes) of storage is required to track bad blocks (1 bit / block).
    - Once a block has been marked bad, it will always be bad.

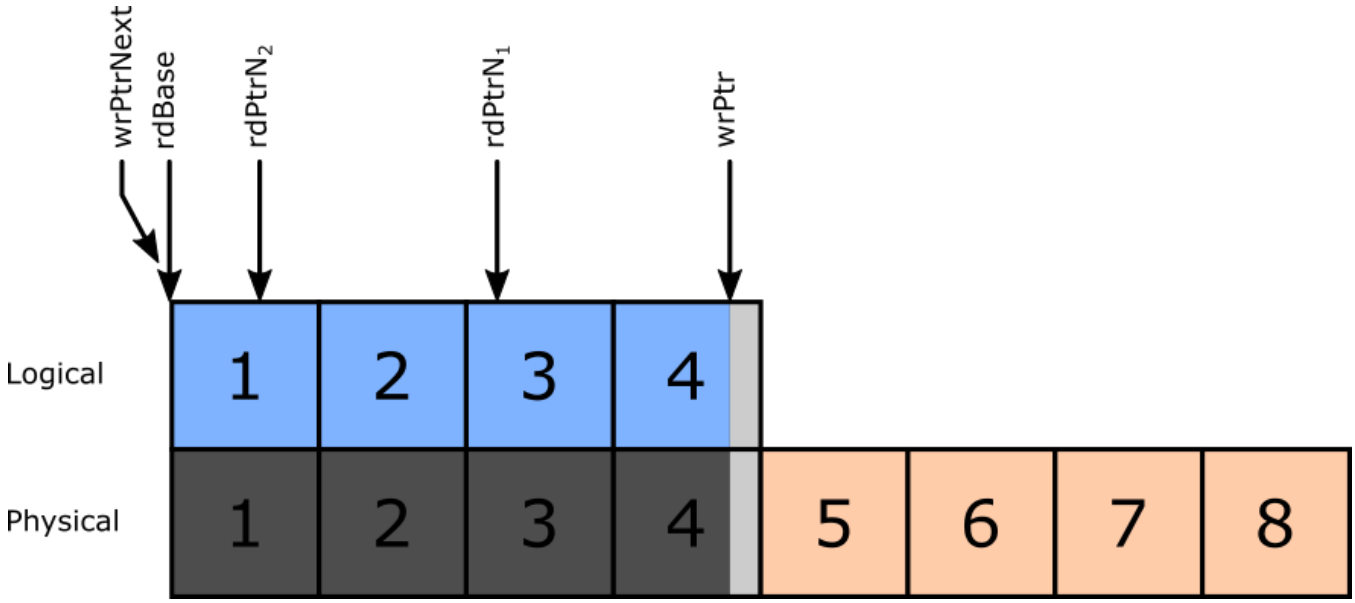


Flash Memory Firmware: Disk Manager

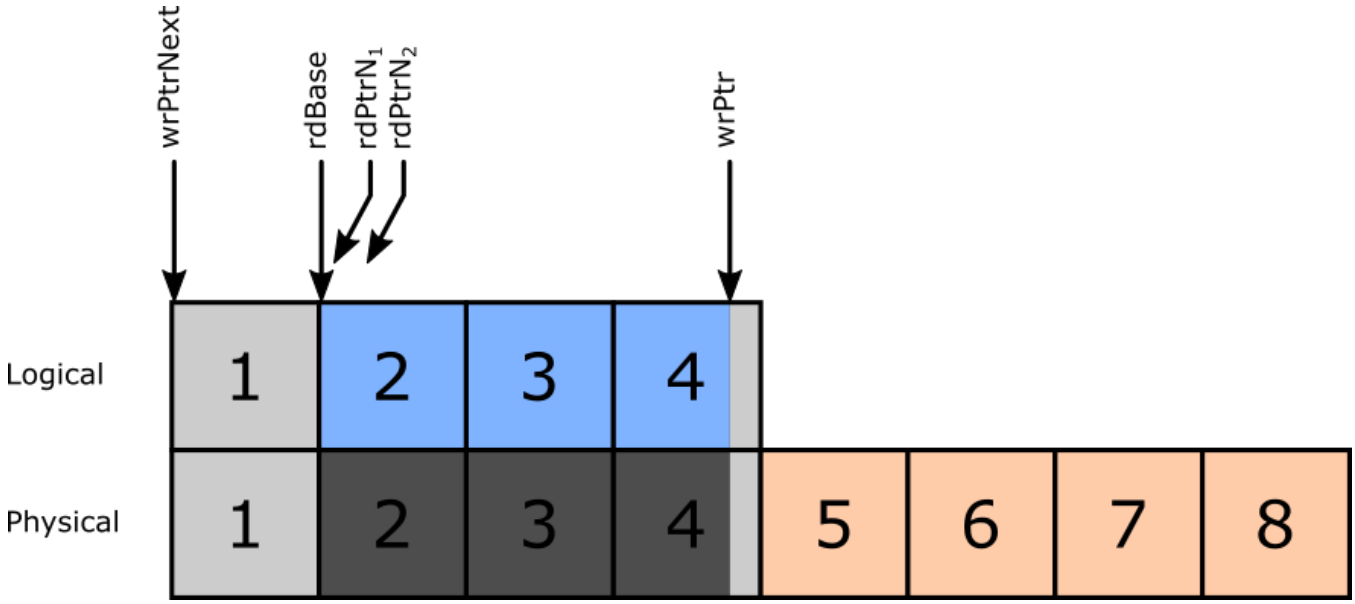
The Disk Manager is a component of the DC Meter which operates on the principal of a contiguous logical address space. Each file that is managed by the Disk Manager is allocated its address range in the form of a logically contiguous file partition. Each file partition contains operating policies (i.e. What to do when the partition is full, overwrite existing data or not).



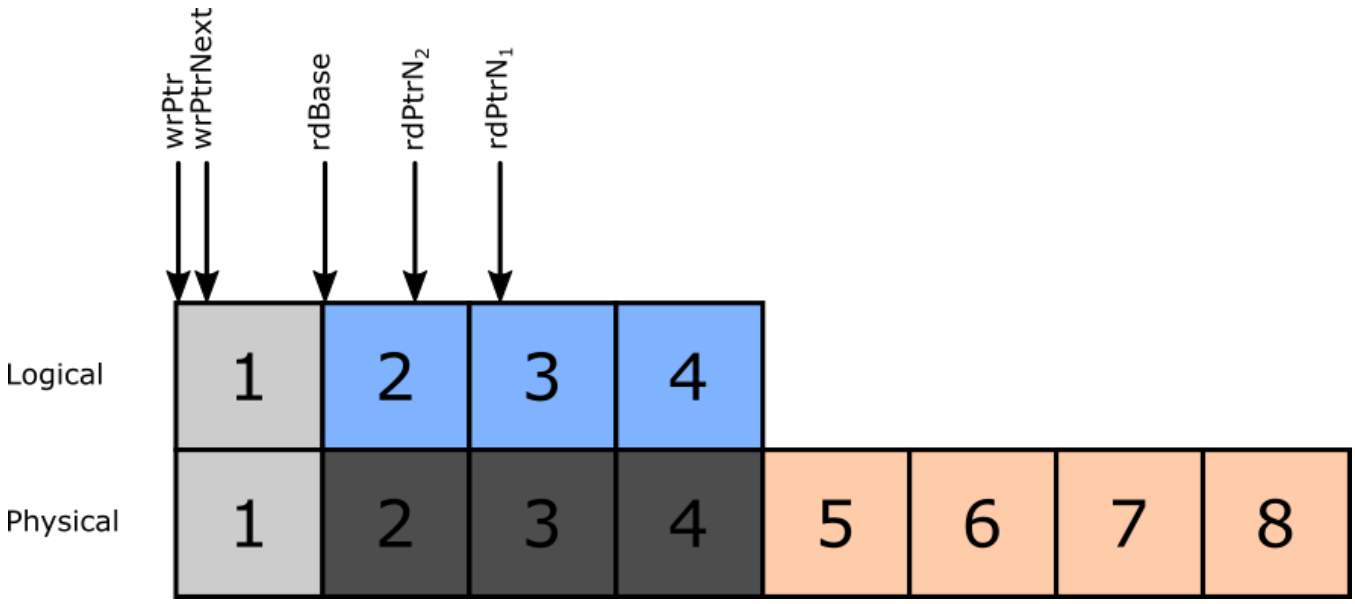
t1 - Normal operation, free space available in user partition.



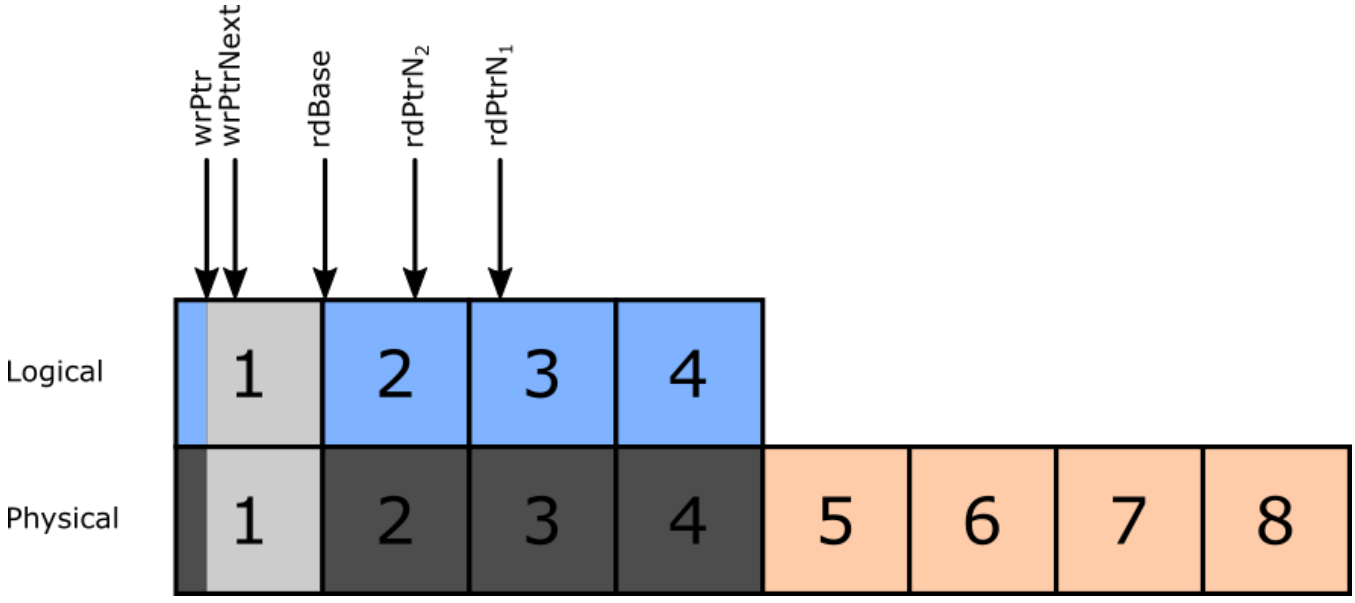
t2 - Normal operation, last page in user partition.



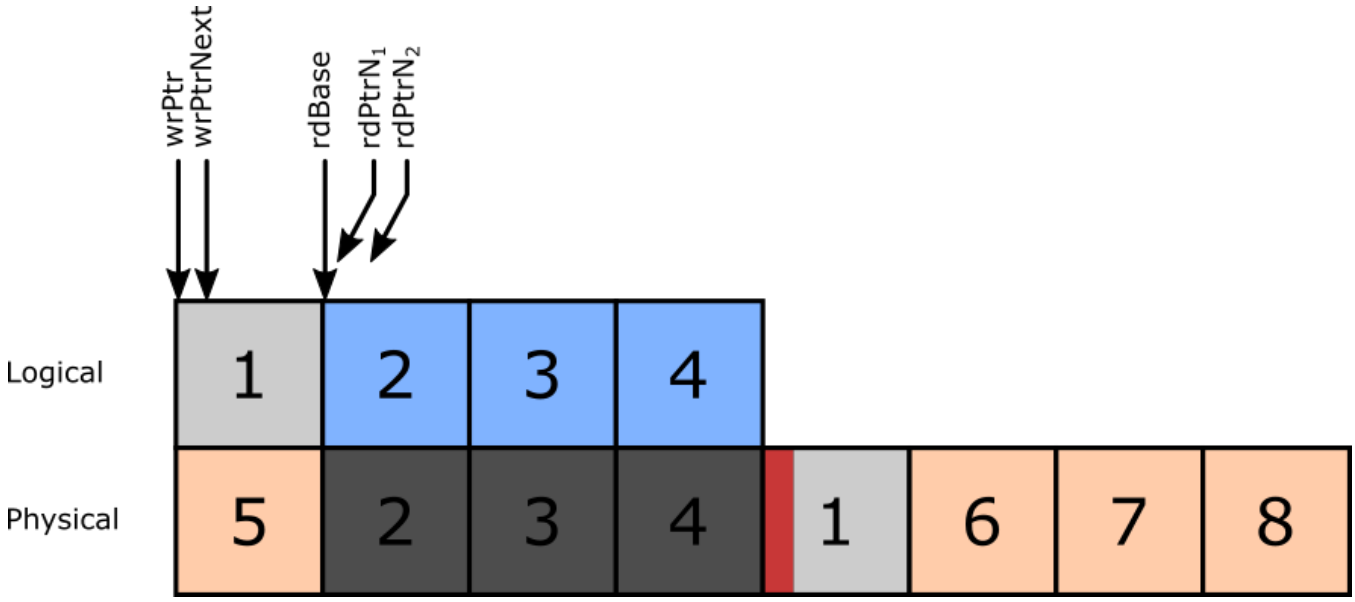
t3 - First block of the user partition is erased to make room for future data. The read base and iterator pointer are reset and relocated to the next logical block.



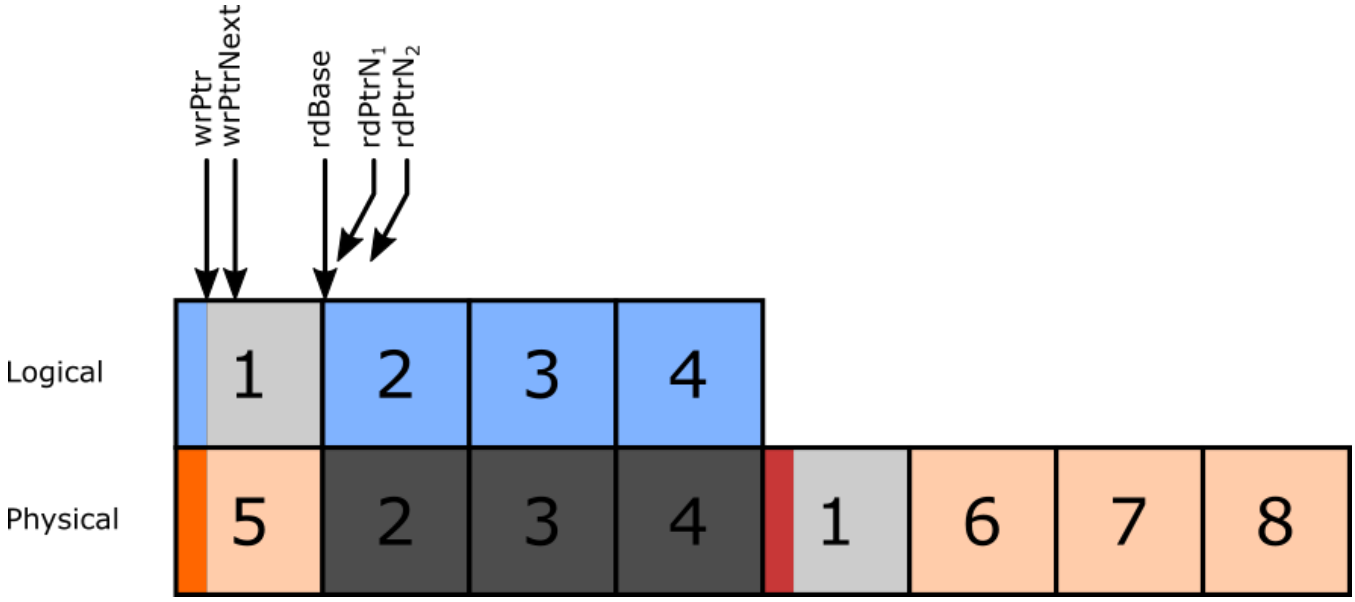
t4 - The user partition has wrapped. The write pointer is moved back to the first block of the user partition.



t5 - Normal operation, free space available in user partition.



t6 - A write operation failure occurs. Logical block 1 is reallocated. The write pointer is logically reset to start of the reallocated block. The contents of physical block 1 is orphaned (not deleted) during the reallocation process. The read iterators pointers are logically reset to protect iterators from pointing to potentially null data.



t7 - Normal operation, free space available in user partition.

Audit Event Log

Overview

The Audit Event Log is required to securely log nominated events. Sufficient memory has been allocated so events can be logged for the meter's entire lifetime.

Audit Event Log is never deleted, if size of memory for events exceeds the allocated space the device goes into out of service permanently.

The Audit Event Log can be extracted from the meter via the TTI. When extracting a hash is calculated on all logs extracted and that hash is cryptographically signed and sent to the TTI to verify the integrity and authenticity.

Event Log Structure

Each event is logged as a fixed 128 byte entry, to align with physical memory, containing:

- Index of event
- Time of event
- Event type
- Optional data (64 bytes)
- SHA256 hash of above
- Protocol version

Optional data contains information relevant to the event logged. i.e. All configuration parameters have an event logged whenever they are changed, data contains a copy of the new parameters.

Some events require multiple event entries due to needing to log more data than a single entry can contain.

Events Logged

The following is a list of the event which are logged by the meter:

- EVENT_LOG_POWER_STARTUP
- EVENT_LOG_POWER_LOSS
- EVENT_LOG_BEGIN_CHARGE_SESSION
- EVENT_LOG_END_CHARGE_SESSION
- EVENT_LOG_AFE_CONFIG_PARAM_CHANGE
- EVENT_LOG_TSOCLIENT_CONFIG_PARAM_CHANGE
- EVENT_LOG_INET_CONFIG_PARAM_CHANGE
- EVENT_LOG_PCBA_TEST_LOG_PARAM_CHANGE
- EVENT_LOG_SUBA_TEST_LOG_PARAM_CHANGE
- EVENT_LOG_HOST_PRIVATE_KEY_PARAM_CHANGE
    - The private key event is explicitly prevented from logging the private key as the optional data field for security reasons
- EVENT_LOG_TRUST_ROOT_KEY_PARAM_CHANGE
- EVENT_LOG_COMMISSION_CONFIG_PARAM_CHANGE
- EVENT_LOG_OUT_OF_SERVICE_CHARGE
- EVENT_LOG_TIME_SYNC
- EVENT_LOG_TIME_DEVIATION
- EVENT_LOG_TECHNICIAN_LOGIN
- EVENT_LOG_TECHNICIAN_LOGOUT
- EVENT_LOG_LOG_EXPORT_OCCURRED
- EVENT_LOG_FIRMWARE_UPDATE_DOWNLOAD
- EVENT_LOG_COMMISSION
- EVENT_LOG_MEAS_STORAGE_USAGE_WARN
- EVENT_LOG_RECORDS_DELETED
- EVENT_LOG_SYSCHECK_RESULT (see Self Tests)

Optional data

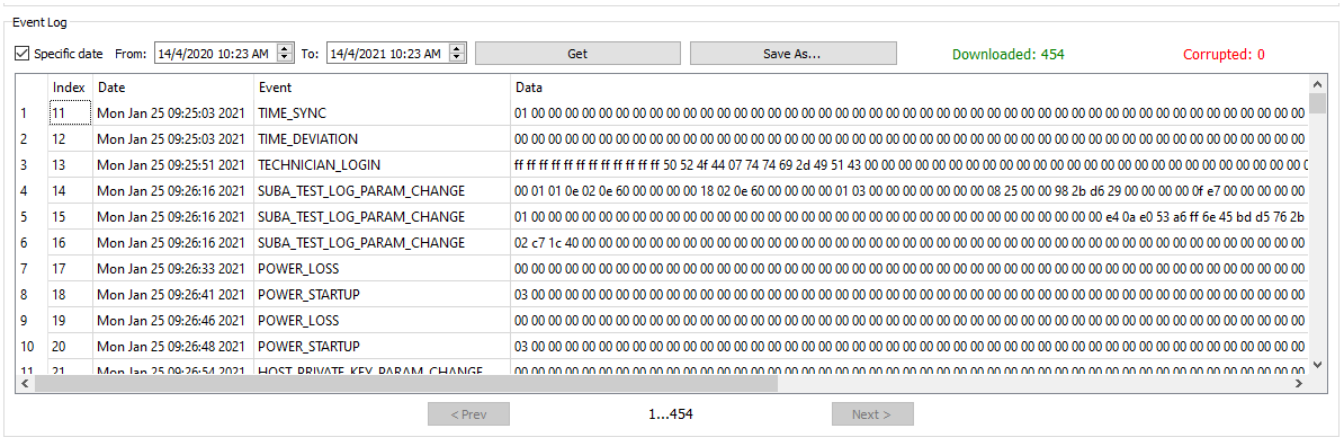
The information in optional data logged is different for different events. Only the events which include optional data logged is included:

Event type	Optional data logged	Notes
EVENT_LOG_POWER_STARTUP	Reset Cause [2 bytes]	MAP_SysCtlResetCauseGet as defined for Texas Instruments TM4C129
EVENT_LOG_BEGIN_CHARGE_SESSION	EDSI [4 bytes]	See Energy Delivery Session Identifier (EDSI)
EVENT_LOG_END_CHARGE_SESSION	EDSI [4 bytes] Charger Session ID [37 bytes]	See Energy Delivery Session Identifier (EDSI) See Supporting the SAFE Initiative

Parameter change EVENT_LOG_AFE_CONFIG_PARAM_CHANGE to EVENT_LOG_COMMISSION_CONFIG_PARAM_CHANGE Excluded: EVENT_LOG_HOST_PRIVATE_KEY_PARAM_CHANGE	Section/chunk number [1 byte] Set successfully (bool) [1 byte] Parameter data [62 bytes]	If parameter type is greater than can fit in one log, it is split into multiple sequential logs (chunks) to fit whole parameter pack. See DC Meter Configuration
EVENT_LOG_OUT_OF_SERVICE_CHARGE	OoS Fault cause [1 byte]	See Charge Session State Machine
EVENT_LOG_TIME_SYNC	hasTimeSyncSucceeded (bool) [1 byte]	
EVENT_LOG_TECHNICIAN_LOGIN EVENT_LOG_TECHNICIAN_LOGOUT	PermissionLevel (read [2 bytes]; write [2 bytes]; execute [2 bytes]; type [4 bytes]) Peer ID [32 bytes]	See Trusted Technician Interface (TTI)
EVENT_LOG_LOG_EXPORT_OCCURRED	Type Log [1 byte]	0=OCMF; 1=Event
EVENT_LOG_FIRMWARE_UPDATE_DOWNLOAD	hasDownloadSucceeded (bool) [1 byte] isServerTrustworthiness (bool) [1 byte] isImageTrustworthiness(bool) [1 byte] isBootloaderImageValid(bool) [1 byte] imageHeaderHash(bool) [32 bytes]	All bools true if successful.
EVENT_LOG_MEAS_STORAGE_USAGE_WARN	Usage Percentage [1 byte]	
EVENT_LOG_RECORDS_DELETED	Erased Block Number [2 bytes]	See External Non Volatile Storage
EVENT_LOG_SYSCHECK_RESULT	Sys check result [4 bytes]	See Self Tests

Event Log Sample

The screenshot below shows a sample of the event log available through the Trusted Technician Interface. The log can be downloaded and saved to disk using the "Save As..." button.



Once downloaded, the log file can be viewed in any plain text editor. An excerpt of a log file is provided below. Each line begins with a timestamp, lines shown are wrapping due to line length. The full contents of this log file can be viewed in A - Full Audit Event Log. The data format can be decoded according to the information described above in "Event Log Structure" and the table "Optional Data".

```
Index Date Event Data Hash Protocol version
11 Mon Jan 25 09:25:03 2021 TIME_SYNC 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

00 00 00 a5c7e4f3d3e78de7ded77bb52e6756913899b3defcd3cebdd7b7d47948abc37f 0
12 Mon Jan 25 09:25:03 2021 TIME_DEVIATION 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 4598e66c3d20389a1f4be83c29b7f6ee83a3937d11e6b133c2196297cc12a18e 0
13 Mon Jan 25 09:25:51 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 50 52 4f 44 07 74 74 69 2d 49
51 43 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 84f416d24ddffc080b76658a71521d3673a7155a2bc94e912565ec541a5f8479 0
14 Mon Jan 25 09:26:16 2021 SUBA_TEST_LOG_PARAM_CHANGE 00 01 01 0e 02 0e 60 00 00 00 00 18 02 0e 60 00 00 00 00
01 03 00 00 00 00 00 00 00 08 25 00 00 98 2b d6 29 00 00 00 00 0f e7 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 cb654c5a020c864aa6dc43c057e5dd977bd54e2bd483a3b61ec50c326d720a27 0
15 Mon Jan 25 09:26:16 2021 SUBA_TEST_LOG_PARAM_CHANGE 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 e4 0a e0 53 a6 ff 6e 45 bd d5 76 2b e2 40 3f 9f 78 c0 85 f8 03
bc 52 6e 16 84 88 7a 4d 8c1964b4a7138aebf4c2db97d9483ca1542e7a445859a3fda803f663c277b8b4 0
16 Mon Jan 25 09:26:16 2021 SUBA_TEST_LOG_PARAM_CHANGE 02 c7 1c 40 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 39990f4d77d0f1915654f785945fc2045ce0a9b3a1060404d16a8292d62d08fa 0
17 Mon Jan 25 09:26:33 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 d46b127bdbcfb3b97821dfb4b864874fd7cce3c6ac746963815995def1b0b81e 0
18 Mon Jan 25 09:26:41 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 e1b162e5f38e7211666db54e2bffc142b4e34e168bb926b197d481f8a57ee966 0
19 Mon Jan 25 09:26:46 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 481d6db29195202326429e392b2c689751ea6ddac94cb8c83273df003922b240 0

Charge Session Log

Charge session information is logged with all information needed to reproduce a receipt. All variable data in the OCMF message is logged to the external non-volatile storage so on request it can be regenerated.

Every charge session uses 1 page of storage. The first 3 sectors of the page are used for OCMF entries with a hash, which are entered once the charge is completed. The last section of that page is reserved for the signature of the entries, written when OCMF message is successfully sent over MQTT. The signature of the data allows authenticity and integrity of the data to be verified.

If the OCMF message can't be sent over MQTT the signature is not written until successful. On power cycle an unsigned OCMF message will be detected, signed and sent over MQTT, once successful the signature is written to the external non-volatile storage.

The memory space reserved for charge session logs should be sufficient for more than 3 months of frequent use. Once the allocated memory space is utilised the oldest block will be erased and reused for new sessions.

The Charge Session Log can be extracted from the meter via the TTI.

Application Package Manager

- Upgrade Binary Package
- Certificate Chaining
- Update Procedure

The DC meter must have the ability to securely update legally relevant firmware while the existing application is running.

To begin a firmware update, a user must sign into the DC Meter with the Trusted Technician Interface (TTI) using a trusted certificate.

Once the firmware update is triggered, the DC Meter locks into out-of-service. The DC Meter connects to and authenticates an update server to be trustworthy. The DC Meter downloads the firmware update, verifies the integrity and the manufacturer signature.

Once the firmware download has been confirmed to be valid, the DC Meter will jump to the bootloader to install the firmware update. The bootloader verifies the downloaded firmware update and then programs as the main image.

On every startup of the device, the bootloader will calculate and verify the hash of the main image. If this hash is incorrect the main application will not be entered and the device will be inoperable.

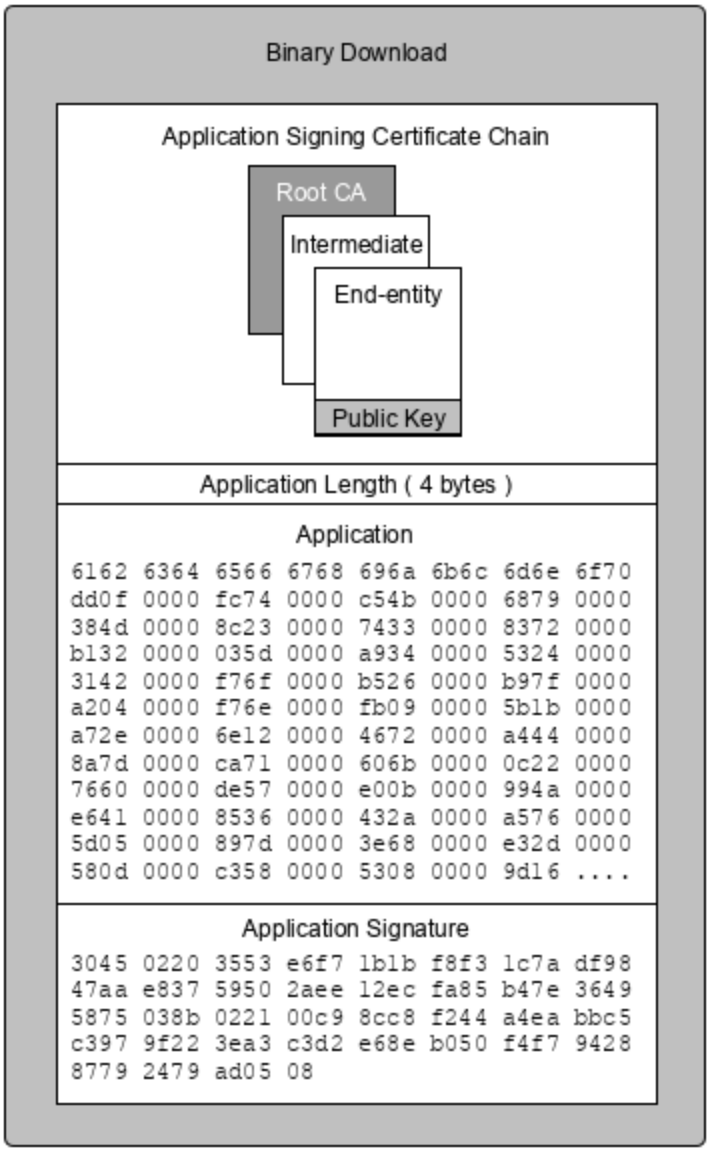
The Application Package Manager task is responsible for;

- Securely connecting to an remote Upgrade Server.
- Verifying that the Upgrade Server is trustworthy.
- Downloading a binary image from a trustworthy Upgrade Server.
- Verifying that the binary image is trustworthy and valid.

Upgrade Binary Package

The binary image contains;

- Application Signing Certification Chain
    - Used to verify the signature of the binary image.
- Application Length (in bytes)
- Application Binary (payload)
- Application Hash
- Application Signature



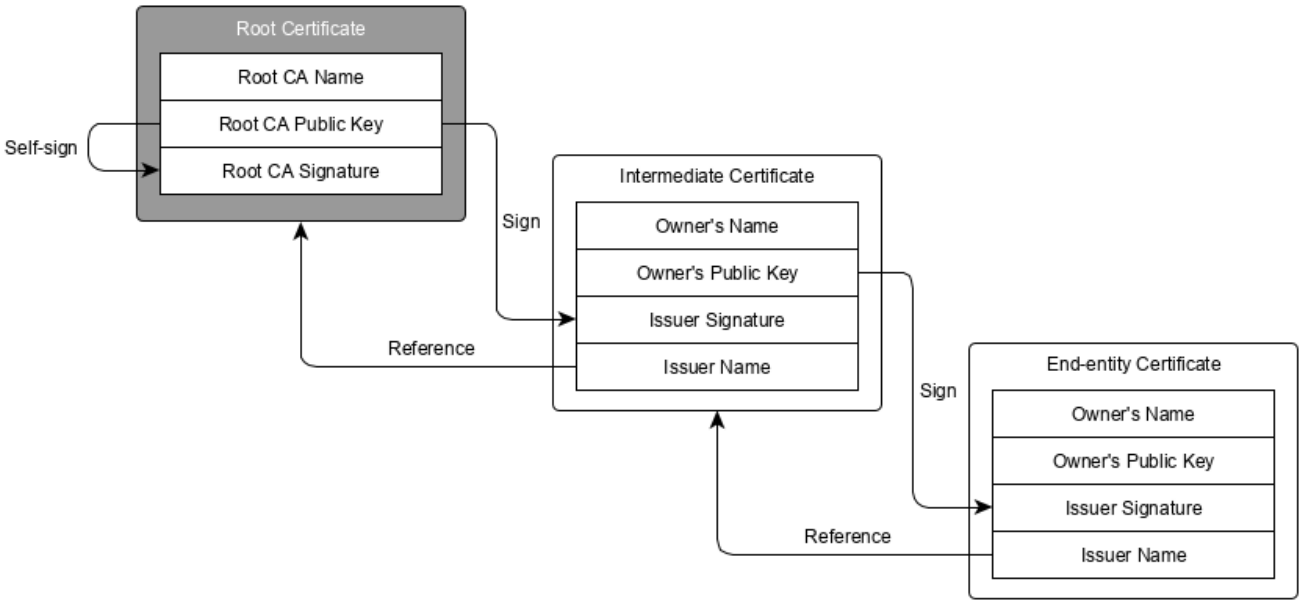
⚠ The binary image may be hosted on an Upgrade Server that is separate to the Binary Signing Server. To allow for this, the binary image is bundled with a Signature and a Signing Certificate Chain to verify the binary image trustworthiness.

Certificate Chaining

To verify the application in the binary image is trustworthy, the Signing Certificate Chain must first be verified. A certificate chain is a list of certificates followed by one or more Certificate Authority (CA) certificates (usually being self-signed).

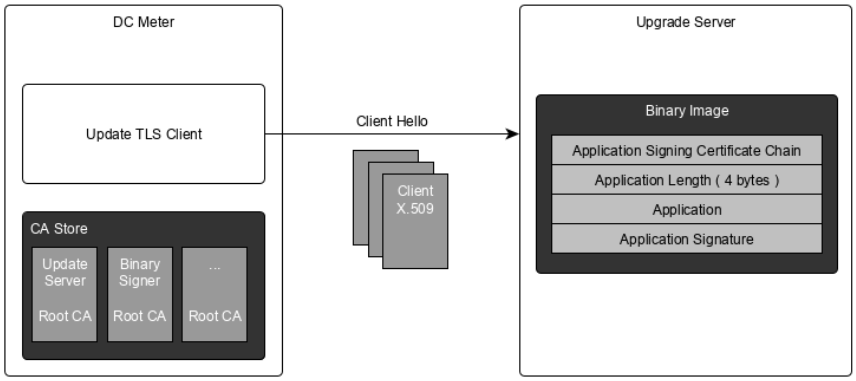
Due to resources on the meter, only a single level of certificates is supported. No intermediate certificates can exist in the chain, the end-entity must link to the root certificate. The root certificate in the list is a trust anchor and can be trusted.

Certificate chains are used in order to check that the Public Key (PK) contained in a target certificate (the first certificate in the chain) and other data contained in it effectively belongs to its subject. In order to ascertain this, the signature on the target certificate is verified by using the PK contained in the following certificate, which is the root certificate. As the last certificate is a trust anchor, successfully reaching it will prove that the target certificate can be trusted.

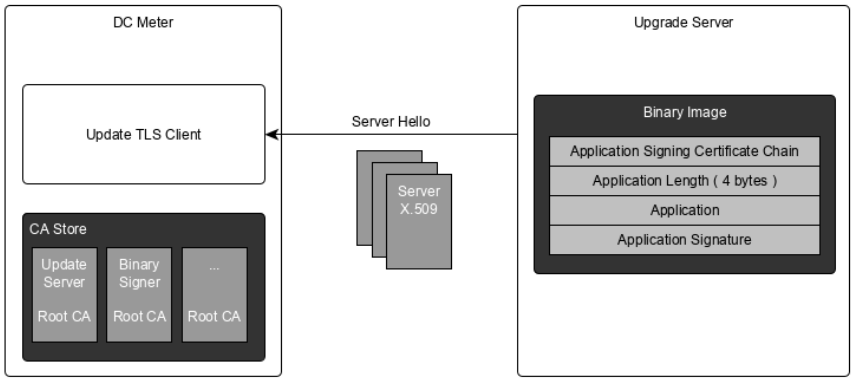


Update Procedure

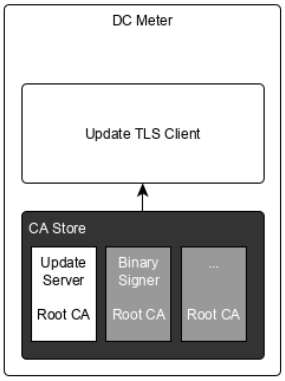
1. Connect to Update Server. Client submits certificate chain for verification.



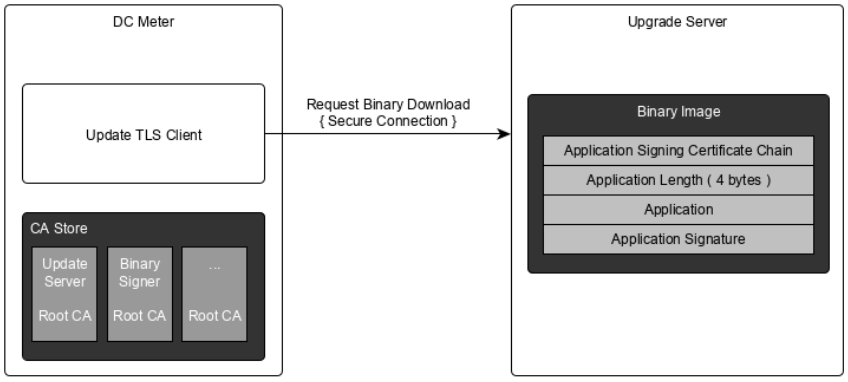
2. Server verifies Client certificate chain. Server responses with Server certificate chain.



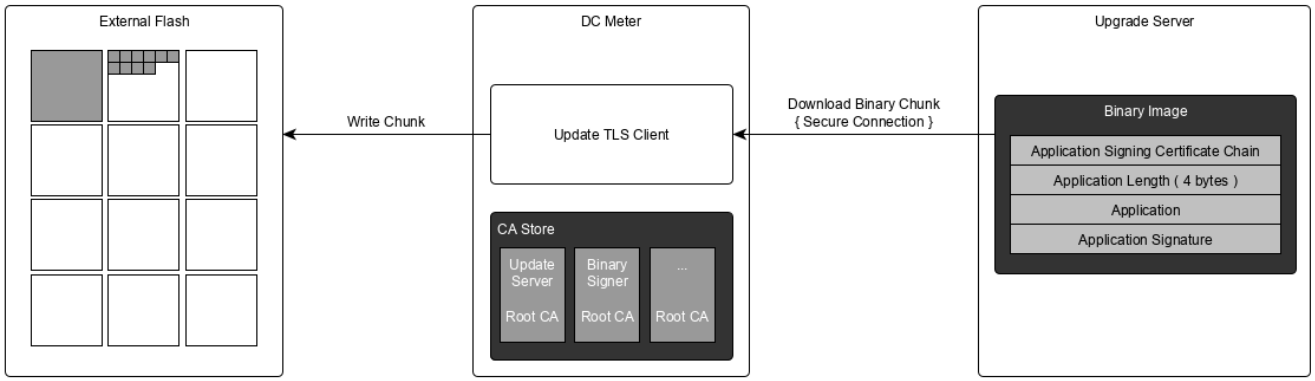
3. Client verifies Server certificate chain; using preloaded CA store as trust anchor.



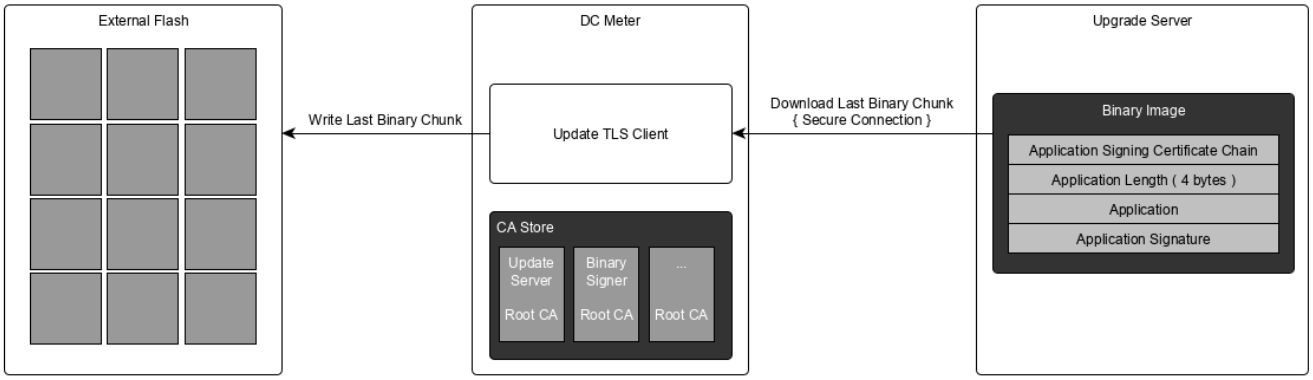
4. Client request binary download.



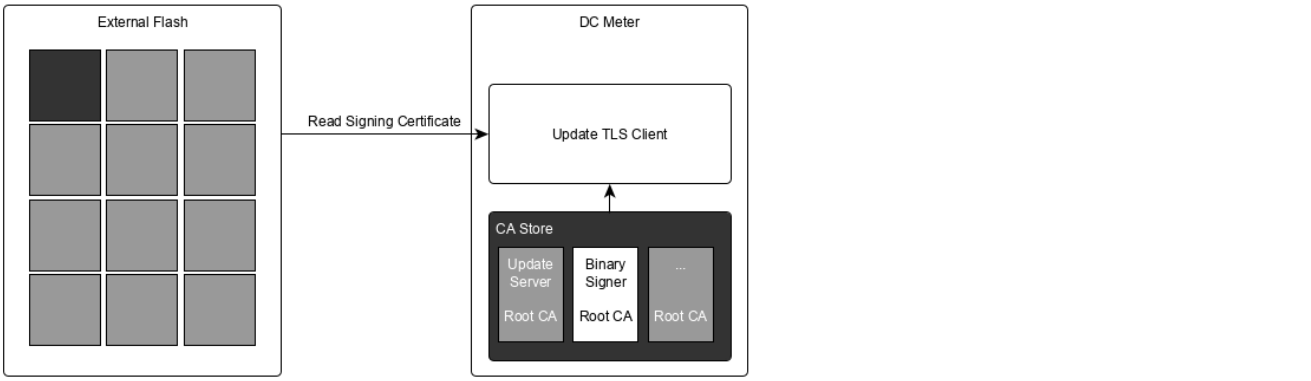
5. Client downloads binary file in chunks. Chunks are written to external flash. The SRAM isn't large enough to hold entire binary.



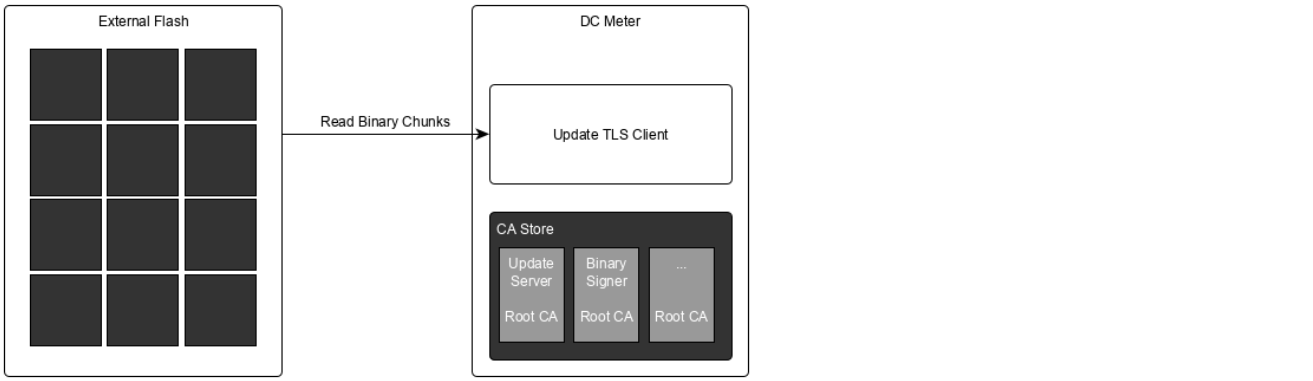
6. After last chunk is downloaded, close connection to Server.



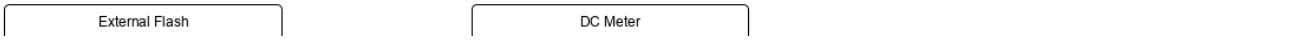
7. Read from external the first chunk containing the Binary Signing certificate chain. Verify that the Binary Signing certificate chain is trustworthy. Extract Public Key from End-Entity (EE) certificate.

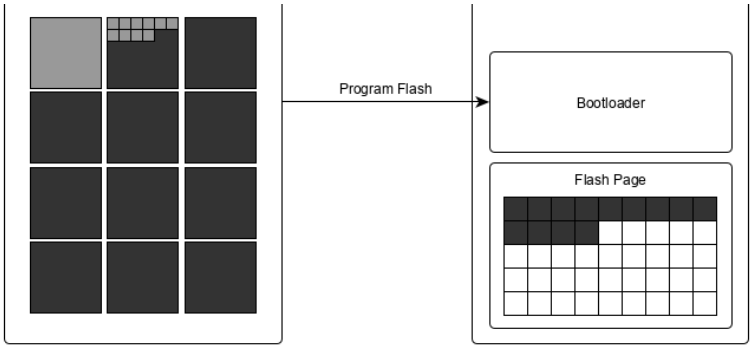


8. Sequentially compute the hash of the application binary. Use the Public Key of the EE certificate with the generated hash to verify the binary signature.



9. Application binary is verified. Jump into Bootloader and program into flash.




Program Flash
Bootloader
Flash Page

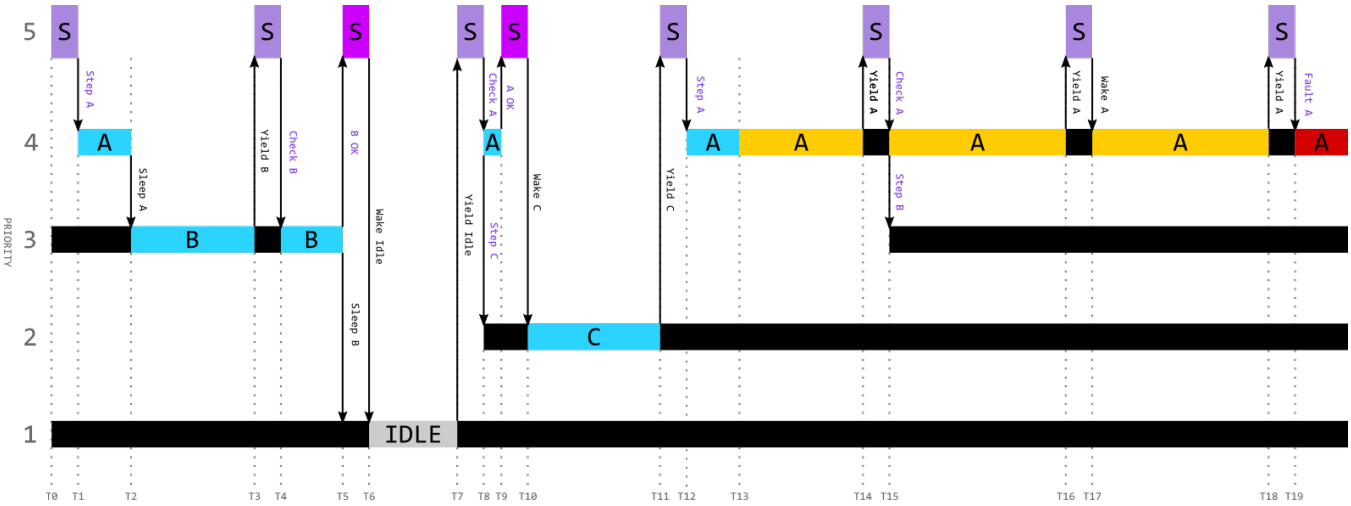
Supervisory State Machine

Introduction

The Supervisory State Machine (SSM) monitors the health of the DC Meter. The SSM resets the hardware watchdog and monitors the state of other RTOS Tasks.

Details

The SSM is the highest priority task in the DC Meter and is responsible for starting, stepping and monitoring other RTOS Tasks. The SSM is implemented using a FreeRTOS Task Notification (https://www.freertos.org/xTaskNotify.html) schema. Periodically, the SSM will issue each task with a "Health Check" Task Notification. The SSM expects that the notified task will send a "Task OK" Task Notification acknowledgement within a specified time window.



The above figure shows an example of the highest priority user task (Task A) becoming unresponsive and detected by the SSM.

Time	Description
T0 - T1	Task B blocked. SSM action; notify Task A to "step".
T1 - T2	Task A woken by "step" notification, executes, completes and sleeps.
T2 - T3	Task B unblocked and executes.
T3 - T4	Task B yields. SSM action; notify Task B to "check".
T4 - T5	Task B unblocked, executes, notifies SSM with "check" acknowledgement and sleeps.
T5 - T6	SSM woken by "check" acknowledgement notification, resets Task B watch-dog counter and sleeps.
T6 - T7	Task Idle woken, executes.
T7 - T8	Task Idle yields. SSM action; notify Task A to "check" and Task C to "step".
T8 - T9	Task A woken by "check" notification, Task A notifies SSM with "check" acknowledgement and sleeps.
T9 - T10	SSM woken by "check" acknowledgement notification, resets Task A watch-dog counter and sleeps.
T10 - T11	Task C unblocked, receives "step" notification and executes.
T11 - T12	Task C yeilds. SSM action; notify Task A to "step".
T12 - T13	Task A executes.
T13 - T14	Task A malfunctions taking CPU time and blocking tasks.
T14 - T15	Task A yields. SSM action; notify Task A to "check" and Task B to "step".
T15 - T16	Task A resumes taking CPU time and blocking tasks. Unable to receive "check" notification.
T16 - T17	Task A yields. SSM action; no action.
T17 - T18	Task A resumes taking CPU time and blocking tasks. Unable to receive "check" notification.

T18 - T19	Task A yields. SSM action; Task watch-dog counter has reached zero. Fault task.

Once a task has been identified as faulty, corrective action is for the Supervisor to software reset the entire microcontroller. This acts as a software watchdog for every individual task, whilst a hardware watchdog oversees the Supervisor task.

Self Tests

Introduction

The DC Meter must conduct self tests on critical components in order to verify the integrity of measurements and the system as a whole.

The self tests run at system start-up or during run time.

Tests during run time are either run periodically every 6 hours or on a continuous interval as required to maintain functionality.

Test	Start-up	Run Time (Periodic)	Run time (Continuous)	Causes CSSM OoS	Logged
Supply voltage - ADC			x	x	x
Supply voltage - UVLO			x	x	x
Ambient temperature			x	x	x
Shunt temperature			x	x	x
Fan state / speed			x		x
Ovenisation temperature			x	x	x
NTP last time synchronisation			x	x	Via 'RTC state' self test
NTP time variance	x	x			x
RTC backup supply voltage			x		x
RTC state	x	x		x	x
Measurement storage integrity	x	x			x
Measurement storage capacity	x		x	x	x
Configuration integrity	x			x	x
Firmware Image integrity	x			x	Bootloader image integrity via 'Firmware Update' task Operational image integrity via 'boot event' log at start of execution
AFE communications		x		x	Via 'CSSM - Out Of Service' event log if failed during charge
External Flash communications		x		x	Via 'CSSM - Out Of Service' event log if failed during charge
Entropy			x		x

Result logging

Logging of self test results is performed periodically every 6 hours.

There are a number of run time tests that independently or indirectly log an event as described in the table above.

Self Test Details

Supply voltage - ADC

The 12V supply voltage to the DC Meter is digitised using the microcontroller's internal ADC.

The measured voltage is checked to be within an acceptable range (Under and Over) for the system to operate.

Supply voltage - UVLO

The 12V supply voltage to the DC Meter is monitored by a voltage comparator circuit internal to the microcontroller.

If the supply voltage level drops below the comparators trigger level (eg: power failure), an Under Voltage Lock Out event is immediately triggered by the system.

Ambient temperature

Ambient temperature in the DC Meter enclosure is digitised using the microcontroller's internal ADC.

The measured temperature is checked to be within an acceptable range (Under and Over) for the system to operate functionally and within accuracy requirements.

Shunt temperature

The temperature of the shunt element used for accurate measurement of the DC Meter high current path is measured using a LMT01 digital sensor.

The shunt temperature is checked to be within an acceptable range (Under and Over) for the system to operate functionally and within accuracy requirements.

Fan state / speed

The fans that cool the shunt element are actively controlled based on shunt temperature. Speed feedback is provided from the fans as a train of pulses that the microcontroller monitors.

The fan control loop checks the speed of the fans to ensure they are within an acceptable range (Under and Over), when determined they should be on, for the shunt temperature to be effectively managed.

Ovenisation temperature

In order to achieve accuracy targets over the wide operating temperature range, the analogue front end circuitry is heated to a constant temperature.

The heating control is independent of the microcontroller, however, the temperature of the circuitry is digitised by the precision ADC (main ovenised component) and provided to the microcontroller.

The measured temperature is checked to be within an acceptable range (Under and Over) for the system to operate functionally and within accuracy requirements.

NTP last time synchronisation

The system time is synchronised on a regular basis to ensure it remains within the required tolerance and to detect any faults.

If a successful synchronisation has not occurred within a defined time frame, a fault is triggered as the time accuracy is not guaranteed.

NTP time variance

The system time is synchronised on a regular basis using the Simple Network Time Protocol to ensure it remains within the required tolerance and to detect any faults.

After a successful synchronisation, the change in time is checked to be within an acceptable range (Under and Over) to ensure the system was within accuracy requirements.

RTC backup supply voltage

In order to preserve system time keeping during a power outage an on board backup energy source is used. The voltage is monitored by a voltage comparator circuit internal to the time keeping module in microcontroller.

If the voltage level drops below the comparators trigger level, an under voltage event is determined implying that time keeping is not guaranteed if the main supply fails.

RTC state

The Real Time Clock is responsible for time keeping on the DC Meter. It is a module internal to the microcontroller.

The configuration of the time keeping module and the present value of time is monitored to ensure the system is operating correctly.

Measurement storage integrity

All power measurements (charge sessions) and events are logged to non volatile memory. All data saved to memory is stored with error detecting information.

The integrity of all saved data is verified to ensure full traceability is preserved.

Measurement storage capacity

The external non volatile memory has a finite capacity which is partitioned to store different types of data.

The remaining storage capacity for event logs is checked against several thresholds in order to help avoid the impact of completely running out of storage capacity.

Configuration integrity

All configuration items are stored in non volatile memory along with error detecting information.

The integrity of all saved configuration is verified to ensure the system will operate functionally and within accuracy requirements.

Firmware Image integrity

The DC Meter firmware is separated into two main functional blocks, the bootloader and operational (main DC Meter logic) firmware. All firmware includes error detecting information to verify its integrity.

The bootloader is run at first power up and during the final stage of a firmware update. The bootloader checks the operational firmware integrity before it is executed.

The operational firmware checks the integrity of the bootloader before a firmware update process is started.

AFE communications

Metrological measurements are performed by an external ADC which are transferred to the microcontroller through an isolated bi-directional communications link.

If new measurements are not received by the microcontroller within a specified time period, an error is determined implying that communications or the ADC have failed.

External Flash communications

Flash memory external to the microcontroller is used for all logging and storage of firmware updates. This test only covers the communications link operation.

Any failure to access the external flash memory triggers an error which will prevent further charge sessions from being metered.

Entropy

The sources of random data are checked and assured to be within entropy limits.

DC Meter Configuration

The configuration of the DC meter is separated into blocks and saved in the internal EEPROM. All configuration is written through the TTI. Each configuration block can be given access to read and/or write independently of other blocks.

On every change of EEPROM block, an event is added to the audit log. All blocks are broken into 128 byte sections (or multiples of) and contains a hash verifying the integrity of the data. Any data that has been determined to be corrupted (i.e. hash verification failed) will not be used and flagged as an error.

- AFE (Calibration) Configuration
- TSO Configuration
- Inet Configuration
- Commissioning Configuration
- Configuration Summary
- Other EEPROM Data

AFE (Calibration) Configuration

This will be written during production in controlled conditions.

This accounts for the individual unit's manufacturing tolerances, and is critical to the metrological accuracy. It contains:

- Current offset (nA) for low gain
- Current offset (nA) for high gain
- Current scale factor (ppm) for low gain
- Current scale factor (ppm) for high gain
- Voltage offset (nV) for channel 1
- Voltage offset (nV) for channel 2
- Voltage scale factor (ppm) for channel 1
- Voltage scale factor (ppm) for channel 2
- Cable resistance (uohm) for channel 1
- Cable resistance (uohm) for channel 2

All parameters are validated that they are within a correct range before they are written. i.e. scale must be 90-110%

TSO Configuration

This will be written during production in controlled conditions.

This is used to configure the parameters need to connect to the TSO, and allow the same DC Meter firmware to support multiple screens. It contains:

- Number of characters in X and Y direction
- Number of pixels in overlay in X and Y direction
- Location of origin of overlay in X and Y pixels
- Location of start of character field in overlay in X and Y pixels
- Border thickness in pixels
- Screen rotation
- Colour map
- Video source: FPGA or HMI
- Selection of screen type: ISD or PK

Parameters for the TSO is validated on the TSO itself.

Inet Configuration

This will be written during production in controlled conditions.

This is used to configure the parameters need to connect to the configurable network interfaces, It contains:

- DNS primary - Resolves network names to IP address, first attempt
- DNS secondary - Resolves network names to IP address, second attempt
- UDP multicast - IP address for debug output
- UDP multicast - port for debug output
- NTP primary - web address for time server
- MQTT broker - web address or IP to connect to
- Update server - web address or IP to connect to download file for boot loading

- Update file path - file name for boot loading
- Static IP address - IP address of the meter to be once commissioned
- Static IP net mask - net mask of the meter to be once commissioned
- Static IP gateway - gateway address for meter to use once commissioned

Commissioning Configuration

This will be written during production in controlled conditions.

This is used to configure the commissioning parameters needed, it contains:

- Commissioned flag - set when all tests have passed in production
- Serial number of device
- 2 x Tamper seal IDs
- CPO public key - used to verify sent tariffs
- Default tariff name used on startup
- Parameters that define the time zone rules applied

The commissioned flag and the time zone parameters are checked to ensure they are in the correct range before they are written.

Configuration Summary

The following table summarises the configuration items specified above.

Subsection	Configuration Item	Type/Unit	Legally Relevant?	Changes Logged?
AFE (Calibration)	Current offset, low gain	nA	Yes	Yes
	Current offset, high gain	nA	Yes	Yes
	Current scale, low gain	ppm	Yes	Yes
	Current scale, high gain	ppm	Yes	Yes
	Voltage offset, channel 1	nV	Yes	Yes
	Voltage offset, channel 2	nV	Yes	Yes
	Voltage scale, channel 1	ppm	Yes	Yes
	Voltage scale, channel 2	ppm	Yes	Yes
	Cable resistance, channel 1	uOhm	Yes	Yes
	Cable resistance, channel 2	uOhm	Yes	Yes
TSO	No. characters X	Number	No	Yes
	No. characters Y	Number	No	Yes
	No. overlay pixels X	Pixels	No	Yes
	No. overlay pixels Y	Pixels	No	Yes
	Overlay origin X	Pixels	No	Yes
	Overlay origin Y	Pixels	No	Yes
	Start character field X	Pixels	No	Yes
	Start character field Y	Pixels	No	Yes
	Border thickness	Pixels	No	Yes
	Screen rotation	Enum	No	Yes
	Colour map	Dict	No	Yes
	Video source	Enum	No	Yes
	Screen type	Enum	No	Yes
Inet	DNS primary	IP	No	Yes
	DNS secondary	IP	No	Yes
	UDP multicast address	IP	No	Yes
	UDP multicast port	Number	No	Yes
	NTP primary	url	No	Yes
	MQTT broker	IP/url	No	Yes

	Update server	IP/url	No	Yes
	Update file path	String	No	Yes
	Static IP	IP	No	Yes
	Static IP net mask	IP	No	Yes
	Static IP gateway	IP	No	Yes
Comissioning	Comissioned Flag	Bool	Yes	Yes
	Serial Number	Number	Yes	Yes
	Tamper Seal ID 1	Number	Yes	Yes
	Tamper Seal ID 2	Number	Yes	Yes
	CPO public key	-	Yes	Yes
	Default tarriff name	String	Yes	Yes
	Time Zone config	Dict	No	Yes

Other EEPROM Data

There is some data stored in EEPROM that is not configuration or set via the TTI:

PCBA Test

This will be indirectly written during outsourced production.

It is used to verify the hardware functionality of the board and the result is checked before commissioning.

Through the TTI, when the PCBAs are in a dedicated jig, this test is initiated and automatically checked whether results are in range.

It checks numerous hardware elements on the board.

The EEPROM contains parameters which are checked by the firmware and given a pass/fail result.

Sub-assembly Test

This will be indirectly written during outsourced production.

It is used to verify the hardware functionality of the DC Meter and the result is checked before commissioning.

Through the TTI, when the DC Meter is assembled, this test is initiated and automatically checked whether results are in range.

It checks the fan speed and shunt temperature.

The EEPROM contains parameters which are checked by the firmware and given a pass/fail result.

Private Key Data

This will be written automatically during production after communicating with the Trust Anchor.

Private key is never displayed, exported, or read externally from the DC Meter at any time but is saved to internal EEPROM. A new private key may be generated, but never seen. The parameter data change for a key generated is not logged to the audit log for security reasons.

Trust Root Key Data

This will be written automatically during production whilst communicating with the Trust Anchor.

It is used to prove the identity of the meter and verify the identity of authentic TSOs. It contains:

- Trust root public key
- Device public key
- Signature of above by the trust root

Time on the DC Meter

Overview

Time and timekeeping on the DC Meter is critical to a number of functions.

Time is kept by an internal accurate RTC with external crystal. Real calendar time (UTC) is obtained by syncing to an NTP server on startup and every 6 hours when not charging.

The accuracy of time passing is critical to the overall metrological accuracy, but is not affected by accuracy to real calendar time.

UTC time is used to timestamp logs for audit events and charge sessions. Time is also displayed to the user on the TSO, corrected for local time.

Validity Period

The onboard crystal's accuracy ensures that the DC Meter time will be within the acceptable tolerance if an NTP sync was successful in the last 30 days.

The onboard super capacitor powers the RTC even when external power has been removed, so time remains valid as long as power is restored within 7 days.

Timezone

To reliably display local time on the TSO, time zones must be accounted for including daylight savings.

The time zone rules are used to determine what the present offset from UTC is in effect.

Time zone rules can be configured so any timezone can be accounted for. 2 rules are configured, standard time and daylight savings time.

Time zone configuration parameters for each rule include:

- Time zone abbreviation
- Hour/Minute offset from UTC
- Type - standard/daylight savings
- Ordinal day rule applies - first/last/always
- Weekday which rule transitions
- Month which rule transitions
- Hour/Minute which rule transitions
- Time Reference when rule transitions - UTC or local

Trusted Technician Interface (TTI)

Overview

The Trusted Technician Interface (TTI) is a tool used to configure, test and commission the DC Meter.

This functionality forms a command interpreter which recognises valid commands and rejects invalid ones.

The TTI has two parts, the embedded firmware running on the meter and the GUI application running on a PC which communicate over TLS 1.2. The meter interacts with the GUI using a defined interface and the meter validates any commands sent to it. The GUI is not considered legally relevant software.

The TTI will authenticate the allowable commands by using extensions within an x.509 certificate. These extensions can be used to limit access to a reduced set of commands,

Certificate Usage

The x.509 certificates passed to the TTI will be verified using a root CA pre-loaded into the image on the DC Meter. A valid certificate must be provided for the TTI to allow access.

By using certificate extensions, access can be allowed or restricted to any of the valid commands. By further use of extensions, two types of certificates will be created: production and customer. Production certificates will only be accepted by the DC Meter if it is not yet commissioned.

Customer certificates will only ever be allowed to be generated as short lived (expiry in hours) and with reduced permissions. This reduces the security risk of a certificate being copied and being obtained by unauthorised personnel.

Production certificates will be long lived, with all permissions but should not be taken outside production facilities. By marking as production certificate it reduces the security risk of being copied and obtained by unauthorised personnel as it will not be valid for units in the field.

X.509 Certificate Extensions

Certificate extensions are embedded into the X509 certificate when the certificate is first generated. Certificate extension may be viewed, along with other certificate properties. Certificate extensions are not secret. Editing a certificate extension will invalidate the certificate signature, and thus render the certificate unusable.

In order to provide the granularity required for the TTI, a methodology called Capability Security Model is being used. It is based on the following principles:

1. There are no user groups or roles. Each "user" is assigned their capabilities individually.
2. There are no stored lists of users.
3. Authentication is granted for each user based on X.509 certificates, signed by Tritium (or a Tritium approved CA). The DC Meter has a preloaded CA and how to check X.509 certificates with them.
4. Capabilities are a combination of:
    a. permitted actions (read/write/execute) and
    b. services to perform the actions on.
5. To ensure capabilities are not shared to people they shouldn't be, the user privileges form part of the certificate.

The figure below (from https://knowledge.digicert.com/solution/SO16297.html) summarises how certificate chaining works.

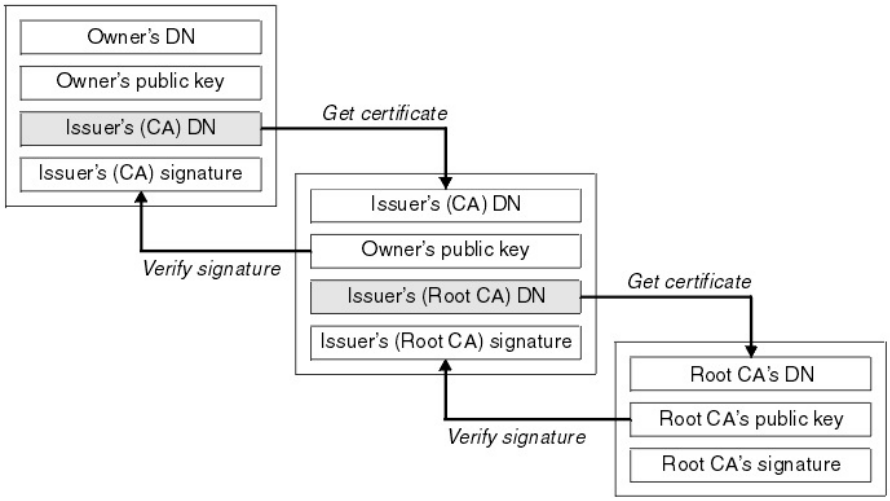


Figure – Certificate chaining example

Commands

The complete list of allowable commands for the TTI interface are as below:

Command	Purpose	Type (rwx)	Mask Offset (for cert gen)
TLSSERVER_COMMAND_GET_TELEMETRY	Gets all telemetry information from DC Meter	r	0
TLSSERVER_COMMAND_GET_CONFIG_AFE	Gets the configuration parameters of the AFE	r	1
TLSSERVER_COMMAND_GET_CONFIG_TSOIF	Gets the configuration parameters of the TSO	r	2
TLSSERVER_COMMAND_GET_CONFIG_INET	Gets the configuration parameters of the Inet	r	3
TLSSERVER_COMMAND_GET_PCB_ASSEMBLY_TEST_LOG	Gets the results of the PCBA test	r	4
TLSSERVER_COMMAND_GET_SUB_ASSEMBLY_TEST_LOG	Gets the results of the Sub-assembly test	r	5
TLSSERVER_COMMAND_GET_OCMF_LOG_INIT	Starts process to get OCMF logs	r	6
TLSSERVER_COMMAND_GET_OCMF_LOG	Gets the OCMF logs	r	7
TLSSERVER_COMMAND_GET_EVENT_LOG_INIT	Starts process to get event logs	r	8
TLSSERVER_COMMAND_GET_EVENT_LOG	Gets the event logs	r	9
TLSSERVER_COMMAND_GET_EVENT_LOG_SIGNATURE	Gets the event log signature	r	10
TLSSERVER_COMMAND_GET_CONFIG_COMMISSION	Gets the configuration parameters of the commissioning	r	11
TLSSERVER_COMMAND_SET_CONFIG_AFE	Sets the configuration parameters of the AFE	w	1
TLSSERVER_COMMAND_SET_CONFIG_TSOIF	Sets the configuration parameters of the TSO	w	2
TLSSERVER_COMMAND_SET_CONFIG_INET	Sets the configuration parameters of the Inet	w	3
TLSSERVER_COMMAND_SET_CONFIG_COMMISSION	Sets the configuration parameters of the commissioning	w	4
TLSSERVER_COMMAND_START_PCB_ASSEMBLY_TEST	Starts the automated PCBA test	x	0
TLSSERVER_COMMAND_START_SUB_ASSEMBLY_TEST	Starts the automated subassembly test	x	1
TLSSERVER_COMMAND_GET_PCB_ASSEMBLY_TEST_STATE	Gets the current status of the PCBA test	x	2
TLSSERVER_COMMAND_GET_SUB_ASSEMBLY_TEST_STATE	Gets the current status of the Subassembly test	x	3
TLSSERVER_COMMAND_UPDATE_START	Starts the automated firmware update process	x	4
TLSSERVER_COMMAND_UPDATE_GET_STATUS	Gets the current status of the firmware update process	x	5

GUI

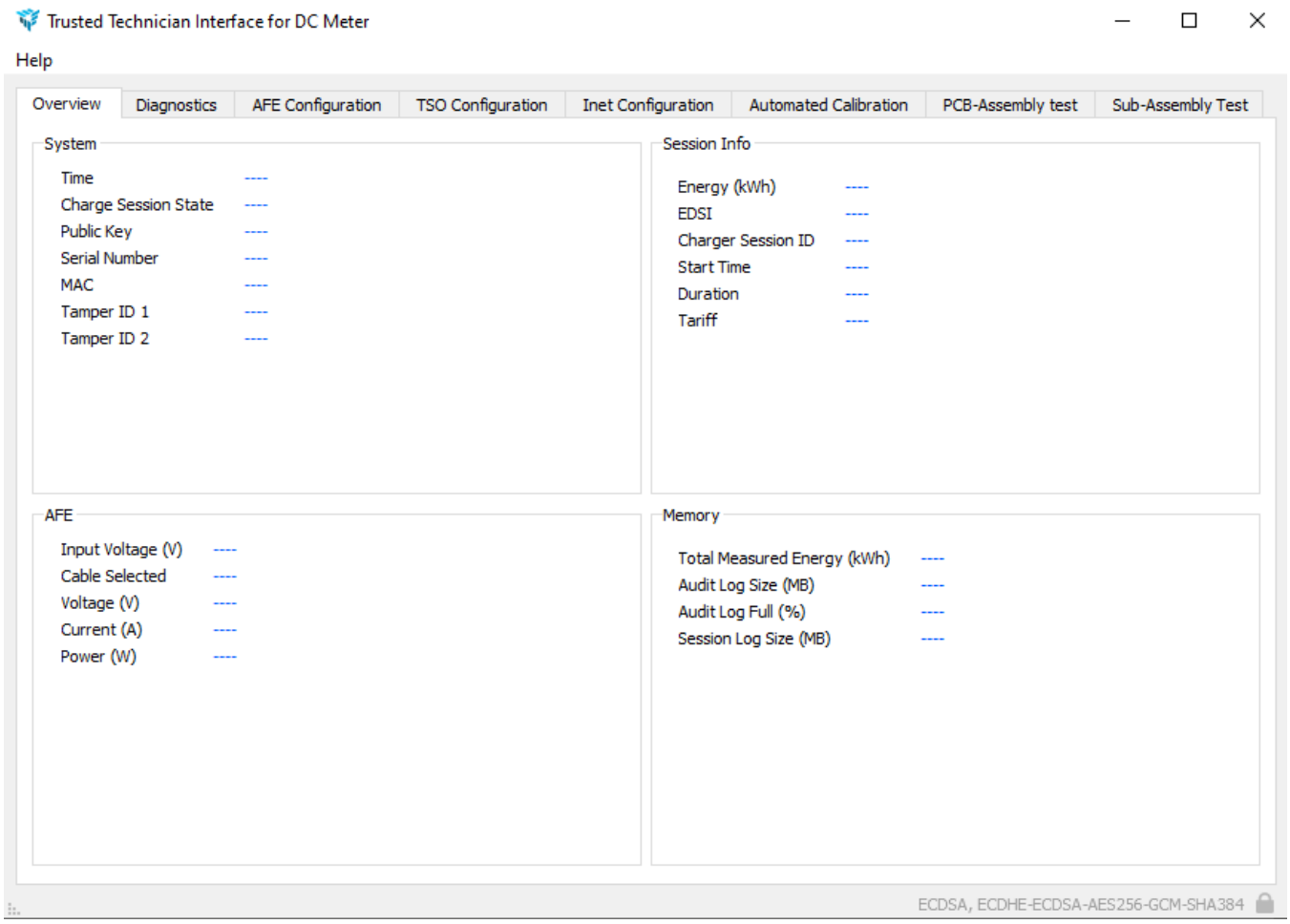
The TTI GUI provides an interface for a user to easily access information from the DC Meter. TTI GUI will dynamically allocate menu based on the certificate permissions provided.

Tabs' Purpose

- Overview and Diagnostics - provide telemetry data
- AFE, TSO, Inet, Commissioning Configuration - all allow configuration to be read and written
- Automated Calibration - allows sequence to be run in a production jig to calibrate the meter
- PCB/Sub Assembly Test - allows sequence to be run in a production jig to verify hardware build of the meter
- Logs - allows extraction of charge session and audit event logs

TTI Screenshots:

Overview:



Diagnostics:

Overview | Diagnostics | AFE Configuration | TSO Configuration | Inet Configuration | Automated Calibration | PCB-Assembly test | Sub-Assembly Test

Task Manager

Heap

Free Size	60304
Minimum Ever Free Size	53168

Tasks

Name	CPU Load	Stack
IDLE	94.7	231
tlsServer	2.8	128
mqttClient	1.0	156
cssm	0.9	404
lwipCore	0.2	762
SAFEMsgTask	0.2	335
demoTask	0.1	108
lwipEmac	0.0	206
udpClient	0.0	80
updateClient	0.0	463
tsoif	0.0	190
ntpClient	0.0	38
assemblyTest	0.0	349
timerService	0.0	100

Thermal

Shunt Temp (°C)	34.4
Amb Temp (°C)	41.1
AFE Temp (°C)	70.0
Fan 1 State	0
Fan 2 State	0
Fan 1 RPM	0
Fan 2 RPM	0

Build Information

Git Branch	TDM-2253
Git Hash	eb5e60eac7e91b6a4b9c1e352d887e920485a6b8
Build Date	Apr 17 2020
Build Time	09:12:58
Application Hash	750c2fe9d7ce7970c534c3d2b43ad012d2673c604e1db39fd41dd159ea20cc4c
Firmware Version	1.005

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End User Manual

General Information

For Electric Vehicle Supply Equipment (EVSE) in some markets, laws exist which require the regulation of billing on public charging facilities.

This manual outlines how a user can check their bill inline with these requirements.

During charging a certified DC meter will generate a digital receipt with a cryptographic signature to ensure it has come from a trusted source. This signature can then be checked with the certified transparency software. Once verified, it is ensured that all information in the receipt was generated by the certified DC meter and has not been tampered with.

The interactions with metering and the user are mostly between the Charge Point Operator (CPO) and the user. For support please contact your CPO.

Transparency Software

Transparency software and instructions for use can be obtained at:

https://www.safe-ev.de/en/transparency_software.php

Activities at the charger

To successfully check the billing receipt you must do the following steps at the charger:

1. Start a charge on your vehicle as usual
    a. Follow the prompts on the charger's LCD screen to begin the charging session.
    b. The DC meter will automatically detect the start of charge based on measured power output to the vehicle. There is no need for any further user input.
2. Either during or after the charging session, take note of the 'Session Signature' displayed on the legally relevant display
    a. It is recommended that a photo is taken to guarantee that the information is correctly recorded.
    b. The three screens shown below will cycle continuously, displaying for 5 seconds each.







Verifying the bill

To verify the billing receipt you must:

- Obtain the billing receipt from the CPO
    - The method of receiving this will vary depending on the provider
- Download and read the instructions of the transparency software
- Check that the 'Session Signature' included in the bill is the same as noted at the charging station
- Use the transparency software to load the billing receipt and verify the data is trusted
- Once the data is confirmed as trusted, all information in the SAFE billing receipt can be used to verify the CPO pricing model
    - 'Energy Delivered' and 'Duration' of charge session is typically used to calculate pricing by the CPO
    - Pricing models will vary depending on the provider

Display Icons

Below is the explanation of the icons on the legally relevant display from the metering device to assist in extracting the relevant information.

Icon	Purpose	Explanation
	Public Key Fingerprint	The public key of the metering device. Can be used to verify the authenticity of the digital receipt
	Play - Charge	Indicates the information about a charge session's start/stop times
	Record - Charge	Indicates the information recorded during a charge session
	Tariff Name	The name of the tariff applied (this can be blank)
	Session Signature (Receipt)	Unique ID of the charging session. Used to link to digital receipt provided by CPO
	Firmware Hash	Can be used to verify the identify of firmware version
	Present Time	Present local time on the metering device
	Start Time	Local time when charge session started
	Stop Time	Local when charge session ended
	Duration	Duration of a charge session

	Energy Delivered	Cumulative total of energy delivered in a session
	Firmware Version	Version of firmware the metering device is running
	Error: No communication	Screen has lost connection with metering device. Note: Will appear on unit when booting up. When on this screen the firmware version displayed is the version of the Trusted Screen Overlay
	Error: Out Of Service	Metering device is out of service

CPO DC Meter Manual

This page defines the interactions that the CPO can have with the DC Meter.

- Requirements
- SAFE Initiative Data
    - OCMF Compatibility
    - Transport Mechanism
        - has.to.be DataTransfer message for initial communication
        - Tritium-specific DataTransfer based protocol for SAFE Initiative Data
- Tariffs
    - Tritium-specific OCPP configuration setting for default Tariff information
    - Tritium-specific OCPP read-only configuration setting for active Tariff information

Requirements

The Charge Point Operator (CPO) must:

- Register DC Meter's public key with German government
- Provide the SAFE data receipt to end user as received from the DC Meter (when requested by user)
- Communicate the calculation of the tariff to the End User.

SAFE Initiative Data

The billing data determined by the DC Meter will be passed to the CPO as a file using the OCMF format to support the SAFE initiative.

This will be passed through OCPP, initially via a Tritium-specific DataTransfer message, and eventually via an OCA-approved enhancement to the main OCPP specification.

CPO must check the signature of received data to use for billing.

OCMF Compatibility

The DC Meter complies with the OCMF v1.0 (Open Charge Metering Format) which can be obtained here:

http://hers.abl.de/SAFE/Datenformat_OCMF/Datenformat_OCMF_v1.0.pdf

To support the flexibility of external billing arrangements, reading entries are created every 15 minutes of charging with energy reading value. The end of transaction reading entry will contain the total session energy usage.

Tritium's DC Meter makes use of two optional manufacturer extension fields:

- VS - in user assignment portion of main data - contains a "Session Signature" that is globally unique to the charge session.
- UT - contains the name of the tariff active in this charge (max 16 characters)

Transport Mechanism

The DC Meter signed SAFE data allows it to be transformed and go through intermediaries whilst still ensuring its authenticity and integrity. The DC Meter will send its SAFE message in base64 to the charger over MQTT which will then pass this message to the CSMS.

has.to.be DataTransfer message for initial communication

https://has-to-be.com/en/development-resources/transparenzsoftware/ocpp-integration/ covers sending initial info to has.to.be on startup

Tritium will add a new type of `SIGNED_OCMF`, that uses the DataTransfer below rather than sending individually signed meter values.

Tritium-specific DataTransfer based protocol for SAFE Initiative Data

As certified meters are required to independently detect the start and stop of energy transfer periods, there may be multiple meter sessions associated with each OCPP transaction.

Tritium will collect all the sessions associated with a transaction on the charger, and then send them to the CSMS as part of a single message at the end of the transaction.

Message from charger to CSMS following completion of charging session:

```
[2, "12341234", "DataTransfer",
  {
    "vendorId": "Tritium",
    "messageId": "CalibratedMeterSessions",
    "data": "{\"transactionId\":\"12345\",\"sessions\":[\"<signed OCMF payload for first energy transfer
period>\",\"<signed OCMF payload for second energy transfer period>\"]}"
  }
]
```

Payload from example DataTransfer message above (i.e. after removing the escape characters and decoding as JSON):

```
{
    "transactionId": "12345",
    "sessions": [
        "<signed OCMF payload for first energy transfer period>",
        "<signed OCMF payload for second energy transfer period>"
    ]
}
```

The CSMS will be expected to respond with a regular OCPP DataTransfer confirmation message with the status "Accepted" and no data payload.

Tariffs

The DC Meter supports a tariff name to be displayed locally on the Trusted Screen Overlay and logged with the charge session information (as the OCMF optional "UT" field in the first record in the signed transaction data). No fiscal calculations are supported.

The DC Meter is configured in production with a default tariff name that will be used on power up.

The CPO can choose to update the active tariff throughout the meter's life through OCPP. Alternatively the Tariffs can be left with the default name.

Tariff name changes are signed with the CPO's private key. The signature is verified with the CPO public key which is pre-written into the meter's memory also during production. The CPO public key will conform to SECP256R1 (Key Curve), ASN.1 DER (Format), RFC-5480 (Standard).

Once tariffs are sent to the meter they will be used until new tariffs are sent or a power cycle. The tariff for a given charge session will be set once power delivery has started and changing the active tariff after this time will only apply to the next charge session.

Tritium-specific OCPP configuration setting for default Tariff information

Whenever power is cycled, the certified DC Meter will always revert to its factory default tariff setting. To update the active tariff, Tritium offer a custom OCPP configuration setting, *"DefaultCalibratedMeterTariff"* that allows a CPO to provide a signed tariff name in the format "<TARIFF NAME>;<TARIFF SIGNATURE>".

The tariff name consists up to 16 alphanumeric ASCII characters. If less than 16 characters long it will be padded for display with leading or trailing spaces as appropriate. To be accepted as a valid tariff by the DC Meter, the tariff name must be followed by a semi-colon, and then a hexadecimal signature generated using the CPOs private key.

If this setting is empty, then the DC Meter will continue to use its existing tariff settings, and will revert to the factory default tariff setting if power cycled. There is not presently any specific mechanism provided to request that the meter revert to the factory default tariff setting.

If this setting is not empty, then its value will be sent to the DC Meter as the active tariff after the DC Meter first reports its serial number and public key to the OCPP client. The DC Meter will only accept the update if the given signature matches the given tariff name.

Tariff changes are not applied immediately if a charging session is in progress. In these cases, the tariff change will be queued internally and applied once the energy delivery session is completed.

Tritium-specific OCPP read-only configuration setting for active Tariff information

Changes to the default tariff may be rejected if the signature was created using a private key that does not match the public key installed in the meter. To assist in debugging these cases, additional read-only settings are published by the Tritium OCPP client:

- *HUx.DCMActiveTariff*: The tariff currently active in the DC Meter for a given user unit
- *HUx.DCMQueuedTariff*: Differs from the active tariff only when a tariff change is queued to become active at the end of the current energy delivery session
- *HUx.DCMCPOPublicKey*: The factory-installed public key that the DC Meter expects to be used to sign tariff changes

These settings will only be published if a user unit has a DC Meter installed.

Concept of Use for DC Meter Tariffs

CPO DC Meter Manual covers how to set the active tariff name via OCPP, but doesn't explain how Tritium expect tariff names to be used.

This supporting document covers those expectations.

Core Operating Concept

Tritium's core operating concept for DC Meter tariffs is that the role of the tariff name is to capture any information that is not otherwise stored as part of the signed OCMF records generated by the DC meter.

Tritium also expect that the tariff name may also capture information that could technically be inferred from the signed OCMF record, but only with difficulty.

The communication of the meaning of tariff names will take place external to the charger (e.g. on the CPOs website, or on a local information sticker). To allow for tariff changes without affecting reconciliation of historical transactions, it is also expected that some kind of version number would be included in the tariff names (at least for the second and subsequent versions of the tariff definition).

Note: not all tariff schemes described in this document are expected to be permitted in all regulatory environments. The use cases cover how to manage Tritium chargers in the event that the local regulatory environment *does* allow for the described tariff model, and a network operator has chosen to use it.

Case 1: Flat Tariff

- Example tariff names: *TRI_FLAT_1*, *TRI_FREE_1*
- Supported in pk.7.0.0 by setting *DefaultCalibratedMeterTariff* via OCPP or else via the factory default DC Meter setting.

This is the simplest possible case: the tariff name simply refers to a particular price per kilowatt hour. The signed OCMF records include the amount of energy delivered, allowing this to be calculated in a straightforward manner.

Free energy delivery is simply the special case where the price per kWh is zero.

Case 2: Time of Day Based Tariff Variation

- Example tariff names: *TRI_AEST_1*, *TRI_CET_1*
- Supported in pk.7.0.0 by setting *DefaultCalibratedMeterTariff* via OCPP.

When tariffs vary by time of day, they vary based on the *local* time where the charger is located, while the signed OCMF records will capture session times in UTC.

In theory, the time zone for the charging session can be inferred from the public key of the DC Meter (since the installation location of the meter must be registered with the appropriate authorities).

In practice, it may be clearer to encode the local time zone information as part of the tariff name.

The signed OCMF records published by the DC Meter capture the energy delivered in 5 minute increments for the first 6 hours, and then in 1 hour increments after that. Tariff definitions will need to cover how increments that span a tariff rate change are handled, with Tritium's expectation being that the most common practice will be to assume constant power delivery across the time interval, and charge for energy accordingly. However, some regulatory environments may require that the entire interval be charged at whichever tariff is lower. Either way, that information is expected to be part of the tariff definition, rather than requiring dynamic changes to the active tariff name.

Case 3: Average Power Based Tariff Variation

- Example tariff names: *TRI_SESSION_1*, *TRI_INTERVAL_1*
- Supported in pk.7.0.0 by setting *DefaultCalibratedMeterTariff* via OCPP or else via the factory default DC Meter setting.

As the signed OCMF records include both energy delivery and duration information, they can be used to calculate the average charging power for both the entire session and for each recorded interval within that session.

This means that billing based on the average power achieved can be handled as part of the tariff definition, making this case equivalent to a flat tariff from an operational perspective.

Case 4: Maximum Power Based Tariff Variation (external billing discounts permitted by regulators)

- Example tariff name: *TRI_PWR_LIMIT_1*
- Supported in pk.7.0.0 by setting *DefaultCalibratedMeterTariff* via OCPP or else via the factory default DC Meter setting.

Regulatory environments are expected to differ in whether or not they allow network operators to apply additional external surcharges and discounts beyond the base tariff based on energy delivered (for example, consider fuel discount vouchers and credit card surcharges that affect the actual price paid by drivers for fuel, but don't affect the nominal price-per-litre of that fuel).

In regulatory environments where at least external discounts are permitted, then maximum power based tariffs can be handled as flat tariffs at the highest rate, combined with a percentage or per kWh discount for accepting a lower power limit on the charging session.

For example, a network operator might charge $1.29 per kWh for a session at the full power a HPC station can deliver, but only $0.79 per kWh if the driver agrees to explicitly cap their session at a maximum charge rate of 50 kW. If acceptable to regulators, this scenario could be modeled as a flat tariff of $1.29 per kWh, with an optional external discount of $0.50 per kWh for accepting the lower power limit.

With this approach, while the signed OCMF data could be used to demonstrate that a particular session had gone above 50 kW for one or more measurement periods, it wouldn't be able to conclusively distinguish sessions that were hard limited to 50 kW on the EVSE side from those where unlimited power was permitted, but the EV simply never requested more than 50 kW.

Case 5: Maximum Power Based Tariff Variation (external billing discounts NOT permitted by regulators)

- Example tariff names: *TRI_HIGH_PWR_1*, *TRI_LOW_PWR_1*
- Not supported in pk.7.0.0
- Expected to be supported in pk.8.0.0 via the *SetDynamicPricing* transaction-specific *DataTransfer* message

If external billing discounts are not permitted in a particular regulatory environment, then dynamic agreements to a different charging rate will presumably need to be reflected in the tariff name set for that charging session. While not expected to be part of the initial DC Meter compatible OCPP client release, Tritium's intention is to support this use case through the *SetDynamicPricing* transaction-specific *DataTransfer* message.

The key operational challenge with this approach is that there's only a limited time window between the Charging Station Management System (CSMS) receiving a *StartTransaction* message indicating that a link has been established to the EV and a charge session successfully authorized, and the DC Meter locking in the active tariff due to the start of energy delivery to the vehicle. While that time window is several seconds long, the failure mode where the *SetDynamicPricing* message arrives too late still needs to be taken into account.

Consider the same example as in case 4, where a network operator charges $1.29 per kWh for a session at the full power a HPC station can deliver, but only $0.79 per kWh if the driver agrees to explicitly cap their session at a maximum charge rate of 50 kW. This scenario can be modeled as two distinct tariffs, one for the low power rate, and one for the high power rate. To ensure customers aren't overcharged for low power sessions, the low power tariff and smart charging profile can be set as default, and then *RemoteStart/SetChargingProfile* and *SetDynamicPricing* used to enable the higher rate when appropriate. If *SetDynamicPricing* is sent second, then an untimely loss of connectivity may lead to a user being undercharged, but it won't lead to them being overcharged.

With this approach, the signed OCMF data explicitly indicates which tariff was in effect at the time of the charger session, and this same tariff name would be displayed on the DC Meter trusted screen overlay while the charge session was in progress.

Case 6: Dynamic Free Mode Tariff Variation

- Example tariff name: *TRI_OFFLINE_1*
- Indicated in pk.7.0.0 by an OCPP transaction ID of "-1" appearing in the signed OCMF records for at least part of the session
- Potentially better supported in pk.8.0.0 by offering an *OfflineCalibratedMeterTariff* setting via OCPP that is used in preference to the default tariff when the OCPP link is down

In the initial implementation, the fact a charge session took place while the charger was offline needs to be inferred from the fact that there's no real OCPP transaction ID set in the signed OCMF records.

A new OCPP setting would allow a specific tariff name to be sent to the DC Meter when the charger determined that the OCPP link is down.

Techniques & Notes for Developers

- 1 Pairing of TSOs & DC Meters
  - 1.1 Trust Root System & Mock Trust Root
  - 1.2 Checking if TSO is Paired/Programmed
  - 1.3 Checking if a DC Meter is Paired/Programmed
- 2 Resetting the JTAG lockout

Pairing of TSOs & DC Meters

Trust Root System & Mock Trust Root

During IQC, there is a step that pre-loads a TSO and DC Meter with cryptographic information that allows them to communicate & verify their "trust" in each other.
The cryptographic pairing process can only be performed by the air-gapped trust root system (which is located in Amsterdam).
If a DC Meter is not paired to the installed TSO, then it will fail to connect (showing error TSO_NO_SESSION).

It is possible to upload firmware to a TSO and DC Meter with a **mock trust root**.
This will bypass the cryptographic trust validation, and can only be used in testing and development.
To create the firmware image for a mock-trusting meter or TSO, the following build options must be set:

DC Meter:

DCMCONFIG_MOCK_TRUST_ANCHOR=1
TSOCLIENT_FULL_SESSION_COUNTER_LIMIT=1

TSO:

MOCK_TRUST_ANCHOR=1

These build options can be modified within the CCS project using the "Project > Properties > Predefined Symbols" window.

Checking if TSO is Paired/Programmed

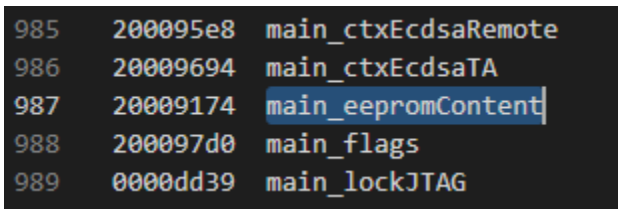
If there is no sticker on the TSO or DC Meter, it is impossible to tell visually if they have been paired or not.
Once the state of a TSO is known, the serial number of its paired DC Meter should be marked on the TSO sticker.
To check if a TSO is programmed, build the PK or ISD TSO release images from CCS, then start a debug session with "Load Symbols Only" enabled.

Once connected to the target, using the memory browser and check the following location in memory:

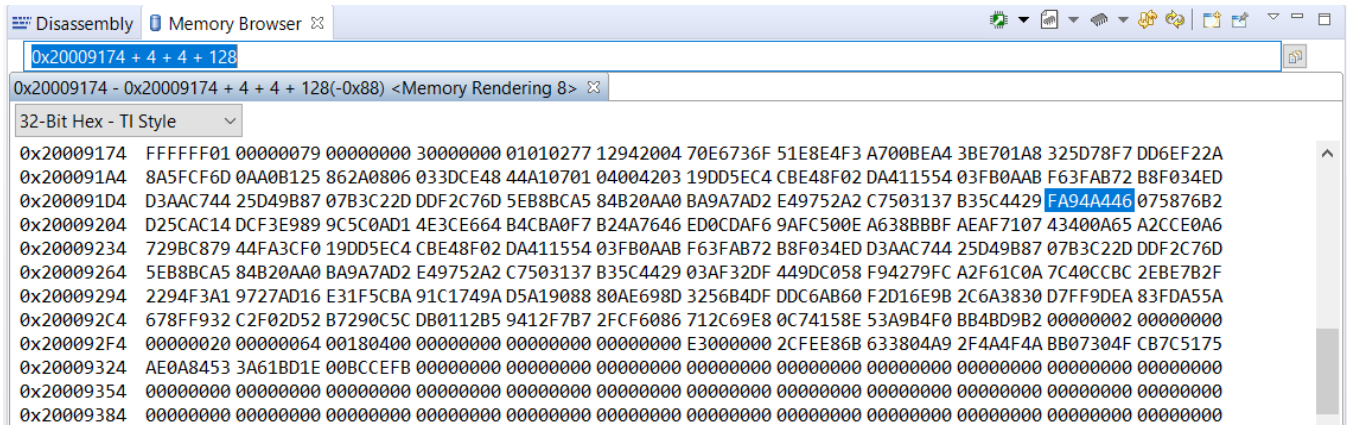
<root> + 4 + 4 + 128

The variable **<root>** is either 0x20009174 for ISD and 0x2000667c for PK at the time of writing.

The exact value of **<root>** can be confirmed by searching the *.map file from the build output for "main_eepromContent".
The address in the left column should show where the variable is mapped in static storage, as seen below:



If the memory looks like the following image, then your TSO is programmed. The next 64 bytes show the public key that was programmed into the EEPROM.
Note that you cannot access the EEPROM directly from CCS, because it's abstracted behind a memory controlled (i.e. not mapped into the CPU memory map).



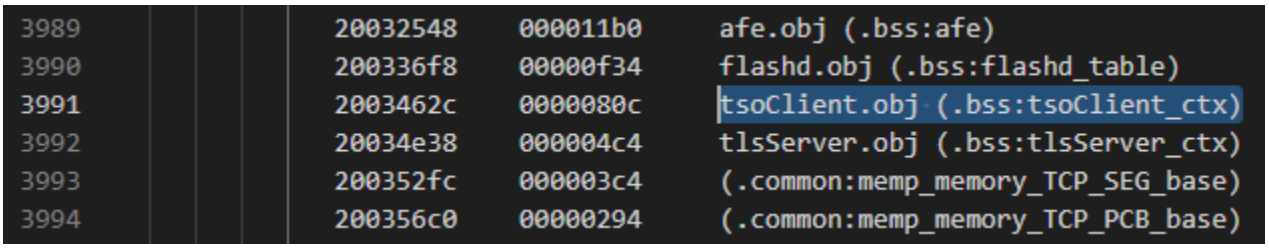
If the memory location contains "FFFFFFFF FFFFFFFF ...", then the EEPROM has not been programmed and the TSO will not work with any DC Meter.

Checking if a DC Meter is Paired/Programmed

The same technique as above is used - connect to the DC meter and use the memory browser to search for this location in memory:

<tsoClient_ctx> + 0x69B

The value of <tsoClient_ctx> can be found in the *.map file of the build by searching for "tsoClient_ctx".
See the below screenshot for an example from firmware version 1.012:

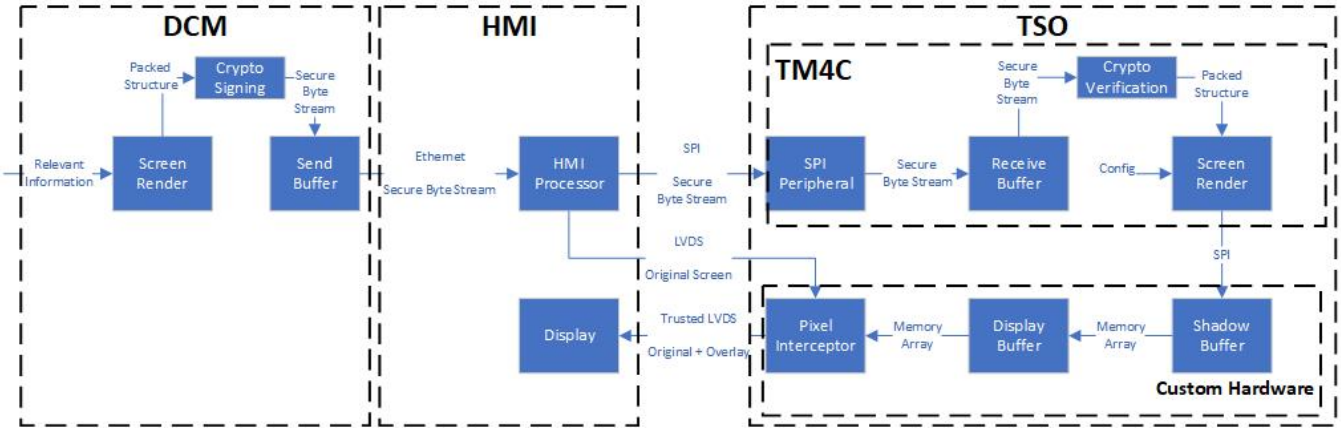


In the case shown above, the root memory address is **0x2003462C**, and 0x80C bytes have been reserved for it.
Note that you will need to allow the DC Meter to run with an active network connection before the public key is filled in.

Resetting the JTAG lockout

Refer to this wiki page: Unlocking JTAG port on TM4C device

Trusted Screen Overlay (TSO)



The TSO receives secure information from the DC meter and forces the legally relevant information onto the screen regardless of any other device /software.

The overlay of the TSO is clearly differentiable from the rest of the display.

Related pages

- Display Format
- FPGA IP core
  - Pixel interceptor to microblaze interface
- Pairing TSO and DCM
  - Trust Anchor Communication
- Rendering of overlay screen

Configuration

See DC Meter Configuration.

Firmware Version

The TSO firmware is programmed in the factory and is never changed. The firmware version is displayed on the screen whenever not connected to the DC Meter including during start-up. The firmware version covers both the TSO's TM4C and the FPGA/microblaze (aka Custom Hardware) and is manually controlled. The version will be incremented if there is any change to the source code. The version will be in a format of:

V[Major].[Minor].[Build]_[Variant] - e.g. V01.00.01_ISD

- Major version is incremented if there is a major change to functionality.
- Minor version is incremented if there is a minor change to functionality, or to reflect many build version level changes.
- Build version is incremented for any change to source code, normally used for insignificant changes/bug fixes.
- Variant represents the build configuration. There can be several build configurations for the same version. Current allowed variants are:
  - PK
  - ISD

The TSO build variant has a corresponding configuration parameter inside the DC Meter (See DC Meter Configuration, section "TSO Configuration"). If the DC Meter configuration is not matched with the firmware variant on the TSO, then the TSO will not display the legally relevant data and instead display a "No Connection" screen with the firmware version printed. The DC Meter will also go to "Out of Service".

Preventing Update

The TSO has several mechanisms to prevent changing firmware.

- TSO is installed in the charger behind a tamper evident seal
- TSO's TM4C
    - Firmware disables the programming header once key is generated and signed by trust root
    - No bootloader present in firmware
- TSO's FPGA
    - Device program is encrypted with security key, FPGA has encrypted key loaded into its memory. Can only be programmed with that key
    - Only communication interface with TM4C which will not bootload the device

Communications Between TSO and DC Meter Controller

Security

Pairing process

See Pairing TSO and DCM.

TSO requires the session to be initiated by the DC meter, otherwise the displayed message cannot be changed.

If TSO does not receive any frame with valid signature for 5 seconds or signature is incorrect it will go to disconnected mode.

Session Sequence

1. DCM will exchange its public key and digital signature provided from the Trust Root to the TSO. The DCM will poll the status of the TSO, which includes the TSO public key and digital signature. Each device will independently verify that their received key can be verified and trusted.
2. DCM will stop an active session (if any). After stopping a session, each device will reset their replay protection counters.
3. DCM will receive a random 32-bit session ID from the TSO. This session ID is to be used in each communication frame, until the session is stopped.
4. DCM will enter an active session with the TSO. The DCM will generate a 32-bit random number and send it to the TSO. The TSO will clear any previous errors and reply with the same random number supplied by the DCM. The DCM will confirm that the number is the same and continue the session.
5. DCM will load the Character Map into the TSO. The Character Map is too large to send in a single frame and is loaded into the TSO in fragments. To improve transfer efficiency, each Character Map fragment is intentionally not digitally signed by the DCM.
6. DCM will digitally sign a frame containing the hash of the Character Map. The TSO will verify the authenticity of the preceding Character Map using the supplied hash. Once the Character Map has been verified, the TSO will set the Character Map loaded bit in the Loaded-Elements byte of the TSO status.
7. DCM will send the Pixel Interceptor configuration. Once the configuration has been verified, the TSO will set the Configuration loaded bit in the Loaded-Elements byte of the TSO status.
8. DCM will send a message overlay to the TSO for rendering to the screen. At this point the session is active, the Character Map and Pixel Interceptor configuration is valid and loaded in the TSO.

Checking signatures

Frames that are checked for signatures:

- COM_STATUS_REQ (when requested in the frame)
- COM_CONFIG
- COM_MESSAGE
- COM_CHARACTER_MAP_HASH
- COM_START_SESSION_DCM
- COM_START_SESSION_TSO
- COM_STATUS_TSO

TSO status:

State	Value
SESSION_STATE_DEFAULT	0x00
SESSION_STATE_BLANK	0x01
SESSION_STATE_NO_SESSION	0x02
SESSION_STATE_SESSION_ACTIVE	0x04

TSO errors (more then one can be set at the same time):

Error	Value
ERROR_SIGNATURE	0x01
ERROR_CNT	0x02
ERROR_SESSION_ID	0x04
ERROR_CONFIG_INCORRECT	0x08
ERROR_HASH	0x10
ERROR_TIMEOUT	0x20
ERROR_STATE_SEQUENCE	0x40
ERROR_CORRUPTED_MEMORY	0x80

TSO loaded elements (more then one can be set at the same time):

Element	Value
CHARACTER_MAP	0x01
CONFIGURATION	0x02
REMOTE_KEY	0x04
CA_KEY	0x08
HOST_KEY	0x10
EEPROM_MEMORY	0x20

Frame format

When sending 16 or 32 bit values in the Data field make sure it is Least Significant Byte First (Little-Endian).

<table>
<tr><td colspan="7">Communication frame, constant length of 512 bytes</td></tr>
<tr><td colspan="5">Data section
444 bytes</td><td>Signature section
64 bytes</td><td>CRC section
4 bytes</td></tr>
<tr><td>Command</td><td>Frame number</td><td>CNT</td><td>Session ID</td><td>Data</td><td rowspan="2"></td><td rowspan="2"></td></tr>
<tr><td>1 byte</td><td>1 byte</td><td>2 bytes</td><td>4 bytes</td><td>436 bytes</td></tr>
</table>

Message Types

#	NAME	Description	Payload description
1	COM_STATUS_REQ	From DC meter to TSO, request for status or readout of last frame. This frame does not change anything in the state machine	Frame #: x Session ID: x CNT: x D.B.0: 0 - TSO will not check the signature, use this to get TSO readout 1 - TSO will check the signature, use it to indicate that there is a communication to DC meter and TSO should not go into out of service mode

<table>
<tr><td>2</td><td>COM_STATUS_TSO</td><td>TSO status frame, signed</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>Current session ID</td></tr>
<tr><td>CNT</td><td>Rolling counter</td></tr>
<tr><td>D.B. 0-3</td><td>Interceptor status</td></tr>
<tr><td>D.B. 4-7</td><td>Microblaze status</td></tr>
<tr><td>D.B. 8</td><td>TSO status</td></tr>
<tr><td>D.B. 9</td><td>TSO errors</td></tr>
<tr><td>D.B. 10</td><td>TSO loaded elements</td></tr>
<tr><td>D.B. 11</td><td>Reserved</td></tr>
<tr><td>D.B. 12-75</td><td>Trust Anchor public key (raw)</td></tr>
<tr><td>D.B. 76-139</td><td>TSO public key (raw)</td></tr>
<tr><td>D.B. 140-203</td><td>Signature of Trust Anchor and TSO public keys (raw)</td></tr>
</table></td></tr>
<tr><td>3</td><td>COM_MESSAGE</td><td>DC meter message data</td><td><table>
<tr><td>Frame #</td><td>Starts from 0, increment until the all the data are send</td></tr>
<tr><td>Session ID</td><td>Current session ID</td></tr>
<tr><td>CNT</td><td>Rolling counter</td></tr>
<tr><td>D.B. 0-247</td><td>Characters in the message</td></tr>
</table></td></tr>
<tr><td>4</td><td>COM_END_SESSION</td><td>End session command from DC meter</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>x</td></tr>
<tr><td>CNT</td><td>x</td></tr>
</table></td></tr>
<tr><td>5</td><td>COM_START_SESSION_DCM</td><td>Start new session, from DC meter</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>Session ID currently signaled by TSO</td></tr>
<tr><td>CNT</td><td>x</td></tr>
<tr><td>D.B. 0-3</td><td>DC meter random number</td></tr>
</table></td></tr>
<tr><td>6</td><td>COM_START_SESSION_TSO</td><td>Start new session, from TSO</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>Current session ID</td></tr>
<tr><td>CNT</td><td>0</td></tr>
<tr><td>D.B. 0-3</td><td>DC meter random number</td></tr>
</table></td></tr>
<tr><td>7</td><td>COM_PARING_REMOVE_DCM</td><td>Command to remove paired DC meter device, from DC meter to TSO, signed either by current DCM or TA</td><td>No payload</td></tr>
<tr><td>8</td><td>COM_PARING_PROMPT</td><td>From TSO signalling that there is no paired DC meter</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>Pairing session ID</td></tr>
<tr><td>CNT</td><td>Rolling counter</td></tr>
</table></td></tr>
<tr><td>9</td><td>COM_PARING_REQ</td><td>Frame from DC meter to TSO to pair the DCM, signed by TA</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>Current session ID</td></tr>
<tr><td>CNT</td><td>x</td></tr>
<tr><td>D.B. 0-63</td><td>TSO public key</td></tr>
<tr><td>D.B. 64-127</td><td>DCM public key</td></tr>
</table></td></tr>
</table>

<table>
<tr><td>10</td><td>COM_CONFIG</td><td>Frame with configuration information for TSO; if configuration is acceptable it triggers the change and communicates changes to interceptor</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>Current session ID</td></tr>
<tr><td>CNT</td><td>Rolling counter</td></tr>
<tr><td>D.B. 0</td><td>Number of characters X</td></tr>
<tr><td>D.B. 1</td><td>Number of characters Y</td></tr>
<tr><td>D.B. 2-3</td><td>Overlay field size X in pix</td></tr>
<tr><td>D.B. 4-5</td><td>Overlay field size Y in pix</td></tr>
<tr><td>D.B. 6-7</td><td>Overlay field position X in pix (absolute)</td></tr>
<tr><td>D.B. 8-9</td><td>Overlay field position Y in pix (absolute)</td></tr>
<tr><td>D.B. 10-11</td><td>Character field position X in pix (relative to overlay field)</td></tr>
<tr><td>D.B. 12-13</td><td>Character field position Y in pix (relative to overlay field)</td></tr>
<tr><td>D.B. 14</td><td>Border thickness in pix</td></tr>
<tr><td>D.B. 15</td><td>Rotation:
0 - native position
1 - rotation (ISD option)</td></tr>
<tr><td>D.B. 16-39</td><td>Color map data</td></tr>
<tr><td>D.B. 40</td><td>LVDS source, 0-FPGA, 1-HMI</td></tr>
<tr><td>D.B. 41</td><td>Screen version: 0 - ISD
1 - PK350</td></tr>
</table></td></tr>
<tr><td>11</td><td>COM_CHARACTER_MAP</td><td>Frames containing character data, signature is ignored</td><td><table>
<tr><td>Frame #</td><td>Start with 0</td></tr>
<tr><td>Session ID</td><td>Current session ID</td></tr>
<tr><td>CNT</td><td>Rolling counter</td></tr>
<tr><td>D.B. 0-248</td><td>Character map data</td></tr>
</table></td></tr>
<tr><td>12</td><td>COM_CHARACTER_MAP_HASH</td><td>Frame with hash of the whole character map data, signed</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>Current session ID</td></tr>
<tr><td>CNT</td><td>Rolling counter</td></tr>
<tr><td>D.B. 0-3</td><td>Number of bytes in character map</td></tr>
<tr><td>D.B. 4-36</td><td>Hash of character map data</td></tr>
</table></td></tr>
<tr><td>13</td><td>COM_KEY_DCM</td><td>Frame with DCM public key</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>x</td></tr>
<tr><td>CNT</td><td>x</td></tr>
<tr><td>D.B. 0-63</td><td>Trust Anchor public key</td></tr>
<tr><td>D.B. 64-127</td><td>DCM public key</td></tr>
<tr><td>D.B. 128-191</td><td>Signature of Trust Anchor public key and DCM public key from Trust Anchor</td></tr>
</table></td></tr>
<tr><td>14</td><td>COM_TRUST_ANCHOR_REQ</td><td>Request from TA to send key for signature</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>x</td></tr>
<tr><td>CNT</td><td>x</td></tr>
</table></td></tr>
</table>

<table>
<tr><td>15</td><td>COM_TSO_TRUST_ANCHOR_RESPONSE</td><td>Response from TSO with its public key, unique ID and signature of those two values</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>x</td></tr>
<tr><td>CNT</td><td>x</td></tr>
<tr><td>D.B. 0-11</td><td>Reserved</td></tr>
<tr><td>D.B. 12-75</td><td>TSO public key</td></tr>
<tr><td>D.B. 76-139</td><td>TSO unique ID</td></tr>
<tr><td>D.B. 140-204</td><td>Signature of public key and unique ID</td></tr>
</table></td></tr>
<tr><td>16</td><td>COM_TRUST_ANCHOR_RESPONSE</td><td>Response of trust anchor with signatures</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>x</td></tr>
<tr><td>CNT</td><td>x</td></tr>
<tr><td>D.B. 0-63</td><td>Trust Anchor public key</td></tr>
<tr><td>D.B. 64-127</td><td>TSO public key</td></tr>
<tr><td>D.B. 128-191</td><td>Trust Anchor signature of TSO public key</td></tr>
</table></td></tr>
</table>

Display Format

Overview

The TSO controlled section of the display will be structured as a fixed character array of X by Y characters. Each fixed size character will be mapped to a pixel map array.

This will allow the DCM to have control of the information displayed, whilst minimising the load on communications.

Iconography can still be used but any icon larger than 1 character will need to be converted into 1 character sections.

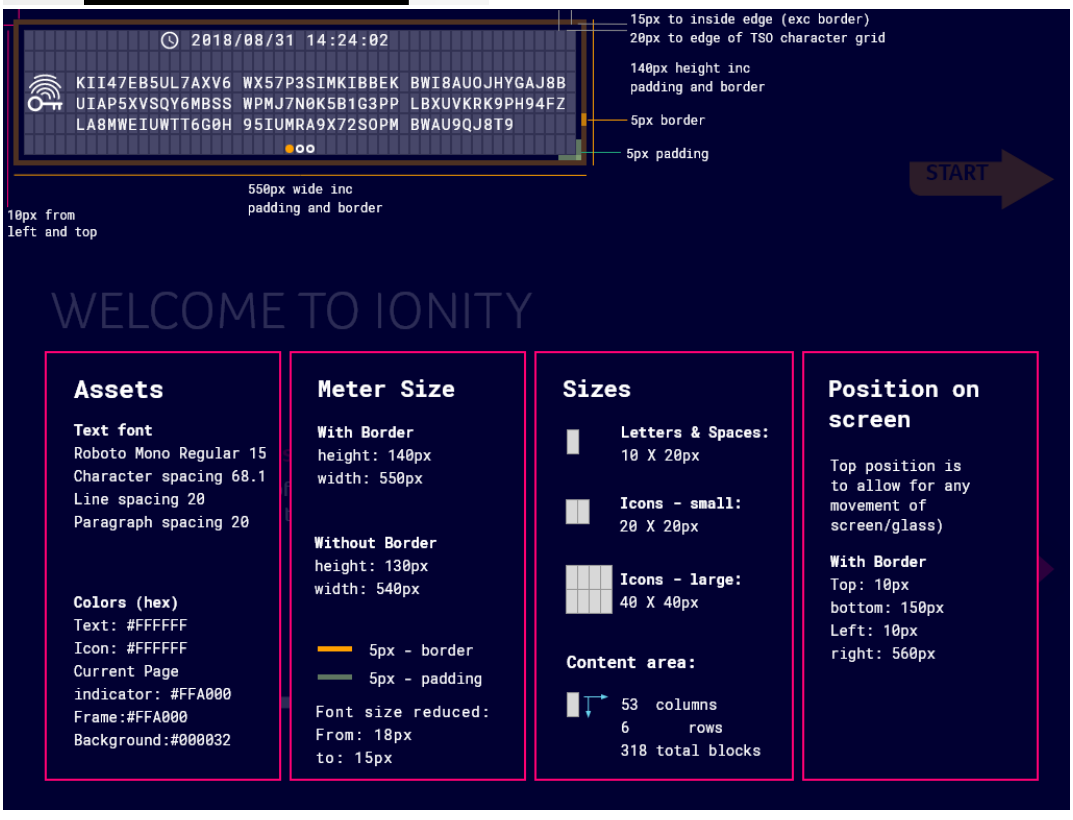
There will be eight colours loaded on startup: foreground colour (shades: 1-4), background colour, accent colour, red and green.

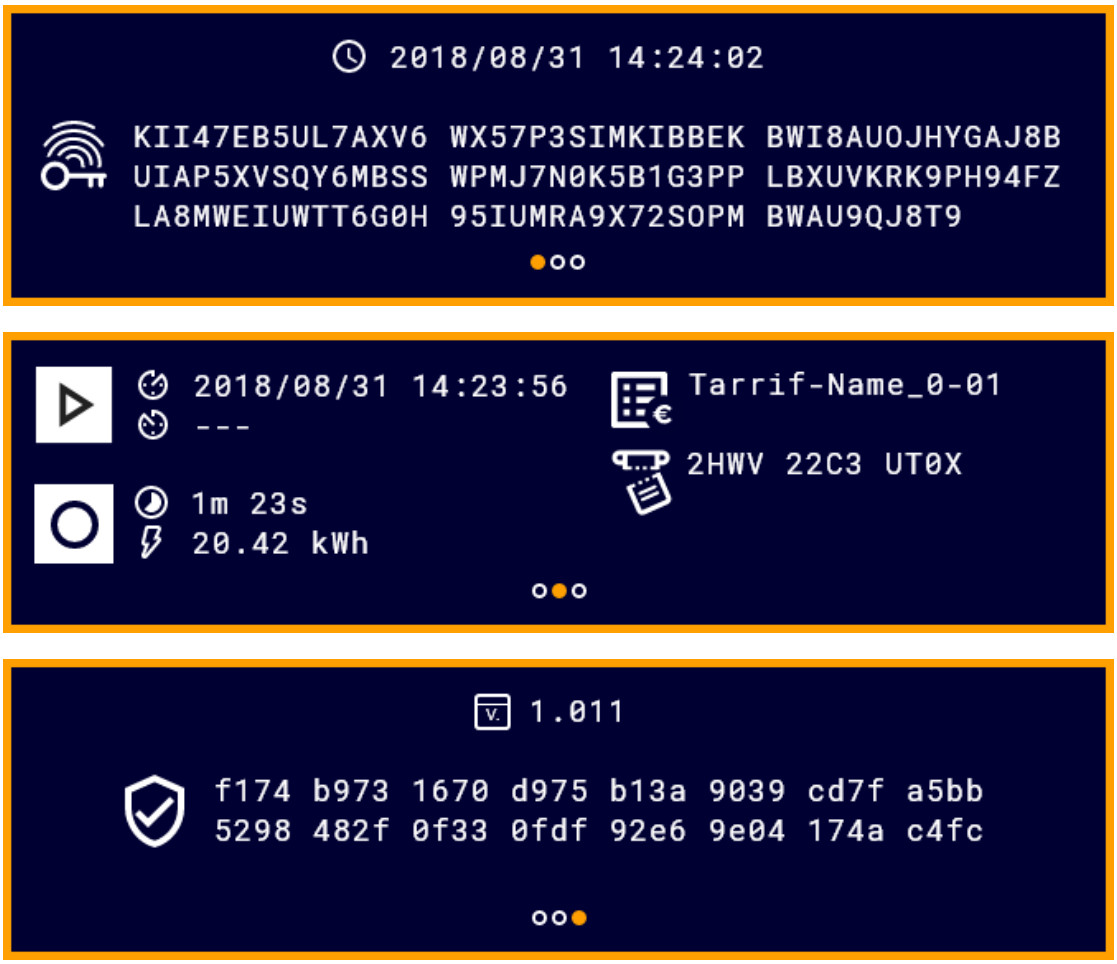
Session energy has sufficient room for more than 7 whole number digits in kWh and 3 decimal place digits.

The three screens shown are cycled through, each screen is displayed for 5 seconds.

Example:


Meter Size
With Border
height: 182px
width: 674px
Without Border
height: 167px
width: 659px
5px - border
Padding:
14px x 14px
Sizes
Letters & Spaces:
12 X 24px
Icons - small:
24 X 24px
Icons - large:
48 X 48px
Content area:
53 columns
6 rows
318 total blocks
Position on screen
Bottom position is to allow for any movement of screen/glass)
With Border
Top: 1091px
bottom: 1265px
Left: 67px
right: 734px
Assets
Text font
Roboto Mono Regular 18
Character spacing 66.67
Line spacing 24
Paragraph spacing 20
Colors (hex)
Text: #FFFFFF
Icon: #FFFFFF
Current Page indicator: #FFA000
Frame:#FFA000
Background:#000032
2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9


2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9
15px to inside edge (exc border)
20px to edge of TSO character grid
140px height inc padding and border
5px border
5px padding
550px wide inc padding and border
10px from left and top
START
WELCOME TO IONITY
Assets
Text font
Roboto Mono Regular 15
Character spacing 68.1
Line spacing 20
Paragraph spacing 20
Colors (hex)
Text: #FFFFFF
Icon: #FFFFFF
Current Page indicator: #FFA000
Frame:#FFA000
Background:#000032
Meter Size
With Border
height: 140px
width: 550px
Without Border
height: 130px
width: 540px
5px - border
5px - padding
Font size reduced:
From: 18px
to: 15px
Sizes
Letters & Spaces:
10 X 20px
Icons - small:
20 X 20px
Icons - large:
40 X 40px
Content area:
53 columns
6 rows
318 total blocks
Position on screen
Top position is to allow for any movement of screen/glass)
With Border
Top: 10px
bottom: 150px
Left: 10px
right: 560px



Responsibilities

DCM:

- Non-volatile storage of pixel map of characters
- Determines characters to display on update of screen
- Manages cycling of different screens

TSO:

- Loads pixel map of characters/colours on startup
- Checks validity of configuration
- When no connection to DCM displays static error screen
- Converts character array to pixel display

Upgrade/Customisation Impact

Change Required	Resultant Change	Involves Re-Certification Involvement
Change colours	Reconfiguration	No
New location for display area	Reconfiguration	No
Change font; within fixed character size	New DCM Image - Bootloadable	Yes?
Change font; outside fixed character size	New DCM Image; New FPGA Image	Yes?
Change information shown within display area	New DCM Image - Bootloadable	Yes?

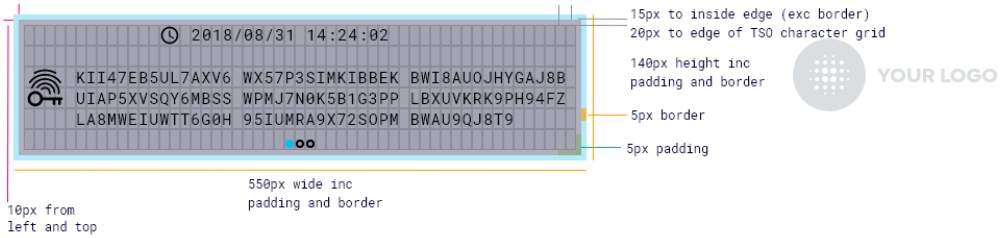
New screen (different pixel ratio/rotation/display area)	New DCM Image; New FPGA Image	Yes

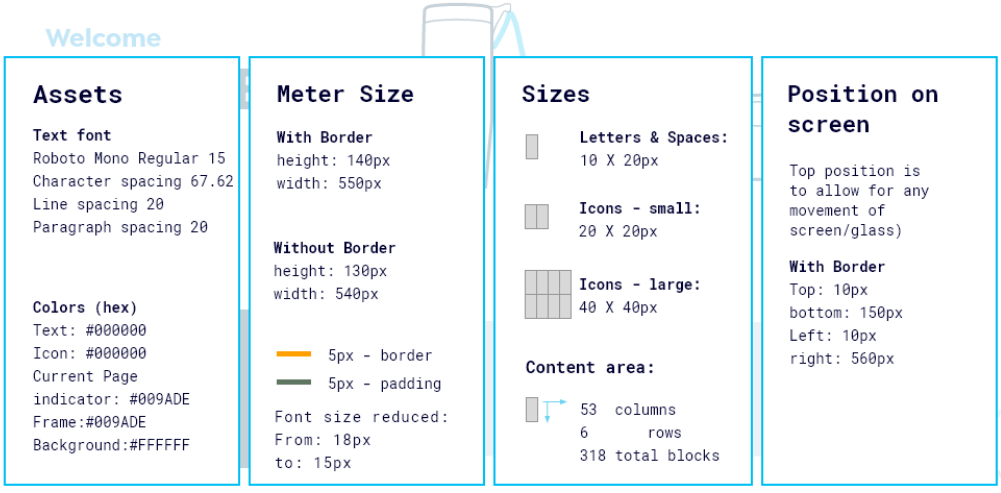
---

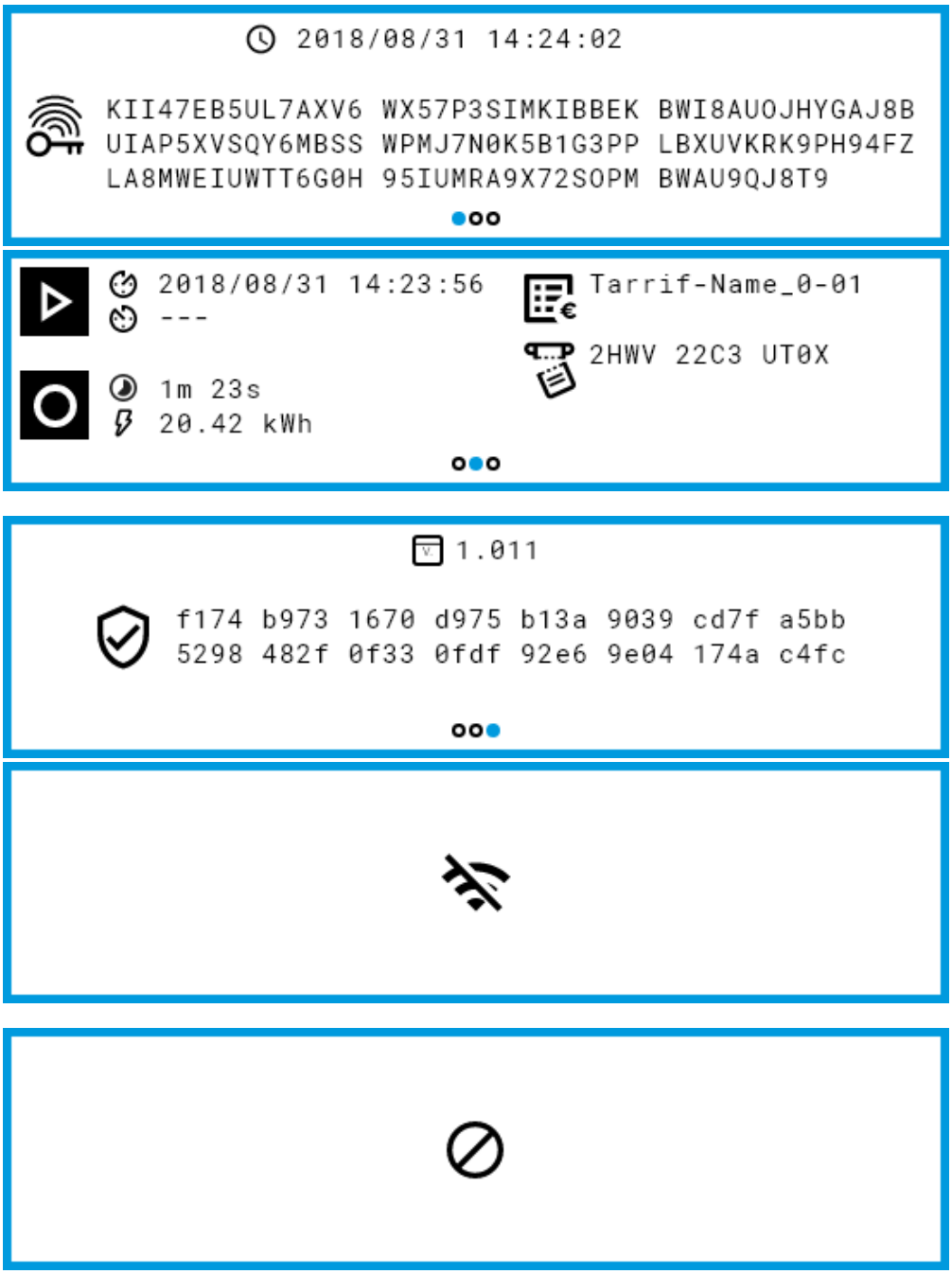
TSO Layout for Tritium's product range

Product group	Tritium PK Ionity PK	ISD	RT175s	MSC
HMI supplier	AiS	Touch Int.	Touch Int.	AiS
Resolution px WxH	1024x768	800x1280	1280x800	1280x800
Size mm WxH	305.2x229.0	136.6x217.0	217.0x136.6	217.0x136.6
Nb of TSO characters	53x6	53x6	53x6	53x6
character size px WxH	10x20	12x24	12x24	12x24
Padding px	5x5	14x14	14x14	0x0 (Tbc)
Border px	5	5	5	3
Overall size px WxH	550x140	674x182	674x182	642x150 (Tbc)

TRITIUM PK

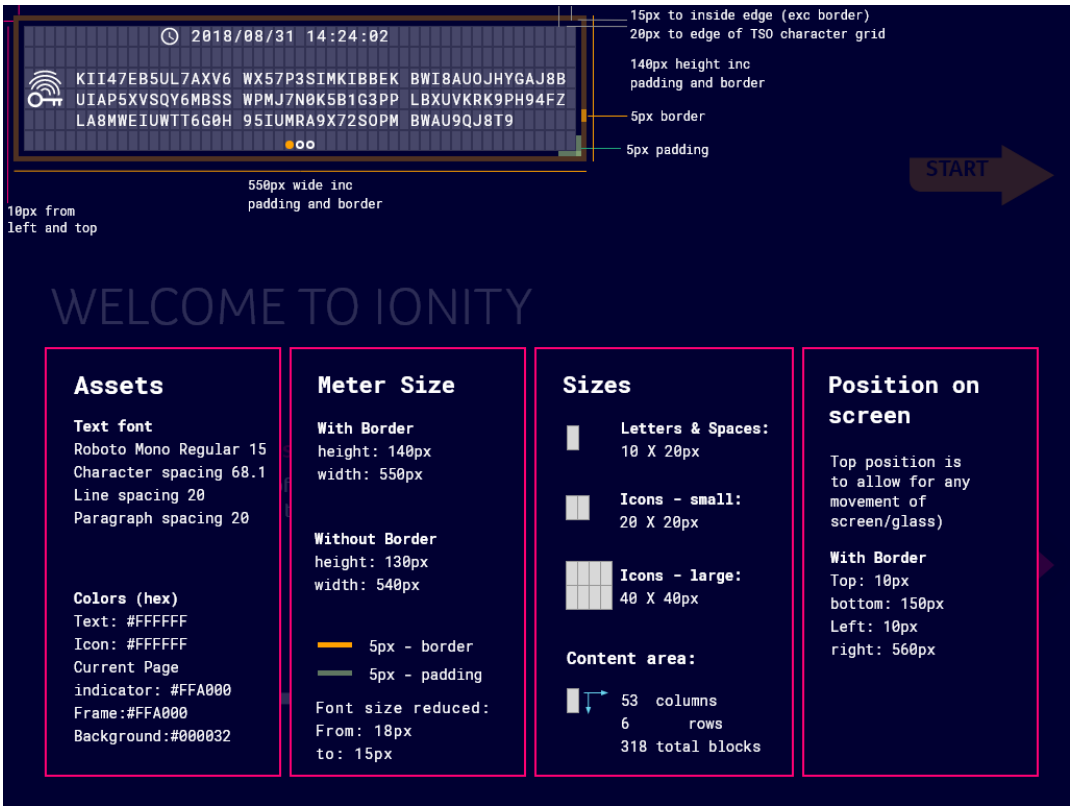


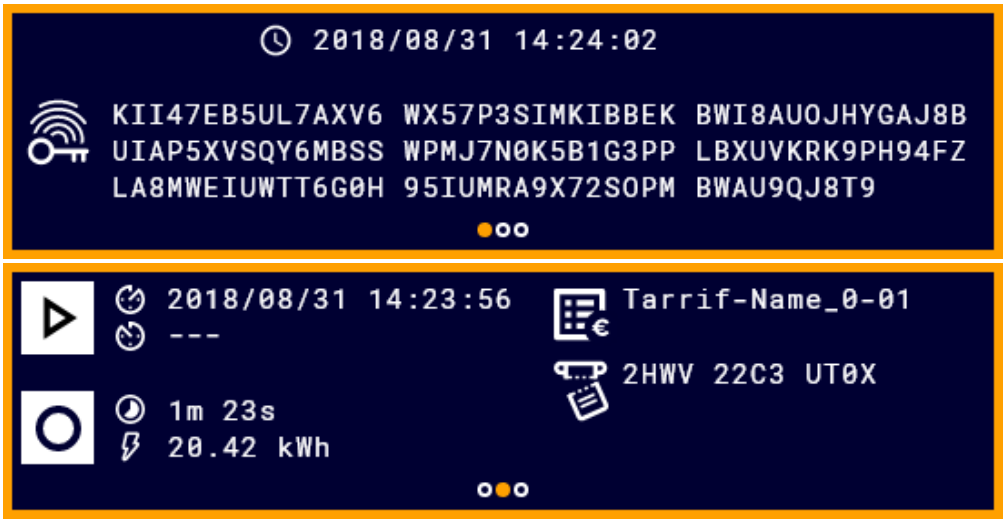



2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9
2018/08/31 14:23:56

Tarrif-Name_0-01
2HWV 22C3 UT0X
1m 23s
20.42 kWh
1.011
f174 b973 1670 d975 b13a 9039 cd7f a5bb
5298 482f 0f33 0fdf 92e6 9e04 174a c4fc

IONITY PK


2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9
15px to inside edge (exc border)
20px to edge of TSO character grid
140px height inc
padding and border
5px border
5px padding
START
550px wide inc
padding and border
10px from
left and top
WELCOME TO IONITY
Assets
Text font
Roboto Mono Regular 15
Character spacing 68.1
Line spacing 20
Paragraph spacing 20
Colors (hex)
Text: #FFFFFF
Icon: #FFFFFF
Current Page
indicator: #FFA000
Frame:#FFA000
Background:#000032
Meter Size
With Border
height: 140px
width: 550px
Without Border
height: 130px
width: 540px
5px - border
5px - padding
Font size reduced:
From: 18px
to: 15px
Sizes
Letters & Spaces:
10 X 20px
Icons - small:
20 X 20px
Icons - large:
40 X 40px
Content area:
53 columns
6 rows
318 total blocks
Position on screen
Top position is to allow for any movement of screen/glass)
With Border
Top: 10px
bottom: 150px
Left: 10px
right: 560px

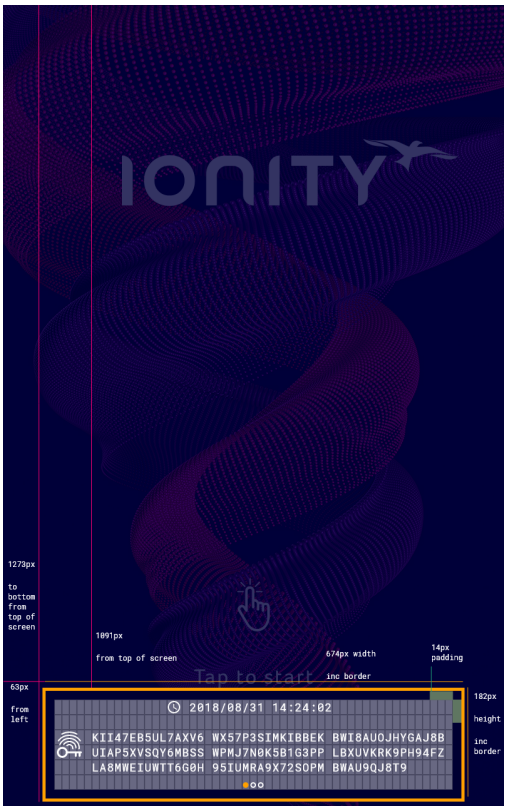

2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9
2018/08/31 14:23:56

Tarrif-Name_0-01
2HWV 22C3 UT0X
1m 23s
20.42 kWh


1.011
f174 b973 1670 d975 b13a 9039 cd7f a5bb
5298 482f 0f33 0fdf 92e6 9e04 174a c4fc

ISD


Meter Size
With Border
height: 182px
width: 674px
Without Border
height: 167px
width: 659px
5px - border
Padding:
14px x 14px
Sizes
Letters & Spaces:
12 X 24px
Icons - small:
24 X 24px
Icons - large:
48 X 48px
Content area:
53 columns
6 rows
318 total blocks
Position on screen
Bottom position is to allow for any movement of screen/glass)
With Border
Top: 1091px
bottom: 1265px
Left: 67px
right: 734px
Assets
Text font
Roboto Mono Regular 18
Character spacing 66.67
Line spacing 24
Paragraph spacing 20
Colors (hex)
Text: #FFFFFF
Icon: #FFFFFF
Current Page indicator: #FFA800
Frame:#FFA800
Background:#000032
Tap to start
2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9

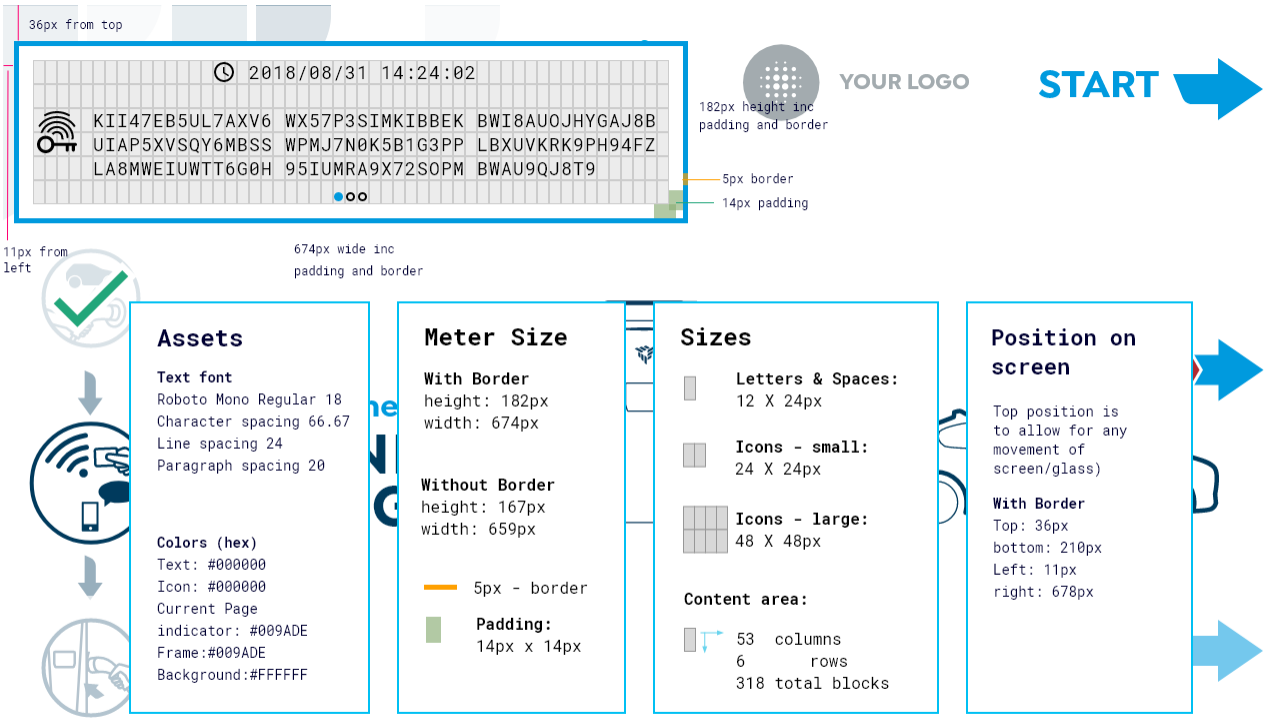

1273px
to bottom from top of screen
1091px
from top of screen
674px width
inc border
14px padding
63px
from left
Tap to start
2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9
182px
height
inc border

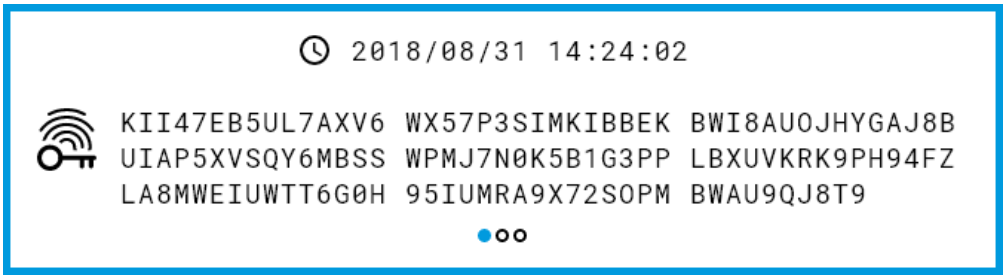

Tap to start
2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9

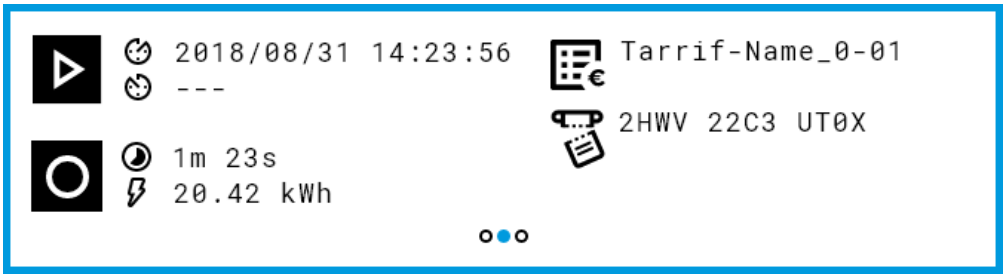

2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9
2018/08/31 14:23:56

Tarrif-Name_0-01
2HWV 22C3 UT0X
1m 23s
20.42 kWh
1.011
f174 b973 1670 d975 b13a 9039 cd7f a5bb
5298 482f 0f33 0fdf 92e6 9e04 174a c4fc

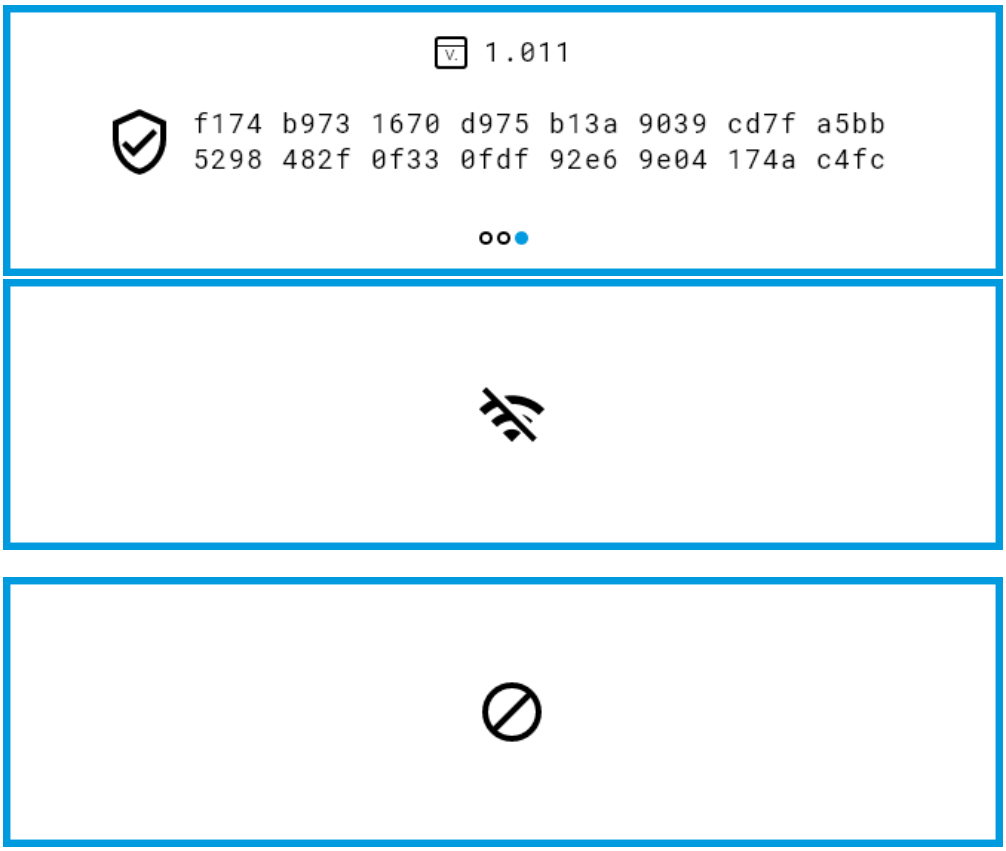
RT175s


36px from top
2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9
182px height inc padding and border
YOUR LOGO
START
5px border
14px padding
11px from left
674px wide inc padding and border
Assets
Text font
Roboto Mono Regular 18
Character spacing 66.67
Line spacing 24
Paragraph spacing 20
Colors (hex)
Text: #000000
Icon: #000000
Current Page indicator: #009ADE
Frame:#009ADE
Background:#FFFFFF
Meter Size
With Border
height: 182px
width: 674px
Without Border
height: 167px
width: 659px
5px - border
Padding:
14px x 14px
Sizes
Letters & Spaces:
12 X 24px
Icons - small:
24 X 24px
Icons - large:
48 X 48px
Content area:
53 columns
6 rows
318 total blocks
Position on screen
Top position is to allow for any movement of screen/glass)
With Border
Top: 36px
bottom: 210px
Left: 11px
right: 678px


2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9

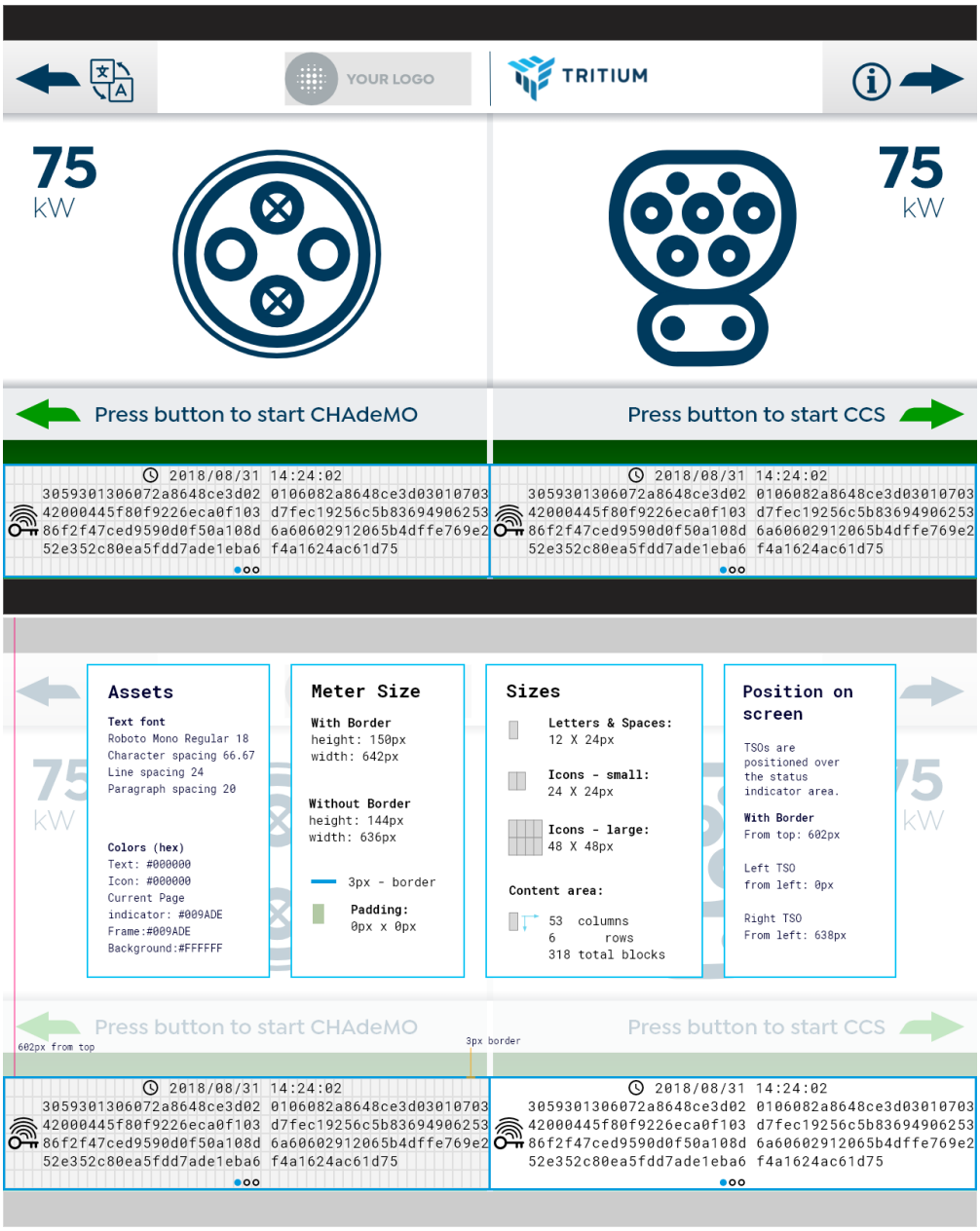

2018/08/31 14:23:56

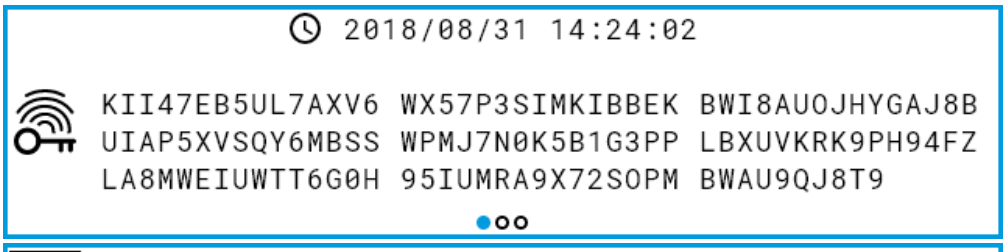
Tarrif-Name_0-01
2HWV 22C3 UT0X
1m 23s
20.42 kWh



MSC (RTM/PKM)

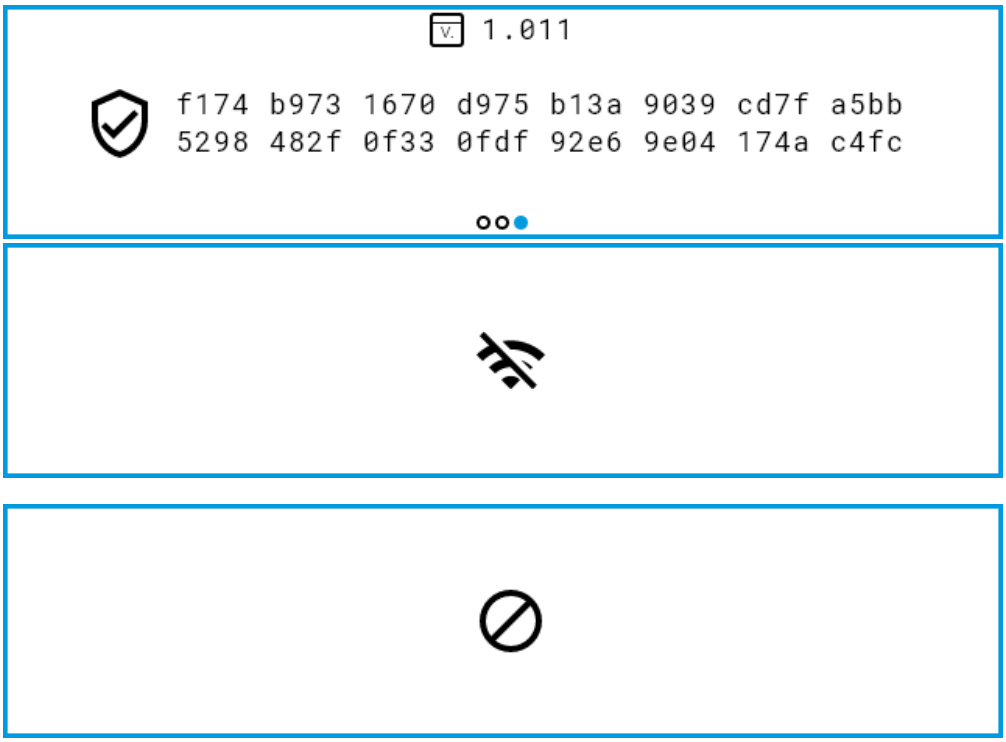
Current implementation (Nov 2021)


YOUR LOGO
TRITIUM
75
kW
75
kW
Press button to start CHAdeMO
Press button to start CCS
2018/08/31 14:24:02
3059301306072a8648ce3d02 0106082a8648ce3d03010703
42000445f80f9226eca0f103 d7fec19256c5b83694906253
86f2f47ced9590d0f50a108d 6a60602912065b4dffe769e2
52e352c80ea5fdd7ade1eba6 f4a1624ac61d75
Assets
Text font
Roboto Mono Regular 18
Character spacing 66.67
Line spacing 24
Paragraph spacing 20
Colors (hex)
Text: #000000
Icon: #000000
Current Page indicator: #009ADE
Frame:#009ADE
Background:#FFFFFF
Meter Size
With Border
height: 150px
width: 642px
Without Border
height: 144px
width: 636px
3px - border
Padding:
0px x 0px
Sizes
Letters & Spaces:
12 X 24px
Icons - small:
24 X 24px
Icons - large:
48 X 48px
Content area:
53 columns
6 rows
318 total blocks
Position on screen
TSOs are positioned over the status indicator area.
With Border
From top: 602px
Left TSO
from left: 0px
Right TSO
From left: 638px
602px from top
3px border

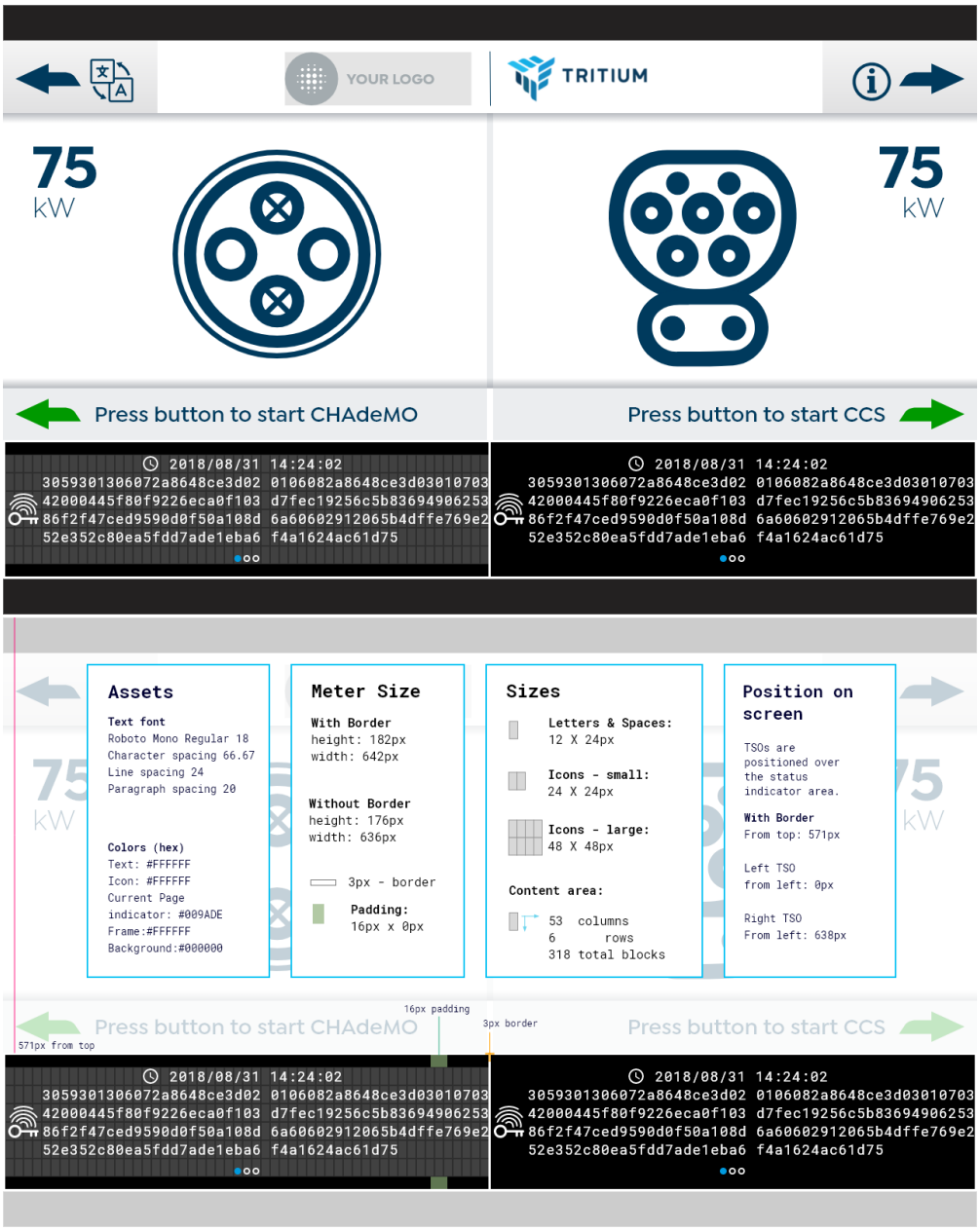

2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9


2018/08/31 14:23:56

Tarrif-Name_0-01
2HWV 22C3 UT0X
1m 23s
20.42 kWh



Alternative proposal with larger padding top and bottom to match place holder zone :


YOUR LOGO
TRITIUM
75
kW
75
kW
Press button to start CHAdeMO
Press button to start CCS
2018/08/31 14:24:02
3059301306072a8648ce3d02 0106082a8648ce3d03010703
42000445f80f9226eca0f103 d7fec19256c5b83694906253
86f2f47ced9590d0f50a108d 6a60602912065b4dffe769e2
52e352c80ea5fdd7ade1eba6 f4a1624ac61d75

Assets
Text font
Roboto Mono Regular 18
Character spacing 66.67
Line spacing 24
Paragraph spacing 20
Colors (hex)
Text: #FFFFFF
Icon: #FFFFFF
Current Page indicator: #009ADE
Frame:#FFFFFF
Background:#000000
Meter Size
With Border
height: 182px
width: 642px
Without Border
height: 176px
width: 636px
3px - border
Padding:
16px x 0px
Sizes
Letters & Spaces:
12 X 24px
Icons - small:
24 X 24px
Icons - large:
48 X 48px
Content area:
53 columns
6 rows
318 total blocks
Position on screen
TSOs are positioned over the status indicator area.
With Border
From top: 571px
Left TSO
from left: 0px
Right TSO
From left: 638px
16px padding
3px border
571px from top
Press button to start CHAdeMO
Press button to start CCS

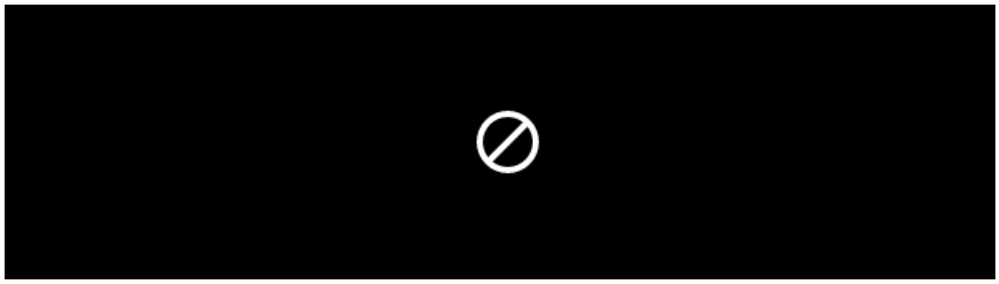

2018/08/31 14:24:02
3059301306072a8648ce3d02 0106082a8648ce3d03010703
42000445f80f9226eca0f103 d7fec19256c5b83694906253
86f2f47ced9590d0f50a108d 6a60602912065b4dffe769e2
52e352c80ea5fdd7ade1eba6 f4a1624ac61d75


2018/08/31 14:23:56

Tarrif-Name_0-01
2HWV 22C3 UT0X
1m 23s
20.42 kWh


1.011
f174 b973 1670 d975 b13a 9039 cd7f a5bb
5298 482f 0f33 0fdf 92e6 9e04 174a c4fc





FPGA IP core

Use

The Interceptor IP core can be used in a standalone VHDL or a Block Diagram project.

Functionality

- Intercept LVDS video data to LCD screen with minimum processing delay
    - Intercept logic runs at DDR, so I/OSERDES is main contributor in system delay
- Selectively override a region of the screen
- Video frame validity checking (V/H Sync, porch, visible region timing; Frame rate)
- Capable of generating LVDS video data if external video source is not present or invalid
- Switch output video source between external or internal source
    - Can be configured to be automatic or manual switching
- Clock Domain Crossing (CDC) safe memory interface to upper level of system
    - Shadow buffered frame memory (pixel colour map)
    - Shadow buffered colour map memory
    - Shadow buffered override locations
    - Interceptor control
    - Interceptor status
- Customisable display parameters
- Monitor backlight
    - Settable minimum brightness
    - Backlight fault status

Limitations

1. LVDS is presently limited to a single channel with four data lanes
2. Backlight is forced to always on
3. Override region presently limited to 122,880 pixels (Due to block RAM availability for frame buffer)
    a. This also limits the interceptor to 8 colours
    b. Depending on available memory could have DCM override area run at reduced resolution. Eg: 1 DCM pixel = 2x2 LCD pixels (4x reduction in required memory)

Interfacing

The Interceptor IP connects directly to the LVDS and backlight IO on the FPGA.

- A 100MHz input clock is required
- An active high reset signal is required

Interfacing with the upper level system can be completed:

- Directly
- Through a supporting IP core
    - Interceptor control AXI IP core to interface with an on-board CPU in the FPGA (Microblaze, ARM) (Pixel interceptor to microblaze interface)
    - *Potentially through an Interceptor control SPI IP core to interface with an external CPU from the FPGA*

To Interceptor

Sub module	Port name	Port width	Description
Frame buffer (Block RAM)	frame_addr	16	Buffer address. Valid address range: 0 - 40959 (No invalid address range checking implemented)
	frame_data	9	Buffer data. Data at each address stores a group of 3 pixels
	frame_wrEn	1	Buffer data write enable. Data is loaded at set address on rising clock clock edge
Colour map (Distributed RAM)	colour_addr	4	Buffer address. Valid address range: 0 - 7 (No invalid address range checking implemented)
	colour_data	24	Buffer data. 24 bit colour depth for each individual colour (RGB)
	colour_wrEn	1	Buffer data write enable. Data is loaded at set address on rising clock clock edge
Intercept X coordinates	ovrd_hStart	12	Start X coordinate (inclusive). Valid range: 0 - display horizontal resolution
	ovrd_hEnd	12	End X coordinate (exclusive). Valid range: 0 - displayhorizontal resolution
Intercept Y coordinates	ovrd_vStart	12	Start Y coordinate (inclusive). Valid range: 0 - display vertical resolution
	ovrd_vEnd	12	End Y coordinate (exclusive). Valid range: 0 - display vertical resolution

Interceptor general control	cpu_control	16	Control of interceptor operation
Read request strobe	read_req	1	Triggers update of 'system status' from FIFO
Backlight minimum duty cycle	bkl_duty_min	12	Backlight minimum display brightness (PWM duty cycle) NOTE: 100% duty = 'PWM period' + 1

Bitmap frame buffer data

The 3, 3bit pixel data group at each memory address of the shadow buffer is stored in the following format.

Each pixel references an element in the colour map which provides a reduced number of defined colors.

(Multi-pixel packing in a single memory address)

Bit	Description
0-2	Pixel n+2
3-5	Pixel n+1
6-8	Pixel n

NOTE: Ensure the buffer address is changed before writing new pixel data. Otherwise the pixels at the old address will be modified

Interceptor general control

The control register of the pixel interceptor operation is presented as follows

Bit	Description
0	HMI LVDS video source select
1	New frame data in shadow buffer
2	New colour data in shadow memory
3	New override X,Y location shadow values
4..15	Reserved

NOTE: The interceptor will commence loading new data at the start of the next video frame

NOTE: Bit-0 is the only signal that will work if HMI source is selected when HMI source has failed

NOTE: Bit-0 is disabled if interceptor configuration is set to AUTO change SOURCE

From Interceptor

Sub module	Port name	Port width	Description
System status	intercept_stat	32	Status of the pixel interceptor

The following status register is presented through a FIFO which the interceptor updates on the first cycle after a new video frame is started.

It is the responsibility of the upper level system to read the status register often enough such that the FIFO never becomes full.

The FIFO has an effective write depth of 63 entires.

- System Status
    - 32bit register
        - Interceptor status
            - 16bits
        - HMI frame validity status
            - 16bits

Interceptor status

Bit	Description
0	HMI LVDS video source valid
1	Frame buffer updated

2	Colour map updated
3	Override X,Y location updated
4	Frame tracking locked
5	Panel backlight fault
6	Panel backlight minimum brightness fault
7	RESERVED
8	RESERVED
9	RESERVED
10	RESERVED
11	RESERVED
12	RESERVED
13	RESERVED
14	RESERVED
15	Updated values present in this register (Excluding 'video source valid' which is always updated) NOTE: This is for the CPU to determine if the read data in this register is updated from the interceptor FIFO or from last read

Frame validity status

Bit	Description
16	hDtcErr - SyncH not detected
17	hDtcErr - SyncEn L-H not detected (Checks HBP. Also inherently VBP and VRES)
18	hDtcErr - SyncEn H-L not detected (Checks HRES)
19	hDtcErr - RESERVED
20	hCntErr - SyncH not at correct pixel location
21	hCntErr - SyncEn L-H not at correct pixel location
22	hCntErr - SyncEn H-L not at correct pixel location
23	hCntErr - HFP / Total row pixel count error (HSYNC + HPORCH[HBP] + HRES + HPORCH[HFP])
24	vDtcErr - SyncV not detected
25	vCntErr - SyncV not at correct row count
26	vCntErr - VFP / Total frame row count error (VSYNC + VPORCH[VBP] + VRES + VPORCH[VFP]) NOTE: This handles VFP detection since frame is checked against generated reference
27	RESERVED
28	RESERVED
29	RESERVED
30	vCntErr - Frame rate over error (Checked against generated reference)
31	vCntErr - Frame rate under error (Checked against generated reference)

NOTE: 'Updated' flags remain set until respective new data flag is cleared by the Microblaze

NOTE: Access to shadow buffer/memory is disabled until respective new data flag is de-asserted

NOTE: The first read of the status register will be invalid

NOTE: Bit-0 is the only signal that will work if HMI source is selected when HMI source has failed

To HW

Interface	Port name	Port width	Description
LVDS	LCD_A_CLK_P/N	1	Differential LVDS clock for Channel A to Display
	LCD_A_TX_P/N	4	Differential LVDS data for Channel Ato Display
Backlight	BKL_ON_OUT	1	Backlight enable to display backlight controller
	BKL_PWM_OUT	1	Backlight brightness PWM to display backlight controller
	BKL_FAULT_OUT	1	Backlight fault output to HMI computer
Colour mode	SEL68_BIST1	1	Display colour mode selection to display. Either 6 or 8bit colour depth
Debug	DBG_P/N	1	RESERVED
	DBG_LED	1	RESERVED

From HW

Interface	Port name	Port width	Description
LVDS	GPU_A_CLK_P/N	1	Differential LVDS clock for Channel A from HMI computer
	GPU_A_RX_P/N	4	Differential LVDS data for Channel A from HMI computer
Backlight	BKL_ON_IN	1	Backlight enable from HMI computer
	BKL_PWM_IN	1	Backlight brightness PWM from HMI computer
	BKL_FAULT_IN	1	Backlight fault output from display backlight controller

Customisation

All parameters required to use the IP core with different LCD panels can be configured through its customisation GUI.

General

Display General

Item	Description
Gen Bg colour	Internally generated frame background colour (RGB). Eg: 0xFF0000 = 100% Red, 0x00007F = 50% Blue
Use 18bit colour	Select between 18bit and 24bit colour modes (In 'general' since most screens support both)
Auto LVDS source	LVDS source is automatically changed to use internally generated signal if LVDS signal is lost

NOTE: Auto LVDS source will not switch based on frame validity. Only LVDS signal loss is used

System Clocking

Item	Description
System VCO Multiplier	Multiplier for MMCM VCO (x1000)
System Clk Divisor	System clock divisor (x1000) - System frequency must be 200MHz
Pixel Clk Divisor	Generated pixel clock divisor - Must closely match frequency of GPU pixel clock
Bit Clk Divisor	Generated bit clock divisor - Must closely match frequency of GPU bit clock
Backlight Clk PreDiv	Generated backlight clock pre-divisor - Used in conjunction with divisor
Backlight Clk Divisor	Generated backlight clock divisor - Used in conjunction with pre-divisor

Display Backlight

Item	Description

PWM period	Backlight brightness PWM period
PWM polarity	Backlight brightness PWM polarity
Fault polarity	Backlight fault polarity

PWM frequency setting

The PWM frequency is set based on the divided system clock and period configuration values.

Backlight divided clock BklClk = (100MHz × 'System VCO Multiplier') / ('Backlight Clk PreDiv' × 'Backlight Clk Divisor')

PWM period = BklClk / 'PWM frequency'

Display specific parameters

LCD General

All of these parameters must match the GPU configuration for the LCD pannel.

Item	Description
VESA colour mapping	Select between VESA and JEIDA colour mapping
Vsync polarity	LCD vertical sync active state polarity
Hsync polarity	LCD horizontal sync active state polarity
Esync polarity	LCD data enable active state polarity
V/Hsync enable	LCD requires V/H sync signals (Some LCD are 'DE only')

NOTE: GPU must always send V/H sync signals even if LCD is 'DE only' mode, due to frame validity checking logic in FPGA.

LCD Timing

All of these parameters must match the GPU configuration for the LCD pannel.

Timings can be calculated using a spreadsheet based of FPGA_G150XNE_LVDS_Calculations.xlsx

Item	Description
LVDS data rate	LVDS data rate in Mbps (in Binary Coded Decimal)
Pixel clock period	Pixel clock period (ns x1000)
Vertical resolution	LCD vertical visible area (lines)
Horizontal resolution	LCD horizontal visible area (pixels)
Vsync duration	LCD vertical sync duration (lines)
Hsync duration	LCD horizontal sync duration (pixels)
VBP duration	LCD vertical back porch duration (lines)
VFP duration	LCD vertical front porch duration (lines)
HBP duration	LCD horizontal back porch duration (pixels)
HFP duration	LCD horizontal front porch duration (pixels)
Frame rate error	Allowed horizontal line count error between selected and reference frame

NOTE: Frame rate error parameter value will depend on how closely the FPGA generated reference frame timing matches the GPU LVDS signal

Pixel interceptor to microblaze interface

The Microblaze CPU will interface with the pixel interceptor as a memory mapped AXI4-Lite peripheral which contains multiple registers than can either be written to or read from

Refer to FPGA IP core for register descriptions and usage

AXI peripheral register address mapping

The following table lists the register offset from the base address of the AXI peripheral that controls the pixel interceptor

Offset	R/W	Register
0	W	Interceptor general control
4	W	Interceptor start pixel - X
8	W	Interceptor stop pixel - X
12	W	Interceptor start pixel - Y
16	W	Interceptor stop pixel - Y
20	W	Frame buffer address
24	W	Frame buffer data
28	W	Colour buffer address
32	W	Colour buffer data
36	R	System status
40	W	Backlight minimum duty cycle

Customisation

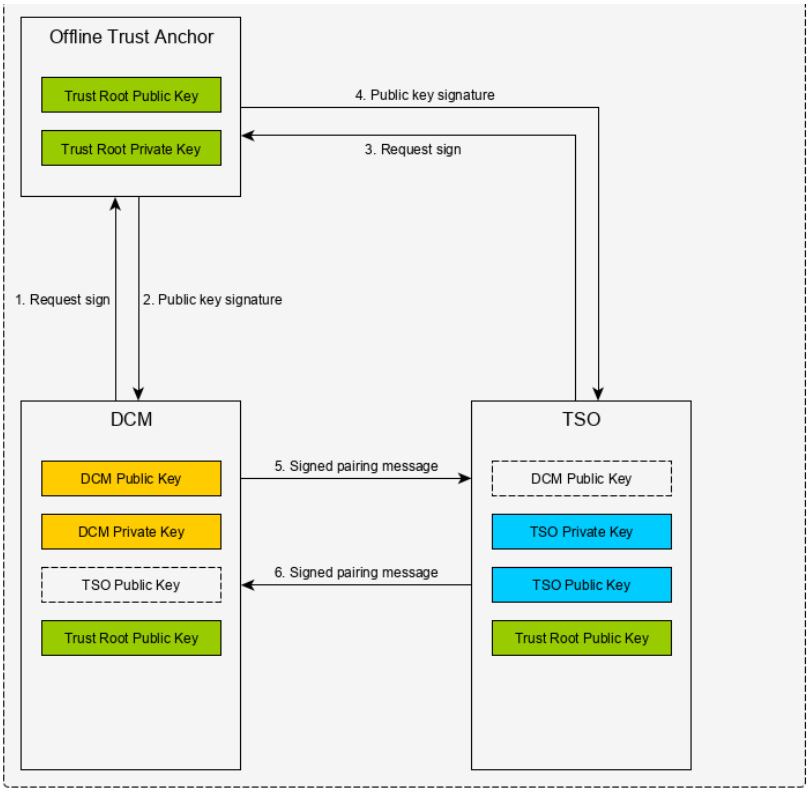
The peripheral allows for some default register values to be defined.

The default values are set after FPGA initialisation and peripheral reset.

Item	Description
Default backlight PWM min duty	Default backlight minimum brightness (PWM duty cycle) NOTE: 100% duty = 'PWM period' + 1

Pairing TSO and DCM

Secure communication is defined as either encrypted or signed.



An offline (air-gapped) trust anchor is used to sign the key of the DCM/TSO and signature is sent back to the DCM/TSO be stored. To pair a DCM and TSO together their public key and signature from trust anchor is exchanged. On verification of signature the units can determine the matching unit can be trusted and all further communication is signed using public key. The DCM and TSO save the trust anchor public key during commissioning. Trust anchor has restricted physical access and signs any device requested.

Exchange Method

- TSO is unpaired and accepts only status read frames and pairing requests. In this states it responds only with COM_PARING_PROMPT frame that contains Pairing session ID.
- DC Meter creates COM_PARING_REQ frame with pairing session ID from TSO. TSO sends its public key and the pre-saved signature.
- DC Meter sends its public key with its signature to TSO.
- DCM and TSO verify the signature with preinstalled trust root public key.
- TSO closes the pairing session and starts normal operation.

Trust Anchor Communication

Overview

The DC Meter (DCM) and Trusted Screen Overlay (TSO) need a method of proving their authenticity to each other in the field.

A simple offline Tritium-managed Trust Anchor (TA) will be used to sign public keys of DCM/TSO devices at time of production.

The public key of the TA will be loaded into all DCM/TSO devices allowing them to be able to verify signatures created by the TA at production.

Physical Interfaces

The DCM has Ethernet and is capable of TLS 1.2 communication using X.509 certificates.

The TSO only has a Serial Peripheral Interface (SPI) and doesn't support TLS 1.2 or X.509 certificates at all.

The TSO will be connected to device that provides Ethernet connectivity.

ECP Curve Type

All keys and signature will be derived from secp256r1 curve (256-bit).

Connection Type

Plain-text TCP/IP.

Port 5000 (Arbitrary unused port number).

DCM/TSO devices will be clients and initiate communication.

Message Format

Elliptic curve public key will be communicated in stripped uncompressed binary format (no prefix).

Elliptic curve public will be exactly 64 bytes.

Elliptic curve signature will be communicated in stripped uncompressed binary format (without ASN.1 encoding).

Elliptic curve signature will be exactly 64 bytes.

Client Outgoing Call

⚠ On power up, the client will generate a Elliptic curve key-pair and permanently save to non-volatile memory once TA signing is successful.

Without X.509 certificates, the Trust Anchor will not be able to verify the authenticity of the connecting client.

The authenticity will be controlled through process and physical restriction.

Version	Acknowledge	Client Public Key	Client Unique ID	Client Signature (SHA256 over Client Public Key + Client Unique ID)
1 Byte, value 0	1 Byte, value 0	64 Bytes	64 Bytes	64 Bytes

The Version value is to distinguish different message formats over time. The above format is identified by the value 0. Future formats will be identified by higher version values.

The Acknowledge value is always set to the value 0 in this request.

Client Incoming Response

Version	Acknowledge	Trust Public Key	Client Public Key	Trust Signature (SHA256 over Trust Public Key and Client Public Key)
1 Byte, value 0	1 Byte, value 0	64 Bytes	64 Bytes	64 Bytes

The Version value is to distinguish different message formats over time. The above format is identified by the value 0. Future formats will be identified by higher version values.

The Acknowledge value is always set to the value 0 in this response.

Client Outgoing Call (Acknowledgement)

Version	Acknowledge	Trust Public Key	Client Public Key	Trust Signature (SHA256 over Trust Public Key and Client Public Key)
1 Byte, value 0	1 Byte	64 Bytes	64 Bytes	64 Bytes

Acknowledge = 0, The client either did not receive or could not successfully store the information from the Trust Anchor. Trust Anchor print failure.

Acknowledge = 1, The client was able to receive and successfully store the information from the Trust Anchor. Trust Anchor print success.

Device Discoverability

The TA will not be connected to the internet.

The TA will communicate to the DCM/TSO via a network router.

The TA/DCM/TSO will communication on a IPv4 network.

The network router will provide a DHCP server.

The DCM/TSO will be capable of acquiring an IP address from a DHCP server.

The DCM/TSO assume that the TA will be at a static IP address. Alternatively the TA could supply a DNS server to provide name resolution facilities to DCM/TSO devices.

The TA will be able to support up to two concurrent connections. To potentially support up to two assemblers, one making DCM and another making TSO.

Connection Flow

1. Client start.
2. Client generates key-pair (if doesn't exist).
3. Client connects to Server using TCP socket on designated port.
4. Client uploads (Client Public Key + Unique ID + Signature).
5. Client downloads (Server Public Key + Client Public Key + Server Signature).
6. Client now has a Trust Anchor signature of it's own public key.
7. If the server encounters any issue in processing a signing request, it logs the nature of the issue and closes the TCP connection with the client.

Cryptographic Primitives and OpenSSL

The cryptographic operations and primitives described above can be emulated and verified with OpenSSL on a regular PC.

Keys

Create a key: `openssl ecparam -name prime256v1 -genkey -noout -out key.pem`

Convert to binary DER format: `openssl ec -in key.pem -outform DER -out key.der`

Extract public key: `openssl ec -in key.pem -pubout -out pub.pem`

The last 64 bytes of the file key.der correspond to the binary elliptic-curve points that go into the message format described above. They can be extracted from the key.der file, e.g., with the command `tail -c 64 key.der > key_msg_format.bin`

Signatures

To sign a message (stored in the file message_to_sign.bin) with a key (stored in the file key.pem) and store the signature in the file signature.der: `openssl dgst -sha256 -sign key.pem -out signature.der message_to_sign.bin`

The signature file contains two binary components that can be inspected with the command `openssl asn1parse -inform DER -in signature.der`:

```
  0:d=0 hl=2 l= 70 cons: SEQUENCE
  2:d=1 hl=2 l= 33 prim: INTEGER :D0423FB2F230ABD6D4F49812C6EC0D577F72E7525250D6D54FD050BD6278B986
 37:d=1 hl=2 l= 33 prim: INTEGER :AFD0A83F3AF7657957D977A15357760F1A1D625F550F525EBFEBEBD7E0CAFA5C
```

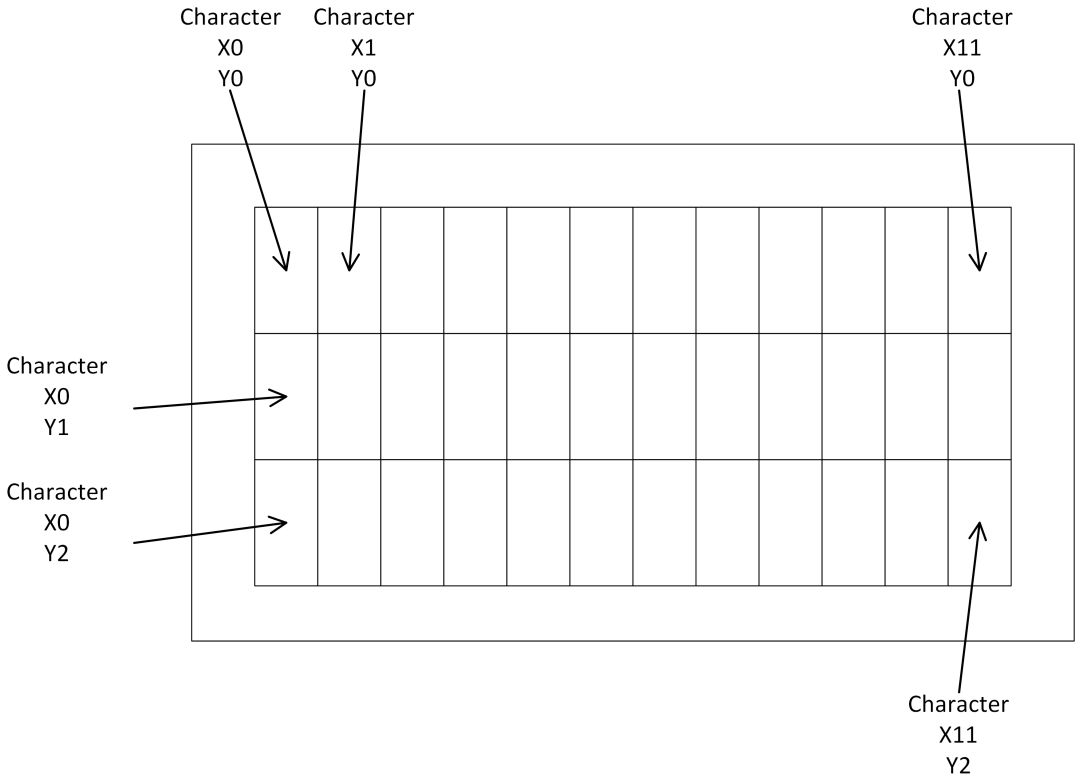
(Ignore the fact that either of them can be reported with length 33 instead of 32. This is merely an artifact of them being DER encoded. As one can see above, each hexadecimal representation is always 64 characters long, which makes for 32 bytes of binary data each.

To generate the 64-byte message format described above, the two 32-byte binary signature components simply need to be concatenated (in the order above).

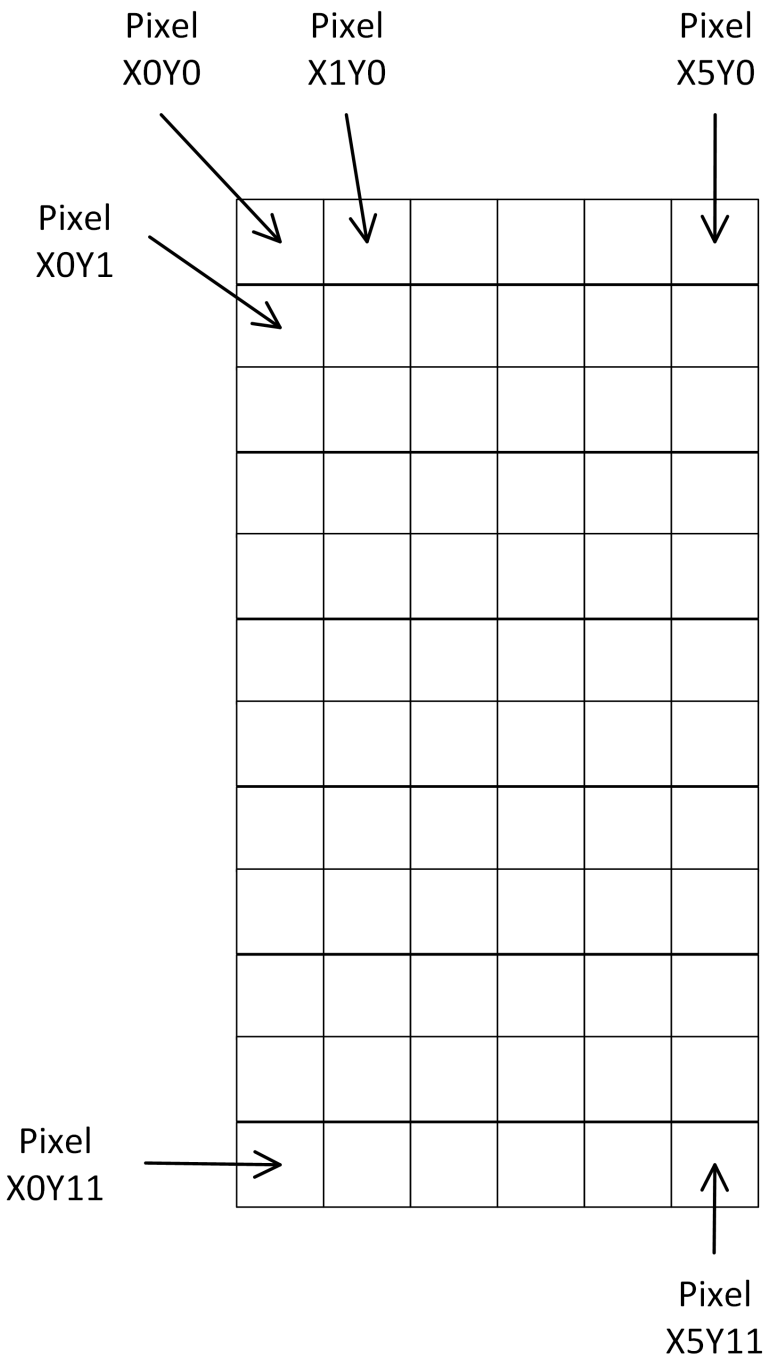
To verify a signature: `openssl dgst -sha256 -verify pub.pem -signature signature.der message_to_sign.bin`

Rendering of overlay screen

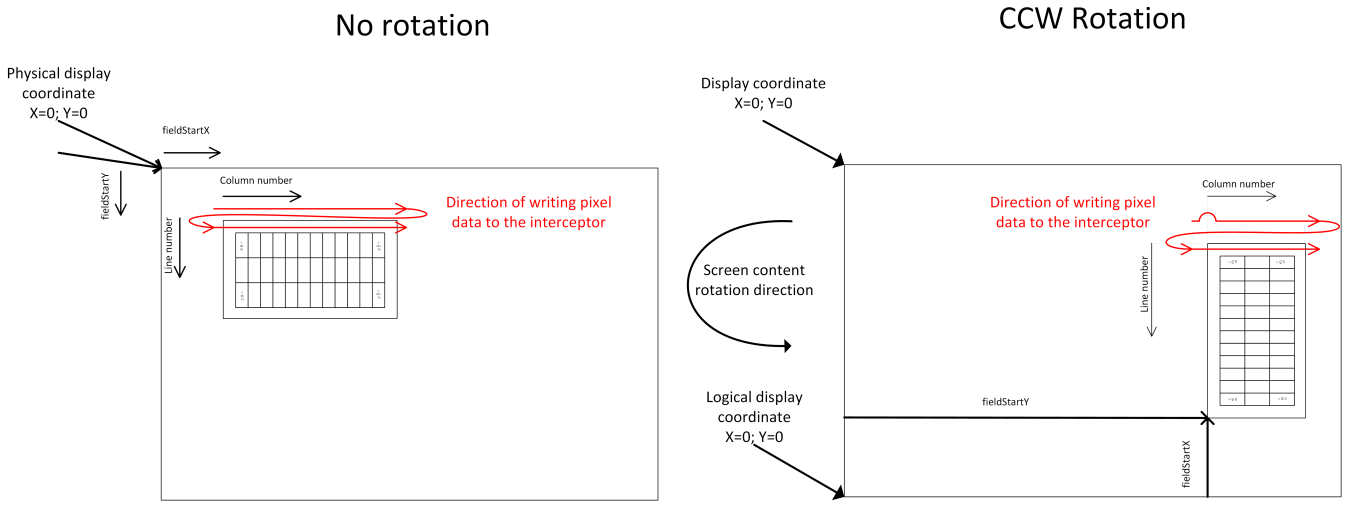
Characters in overlay field (example with character field size 12x3)



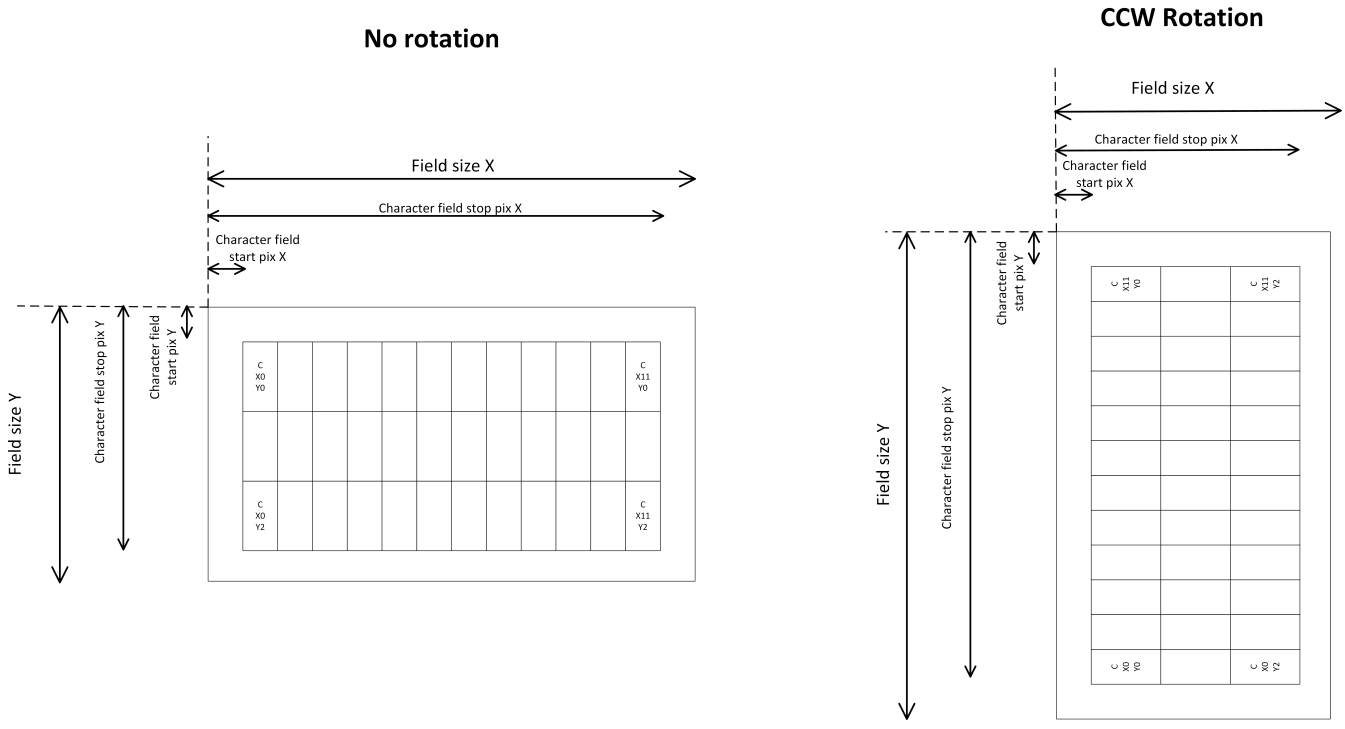
Pixels in character (example with character size 6x12)



Overlay position in normal and rotated mode



Variables used for pixel extraction (in bufferDriver_pixelExtractionContext)



DC Meter Approved Firmware Version

This page contains the firmware versions approved (or in approval) by the notified/certifying bodies.

DC Meter

Firmware version displayed in normal operation on the screen.

The application hash of the firmware is automatically generated by the firmware building pipeline and written into the image. The version number is manually controlled and incremented on release.

Status	Date Approved	Version Purpose Summary	Notified /Certification Body	Standard	Market	Version	Application Hash
In assessment	-	Original submission - overall certification	VDE	EN50470-1/-3	Germany	1.011	`f174b9731670d975b13a9039cd7fa5bb5298482f0f330fdf92e69e04174ac4fc`
Approved	TBD	PK Chargers	VDE	EN50470-1/-3	Germany	1.010	`a0f41f2ebf3ce9cdcb0833f75649627b3484321f40e422302cba12cdca565c0c`

TSO

Firmware version displayed on start-up on the screen.

The version number is manually controlled and incremented on release (more details see Trusted Screen Overlay (TSO)).

Status	Date Approved	Version Purpose Summary	Notified/Certification Body	Standard	Market	Approved Firmware Version (s)
In assessment	-	Original submission - overall certification	VDE	EN50470-1/-3	Germany	01.00.01_PK 01.00.01_ISD

Charger power derate on TCU

Overview

In PK and ISD chargers, Thermal Control Unit (TCU) managers the internal cooling system. The primary goal for internal liquid cooling is good thermal management to maintain performance and reliability.

The functionality of TCU is to:

- monitor the temperature using thermal sensors
- control fans(heat ex-changer)
- passive circulation of liquid coolant through the pumps
- heat dissipation of the coolant through radiators

Power Derate

When ambient temp is high or cooling component malfunctions, the thermal gain during high current charge may exceed the cooling capacity, in such cases TCU requests Charge Control Unit (CCU) CCU to derate the power delivered to vehicle. Otherwise, high temperature inside the charger may damage certain components, especially the power electronic compoents. The power derate is calculaed based on the temperature of internal air and liquid coolant.

$derate_{TCU}$ is based on two separate calculation: $derate_{air}$ and $derate_{coolant}$.

$derate_{TCU} = min(derate_{air} , derate_{coolant})$

$derate_{air}$ is calculated based on internal Air temperature. The derate starts from 70°C (100%) and reaches full derate 5% at 85°C, as illustrated in Figure 1. The slope of the line is 6.667.

$derate_{air}(\%) = 100 – ((temp - 70) * 6.667)$

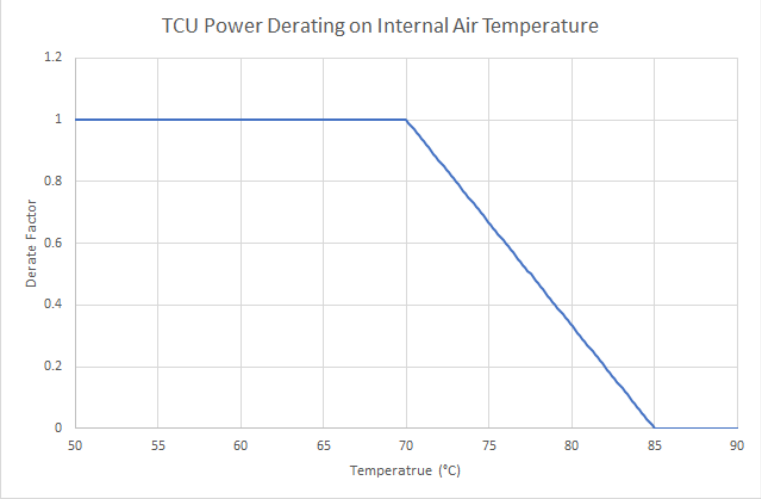


Figure 1 TCU power derating on Internal Air Temperature

$derate_{coolant}$ is calculated based on Coolant temperature. The derate starts from 65°C (100%) and reaches full derate 5% at 70°C, as illustrated in Figure 2. The slope of the line is 20.

$derate_{coolant}(\%)= 100 – ((temp - 65) * 20)$

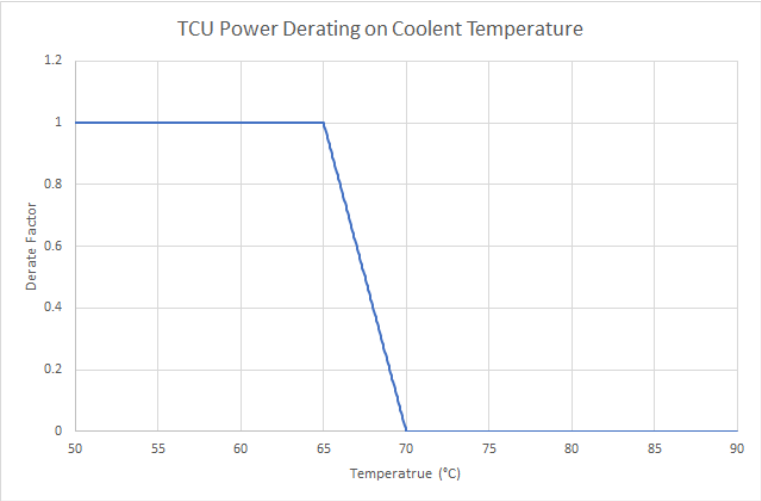


Figure 2 TCU power derating on Coolant Temperature

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Appendix

A - Full Audit Event Log

```
Index Date Event Data Hash Protocol version
11 Mon Jan 25 09:25:03 2021 TIME_SYNC 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 a5c7e4f3d3e78de7ded77bb52e6756913899b3defcd3cebdd7b7d47948abc37f 0
12 Mon Jan 25 09:25:03 2021 TIME_DEVIATION 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 4598e66c3d20389a1f4be83c29b7f6ee83a3937d11e6b133c2196297cc12a18e 0
13 Mon Jan 25 09:25:51 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 50 52 4f 44 07 74 74 69 2d 49
51 43 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 84f416d24ddffc080b76658a71521d3673a7155a2bc94e912565ec541a5f8479 0
14 Mon Jan 25 09:26:16 2021 SUBA_TEST_LOG_PARAM_CHANGE 00 01 01 0e 02 0e 60 00 00 00 00 18 02 0e 60 00 00 00 00
01 03 00 00 00 00 00 00 00 08 25 00 00 98 2b d6 29 00 00 00 00 0f e7 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 cb654c5a020c864aa6dc43c057e5dd977bd54e2bd483a3b61ec50c326d720a27 0
15 Mon Jan 25 09:26:16 2021 SUBA_TEST_LOG_PARAM_CHANGE 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 e4 0a e0 53 a6 ff 6e 45 bd d5 76 2b e2 40 3f 9f 78 c0 85 f8 03
bc 52 6e 16 84 88 7a 4d 8c1964b4a7138aebf4c2db97d9483ca1542e7a445859a3fda803f663c277b8b4 0
16 Mon Jan 25 09:26:16 2021 SUBA_TEST_LOG_PARAM_CHANGE 02 c7 1c 40 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 39990f4d77d0f1915654f785945fc2045ce0a9b3a1060404d16a8292d62d08fa 0
17 Mon Jan 25 09:26:33 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 d46b127bdbcfb3b97821dfb4b864874fd7cce3c6ac746963815995def1b0b81e 0
18 Mon Jan 25 09:26:41 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 e1b162e5f38e7211666db54e2bffc142b4e34e168bb926b197d481f8a57ee966 0
19 Mon Jan 25 09:26:46 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 481d6db29195202326429e392b2c689751ea6ddac94cb8c83273df003922b240 0
20 Mon Jan 25 09:26:48 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 947c68de97b68c0584c99e7a89e25d10445278f0ca3ddbdfb5f5fdf6f4f9b19d 0
21 Mon Jan 25 09:26:54 2021 HOST_PRIVATE_KEY_PARAM_CHANGE 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 f9173a823c2a8c141db482a423ad774aa8de572846525a0f16016a71cc8efe20 0
22 Mon Jan 25 09:26:54 2021 TRUST_ROOT_KEY_PARAM_CHANGE 00 01 96 7a 34 2a a7 10 8a 68 b9 c0 ab 5f 00 0b 9c d7
c4 c5 2c 2b 9c 94 ad a6 28 b3 14 5f 6e 89 8e aa 17 37 43 e7 d7 b6 60 e7 cb f8 d1 db a0 bf d9 b2 25 0e 26 c6 0b
d4 f5 ad 23 ec b7 24 f6 7a ac36bdfad55e6d4c01a6ba2262617c68b6c9ae148d944921c5004a5f4244954e 0
23 Mon Jan 25 09:26:54 2021 TRUST_ROOT_KEY_PARAM_CHANGE 01 49 8f cc 54 13 84 13 f0 7e 0a aa 08 f3 21 e3 da b7
d6 d1 60 c3 2b b7 07 03 06 9c 0b 4d 4c 5a c3 f1 be 83 50 30 2b 41 9f c4 97 37 8d d1 e3 33 92 1b 5f 1f b3 c6 4f
24 9b 15 23 93 b1 52 e3 97 38b0d297aba8cc3153bc4f6b2d376487ca754f078a75f95e33465c630a619e32 0
24 Mon Jan 25 09:26:54 2021 TRUST_ROOT_KEY_PARAM_CHANGE 02 c2 84 f0 76 43 51 1a 21 0d fd ef 1e 24 a5 1a e0 15
44 6d 1c 3f ec e7 4a 1a f2 4a 94 a7 cb 5d 60 40 0d 60 6d 9b ed ac 2c fd 37 bf f2 9e 6b f7 a9 20 16 39 86 7c 29
72 fb 6e fe 53 1d 0f 1e c2 5ee7272bf298f1011e5acb9d1f36526d774dc42def515fd4a5ed3be711989cc1 0
25 Mon Jan 25 09:26:54 2021 TRUST_ROOT_KEY_PARAM_CHANGE 03 51 e7 00 c8 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 88 56 a3 9c ac 63 a8 1b ea 4a a6 49 af 30 85 e2 02 5b
3f 74 53 22 93 cb ab 43 5c 83e696e676631b1547e1ed8bdedf15d0edfae3db1f0130b19e31f93dc3585ad0 0
26 Mon Jan 25 09:26:54 2021 TRUST_ROOT_KEY_PARAM_CHANGE 04 6a 49 b6 b8 28 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 e70b94d40cc604c236836c26e5c380f1a4614a4c7a5f5f07f649adfd8a26c571 0
27 Mon Jan 25 09:27:16 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 bf52c205994bf7d15800aafb2c5b5af4410022620cede05216ee8a3410f31815 0
28 Mon Jan 25 09:28:35 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 14343a2fac3f4a9ee245fd9e59cc070df5eedac3357e232289feaa2872c5769d 0
29 Mon Jan 25 09:28:48 2021 TIME_SYNC 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 474dea80656b98cb45b9142d10c703fde70db2c5b8b34d8c1ed5fb5b72a1908c 0
30 Mon Jan 25 09:31:29 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 50 52 4f 44 07 74 74 69 2d 49
51 43 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 b47243f0745ea3b54d8645c219eaef24574f962d916b3f26d6e598fdc311aae3 0
31 Mon Jan 25 09:32:56 2021 TSOIF_CONFIG_PARAM_CHANGE 00 01 35 06 a2 02 b6 00 3f 00 3e 04 13 00 13 00 05 01 00
02 34 58 58 78 98 98 ac c6 c5 d0 fa fb f9 f7 9a 03 d6 33 3b 57 e4 55 01 00 79 00 12 0a e8 03 00 00 00 00 00 00
00 00 00 00 00 00 00 00 e6c3fb2e90352276b204601a732a67200722b90d5c6c6b8a996efab827f009e9 0
32 Mon Jan 25 09:32:56 2021 TSOIF_CONFIG_PARAM_CHANGE 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 1a 28 c8 27 95 6c c0 49 67 8b 89 fe 8a 5a b9 e4 e0 7d e8 e2 2b
3b f9 75 dd b2 c2 e0 2c f5a4e93bc1148d3e67e04623edcfa1939913d497bd82d4a0680e13c0e4462738 0
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33 Mon Jan 25 09:32:56 2021 TSOIF_CONFIG_PARAM_CHANGE 02 96 6b 50 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 6339f9a6740758130be2c83bda94a399e84becd2eb7684b1c6648a64cf89dea3 0
34 Mon Jan 25 09:32:59 2021 INET_CONFIG_PARAM_CHANGE 00 01 08 08 08 08 04 04 08 08 46 3c ff ef d5 07 61 75 2e
70 6f 6f 6c 2e 6e 74 70 2e 6f 72 67 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 c51a6d50dbce10bf680088d356e5f259a3096ad2d7b21f99021c4a18a583e3d6 0
35 Mon Jan 25 09:32:59 2021 INET_CONFIG_PARAM_CHANGE 01 31 30 2e 31 38 2e 30 2e 31 30 30 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 64 63 6d 75 70 64 61
74 65 2e 74 72 69 74 69 18dc49eeffa2088f7987304b38ebc89111ab3d2fce279f21ae950d4251ea9820 0
36 Mon Jan 25 09:32:59 2021 INET_CONFIG_PARAM_CHANGE 02 75 6d 2e 63 6f 6d 2e 61 75 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 2f 64 63 6d 65 74 65 72 2e 69 6d 67 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 8a7da4cd93da561b6dfae9fa4711183088a7214639110bca4d617547c9658436 0
37 Mon Jan 25 09:32:59 2021 INET_CONFIG_PARAM_CHANGE 03 00 00 78 00 12 0a 00 ff ff ff 01 00 12 0a 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 2c 65 ee d5 c1 3c 67 51 64 e6 8a 9e 84 04 ab 6c 90 1a 01
cd 0b 50 d1 97 70 d2 00 2e28d7dfd9c11f18536daed879477be9f2a7fc1a0c934c7c7938f5a0fda6d103 0
38 Mon Jan 25 09:32:59 2021 INET_CONFIG_PARAM_CHANGE 04 c2 e7 18 9f e1 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 21b1764a4fdc45e6bef64989985050379e1a1ddac7b6de5ee0eb865b271fa5d0 0
39 Mon Jan 25 09:33:02 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 00 00 00 00 00 20 00 00 00 71 00 74 00 20 00
30 00 30 00 30 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 5db343928aaef030d56944b9d99c10a21f5079788a6153d3f8e3ca2db1041763 0
40 Mon Jan 25 09:33:02 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 30 00 30 00 30 00 30 00 30 00 31 00 30 00 43 45 54 00 30 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 2b1652b2636c716c902814395576888b7830840ac596464a95b3d8263301016d 0
41 Mon Jan 25 09:33:02 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 b3936158e4017d96222b574823f2c0c4611e6dc67e594552b34afdc5d24304db 0
42 Mon Jan 25 09:33:02 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 f6 b6 ce e9 27 ac ce 91 9d ba 51 82 f3 03 6d bc 48
12 7a 69 86 9a d4 28 1d 5a 27 7769094c48e2f6ed35c87a3592e6b8f778eb580d25f43723c1b9751c54a8abd9 0
43 Mon Jan 25 09:33:02 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 6c 39 89 1a 3b 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 7396637b57fa8646affa9f92e40c869dc6207efaf615d1a40a7a3947a650be51 0
44 Mon Jan 25 09:39:29 2021 AFE_CONFIG_PARAM_CHANGE 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 40 42
0f 00 40 42 0f 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 40 42 0f 00 40 42 0f 00 00 00 00 00 00 00
00 00 00 00 00 00 00 65e8bed0c0bffb366fa55658a63f097de3d24a231ddcd14d8137b373689cdf5b 0
45 Mon Jan 25 09:39:29 2021 AFE_CONFIG_PARAM_CHANGE 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 2f 76 de cc 23 8c b7 86 74 50 ec e8 9d 76 61 15 54 d1 d7 83 a8 6a
83 12 26 56 11 03 df 0b4c40439dbd44368f8240a5642e97e10bae822654ce2c5c9f207838af3db551 0
46 Mon Jan 25 09:39:29 2021 AFE_CONFIG_PARAM_CHANGE 02 9c 70 95 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 f036cd3629c8d132762345aaccb3c4165962d06f186103e37b9d9584da698725 0
47 Mon Jan 25 09:39:42 2021 AFE_CONFIG_PARAM_CHANGE 00 01 40 ce 59 07 00 00 00 00 40 ce 59 07 00 00 00 00 40 42
0f 00 40 42 0f 00 0b be 75 ff ff ff ff ff b8 d0 a6 ff ff ff ff ff 40 42 0f 00 40 42 0f 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 d644970d70c3052e3a6af74208d86d0202baf395e617cc32e58184a9e19c375e 0
48 Mon Jan 25 09:39:42 2021 AFE_CONFIG_PARAM_CHANGE 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 92 77 66 b7 53 3c cc 47 31 01 8e ca 51 85 ee 91 e1 7c 64 f4 76 04
e0 e5 6a 0a f3 e2 99 464988b3e49ec40f85a8e923de888af6614c5e28cf13bca0441e1063da2b7182 0
49 Mon Jan 25 09:39:42 2021 AFE_CONFIG_PARAM_CHANGE 02 9a 6f ae 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 870a7c8986909b89dacdaaa46eadf2655303e26bb5872b373a02dae010811332 0
50 Mon Jan 25 09:41:40 2021 AFE_CONFIG_PARAM_CHANGE 00 01 40 ce 59 07 00 00 00 00 40 ce 59 07 00 00 00 00 10 f1
0e 00 7e f1 0e 00 0b be 75 ff ff ff ff ff b8 d0 a6 ff ff ff ff ff f5 42 0f 00 05 43 0f 00 00 00 00 00 00 00 00
00 00 00 00 00 00 b32eba2f70f044fbe03a2a34018de203f9d60b71a53510776e90a75f118a545a 0
51 Mon Jan 25 09:41:40 2021 AFE_CONFIG_PARAM_CHANGE 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 07 ee a7 41 7b 8b 1d 95 a9 41 f4 e0 b0 3a 75 b9 88 78 df 26 42 7a
cd ad 79 de 5f 96 a4 1e266b2d2fd9f06429265b711f52c02936773e794051d7dad68cfe8a9efabd64 0
52 Mon Jan 25 09:41:40 2021 AFE_CONFIG_PARAM_CHANGE 02 01 a3 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 984d5088a3ffe35e2c03b7df48a2dd0b70b8c4aa5a4ca6e6baad322791f67ec1 0
53 Mon Jan 25 09:42:05 2021 BEGIN_CHARGE_SESSION 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 23aa820f25391c021b824b9073f3580cb41e0b8551ad518c804bc392554677b4 0
54 Mon Jan 25 09:43:47 2021 END_CHARGE_SESSION 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 4bda85c6d4dba33ce3827fc837862ffc37e7afdec694748dd5e53203af7ad306 0
55 Mon Jan 25 09:48:15 2021 BEGIN_CHARGE_SESSION 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 c9d224d32439581fc6f8a0a47ad7b416f646a2560a4ed12023a9e4421e25bdf8 0
56 Mon Jan 25 09:48:17 2021 END_CHARGE_SESSION 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

00 00 00 00 00 00 75b9dc08ca7e8e59ae4d729a1e719126c42e1a5db543bd20368eada59b0d7513 0
57 Mon Jan 25 09:48:20 2021 AFE_CONFIG_PARAM_CHANGE 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 40 42
0f 00 40 42 0f 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 40 42 0f 00 40 42 0f 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 1cf2c05158f42eb064ece31fe2fb211ef062963760b9d05470adff5ff4545c4e 0
58 Mon Jan 25 09:48:20 2021 AFE_CONFIG_PARAM_CHANGE 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 2f 76 de cc 23 8c b7 86 74 50 ec e8 9d 76 61 15 54 d1 d7 83 a8 6a
83 12 26 56 11 03 df 2639643e77ad6473bf8a8121d9e217840d399b706fb27024e838e8a7d5699da5 0
59 Mon Jan 25 09:48:20 2021 AFE_CONFIG_PARAM_CHANGE 02 9c 70 95 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 9547ebf5c2c38ddaa5405ffac2adce753932ba472d0a9b9bccb911527117baaa 0
60 Mon Jan 25 09:48:33 2021 AFE_CONFIG_PARAM_CHANGE 00 01 b0 51 98 07 00 00 00 00 b0 51 98 07 00 00 00 00 40 42
0f 00 40 42 0f 00 eb 31 70 ff ff ff ff ff 8f 22 3c ff ff ff ff ff 40 42 0f 00 40 42 0f 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 5f04463cfd89b1c497de8c24fd75e2f3d1a161e4a1087f554d9b4610ee41be6e 0
61 Mon Jan 25 09:48:33 2021 AFE_CONFIG_PARAM_CHANGE 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 68 de fb 92 e0 01 ea f7 95 b2 cf d3 ff 07 3b 5c 9a 83 a0 d8 02 f0
b2 9d c9 72 e7 a2 23 fea20905ea72c3f59644983bb5e16fc189c21c8abf20050b4347bbbb35127398 0
62 Mon Jan 25 09:48:33 2021 AFE_CONFIG_PARAM_CHANGE 02 60 97 d9 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 b716137aea29193091a0dbcdc8b2261b0803242d7a4bbb1618d6e12e2cb47830 0
63 Mon Jan 25 09:50:27 2021 AFE_CONFIG_PARAM_CHANGE 00 01 b0 51 98 07 00 00 00 00 b0 51 98 07 00 00 00 00 d4 f0
0e 00 bd f1 0e 00 eb 31 70 ff ff ff ff ff 8f 22 3c ff ff ff ff ff f2 42 0f 00 f0 42 0f 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 59b9f5235f8226779de5f4e31a02bdcfcf197ad0d40eb3fb8f7ac772f2b84508 0
64 Mon Jan 25 09:50:27 2021 AFE_CONFIG_PARAM_CHANGE 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 50 1d d9 bb 40 be 4a 1b 1f 19 89 f0 d2 5b 07 de b1 58 9e d1 4a 28
c0 3e 32 08 6b 7b 2a 932702235d90caf8733881a26b3cab3eb9cdad385e4249ce1ee0c0e9560fb5b7 0
65 Mon Jan 25 09:50:27 2021 AFE_CONFIG_PARAM_CHANGE 02 a3 f1 90 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 8ba032c5c1d50a56992cdab15346044d300e8f8ed742343b308a6ae5a199201d 0
66 Mon Jan 25 09:50:51 2021 BEGIN_CHARGE_SESSION 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 6935ba58ff766146259475d0faad7f0d8d47afd5789d6fe94fc350c36e90051f 0
67 Mon Jan 25 09:52:33 2021 END_CHARGE_SESSION 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 196b999466f1a963be116451cd676edcee88fcf33592a98ae8318a52cde3df18 0
68 Mon Jan 25 09:54:39 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 151654ab8f1f793f77fed1a789b45ba70ae439912867eb2767d22304d472cd80 0
69 Wed Jan 27 11:14:28 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 23ff21142b5b53066426f55d5c514ff29cdf830f17a7186bde5c6e850d13c7c6 0
70 Wed Jan 27 11:14:43 2021 TIME_SYNC 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 0d8697253cd926a622392587ac1bd86f0623336f6a70dfee4699bb87a68e37f3 0
71 Wed Jan 27 11:17:00 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 50 52 4f 44 07 74 74 69 2d 49
51 43 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 958d15a2816e5f1c5d9661b05bc0c439ab6d5c29592c53b225c12f2692e42eec 0
72 Wed Jan 27 11:18:13 2021 BEGIN_CHARGE_SESSION 04 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 809b4bda1ddc971a5c3acf054f78ad8f1645ccec1af396376bc08678bf088b57 0
73 Wed Jan 27 11:18:16 2021 END_CHARGE_SESSION 04 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 d21985e174f9aeeff63d83919f0853cbdbc8aeb21d24b2dcd4973cdc1d10b49b 0
74 Wed Jan 27 11:18:27 2021 AFE_CONFIG_PARAM_CHANGE 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 40 42
0f 00 40 42 0f 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 40 42 0f 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 49e793957751b9c1aa4496739cc4b69613c48567bd400f78265c1432cf6b13bc 0
75 Wed Jan 27 11:18:27 2021 AFE_CONFIG_PARAM_CHANGE 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 2f 76 de cc 23 8c b7 86 74 50 ec e8 9d 76 61 15 54 d1 d7 83 a8 6a
83 12 26 56 11 03 df 59b1d2ec9a1bc218b24f59f07b6256e8e6cf8cf899c4b528f1b71ec108b71953 0
76 Wed Jan 27 11:18:27 2021 AFE_CONFIG_PARAM_CHANGE 02 9c 70 95 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 883adfda0e522ce776dd715a0a9ddde4245737ed9e96fa72d1f59c6bfc45773d 0
77 Wed Jan 27 11:18:39 2021 AFE_CONFIG_PARAM_CHANGE 00 01 bc 7a aa 07 00 00 00 00 bc 7a aa 07 00 00 00 00 40 42
0f 00 40 42 0f 00 04 29 99 ff ff ff ff ff 40 14 28 ff ff ff ff ff 40 42 0f 00 40 42 0f 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 8a6cd750dbe636b6494704f26ad5491253a173885bdd435d3ecf3e14d6ec2b39 0
78 Wed Jan 27 11:18:39 2021 AFE_CONFIG_PARAM_CHANGE 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 7a a3 3b d7 8d 08 7a d8 1c 1d d4 34 9f 37 8b 2b da b2 26 c2 57 fa
50 92 75 17 73 b5 d6 2ecd39ca10866485e4a410b6f40e72e9d94638675afb6e866644a626428154f7 0
79 Wed Jan 27 11:18:39 2021 AFE_CONFIG_PARAM_CHANGE 02 f6 58 78 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 6821d950f1182bffc62af7d07fd28681c7d72a6538a6e4da3c3daee40f110136 0
80 Wed Jan 27 11:20:54 2021 AFE_CONFIG_PARAM_CHANGE 00 01 bc 7a aa 07 00 00 00 00 bc 7a aa 07 00 00 00 00 0b f3

0e 00 3a f3 0e 00 04 29 99 ff ff ff ff ff 40 14 28 ff ff ff ff ff fc 42 0f 00 ee 42 0f 00 00 00 00 00 00 00
00 00 00 00 00 00 00 20b67cf348f4fe025509878894a65d7f20330ea931cda27e20ef7e0c29c63e2b 0
81 Wed Jan 27 11:20:54 2021 AFE_CONFIG_PARAM_CHANGE 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 18 d3 84 34 a5 f4 de b7 7c 05 9b 1b 5d ee 57 7d 61 96 8e 3f 59 41
b6 0c 4b cc 84 6c 2c d309db23d49dbbdd01700226a58fbab465ccd9ec87ea709379d90599365b58fe 0
82 Wed Jan 27 11:20:54 2021 AFE_CONFIG_PARAM_CHANGE 02 d8 ba 53 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 919f36021446b2a44652d0bec1f730764c1b238873b33d1d2a9c6e95af8e69b9 0
83 Wed Jan 27 11:21:21 2021 BEGIN_CHARGE_SESSION 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 dd4cf162ab15c87f06bd3b4d2564e3f7654f847d4c912e51a121e9179b40ae5a 0
84 Wed Jan 27 11:23:03 2021 END_CHARGE_SESSION 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 98424319a51c893ebc66c60997436b679522569770dd772209aa0727a4a91f3a 0
85 Wed Jan 27 11:23:30 2021 BEGIN_CHARGE_SESSION 06 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 cf415a1e1e23bc041b894bb3aeb1cbb333c5daeef5b740521c38f6dbe116af0b 0
86 Wed Jan 27 11:23:32 2021 END_CHARGE_SESSION 06 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 a70fac83e1c9ca6b5c9f4ffc97f62855842842923d61ae52f991be023d95834e 0
87 Wed Jan 27 11:23:37 2021 AFE_CONFIG_PARAM_CHANGE 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 40 42
0f 00 40 42 0f 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 40 42 0f 00 40 42 0f 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 29f7b10c202274aea3d6777fa99d875dea6eab15922026413a3dc9d8519441c1 0
88 Wed Jan 27 11:23:37 2021 AFE_CONFIG_PARAM_CHANGE 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 2f 76 de cc 23 8c b7 86 74 50 ec e8 9d 76 61 15 54 d1 d7 83 a8 6a
83 12 26 56 11 03 df b756068128894cac45ae5e19be01aa2f10003102abfa5a67eda75619adebb013 0
89 Wed Jan 27 11:23:37 2021 AFE_CONFIG_PARAM_CHANGE 02 9c 70 95 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 d08920221745271a52de23161c7da2bcba1a3f039d7091f27fdf1589cbef7276 0
90 Wed Jan 27 11:23:49 2021 AFE_CONFIG_PARAM_CHANGE 00 01 ce 8b 56 07 00 00 00 00 ce 8b 56 07 00 00 00 00 40 42
0f 00 40 42 0f 00 a3 79 71 ff ff ff ff ff 8b 43 81 ff ff ff ff ff 40 42 0f 00 40 42 0f 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 414b7ea34aac1114d6e3cdc6970a5c4dfc744f52a7876638b9ea362f84a7ee30 0
91 Wed Jan 27 11:23:49 2021 AFE_CONFIG_PARAM_CHANGE 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 1e d5 12 d1 34 5d 1d f7 8d 44 94 63 07 38 7d b7 90 5a cb a2 a0 b1
1e 25 cd 69 f0 11 a2 f72552808942bdcf191c4105191ba451d9141f513e382e977cab78a174e7418f 0
92 Wed Jan 27 11:23:49 2021 AFE_CONFIG_PARAM_CHANGE 02 3f 30 a2 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 742b2d5383eb6f5c3d22d31ac93813a3b24c74a224885b73e253346dfccfdae0 0
93 Wed Jan 27 11:25:58 2021 AFE_CONFIG_PARAM_CHANGE 00 01 ce 8b 56 07 00 00 00 00 ce 8b 56 07 00 00 00 00 d4 f2
0e 00 05 f3 0e 00 a3 79 71 ff ff ff ff ff 8b 43 81 ff ff ff ff ff f6 42 0f 00 fc 42 0f 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 2a5b0f1efc64cfe356d00c38d4117a620f2fd605c167b89f6b978ad123358851 0
94 Wed Jan 27 11:25:58 2021 AFE_CONFIG_PARAM_CHANGE 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 05 fd d7 88 ff ec 3f fb f8 2f 3b 76 6f 90 5f b8 dd 65 a9 03 23 9d
cb 20 e7 cd 61 ce 74 da9eae0b332bfaa8ddfe51bd54e86597009718740d3eeb6d2385466406604e4e 0
95 Wed Jan 27 11:25:58 2021 AFE_CONFIG_PARAM_CHANGE 02 81 92 53 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 de9658dfd9ad990768882b9dc59df8aca67e1ad761fd4f89d2d1ba6e1ff457ea 0
96 Wed Jan 27 11:27:33 2021 BEGIN_CHARGE_SESSION 07 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 28aa3795f45f7a994a56d58a246ed508e46d597d281bb953522027262383748e 0
97 Wed Jan 27 11:29:15 2021 END_CHARGE_SESSION 07 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 38ac1fe1825c48942859bd69f74aa9d5e7c5d4f4b825a9c1faf85584b3f01810 0
98 Wed Jan 27 11:32:30 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 01 a4 e4 bf 38 73 32 34 34 30 34 30 00 73 32
34 34 30 33 39 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 0f333ad14bb8063089d41a0c1f358a4101600bb0af697bb129d3b6c174fa3524 0
99 Wed Jan 27 11:32:30 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 07 00 a0 61 7b 00 48 65 7b 00 f0 a8 7a 00 43 45 54 00 38 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 349293f312b02f41c71b4c518e71c2ae04ca2c5b897429d62a2d071e87eadbbf 0
100 Wed Jan 27 11:32:30 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 9b1692e43e951f3e94f3d895e71047ab3593a378f65892fed012142d5ef12754 0
101 Wed Jan 27 11:32:30 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 16 60 b7 07 53 7e 5b 1d d5 bb c3 35 28 02 20 23 54
c2 d1 0e 2b 90 6f 1a 57 48 08 f57c69a3c6041340264cc815fd22c7ff07fb926ac96f3c208d5a9975588d81b4 0
102 Wed Jan 27 11:32:30 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 03 5a 5f 71 6c 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 4b2af455c53f6e77602083ae1eab335ed972ef8e9bbb0c4ee43a15714a59a067 0
103 Wed Jan 27 11:32:30 2021 COMMISSIONED 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 1347bec4b0906e71e80e4797ca53595d2ce7461801984241fef5b9aa35154e3d 0

104 Wed Jan 27 11:33:57 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 c8a099dbb6c456f58455c2118e2f53a319e0b982cfb6b0141a69ea08dec119e0 0
106 Fri Mar 19 11:13:31 2021 TIME_SYNC 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 7e05f37b506ebe7291668d20b5206f87918de14876c0d02e13d8d678b6a35ce3 0
107 Fri Mar 19 11:13:31 2021 TIME_DEVIATION 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 14a8c05102e9f4679deff7502f7eb04584d50c58d2d08b67b734f8cd1876aec4 0
108 Fri Mar 19 11:13:38 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 43 55 53 54 0f 64 63 6d 2d 66
75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 c7ce34ba8e7f552a898064c90d1d05deb48e37687dd64af3ca4cce32e3146a8a 0
109 Fri Mar 19 11:16:10 2021 FIRMWARE_UPDATE_DOWNLOAD 01 01 01 01 7a ba c5 3f 83 2f 84 86 5f 0f c6 73 74 a0 14
a6 67 1d 8c 04 33 9b ad 6b 89 8a 17 7e 5b a1 fc 24 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 bfff7b4bd2717ae08c21b22f9bbf9410ec0de2f2260a42476c100b90b71546ba 0
110 Fri Mar 19 11:16:11 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 d4228b97a6441a0e1a94285ce083d13e753ede80fde6b99c79ff7546e01f95c9 0
111 Fri Mar 19 11:16:20 2021 POWER_STARTUP 10 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 b9f3bb884383e1833677578a8134c418922877e1fd0fe77a2d1bbeaebd5ae8e9 0
112 Fri Mar 19 11:16:28 2021 TIME_SYNC 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 61800c7f5173ecd5c0c8a733dde83f19c5231317ea9a19c09d569c214569e08d 0
113 Fri Mar 19 11:17:20 2021 TECHNICIAN_LOGIN ff ff ff ff 00 00 00 00 ff ff ff ff 43 55 53 54 0b 44 43 4d 5f 75
70 64 5f 56 44 45 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 4b3d020028ca1f7f160d60d9f47094b01146f515a64245df4042119db35b4b66 0
114 Fri Mar 19 11:20:39 2021 TECHNICIAN_LOGOUT ff ff ff ff 00 00 00 00 ff ff ff ff 43 55 53 54 0b 44 43 4d 5f
75 70 64 5f 56 44 45 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 2c1e054c5349d5dfeffdbedb68c56b4ee92115ee2148bd2e51385fdd5db98413 0
115 Fri Mar 19 11:22:09 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 43 55 53 54 0f 64 63 6d 2d 66
75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 14aff89aec6101b140939a0d1e3d9a7828c37e1946cbf3d4431ec60964fe6d01 0
116 Fri Mar 19 11:23:03 2021 TSOIF_CONFIG_PARAM_CHANGE 00 01 35 06 82 02 96 00 00 00 67 02 03 00 03 00 03 00 fd
ff fc ca cd ca 96 99 95 56 58 56 13 15 12 00 6a 89 d6 33 3b 57 e4 55 01 00 6e 00 a8 c0 e8 03 00 00 00 00 00 00
00 00 00 00 00 00 00 00 77901f9fe5906cf3fc060834223d9091f48e2f0904694fe36786647a294c1dff 0
117 Fri Mar 19 11:23:03 2021 TSOIF_CONFIG_PARAM_CHANGE 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 9f f6 8f de 7a af 27 9d 61 a0 4f 96 77 a9 b2 70 55 ff a7 df 79
d0 f2 89 f3 52 5b c7 0b a63a6fc73950b05c341dc952f78a5ad6f7fd259c8614b9e00a6612453565c644 0
118 Fri Mar 19 11:23:03 2021 TSOIF_CONFIG_PARAM_CHANGE 02 ef 7a 21 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 7e573ff553704afd34eca8db7b7aa91ad4d9d94fd86c5cc1361f67707886e604 0
119 Fri Mar 19 11:23:07 2021 INET_CONFIG_PARAM_CHANGE 00 01 08 08 08 08 04 04 08 08 46 3c ff ef d5 07 61 75 2e
70 6f 6f 6c 2e 6e 74 70 2e 6f 72 67 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 723b603d48ddf971496f02b6085b7f7ec0cf8cafabf675315952665827cd5c36 0
120 Fri Mar 19 11:23:07 2021 INET_CONFIG_PARAM_CHANGE 01 31 30 2e 31 38 2e 30 2e 31 30 30 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 64 63 6d 75 70 64 61
74 65 2e 74 72 69 74 69 14a1e66c0a1758dcd675d3712a226825d91029627e1147e3d60ef02b4748740c 0
121 Fri Mar 19 11:23:07 2021 INET_CONFIG_PARAM_CHANGE 02 75 6d 2e 63 6f 6d 2e 61 75 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 2f 64 63 6d 65 74 65 72 2e 69 6d 67 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 3e1f40d6e12519934228976dbbf4f5ae6e5f20dc3565dcd4e7571082b161ea14 0
122 Fri Mar 19 11:23:07 2021 INET_CONFIG_PARAM_CHANGE 03 00 00 78 00 12 0a 00 ff ff ff 01 00 12 0a 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 2c 65 ee d5 c1 3c 67 51 64 e6 8a 9e 84 04 ab 6c 90 1a 01
cd 0b 50 d1 97 70 d2 00 46b8cdc900cf5f1924eadf174385eb1ae0a4cc801a82fc1c2c22ae3d3f084f3f 0
123 Fri Mar 19 11:23:07 2021 INET_CONFIG_PARAM_CHANGE 04 c2 e7 18 9f e1 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 35c78f4579afd75aa360b45249e3704c4c0908db87975e71a3fdd020ecf29b51 0
124 Fri Mar 19 11:23:10 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 00 00 00 00 00 20 00 65 00 5c 00 4d 00 20 00
63 00 72 00 6f 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 b6c6c170debf61a86c9bdfa31eedd408b4954f8a8dce2cad47e19b167103c468 0
125 Fri Mar 19 11:23:10 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 5c 00 44 00 61 00 74 00 61 00 50 00 72 00 43 45 54 00 65 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 1fe389ff50129e9668627fc7cef0ad9c8f7014056c32ddba550a718413c4e9b7 0
126 Fri Mar 19 11:23:10 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 ca751226fc973a40a95c6a39980d4518268014d07ba59c1c487afeca991802f9 0
127 Fri Mar 19 11:23:10 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 c1 80 58 ee 57 b2 86 22 d6 0d 35 c4 aa e2 d1 3f c3
8e 97 ef 8e 00 9d b7 6f 52 4f 4451ca6515f4b8422afbd82d11f6d758c87eace68b5f8c91f3d35a6c4d6995a5 0
128 Fri Mar 19 11:23:10 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 0d b9 b4 e3 b5 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

00 00 00 00 00 00 00 00 00 00 43b5490070505902c39a2239ed4605101ba80d595a7dff51ab0411aa4b11d241 0
129 Fri Mar 19 11:23:49 2021 TSOIF_CONFIG_PARAM_CHANGE 00 01 35 06 82 02 96 00 00 00 67 02 03 00 03 00 03 00 fd
ff fc ca cd ca 96 99 95 56 58 56 13 15 12 00 6a 89 d6 33 3b 57 e4 55 01 00 79 00 12 0a e8 03 00 00 00 00 00 00
00 00 00 00 00 00 00 00 9268d0373b10d95bf7da0809b4c480f30350ba4387341d868a8bc234056af54f 0
130 Fri Mar 19 11:23:49 2021 TSOIF_CONFIG_PARAM_CHANGE 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 59 15 2e 40 f2 dd 83 55 37 ee e7 66 49 fa b7 ac db 5c d2 de 41
f0 e2 46 1f bd a8 78 62 e9e30d21afdd6e15affda3ada555e10072ef8b482fa32984687959191f02f1cb 0
131 Fri Mar 19 11:23:49 2021 TSOIF_CONFIG_PARAM_CHANGE 02 15 5c 33 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 27809b92f320a4f83870edc83fbfa8405bc180a4fd39df0faa6cc55467c7a5de 0
132 Fri Mar 19 11:24:40 2021 TSOIF_CONFIG_PARAM_CHANGE 00 01 35 06 82 02 96 00 00 00 67 02 03 00 03 00 03 00 fd
ff fc ca cd ca 96 99 95 56 58 56 13 15 12 00 6a 89 d6 33 3b 57 e4 55 01 02 79 00 12 0a e8 03 00 00 00 00 00 00
00 00 00 00 00 00 00 00 9263c531d5b0afed514c19b9a99143d1476af033a16e3e4fbc5f249b3944ca25 0
133 Fri Mar 19 11:24:40 2021 TSOIF_CONFIG_PARAM_CHANGE 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 e6 7d 9c 77 96 68 9b 82 3d 9e 7b dd d0 f0 f4 60 0b 39 86 83 80
c6 f5 da e7 62 f8 a9 42 0dfa2a3e248550242851b5a6d9bc7fcd2cda5a12135392ca867b0a52e8605720 0
134 Fri Mar 19 11:24:40 2021 TSOIF_CONFIG_PARAM_CHANGE 02 ba b2 a9 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 a23a96299121f4ec44d2395da66b0c75343554abb754f1252933450c59b76088 0
135 Fri Mar 19 11:26:32 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 dbe4c15a32039e867551a8ed639a2ae2c7335ebfff484b38fce51426209cedfb 0
136 Fri Mar 19 11:27:04 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 9ba794e052e215f50ae9c6bf6db290448d59025452d43c6638e753aefaa33c94 0
137 Fri Mar 19 11:27:11 2021 TIME_SYNC 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 e9bcaefa4281d574e2f3d8b40f9590947a148f92ed21b29f64001d71966f9a6b 0
138 Fri Mar 19 11:38:15 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 f6849d99494af59a789cc5fe22cb48d202f7bc7b9f82ec077b9e3fedb63c384b 0
139 Fri Mar 19 11:38:16 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 741a26558e92e92d70000610fcdf98a67e432ff51aea7451b77b6ade9322829f 0
140 Fri Mar 19 11:38:24 2021 TIME_SYNC 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 097a7015c64b0d1c1b734f03c761f04106d698119f594c193c48c353fce02bdd 0
141 Fri Mar 19 11:40:01 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 43 55 53 54 0f 64 63 6d 2d 66
75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 3ff1e78553896aec7433579446be0d7cd6ab2c4a69ecf40bfe0148c1586bb096 0
142 Fri Mar 19 11:40:40 2021 TSOIF_CONFIG_PARAM_CHANGE 00 01 35 06 82 02 96 00 7e 02 67 02 03 00 03 00 03 00 fd
ff fc ca cd ca 96 99 95 56 58 56 13 15 12 00 6a 89 d6 33 3b 57 e4 55 01 02 79 00 12 0a e8 03 00 00 00 00 00 00
00 00 00 00 00 00 00 00 19f1658fad5f5f4f7b6f03602509f869b28d460f265f98b6350b075607fc0571 0
143 Fri Mar 19 11:40:40 2021 TSOIF_CONFIG_PARAM_CHANGE 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 53 15 b3 2a 81 e5 94 90 58 e8 0b 8b 64 96 67 c7 91 e3 65 d7 1a
d4 aa 98 d8 16 ae 7d f4 f29d5a2b4fed4ac078acf6c7550c71d3b2b102336e5c12f22389525f90e88afb 0
144 Fri Mar 19 11:40:40 2021 TSOIF_CONFIG_PARAM_CHANGE 02 7d 27 8b 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 9b3142323bc7b3fa8e2a455e561f773052ed9d7b5cbc102f98386d5e18d24ea5 0
145 Fri Mar 19 11:40:54 2021 TSOIF_CONFIG_PARAM_CHANGE 00 01 35 06 82 02 96 00 7e 02 67 02 03 00 03 00 03 00 fd
ff fc ca cd ca 96 99 95 56 58 56 13 15 12 00 6a 89 d6 33 3b 57 e4 55 01 02 79 00 12 0a e8 03 00 00 00 00 00 00
00 00 00 00 00 00 00 00 fd115b290a556b71bd0fece28ba7186ee00cacfef296436603706783b5a8934a 0
146 Fri Mar 19 11:40:54 2021 TSOIF_CONFIG_PARAM_CHANGE 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 53 15 b3 2a 81 e5 94 90 58 e8 0b 8b 64 96 67 c7 91 e3 65 d7 1a
d4 aa 98 d8 16 ae 7d f4 f025734a61fa6a906881dde714ab49981f0d50d2d82a44d636804e41f1c8da4f 0
147 Fri Mar 19 11:40:54 2021 TSOIF_CONFIG_PARAM_CHANGE 02 7d 27 8b 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 b444de89c88532e8f663b0c03739e127c58f23fe837af5aee832d3173b4881e8 0
148 Fri Mar 19 11:41:02 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 0ae04e2641fa7c6005b9fc91bfcf7bd2ca552e784dd2d6c29be9fbf856b54af5 0
149 Fri Mar 19 11:41:03 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 1fe1f079a21df2726ca9dc19d74aab7809b6c2bdaf445dd0f14c22e3a5ddc1d3 0
150 Fri Mar 19 11:41:10 2021 TIME_SYNC 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 47b5df37a9f7e285263204bcb30577e9722b1d6af2de4a21ca6651816089dd87 0
151 Fri Mar 19 11:45:10 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 d6d00427e4d8c42b967265d6bf81387a6655f9ba13d21819fbf9115a2c6a35dc 0
152 Fri Mar 19 11:45:17 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 2f1f2d21f7a89afb86cde04ae79cfbb750a720ef6808d60033469e0628a1dd40 0
153 Fri Mar 19 11:45:24 2021 TIME_SYNC 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 7bdb183e963d908110c4a7c4e7c59044f54bf5506bd3bc3559bb0c281e3fcccf 0
154 Fri Mar 19 11:50:02 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 66d0efc991541884e0d447017f21746e8208bce701914a650f66f3dd8f296512 0
155 Mon Mar 22 13:52:43 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 015e0fc641e3c77ce9c7d5db0c5998b58f66817a3500533c397478273c54a08f 0
156 Mon Mar 22 13:57:28 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 5653bfdd03f02dcebe3ffc9ec9c5ea0d8739e8ae5f3166eb0f8b3ba5250c2ce2 0
157 Mon Mar 22 13:58:25 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 10bcf58b6b49a27713dbdfed348a8d101be1973004c5901389be1c65a4371b65 0
158 Mon Mar 22 13:59:59 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 52b81a3b566faa833f185222e1ee6edc219d00e4ed4f109789e34b4d1c2f3d40 0
159 Mon Mar 22 14:00:28 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 0df47f526a28c9e197f409ab10f9c16765b2101c1f9fb41510b8e0e884a7ded4 0
160 Mon Mar 22 14:03:58 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 ab73e99f6520e96109462fc0b847072b540854efe9514ba86ae1b7cc6195782a 0
161 Mon Mar 22 14:08:47 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 c79d1f6519e1a8c557d706cf0299cd665a037e6d97969fe7cbf6017a90b3e9f9 0
162 Mon Mar 22 14:13:01 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 6d77a6723c20002ef8f248fee83864bdf1e46d098ddf817c7406d2ab7a977180 0
163 Mon Mar 22 14:13:53 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 0fb6c9cdaca170e20d745c9904209c9402d6d2d49e2be32d69c63e9225a31178 0
164 Mon Mar 22 14:14:01 2021 TIME_SYNC 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 1c8df308a24286b67336084b8a72bd8f4399fb6a044db94ab2bdbcc4664a8286 0
165 Mon Mar 22 14:14:37 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 43 55 53 54 0f 64 63 6d 2d 66
75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 fdaee10349ef89b558990983be0e62b2f042ca0989457f8ec1b5e691a2e6cfe0 0
166 Mon Mar 22 14:15:46 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 43 55 53 54 0f 64 63 6d 2d 66
75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 16a9d6e3bb9e711609339084adb1f3d8cac6db4cac9a8a7675540a5d064aaab3 0
167 Mon Mar 22 14:17:25 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 44d52e2fc39f14b03df2fb56612271e85d9319e4a916ab8bd63cf29b05dc5e7c 0
168 Mon Mar 22 15:02:05 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 68c110fc2bd217dde316dd6dcfae3d7827b030ba089332661aa9970a6f6721f4 0
169 Mon Mar 22 15:10:10 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 62d67171acf15efb3bd49b73e7335595da7113922fdff5369e5e4181812346ec 0
170 Mon Mar 22 15:10:50 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 12bc3eb75ce1565cc2645f620c652cf66513b49786321d0dbbd154e9f44e9346 0
171 Mon Mar 22 15:17:58 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 920348a1e1f1253f64735066f650fd0f893843df3c59d100d810c1e7c50ccb9f 0
172 Mon Mar 22 15:20:48 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 afe9048247520280bdb7971cbdfd807da52d5836bcd17296e43970b25bb84b44 0
173 Mon Mar 22 15:20:55 2021 TIME_SYNC 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 7ffa65616d79017dfad55576948a0b355fb1b0e15ed0db22cb7dc16c60f33ded 0
174 Mon Mar 22 15:21:18 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 e5b2b9020dfe978b182e6a29f5c0676a079ca3d9ef9056f8c4f7b4668a5e60a3 0
175 Mon Mar 22 15:23:34 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 93e6cb5429e7c42ec45b0ddeccc700e3a91f0ac7760f72f9486a123a75d7ac8d 0

176 Mon Mar 22 15:28:46 2021 BEGIN_CHARGE_SESSION 08 00 00 00

00 00 00 00 00 00 00 00 00 00 dd6e391973bdb1bcb50edff3147fae9325e673666085783436ea4283dadefd86 0
200 Mon Mar 22 16:32:58 2021 COMMISSIONED 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 7d56a6f59aa138d2cb415dd52f0e3a01a285b0eb1c3966f5cbec04a778e62432 0
201 Mon Mar 22 16:33:00 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 01 00 00 00 00 20 00 00 00 a8 05 94 00 20 00
49 f4 00 b7 94 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 250fd4c9c01f6b13cf6ce974f3ba793162dc8a67cadd20b7a77a0ce18c79194e 0
202 Mon Mar 22 16:33:00 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 08 00 d8 40 a3 00 c0 45 a3 00 80 9c a2 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 8809ced3e4ff20bdd4768986667d910f55a8a45023f87c71e4d020b489fca598 0
203 Mon Mar 22 16:33:00 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 55db60c64d8282473eed4136aac2468143e34ab7cc8c5f71b9e6e2f815e8955d 0
204 Mon Mar 22 16:33:00 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 c9 85 71 17 e8 37 75 76 e3 52 1b 10 4f 9b 49 d6 63
15 ad 38 62 b8 85 bc 1c ee c0 345de0f0cea3824d91cb007095113485cd37dcd2ef733a3cc90126f3d09dc427 0
205 Mon Mar 22 16:33:00 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 b6 8f d9 92 cb 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 5669289343b97325d7b95eafb81aa2f38d5a608666c35e459baf3e8c9a348d8c 0
206 Mon Mar 22 16:40:18 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 43 55 53 54 0f 64 63 6d 2d 66
75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 9ed0ae33381d5152ad096363f19300b9f422453ff400795300c7974360a398b5 0
207 Mon Mar 22 16:40:25 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 00 00 00 00 00 20 00 00 00 a8 05 94 00 20 00
49 f4 00 b7 94 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 4ec1d24a0ab89ba7f363f6cee4fd99e4ec4ab87a62cd914be9e6d8f32fe89939 0
208 Mon Mar 22 16:40:25 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 08 00 d8 40 a3 00 c0 45 a3 00 80 9c a2 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 29dcbabf8efa8a2cbe5da59c7eac956cb954d19d6cae365dc4e7d00c23752e4e 0
209 Mon Mar 22 16:40:25 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 f3433f0d7e7b496af52dc42346d73be3ddedbdd9a508a9c425e57b16efae2f99 0
210 Mon Mar 22 16:40:25 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 61 0f 56 5f 07 e2 c7 fc 26 a5 37 34 36 f7 65 8d ce
65 28 40 13 f4 81 40 58 88 63 3070de3993c613738f3567bc8a2d2b51fd18b6abcdfb0191b3047e30cbf96412 0
211 Mon Mar 22 16:40:25 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 ae 24 d6 89 73 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 80f44da94108563d8cc14a2da83fda3c8dc363f00f905c7b06f703898bc072b1 0
212 Mon Mar 22 16:40:27 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 00 00 00 00 00 20 00 00 00 a8 05 94 00 20 00
49 f4 00 b7 94 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 f1cf5576ce87ad792db0ed1d25b66712b90d40b04b80fb287c26b1ebd10b0a6c 0
213 Mon Mar 22 16:40:27 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 08 00 d8 40 a3 00 c0 45 a3 00 80 9c a2 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 7a2bf42f3dd4c7e396a1821fa095e4942f9c93077cb40524eb9920f9320046e3 0
214 Mon Mar 22 16:40:27 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 208aebc84b321b882f0660102d098aceea1eb0f16b71fed0d320665ddf2654bf 0
215 Mon Mar 22 16:40:27 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 61 0f 56 5f 07 e2 c7 fc 26 a5 37 34 36 f7 65 8d ce
65 28 40 13 f4 81 40 58 88 63 d831b05851bb60930636c1b4340e9398fc83120187570c55ee9c34a09db330ac 0
216 Mon Mar 22 16:40:27 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 ae 24 d6 89 73 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 be39754ab738bb9745fd04d38ac2a50b2d9614a69b0a6f846440e7db08fc91ce 0
217 Mon Mar 22 16:40:27 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 00 00 00 00 00 20 00 00 00 a8 05 94 00 20 00
49 f4 00 b7 94 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 4325b379143fc79976850f6352d783fb13a6c8b5e8366e5dcf796cae122af339 0
218 Mon Mar 22 16:40:27 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 08 00 d8 40 a3 00 c0 45 a3 00 80 9c a2 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 5e9ebc4d332ac538445535d0d674dcf68c7a5fc857de9243487202e01284cf92 0
219 Mon Mar 22 16:40:27 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 0ffddd03137ba5139a54d42d69450bca2ccccbfa2c44b03b96fba9f76b449f84 0
220 Mon Mar 22 16:40:27 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 61 0f 56 5f 07 e2 c7 fc 26 a5 37 34 36 f7 65 8d ce
65 28 40 13 f4 81 40 58 88 63 559644d022d06a7ec7bad5e66459dd1e573d388d57c29ef30dc2f162c65d7d68 0
221 Mon Mar 22 16:40:27 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 ae 24 d6 89 73 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 f4d3b4c4223f2171b6bbac995a6e90cf07f4f5a9e15f06727d077d328d7b4be1 0
222 Mon Mar 22 16:40:28 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 00 00 00 00 00 20 00 00 00 a8 05 94 00 20 00
49 f4 00 b7 94 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 b9e46e299afa39e435c24bcff4afaab399f3223b91c4efd3aba293d2f7b139a4 0
223 Mon Mar 22 16:40:28 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86

2c d4 aa a7 f2 e2 65 20 00 08 00 d8 40 a3 00 c0 45 a3 00 80 9c a2 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 49bf1f28edc995e0568e3bc1bfa2d7d83e1ec2f54b11e453a8909ccd66dd8560 0
224 Mon Mar 22 16:40:28 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 61e8d62f6767611c26225852fc187d4118de5bb7d9096b6ceaf3b20dd5fd810d 0
225 Mon Mar 22 16:40:28 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 61 0f 56 5f 07 e2 c7 fc 26 a5 37 34 36 f7 65 8d ce
65 28 40 13 f4 81 40 58 88 63 7c0c92c79a97783f82f369acd322fa0c1232db156f39c65027931a66d9b34106 0
226 Mon Mar 22 16:40:28 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 ae 24 d6 89 73 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 6ad6ee9afd80c2c89de4045771c104415e3d2ccc40de8441b7a0a08ea407abf6 0
227 Mon Mar 22 16:40:37 2021 TECHNICIAN_LOGOUT ff ff ff ff ff ff ff ff ff ff ff ff 43 55 53 54 0f 64 63 6d 2d
66 75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 baf29f73cba506f48b394b0663c2574604a2a68dd18ebcc9ea4bcc10df3cc1ca 0
228 Mon Mar 22 16:41:14 2021 TIME_SYNC 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 3746b8d0726347014cb6743771e4be77229b5ca393152d748cfa87ff19907abb 0
229 Mon Mar 22 16:41:20 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 43 55 53 54 0f 64 63 6d 2d 66
75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 0d30b9e90f719a15282f8b7d4b80e7ec1891215fabe6689f8f97ffc8b854d983 0
230 Mon Mar 22 16:42:12 2021 INET_CONFIG_PARAM_CHANGE 00 01 08 08 08 08 04 04 08 08 46 3c ff ef d5 07 61 75 2e
70 6f 6f 6c 2e 6e 74 70 2e 6f 72 67 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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231 Mon Mar 22 16:42:12 2021 INET_CONFIG_PARAM_CHANGE 01 31 30 2e 31 38 2e 30 2e 31 30 30 00 00 00 00 00 00 00
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74 65 2e 74 72 69 74 69 74493504630773ab90c6a4de29fa974d77a75a6e55717ca7469734a81b4d4d2e 0
232 Mon Mar 22 16:42:12 2021 INET_CONFIG_PARAM_CHANGE 02 75 6d 2e 63 6f 6d 2e 61 75 00 00 00 00 00 00 00 00
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233 Mon Mar 22 16:42:12 2021 INET_CONFIG_PARAM_CHANGE 03 00 00 78 00 12 0a 00 ff ff ff 01 00 12 0a 00 00 00 00
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cd 0b 50 d1 97 70 d2 00 99956c4326aa92b2e20cd3e71839aabbb335e574d36bb4539ea6d01b743008ce 0
234 Mon Mar 22 16:42:12 2021 INET_CONFIG_PARAM_CHANGE 04 c2 e7 18 9f e1 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 71bbaa914d22074bcb5475ca38822b54a8bfc8b42417838613fd6ce9949ceeae 0
235 Mon Mar 22 16:51:59 2021 INET_CONFIG_PARAM_CHANGE 00 01 08 08 08 08 04 04 08 08 46 3c ff ef d5 07 61 75 2e
70 6f 6f 6c 2e 6e 74 70 2e 6f 72 67 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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236 Mon Mar 22 16:51:59 2021 INET_CONFIG_PARAM_CHANGE 01 31 30 2e 31 38 2e 30 2e 31 30 30 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 64 63 6d 75 70 64 61
74 65 2e 74 72 69 74 69 7fbcd4d74e35278153e5ec66171af0b3a17b925e928e3471505e50443061299e 0
237 Mon Mar 22 16:51:59 2021 INET_CONFIG_PARAM_CHANGE 02 75 6d 2e 63 6f 6d 2e 61 75 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 2f 64 63 6d 65 74 65 72 2d 54 44 4d 2d 32 34 37 31 2e 69 6d 67 00
00 00 00 00 00 00 00 00 fa1f592ee6e5e0a391c9fd4dad103f7d8019b677a5813f962b004be76d8204e6 0
238 Mon Mar 22 16:51:59 2021 INET_CONFIG_PARAM_CHANGE 03 00 00 78 00 12 0a 00 ff ff ff 01 00 12 0a 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 1f 72 cb f0 38 9c 0f 30 0e 7c 26 07 54 15 90 e4 c7 7b c9
5e 95 0a 34 14 7a ee bb d2b4fd1200537ff10a682e5dae4e7def944a2a2e26291f66d4f637bc950fd132 0
239 Mon Mar 22 16:51:59 2021 INET_CONFIG_PARAM_CHANGE 04 96 e5 ab 40 53 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 d35bb4002f9e996cd6665423b3aa5251cf361dc293f66a1b0d1c5f85c45db802 0
240 Mon Mar 22 16:52:23 2021 FIRMWARE_UPDATE_DOWNLOAD 00 01 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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241 Mon Mar 22 16:52:44 2021 FIRMWARE_UPDATE_DOWNLOAD 00 01 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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242 Mon Mar 22 16:53:04 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 01 00 00 00 00 20 00 00 00 a8 05 55 01 20 00
35 0e 00 b7 55 01 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 74c9d4ce23d2863091de754dad984827a6f3d9a567fd2f3493ecd93ae629b06f 0
243 Mon Mar 22 16:53:04 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 05 00 38 dc 72 00 58 df 72 00 f0 96 72 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 eedfa3bcebadd05ba51a278111dae33bbe982c271ba6463dc8d0d59a729ab310 0
244 Mon Mar 22 16:53:04 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 f25962a061540b197e98b11b4803e70cf05cbf625b6bac13fc4a525c8b455031 0
245 Mon Mar 22 16:53:04 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 5a d6 16 bc 42 3b cd d9 d6 1b 6f 77 03 17 2e f1 c0
66 57 eb 76 d6 77 69 2f 70 08 689b17ce598a6415f89a49c8d6ea2988799a51fa531d9aa8572e65bae23356b8 0
246 Mon Mar 22 16:53:04 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 07 a3 10 9a c4 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 a086d175b381da35b0d58298d9864dbcf4bcde5cc7c3cf5ab3085787f61166f9 0

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247 Mon Mar 22 16:53:04 2021 COMMISSIONED 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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00 00 00 00 256f1f6379ccccb56dadbf4ad50b1e0c437eeea19eb7a83d50f5e9bd9c6c117b 0
248 Mon Mar 22 16:53:28 2021 FIRMWARE_UPDATE_DOWNLOAD 00 01 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 2642ff7d45f6879c149a0e193f944d71b06d379309c4286ef1feff9d2828ac88 0
249 Mon Mar 22 16:54:09 2021 INET_CONFIG_PARAM_CHANGE 00 01 08 08 08 08 04 04 08 08 46 3c ff ef d5 07 61 75 2e
70 6f 6f 6c 2e 6e 74 70 2e 6f 72 67 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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250 Mon Mar 22 16:54:09 2021 INET_CONFIG_PARAM_CHANGE 01 31 30 2e 31 38 2e 30 2e 31 30 30 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 64 63 6d 75 70 64 61
74 65 2e 74 72 69 74 69 359d6c99c1fb25599d63e06b5d328a9f8d7fb304e54d300db1831cb99db8f660 0
251 Mon Mar 22 16:54:09 2021 INET_CONFIG_PARAM_CHANGE 02 75 6d 2e 63 6f 6d 2e 61 75 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 2f 64 63 6d 65 74 65 72 2e 69 6d 67 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 825000f28a84d30688b44270106da9995aaab3e583f21b2bf0218c5ffb8fbdec 0
252 Mon Mar 22 16:54:09 2021 INET_CONFIG_PARAM_CHANGE 03 00 00 78 00 12 0a 00 ff ff ff 01 00 12 0a 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 2c 65 ee d5 c1 3c 67 51 64 e6 8a 9e 84 04 ab 6c 90 1a 01
cd 0b 50 d1 97 70 d2 00 29e6f947b732a06748acf0151cff486892ccf25f4dd17d7ad6c91ee21f72545d 0
253 Mon Mar 22 16:54:09 2021 INET_CONFIG_PARAM_CHANGE 04 c2 e7 18 9f e1 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 7b3d758f80efd7240dda6fe9ef023c6fec7994a4db31e0fbe3e4ccd78b738aa1 0
254 Mon Mar 22 16:54:26 2021 FIRMWARE_UPDATE_DOWNLOAD 00 01 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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255 Mon Mar 22 16:54:37 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 01 40 e2 01 00 20 00 00 00 a8 05 55 01 20 00
35 0e 00 b7 55 01 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 1295d888b13648b8a598ca19b3e3ea30d8bb0233976bfb24ee20cff3570f3316 0
256 Mon Mar 22 16:54:37 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 05 00 38 dc 72 00 58 df 72 00 f0 96 72 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 bdd5c62ebdff95aebd8ef3733907964c3a3727e4cf3f8cbda5f06c5cf2784f8d 0
257 Mon Mar 22 16:54:37 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00 00
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258 Mon Mar 22 16:54:37 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 cd b3 72 f2 a7 e9 7c ac 56 c4 8b 66 32 02 38 40 63
54 76 89 88 a5 2e d6 16 5b 16 360a977d644cc5d084d4ad914678106844429b400ec069b95dd5f4e1bd51d30f 0
259 Mon Mar 22 16:54:37 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 bd 45 a8 dd b1 00 00 00 00 00 00 00 00 00 00 00
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260 Mon Mar 22 16:54:56 2021 FIRMWARE_UPDATE_DOWNLOAD 00 01 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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261 Mon Mar 22 16:55:14 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 01 40 e2 01 00 20 31 32 33 34 00 55 01 35 36
37 38 00 b7 55 01 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 c204e4b0714d232ab73cf84c8e1bca17d173fa456b70605b795798ba0d320d0c 0
262 Mon Mar 22 16:55:14 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 05 00 38 dc 72 00 58 df 72 00 f0 96 72 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 2767ef6557aac1bb753d829e602c923b394d09807b00ebfc1265b522d6eaf922 0
263 Mon Mar 22 16:55:14 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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3b c8 13 33 16 61 ac e6 dc e1 5b910d220ca3fef793ac189d2b80859936d46f9873707051682a24cc1e2500df 0
265 Mon Mar 22 16:55:14 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 eb 4d 6f ad b0 00 00 00 00 00 00 00 00 00 00 00
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266 Mon Mar 22 16:55:33 2021 FIRMWARE_UPDATE_DOWNLOAD 00 01 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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00 00 00 00 00 00 00 00 8e99f68ac6f5e826a5374c749d3c4e0263e1e3d9bcda0b18209c5ea514815b7f 0
267 Mon Mar 22 16:58:34 2021 INET_CONFIG_PARAM_CHANGE 00 01 08 08 08 08 04 04 08 08 46 3c ff ef d5 07 61 75 2e
70 6f 6f 6c 2e 6e 74 70 2e 6f 72 67 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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268 Mon Mar 22 16:58:34 2021 INET_CONFIG_PARAM_CHANGE 01 31 30 2e 31 38 2e 30 2e 31 30 30 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 64 63 6d 75 70 64 61
74 65 2e 74 72 69 74 69 3d3e291872d8fa687665977b96d6354b29b7205dbe67a416fccc5dbe22710f7a 0
269 Mon Mar 22 16:58:34 2021 INET_CONFIG_PARAM_CHANGE 02 75 6d 2e 63 6f 6d 2e 61 75 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 2f 64 63 6d 65 74 65 72 2e 69 6d 67 00 00 00 00 00 00 00 00 00 00 00
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270 Mon Mar 22 16:58:34 2021 INET_CONFIG_PARAM_CHANGE 03 00 00 78 00 12 0a 00 ff ff ff 01 00 12 0a 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 2c 65 ee d5 c1 3c 67 51 64 e6 8a 9e 84 04 ab 6c 90 1a 01
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271 Mon Mar 22 16:58:34 2021 INET_CONFIG_PARAM_CHANGE 04 c2 e7 18 9f e1 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 0710ddbe5878a6cde6536e6e6f1f167a957cca5f4c90a8f1303f35026df9ab9d 0
272 Mon Mar 22 16:59:28 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 01 a3 e4 bf 38 53 32 34 34 30 33 36 00 53 32
34 34 30 33 35 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 fa027b488babb91c482dac41ed5f6567e87a3da6f93f5a4711429ee240a67a66 0
273 Mon Mar 22 16:59:28 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 05 00 38 dc 72 00 58 df 72 00 f0 96 72 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 02911f461c54a647ea8916e8dd11ec6f9ff5200ade97ca66b9644adee5553aa0 0
274 Mon Mar 22 16:59:28 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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275 Mon Mar 22 16:59:28 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 0d 3d cc 0a 5c 15 9f 6f ca 90 17 c6 f7 ff 3e f9 b7
e8 17 53 80 be 14 97 bb 60 13 232591dece6651763ddd62573a209dc213cbeaea697fdc03830e24e4ae500fa0 0
276 Mon Mar 22 16:59:28 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 62 3a 6c 74 51 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 5bc03b7cafb1f2342329c212e0b89db30d002500cbbbf0de66fadc8907dc10fb 0
277 Mon Mar 22 16:59:29 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 01 a3 e4 bf 38 53 32 34 34 30 33 36 00 53 32
34 34 30 33 35 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 99d9e88fa354854d409f2b82632c579e73cbb352ea0621f4a12ed6c30fb64375 0
278 Mon Mar 22 16:59:29 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 05 00 38 dc 72 00 58 df 72 00 f0 96 72 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 6949b5dd0a981412dcafefa0c6f909fda15c42706cfd8f2bf5951b68236ca4f2 0
279 Mon Mar 22 16:59:29 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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280 Mon Mar 22 16:59:29 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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281 Mon Mar 22 16:59:29 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 62 3a 6c 74 51 00 00 00 00 00 00 00 00 00 00
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282 Mon Mar 22 16:59:30 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 01 a3 e4 bf 38 53 32 34 34 30 33 36 00 53 32
34 34 30 33 35 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 a31ea5da440f673271a1aaa2618dbb09a10c91cfa34001c0576629c144b2faad 0
283 Mon Mar 22 16:59:30 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 05 00 38 dc 72 00 58 df 72 00 f0 96 72 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 e437e042ee546c6f6b121bafbaa6dfa85ed6ce9e53eef1015d80ffece25aa371 0
284 Mon Mar 22 16:59:30 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00
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286 Mon Mar 22 16:59:30 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 62 3a 6c 74 51 00 00 00 00 00 00 00 00 00 00
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287 Mon Mar 22 16:59:34 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 01 a3 e4 bf 38 53 32 34 34 30 33 36 00 53 32
34 34 30 33 35 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
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288 Mon Mar 22 16:59:34 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 05 00 38 dc 72 00 58 df 72 00 f0 96 72 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 4a0fdc25085939c90ffacf13e01ef19dbc56f3f4f3e528f06bf630415fdc871e 0
289 Mon Mar 22 16:59:34 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00
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292 Mon Mar 22 16:59:34 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 01 a3 e4 bf 38 53 32 34 34 30 33 36 00 53 32
34 34 30 33 35 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
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293 Mon Mar 22 16:59:34 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 05 00 38 dc 72 00 58 df 72 00 f0 96 72 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 4df460f71cf53eeb59729adcc84d90690de1511628e09d332ab868763a8d2cf0 0
294 Mon Mar 22 16:59:34 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00
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296 Mon Mar 22 16:59:34 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 62 3a 6c 74 51 00 00 00 00 00 00 00 00 00 00
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297 Mon Mar 22 17:01:10 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 43 55 53 54 0f 64 63 6d 2d 66
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304 Tue Mar 23 04:30:51 2021 SYSCHECK_RESULT 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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305 Tue Mar 23 07:11:31 2021 TIME_SYNC 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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307 Tue Mar 23 08:58:54 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 00 a3 e4 bf 38 53 32 34 34 30 33 39 00 53 32
34 34 30 34 30 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 0efb54a7aa64f83c53fcf700492004f0ed74389fcf227846ed7f72931c4348f5 0
308 Tue Mar 23 08:58:54 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 14 00 e0 d7 a4 00 e8 e3 a4 00 28 c2 a3 00 43 45 54 00 00 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 02f95800627dd2d5f485a15a761ea6f7b188adae0e9315784c7d8c1eed46898d 0
309 Tue Mar 23 08:58:54 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00
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310 Tue Mar 23 08:58:54 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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bb 6a fd 6a 25 eb a7 9f 89 58 f6b5e5d0b96eea20393d95803c4eff22a6275e6cacd9d5c9705c445083d69278 0
311 Tue Mar 23 08:58:54 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 16 99 ee 8c df 00 00 00 00 00 00 00 00 00 00
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312 Tue Mar 23 08:59:09 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 00 a4 e4 bf 38 53 32 34 34 30 33 39 00 53 32
34 34 30 34 30 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 fe1e484378f5cec9e76faa0e44ff7e214eca9ab483e2392d39e5080cb3ad7ab6 0
313 Tue Mar 23 08:59:09 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 14 00 e0 d7 a4 00 e8 e3 a4 00 28 c2 a3 00 43 45 54 00 00 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 0686258f39a4d84e7c20a9afa131fa745c3fbbb9e0ed442c809354d6f8365d8e 0
314 Tue Mar 23 08:59:09 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00
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315 Tue Mar 23 08:59:09 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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71 88 34 0f ba b7 42 09 30 4c 9d907fe1ad62c0e5152b97b4ec1c70ba4a8bee8c32df8a092fd77f60899fdb1f 0
316 Tue Mar 23 08:59:09 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 b6 cd 20 bf be 00 00 00 00 00 00 00 00 00 00
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317 Tue Mar 23 08:59:14 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 01 a4 e4 bf 38 53 32 34 34 30 33 39 00 53 32
34 34 30 34 30 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 b3b3dc16bf605d26aeb07b561f40e667b3f1b4ef8cadb90cde21dc69d7220e95 0

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318 Tue Mar 23 08:59:14 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 14 00 e0 d7 a4 00 e8 e3 a4 00 28 c2 a3 00 43 45 54 00 00 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 b036416f708bec27c5be51685ffe93f339ceec6a25be8b2cb326446e25261a0d 0
319 Tue Mar 23 08:59:14 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00
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3a 13 4f dd 19 80 b2 e0 01 e8 2f8171294bdfd1c40e3e21eab407763d43c1ef1aab3d70cce21090f7bbf00138 0
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322 Tue Mar 23 08:59:14 2021 COMMISSIONED 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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323 Tue Mar 23 08:59:51 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 01 a4 e4 bf 38 53 32 34 34 30 33 39 00 53 32
34 34 30 34 30 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 ecd0ab0323ff773278d5a860659c7c6b878b5a8f59840cf4caa952e23f1cab84 0
324 Tue Mar 23 08:59:51 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 14 00 e0 d7 a4 00 e8 e3 a4 00 28 c2 a3 00 43 45 54 00 00 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 b5c077dae2d935f8b0d5598a20b7dc1fc097ad15d37a7a69614caf8e7c8dee9d 0
325 Tue Mar 23 08:59:51 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00
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327 Tue Mar 23 08:59:51 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 c7 bf 7d 4a eb 00 00 00 00 00 00 00 00 00 00
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328 Tue Mar 23 08:59:59 2021 FIRMWARE_UPDATE_DOWNLOAD 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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329 Tue Mar 23 09:11:44 2021 TECHNICIAN_LOGOUT ff ff ff ff ff ff ff ff ff ff ff ff 43 55 53 54 0f 64 63 6d 2d
66 75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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330 Tue Mar 23 09:12:05 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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331 Tue Mar 23 11:11:38 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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333 Tue Mar 23 11:12:46 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 43 55 53 54 0f 64 63 6d 2d 66
75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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334 Tue Mar 23 11:14:22 2021 FIRMWARE_UPDATE_DOWNLOAD 01 01 01 01 7a ba c5 3f 83 2f 84 86 5f 0f c6 73 74 a0 14
a6 67 1d 8c 04 33 9b ad 6b 89 8a 17 7e 5b a1 fc 24 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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336 Tue Mar 23 11:14:31 2021 POWER_STARTUP 10 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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338 Tue Mar 23 11:14:50 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 43 55 53 54 0f 64 63 6d 2d 66
75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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339 Tue Mar 23 11:15:13 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 00 a4 e4 bf 38 53 32 34 34 30 33 39 00 53 32
34 34 30 34 30 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
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340 Tue Mar 23 11:15:13 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 15 00 f8 32 84 00 98 3f 84 00 58 9b 83 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 ec2884a0dfde14217fe0e276ad1a40ba5579fe3214dcff14ccc002a0a97f2597 0
341 Tue Mar 23 11:15:13 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00
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343 Tue Mar 23 11:15:13 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 e9 49 ab f8 0b 00 00 00 00 00 00 00 00 00 00
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344 Tue Mar 23 11:15:16 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 00 a4 e4 bf 38 53 32 34 34 30 33 39 00 53 32
34 34 30 34 30 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
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345 Tue Mar 23 11:15:16 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 15 00 f8 32 84 00 98 3f 84 00 58 9b 83 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 19b8d3d81a0f2b85e19fb9949fb2357296e49330d6be3bbeee7a9935fec994a5 0
346 Tue Mar 23 11:15:16 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00
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348 Tue Mar 23 11:15:16 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 e9 49 ab f8 0b 00 00 00 00 00 00 00 00 00 00
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349 Tue Mar 23 11:15:17 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 00 a4 e4 bf 38 53 32 34 34 30 33 39 00 53 32
34 34 30 34 30 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
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350 Tue Mar 23 11:15:17 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 15 00 f8 32 84 00 98 3f 84 00 58 9b 83 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 3339061c0affc774c2383c0428a1d917624e5e123e62eb25eda15a4a5613bdc0 0
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353 Tue Mar 23 11:15:17 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 e9 49 ab f8 0b 00 00 00 00 00 00 00 00 00 00
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354 Tue Mar 23 11:15:18 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 00 a4 e4 bf 38 53 32 34 34 30 33 39 00 53 32
34 34 30 34 30 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 f3e445efbd4b4886c4a01f79eb53b411e2b2bc0be10b3976b09a3df52f12a7a2 0
355 Tue Mar 23 11:15:18 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 15 00 f8 32 84 00 98 3f 84 00 58 9b 83 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 46456b4bc690485547d7c0b4e8e749212343c5fcb95296a36d19dfaa9284a0b7 0
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358 Tue Mar 23 11:15:18 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 e9 49 ab f8 0b 00 00 00 00 00 00 00 00 00 00
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359 Tue Mar 23 11:15:19 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 00 a4 e4 bf 38 53 32 34 34 30 33 39 00 53 32
34 34 30 34 30 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
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360 Tue Mar 23 11:15:19 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 15 00 f8 32 84 00 98 3f 84 00 58 9b 83 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 e1280465cb71499fb8bc872c63d892920d80802f1f8524ec77c0a42b89f5d89f 0
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362 Tue Mar 23 11:15:19 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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363 Tue Mar 23 11:15:19 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 e9 49 ab f8 0b 00 00 00 00 00 00 00 00 00 00
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364 Tue Mar 23 11:15:20 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 00 a4 e4 bf 38 53 32 34 34 30 33 39 00 53 32
34 34 30 34 30 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
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365 Tue Mar 23 11:15:20 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86

2c d4 aa a7 f2 e2 65 20 00 15 00 f8 32 84 00 98 3f 84 00 58 9b 83 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 a4f6bb314ff93fc369192e6b616f1d8597ff309193b63fd4e320fb42a7718934 0
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367 Tue Mar 23 11:15:20 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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368 Tue Mar 23 11:15:20 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 e9 49 ab f8 0b 00 00 00 00 00 00 00 00 00 00
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369 Tue Mar 23 11:15:21 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 00 a4 e4 bf 38 53 32 34 34 30 33 39 00 53 32
34 34 30 34 30 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 98a9ff7f8b33b22300ccaeabd4602eac26b5a21071999aecfb05b4def267b495 0
370 Tue Mar 23 11:15:21 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 15 00 f8 32 84 00 98 3f 84 00 58 9b 83 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 29325b24e1769568d2051242b5b85a5defa179a0fe6e3b3605794a6b89dd5c9e 0
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373 Tue Mar 23 11:15:21 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 e9 49 ab f8 0b 00 00 00 00 00 00 00 00 00 00
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374 Tue Mar 23 11:15:24 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 00 a4 e4 bf 38 53 32 34 34 30 33 39 00 53 32
34 34 30 34 30 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
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375 Tue Mar 23 11:15:24 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 15 00 f8 32 84 00 98 3f 84 00 58 9b 83 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 e2c7880ed9b42661543b7c018654f4b2ec353a95f52b9dd2c3fc011ea1dd5d49 0
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378 Tue Mar 23 11:15:24 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 e9 49 ab f8 0b 00 00 00 00 00 00 00 00 00 00
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379 Tue Mar 23 11:15:25 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 00 a4 e4 bf 38 53 32 34 34 30 33 39 00 53 32
34 34 30 34 30 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
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380 Tue Mar 23 11:15:25 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 15 00 f8 32 84 00 98 3f 84 00 58 9b 83 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
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383 Tue Mar 23 11:15:25 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 e9 49 ab f8 0b 00 00 00 00 00 00 00 00 00 00
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384 Tue Mar 23 11:15:26 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 00 a4 e4 bf 38 53 32 34 34 30 33 39 00 53 32
34 34 30 34 30 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 df54fd395d0cdbf7e7eb477558290ab57c65fb5b19afceb5232955b3bf8a581d 0
385 Tue Mar 23 11:15:26 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 15 00 f8 32 84 00 98 3f 84 00 58 9b 83 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 90ced9acf13b958c42946c85556f1ee95584bd2a4badb4b9db76bed4f5e25f68 0
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387 Tue Mar 23 11:15:26 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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388 Tue Mar 23 11:15:26 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 e9 49 ab f8 0b 00 00 00 00 00 00 00 00 00 00
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389 Tue Mar 23 11:16:53 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 01 a4 e4 bf 38 53 32 34 34 30 33 39 00 53 32
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f8 42 1b e4 98 f2 75 74 09 35 061ea796f9a83fb973ca0f24d80df99e7a6cb300f02371ca6f8cd72268480eb2 0
390 Tue Mar 23 11:16:53 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 15 00 f8 32 84 00 98 3f 84 00 58 9b 83 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 c3c7d15690a124d65492e33944c4a7a443394acc3c6353e580994b1c24ef9485 0
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392 Tue Mar 23 11:16:53 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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48 71 24 8b 6c 9c b9 2a d0 38 783c084491d8fd41acba0636ca4753211c6a7f87f3c96fd22e3b3f64c0a41442 0
393 Tue Mar 23 11:16:53 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 f3 e0 3e 10 56 00 00 00 00 00 00 00 00 00 00
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394 Tue Mar 23 11:16:53 2021 COMMISSIONED 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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00 00 00 00 ac3061823b8298e03a2fd2f5bec52a963bb93c0207a71b0d15734364d0ed2880 0
395 Tue Mar 23 11:16:54 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 01 a4 e4 bf 38 53 32 34 34 30 33 39 00 53 32
34 34 30 34 30 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 23caface65ec94cc2b106fb46cf6e2d0c294de54bd95a1c4fc2c150891645f7b 0
396 Tue Mar 23 11:16:54 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 15 00 f8 32 84 00 98 3f 84 00 58 9b 83 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 86c4988e1824f27e2994c35bec5bff9506021775a36adb2cbdce6e06c7819431 0
397 Tue Mar 23 11:16:54 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 55482ee7c8fcb9fe9617156463589e55d11726bc7f170993e90220dae252d84c 0
398 Tue Mar 23 11:16:54 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 45 ed 0d e1 a7 e5 1b 81 30 f1 fc 79 87 05 61 f3 e1
48 71 24 8b 6c 9c b9 2a d0 38 bf2395fd3c4182aac6b1ae67844bfbb008c14b40f094918a7ff4776b5a002bbc 0
399 Tue Mar 23 11:16:54 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 f3 e0 3e 10 56 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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400 Tue Mar 23 11:17:22 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 01 a4 e4 bf 38 53 32 34 34 30 33 39 00 53 32
34 34 30 34 30 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 12c6e72cfc4e4bf63b3195851bff7156e549bde6bc75ec9c2d654684c7603c06 0
401 Tue Mar 23 11:17:22 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 15 00 f8 32 84 00 98 3f 84 00 58 9b 83 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 ce4055f7ff0f5a755d8e4b6ffcccf40c72c6f0f193e7cb1d7513885d2d7ec657 0
402 Tue Mar 23 11:17:22 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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403 Tue Mar 23 11:17:22 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 45 ed 0d e1 a7 e5 1b 81 30 f1 fc 79 87 05 61 f3 e1
48 71 24 8b 6c 9c b9 2a d0 38 1735efc52cbcf711e775bd522a6e0dadadad641337ec1ffaa5b6bb885f21741e 0
404 Tue Mar 23 11:17:22 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 f3 e0 3e 10 56 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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405 Tue Mar 23 11:17:24 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 01 a4 e4 bf 38 53 32 34 34 30 33 39 00 53 32
34 34 30 34 30 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 7d7debf41e8579e2fc37734c151c9c6231e4d2a830e292cf5f65abf0ef37959f 0
406 Tue Mar 23 11:17:24 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 15 00 f8 32 84 00 98 3f 84 00 58 9b 83 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 68c0ab91d85d1b00b9b14fbc6b33d63b3b4cd8ab88b2b73f59cda6e24ab1fdb9 0
407 Tue Mar 23 11:17:24 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 3b967e6f8bc8d8561ca321ea6cf0b60c2d1f520b04e12806b0c9fb64919a1290 0
408 Tue Mar 23 11:17:24 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 45 ed 0d e1 a7 e5 1b 81 30 f1 fc 79 87 05 61 f3 e1
48 71 24 8b 6c 9c b9 2a d0 38 e3ed63cdb9ceec414f92e37d4f9c478c3d0d4af267cedcde455dc843f00490c6 0
409 Tue Mar 23 11:17:24 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 f3 e0 3e 10 56 00 00 00 00 00 00 00 00 00 00
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410 Tue Mar 23 11:23:32 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 01 a4 e4 bf 38 53 32 34 34 30 33 39 00 53 32
34 34 30 34 30 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 618b7c00951086958671b004ebc466a882f2dfb148ce5d05f64c7d54c32c641f 0
411 Tue Mar 23 11:23:32 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 15 00 f8 32 84 00 98 3f 84 00 58 9b 83 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 88085f63e2f51b6e1fba8f34f60197cf6f417e1a28071dfd71215831cefa014c 0
412 Tue Mar 23 11:23:32 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00
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00 00 00 00 00 00 00 00 00 00 fb035d9d41ea54c1cddb5f9ed33681e5d13a5989b18d71161b44987ed25c666d 0
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00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 45 ed 0d e1 a7 e5 1b 81 30 f1 fc 79 87 05 61 f3 e1
48 71 24 8b 6c 9c b9 2a d0 38 8a2a6d961c35ffd191ad0877b7402078198e7fa098e88488a3f29644bbaab230 0
414 Tue Mar 23 11:23:32 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 f3 e0 3e 10 56 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 1842f7a0f9c2e58c5a856dcad82b87e9a8868b6409afa0f9b286e42aea05ea2b 0
415 Tue Mar 23 11:23:33 2021 COMMISSION_CONFIG_PARAM_CHANGE 00 01 01 a4 e4 bf 38 53 32 34 34 30 33 39 00 53 32
34 34 30 34 30 00 7c ee 1c ea 53 0b 56 2e 08 b4 37 e4 b5 7b 06 e5 a9 4e 20 f2 82 bd 52 f4 84 de 29 d9 7d 13 f6
f8 42 1b e4 98 f2 75 74 09 35 981a472312a62dbe861beef698e9835afe1a428216f3c6f8e0eac6ed1fc0ac81 0
416 Tue Mar 23 11:23:33 2021 COMMISSION_CONFIG_PARAM_CHANGE 01 8b 10 12 a1 4f 9f 46 bc 02 cd 40 2b 93 f3 7c 86
2c d4 aa a7 f2 e2 65 20 00 15 00 f8 32 84 00 98 3f 84 00 58 9b 83 00 43 45 54 00 42 01 00 01 06 07 0a 01 00 02
43 45 53 54 00 02 00 01 06 07 dd754a6c44ca7cb64fa24664a2daf205ea49a46b985b79a7ecc3ddf7a3459987 0
417 Tue Mar 23 11:23:33 2021 COMMISSION_CONFIG_PARAM_CHANGE 02 03 01 00 02 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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418 Tue Mar 23 11:23:33 2021 COMMISSION_CONFIG_PARAM_CHANGE 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 45 ed 0d e1 a7 e5 1b 81 30 f1 fc 79 87 05 61 f3 e1
48 71 24 8b 6c 9c b9 2a d0 38 f9cca18151f9ae75d6c59ce841f44534149b6763f2030bf749065fc2d3e60df1 0
419 Tue Mar 23 11:23:33 2021 COMMISSION_CONFIG_PARAM_CHANGE 04 f3 e0 3e 10 56 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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420 Tue Mar 23 11:23:53 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 6cb0a8b86d79a64fb2c2da77e8ceaaed14f5cc0619ec03388a11c1ad2d283078 0
421 Tue Mar 23 11:23:58 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 1699af1c990b0c964410648e564e28e799b4e9a148698d3e17eff8ed04e6dc86 0
422 Tue Mar 23 11:24:05 2021 TIME_SYNC 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 c791181d552845c6cf9818f93a94b5b8b1d7977dfe004ae4bb1d35968e531720 0
423 Tue Mar 23 11:24:12 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 43 55 53 54 0f 64 63 6d 2d 66
75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 c370fa1adc15382bd349c8d2eb8e515d7a797dbaf6d8273ce0d26156d0b3d4bb 0
424 Tue Mar 23 11:32:02 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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425 Tue Mar 23 11:48:41 2021 POWER_STARTUP 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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426 Tue Mar 23 11:48:47 2021 TIME_SYNC 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 961b13edce5b80521a1f4148acbc7f31088327a2d143b8d203961cdbc97c7dc6 0
427 Tue Mar 23 11:48:49 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 43 55 53 54 0f 64 63 6d 2d 66
75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 261d71744dd161646febdb10fdc3fad6bcf8a83abb3b630cc0c6420b2d133588 0
428 Tue Mar 23 12:32:38 2021 FIRMWARE_UPDATE_DOWNLOAD 01 01 01 01 7a ba c5 3f 83 2f 84 86 5f 0f c6 73 74 a0 14
a6 67 1d 8c 04 33 9b ad 6b 89 8a 17 7e 5b a1 fc 24 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 e87fd4fe3cc3060dac40a1b08e8a4a371918225a97d0757074bd1f5a1dfb3f29 0
429 Tue Mar 23 12:32:38 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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430 Tue Mar 23 12:32:47 2021 POWER_STARTUP 10 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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433 Tue Mar 23 12:33:27 2021 FIRMWARE_UPDATE_DOWNLOAD 01 01 01 01 7a ba c5 3f 83 2f 84 86 5f 0f c6 73 74 a0 14
a6 67 1d 8c 04 33 9b ad 6b 89 8a 17 7e 5b a1 fc 24 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 0beb15dcdf819d80f436ee6f042f1a75890eb3d202986dd4cc34c399e17db955 0
434 Tue Mar 23 12:33:27 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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435 Tue Mar 23 12:33:36 2021 POWER_STARTUP 10 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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436 Tue Mar 23 12:33:43 2021 TIME_SYNC 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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437 Tue Mar 23 12:34:26 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 43 55 53 54 0f 64 63 6d 2d 66
75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 ecf6a646dd5e50f85be5de35fd9b172d459d686f32c9c2a90cce227a542355dc 0
438 Tue Mar 23 12:35:04 2021 FIRMWARE_UPDATE_DOWNLOAD 01 01 01 01 7a ba c5 3f 83 2f 84 86 5f 0f c6 73 74 a0 14
a6 67 1d 8c 04 33 9b ad 6b 89 8a 17 7e 5b a1 fc 24 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 2a4166663b55250228973f383d0ee217ad04301992ef7b2ae90c72bc35e96250 0
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00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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440 Tue Mar 23 12:35:14 2021 POWER_STARTUP 10 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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442 Tue Mar 23 12:35:40 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 43 55 53 54 0f 64 63 6d 2d 66
75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 166561933a1290618453948db21308fdd30ef249af804d41ceb7da0177ee605a 0
443 Tue Mar 23 12:36:00 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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447 Tue Mar 23 12:37:24 2021 FIRMWARE_UPDATE_DOWNLOAD 01 01 01 01 7a ba c5 3f 83 2f 84 86 5f 0f c6 73 74 a0 14
a6 67 1d 8c 04 33 9b ad 6b 89 8a 17 7e 5b a1 fc 24 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 4ef25aa3a9d2c54f605f9c822ff39a7702806c5fd22a0f074b244f5ec46a48a7 0
448 Tue Mar 23 12:37:25 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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449 Tue Mar 23 12:37:34 2021 POWER_STARTUP 10 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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450 Tue Mar 23 12:37:41 2021 TIME_SYNC 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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451 Tue Mar 23 12:37:47 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 43 55 53 54 0f 64 63 6d 2d 66
75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 ac1a06853f131f9b33d245a9c33500e0ad026d2ab4dc46c7860622071b82bfc6 0
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00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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455 Tue Mar 23 14:38:29 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 43 55 53 54 0f 64 63 6d 2d 66
75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 3916f4b0ea32b875d4b74c897c35bbdf1f4d5e695769b34c9d451884118016d3 0
456 Tue Mar 23 14:40:56 2021 FIRMWARE_UPDATE_DOWNLOAD 01 01 01 01 7a ba c5 3f 83 2f 84 86 5f 0f c6 73 74 a0 14
a6 67 1d 8c 04 33 9b ad 6b 89 8a 17 7e 5b a1 fc 24 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 b83bfe665567bee258c8b40c5a5a31bf249faba1dfef9594a5b98e8dbcc4a3f5 0
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00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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458 Tue Mar 23 14:41:06 2021 POWER_STARTUP 10 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 8d3954694f97b49b940e72ec92982439e370294c3d123841b0d8b53b4cb6b565 0
459 Tue Mar 23 14:41:13 2021 TIME_SYNC 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 db34879f85d62e7303f0acf8e3d2dc64a8d886c4d2bd949df870f24dd6b6f788 0

460 Tue Mar 23 14:41:24 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 43 55 53 54 0f 64 63 6d 2d 66
75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 5cf376f2494eac92beca801e728558aab8de481b59b017ba24623ab393493255 0
461 Tue Mar 23 14:44:08 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 122cebe7facc8b78550f2ebe6965958d9422529a34f4dbcf41bdced3b264a9b1 0
467 Wed Apr 14 10:01:50 2021 TIME_SYNC 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 cc6108c735b7bb7fc1d4662f58be13eee1eb6abe4a1317d6cecf00dcd861409c 0
468 Wed Apr 14 10:01:50 2021 TIME_DEVIATION 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 99ca74a49a92efd265eaad1eb1e0e1796a234d3858acf5474f9276c05b57a89d 0
469 Wed Apr 14 10:01:54 2021 TECHNICIAN_LOGIN ff ff ff ff ff ff ff ff ff ff ff ff 43 55 53 54 0f 64 63 6d 2d 66
75 6c 6c 2d 61 63 63 65 73 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 5eb063568eea7221b2c56f21572748c4cdffd0142945ddc2bf920b01a6c54b5e 0
470 Wed Apr 14 10:17:08 2021 POWER_LOSS 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 a0aa36ebe23164932b00ccd8e16b6acfa4d0511ec38c1c7092e4f03dbd4eb6f7 0

Signature:
30440220572a5b8b73d7cd4234f1acfadc36bfc8326cb3617d27d822de2dd71f0a833a59022049de7c365b59e537a770a3223e46b57af1d1
1132057719c490a7a88c8c82bfd8

_Resuable Library

Everything on this page and below can be reference elsewhere in Confluence using various macros such as "Excerpt Include", "Page Include", "Children Display", and "Content by Label".

- _DCM_SDD_Attachments
- _Design_Calculations
- _Reusable Content
- _Reusable Glossary
- _Reusable Images
- _Reusable References

_DCM_SDD_Attachments

This page contains a list of all attachments such as images/diagrams referenced in the DC Meter Software Design Document.
All attachments are stored and version controlled here but referenced either as a link or thumbnail/preview.
Using the underscore in the page title allows exports to Word/PDF to exclude this page more easily.

File
PNG File UserPermissionsUML.png Re-Attached diagram
PNG File DCMeterDeploymentDiagram.png DC Meter Software deployment diagram. Gives an overview of the software modules with the DC Meter and what the DC meter is communicating to.
File DCMeterDeploymentDiagram.dia The project file for modifying the DC Meter Deployment diagram. Currently this is still in progress and the 'master' copy is on my laptop. Once I've releas
PNG File DC Meter Software Design.png

Drag and drop to upload or browse for files

Download All

_Design_Calculations

This page contains a list of all design calculations for the DC Meter.

All attachments are stored and version controlled here but can be referenced either as a link or thumbnail/preview.

Using the underscore in the page title allows exports to Word/PDF to exclude this page more easily.

File	Modified
Microsoft Excel Spreadsheet Tritium_DCmeter_ISA_35u.xlsx Analog front end design calculations - using Isabellenhutte 35uR shunt	Oct 25, 2018 by Alex Rudzki
Microsoft Excel Spreadsheet afe-error-calculations.xlsx Created by James Sneyd to calculate error in results based on signed integer math in the AFE.	Jul 11, 2019 by Jeremy Kincaid
Microsoft Excel Spreadsheet StackCalculations.xlsx Used to calculate how to allocate memory for task stacks	Nov 04, 2019 by Jeremy Kincaid
Microsoft Excel Spreadsheet FPGA_Memory_Calculations.xlsx Separated FPGA memory calculations from display dependent LCDS calculations	Nov 21, 2019 by Alex Rudzki
Microsoft Excel Spreadsheet FPGA_IONITY_LVDS_Calculations.xlsx Removed memory calculations	Nov 21, 2019 by Alex Rudzki
Microsoft Excel Spreadsheet FPGA_G150XNE_LVDS_Calculations.xlsx Corrected cell calculation that was broken after re-arranging cells	Dec 12, 2019 by Alex Rudzki
Microsoft Excel Spreadsheet Tritium_DCmeter_Vishay_50u.xlsx Addition of Overview Tab with explanation of terms of conditions applied	Feb 22, 2021 by William Chen
Microsoft Excel Spreadsheet StaticMemoryRequirements.xlsx Cleaned to include both audit and charge session logs. Description added.	Feb 25, 2021 by Jeremy Street-Thomas

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_Reusable Content

- DC Meter Identification
- DC Meter JIRA Ticket Labelling Guide
- Relationship to Other Plans
- Requirement Levels Definition
- Requirements Traceability Guide
- System Overview
- Tariff Information
- Traceability Matrix

DC Meter Identification

This document contains details about the software development process for the DC Meter project identified by the Tritium internal project code: TRI110. In the Tritium issue management system known as JIRA, this project has the code: TDM and can be found at https://jira.tritium.com.au/projects/TDM/summary. In Tritium's Wiki, documentation relating to this project can be found at https://wiki.tritium.com.au/display/RND/DC+Meter.

DC Meter JIRA Ticket Labelling Guide

JIRA tickets (especially requirements and design elements) should be labelled appropriately so they can be easily filtered using Confluence/JIRA macros or JQL statements. To help provide consistency The DC Meter project has identified the labels listed below with a hint of when to use it. This list will grow with time as more definitive filtering needs are realised. More than one label can be applied to the one JIRA ticket.

Label	Definition
accuracy	A specific quality type. For example calibration tables improve accuracy.
adaptation	the JIRA ticket relates to adaptation functionality such as configuration
communication	the JIRA ticket relates to communication functionality such as physical lines, protocols, message data, etc.
constraint	the JIRA ticket is a requirement that constrains design options typically by specifying a solution
construction	the JIRA ticket relates to manufacturing criteria
documentation	The JIRA ticket relates to a requirement on documentation instead of functionality
environmental	the JIRA ticket relates to specific performance criteria based on environmental needs
events	the JIRA ticket relates to functionality surrounding events on the DC Meter, such as reset, change in configuration, storing, erasing, etc.
export	functionality relating to retrieving data from the DC Meter to be stored in a file external to the DC Meter
functional	the JIRA ticket relates to functionality that is directly required to complete the product
HRS	the label used to identify a JIRA ticket as being part of the Hardware Requirement Specification. This ticket should also be of type 'New Feature' to indicate it's a requirement as opposed to a task.
Maintenance	activities relating to maintenance such as replacing worn parts, servicing, etc.
metering	identifies information relating to measuring, recording of measured data, calculations, etc.
non-functional	the JIRA ticket relates to functionality that is indirectly required to complete the product
parent-customer-contract	origin of the requirement is from the customer contract
parent-PRD	origin of the requirement is from the Product Requirement Document (internal to Tritium)
parent-VDE	the origin of the requirement is from the VDE specification/standard
performance	the JIRA ticket relates to performance criteria of the product
process	The requirement may not directly relate to functionality or product, but instead focuses on discussion, agreement, behaviour between Tritium and a 3rd party.
power	requirement relates to power loss, sleep mode, startup, etc.
quality	the JIRA ticket contains information that directly relates to perceived quality such as robustness.
requirement	the JIRA ticket type is of type requirement.
risk	The JIRA ticket identifies a risk to the success of the product (either certification, in-use).
safety	The requirement relates to a safety concern.
SDD	indicates this JIRA ticket is part of the Software Design Document.

security	relates to a security measure such as encryption, signing, authentication, etc.
SRS	the label used to identify a JIRA ticket as being part of the Software Requirement Specification. This ticket should also be of type 'New Feature' to indicate it's a requirement as opposed to a task.
SS	the JIRA ticket is a requirement at level 'System Specification'
state	the JIRA ticket relates to 'state' behaviour
status-reporting	the JIRA ticket relates to status reporting functionality such as display, email, etc
storage	the JIRA ticket relates to storing to persistent memory. This may be paired with the 'event' label to further add clarification.
support	the JIRA ticket relates to functionality that specialises in support infrastructure
tariff	label used to identify requirements/functionality/design is related to supporting tariffs.
TTI	Used to help filter JIRA issues relating to the TTI application that runs on the laptop/pc and is written in Python
UI	User Interface related functionality/requirement such as language
VDE-CertificationQuestion	Helps filter out JIRA tickets that are only used to manage communications with the certification house. These JIRA issues are not directly implemented, any result from a conclusion of the VDE certification question should result in a new JIRA issue of the correct type to be acted upon. Another way to filter these out would be: and parent not in("TDM-905").

Relationship to Other Plans

Document	Subordinate/Equal/Superior
Project Management Plan	Subordinate
DC Meter Software Test Plan	Superior
Certification Plan	Equal
Installation/Support Plan	Equal
DC Meter System Specification	Subordinate
DC Meter Firmware Design	Superior
DC Meter Software Requirement Specification	Superior

Requirement Levels Definition

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in RFC-2119. The definition of these words has been excerpted below for convenience in reading a printed version of this document.

MUST - This word, or the terms "REQUIRED" or "SHALL", mean that the definition is an absolute requirement of the specification.

MUST NOT - This phrase, or the phrase "SHALL NOT", mean that the definition is an absolute prohibition of the specification.

SHOULD - This word, or the adjective "RECOMMENDED", mean that there may exist valid reasons in particular circumstances to ignore a particular item, but the full implications must be understood and carefully weighed before choosing a different course.

SHOULD NOT - This phrase, or the phrase "NOT RECOMMENDED" mean that there may exist valid reasons in particular circumstances when the particular behavior is acceptable or even useful, but the full implications should be understood and the case carefully weighed before implementing any behavior described with this label.

MAY - This word, or the adjective "OPTIONAL", mean that an item is truly optional. One vendor may choose to include the item because a particular marketplace requires it or because the vendor feels that it enhances the product while another vendor may omit the same item. An implementation which does not include a particular option MUST be prepared to interoperate with another implementation which does include the option, though perhaps with reduced functionality. In the same vein an implementation which does include a particular option MUST be prepared to interoperate with another implementation which does not include the option (except, of course, for the feature the option provides.)

Requirements Traceability Guide

Requirements Traceability Introduction

The following paragraphs describe the mechanism for performing requirements traceability in the DC Meter project. This process leverages JIRA and Confluence (a.k.a. the Wiki) and the 'SQL Play' plug-in in Confluence to provide a table/matrix that is automatically filled in as JIRA tickets are created /modified/linked. At the moment linking occurs from external requirements to design elements. Later the queries can be updated to add test suites.

The DC Meter Requirements Traceability Matrix can be found here: https://wiki.tritium.com.au/x/kYvNAg

Requirements exist either in JIRA with the type "New Feature" or externally in various standards that are imposed upon a system.

For the DC Meter project the following types of requirements exist in JIRA:

1. System Specification (SS)
2. Software Requirement Specification (SRS)
3. Hardware Requirement Specification (HRS)

Also captured in JIRA are design elements:

1. Software Design Document elements (SDD)
2. Hardware Design Document elements (HDD)

How To Identify and Map JIRA Tickets

Mapping external requirements to internal requirements to design elements is automated inside JIRA and the Wiki if you use the correct JIRA type, labels, and linking.

To identify a JIRA ticket as a System Specification set the following fields:

- JIRA type = "New Feature"
- label = SS

To identify a JIRA ticket as a Software Requirement Specification set the following fields:

- JIRA type = "New Feature"
- label = SRS

To identify a JIRA ticket as a Hardware Requirement Specification set the following fields:

- JIRA type = "New Feature"
- label = HRS

To map an internal requirement to an external requirement (or external requirement to a design element):

- View the JIRA ticket to create the mapping in
- Add a comment that
    - names the document containing the external requirement(s) to map to
    - list the unique requirement numbers as specified in the external requirement document (make sure each requirement is separated somehow, space, comma, new line, etc). For now, this works for VDE and WELMEC documents.

To identify a JIRA ticket as a Software Design Element set the following fields:

- JIRA type in (Task, Sub-task)
- label = SDD

To identify a JIRA ticket as a Hardware Design Element set the following fields:

- JIRA type in (Task, Sub-task)
- label = HDD

To link a System Requirement to a sub-requirement (software SRS, hardware HRS, etc), or sub-requirement to a Design Element:

- View the lowest-level JIRA ticket to create the link from
- Click on 'More'
- Select 'Link'
- Select link type 'relates to'
- Type in the JIRA number (e.g. TDM-123) into the 'Issue' field (note, this field can contain multiple ticket numbers. Choose a JIRA ticket number that is one level higher than the JIRA ticket you're linking from)
- Click 'Link'

Adding Clarity to JIRA Requirement and Design Tickets

The 'label' field on JIRA tickets has been utilised to help filter requirements and design tickets into related categories. To keep this labeling consistent an official DC Meter label list has been created and is maintained here https://wiki.tritium.com.au/x/s4-NAg.

The labels allow the JIRA Issues Filter Widget in the Wiki to list only specific topic related JIRAs. This is useful for documentation and wanting to discuss further details on a topic and relate to requirements. It can also help with dividing requirements/design to different teams.

Behind the Scenes

There are three SQL queries that interact directly with the JIRA PostgreSQL database.

1. RequirementsTraceabilityMatrix_withSummary – Performs the mapping between external requirements to system specification to sub-system specification (SRS, HRS, etc) to design elements (SDD, HDD, etc).
2. ExternalVDE_RequirementsMatrix – Performs the mapping between external requirements in the VDE to sub-system requirements (SRS, HRS, etc) to design.
3. WELMEC7.2_Requirements_To_Design – Performs the mapping between external requirements in the WELMEC 7.2 to sub-system requirements (SRS, HRS, etc) to design.

Query for mapping VDE requirements to system and then subsystem requirements to design

```
SELECT system_req_list."$docType Requirement ID", system_req_list."$docType Summary"
 , ss_srs_link."$cd Requirement ID", ss_srs_link."$cd Summary"
 , srs_design_link."$designLabel Design ID", srs_design_link."$designLabel Status",
srs_design_link."$designLabel Summary"
FROM
 (SELECT CONCAT('$projectKey', '-', '$docType', '-', jiraissue.issuenum) AS "$docType Requirement ID",
jiraissue.id AS "$docType ID", jiraissue.summary AS "$docType Summary"
 FROM jiraissue
 WHERE
 project = (SELECT project_key.project_id FROM project_key WHERE project_key.project_key = '$projectKey') AND
-- Filter on project
 issuetype = (SELECT issuetype.id FROM issuetype WHERE issuetype.pname = 'New Feature') AND -- Filter on
requirement types
 -- jiraissue.id IN (SELECT issueid FROM jiraaction WHERE actionbody LIKE '%$labelFilter%') OR (description
LIKE '%$labelFilter%') AND -- Filter on external requirement id
 jiraissue.id IN (SELECT issue FROM label WHERE label in ('$docType')) -- Filter on requirement type
 ) system_req_list
LEFT OUTER JOIN -- We want all rows from the above table. This way we get <null> result for any System
Specification Requirements addressing the specified external requirement that don't have a derived requirement
 (SELECT issuelink.source, issuelink.destination FROM issuelink WHERE issuelink.linktype = (SELECT
issuelinktype.id FROM issuelinktype WHERE linkname = 'Relates')) issue_link
 ON system_req_list."$docType ID" = issue_link.destination AND issue_link.source in ( -- only want ids for
issues of type SRS and
 (SELECT jiraissue.id
 FROM jiraissue
 WHERE
 project = (SELECT project_key.project_id FROM project_key WHERE project_key.project_key = '$projectKey') AND
-- Filter on project
 issuetype = (SELECT issuetype.id FROM issuetype WHERE issuetype.pname = 'New Feature') AND -- Filter on
requirement types
 jiraissue.id IN (SELECT issue FROM label WHERE label in ('$cd')) -- Filter on requirement type
 )
 )
LEFT JOIN
 (SELECT CONCAT('$projectKey', '-', '$cd', '-', jiraissue.issuenum) AS "$cd Requirement ID", jiraissue.id AS
"$cd ID", jiraissue.summary "$cd Summary"
 FROM jiraissue
 WHERE
 project = (SELECT project_key.project_id FROM project_key WHERE project_key.project_key = '$projectKey') AND
-- Filter on project
 issuetype = (SELECT issuetype.id FROM issuetype WHERE issuetype.pname = 'New Feature') AND -- Filter on
requirement types
 jiraissue.id IN (SELECT issue FROM label WHERE label in ('$cd')) -- Filter on requirement type
 ) ss_srs_link ON issue_link.source = ss_srs_link."$cd ID"
LEFT OUTER JOIN -- We want all rows from the above table. This way we get <null> result for any System
Specification Requirements addressing the specified external requirement that don't have a derived requirement
 (SELECT issuelink.source AS "$cd_$designLabel_src", issuelink.destination AS "$cd_$designLabel_dst" FROM
issuelink WHERE issuelink.linktype = (SELECT issuelinktype.id FROM issuelinktype WHERE linkname = 'Relates'))
```

```
design_link
 ON ss_srs_link."$cd ID" = design_link."$cd_$designLabel_dst" AND design_link."$cd_$designLabel_src" in ( --
only want ids for issues of type SRS and
 (SELECT jiraissue.id
 FROM jiraissue
 WHERE
 project = (SELECT project_key.project_id FROM project_key WHERE project_key.project_key = '$projectKey') AND
-- Filter on project
 issuetype in (SELECT issuetype.id FROM issuetype WHERE issuetype.pname in ('Task', 'Sub-task')) AND -- Filter
on requirement types
 jiraissue.id IN (SELECT issue FROM label WHERE label in ('$designLabel')) -- Filter on requirement type
 )
 )
LEFT JOIN
 (SELECT CONCAT('$projectKey', '-', '$designLabel', '-', jiraissue.issuenum) AS "$designLabel Design ID",
(SELECT issuestatus.pname FROM issuestatus WHERE issuestatus.id = jiraissue.issuestatus) AS "$designLabel
Status", jiraissue.id AS "$designLabel ID", jiraissue.summary "$designLabel Summary"
 FROM jiraissue
 WHERE
 project = (SELECT project_key.project_id FROM project_key WHERE project_key.project_key = '$projectKey') AND
-- Filter on project
 issuetype in (SELECT issuetype.id FROM issuetype WHERE issuetype.pname in ('Task', 'Sub-task')) AND -- Filter
on requirement types
 jiraissue.id IN (SELECT issue FROM label WHERE label in ('$designLabel')) -- Filter on requirement type
 ) srs_design_link ON design_link."$cd_$designLabel_src" = srs_design_link."$designLabel ID"
```

Query for mapping VDE requirements to subsystem requirements to design

```
SELECT external_requirements.vde_req AS "External Requirements",
        internal_requirements."$cd Requirement ID", internal_requirements."$cd Summary",
internal_requirements."$designLabel Design ID", internal_requirements."$designLabel Status",
internal_requirements."$designLabel Summary" FROM
    (SELECT * FROM (
        VALUES -- the following list of VDE requirement id's are not all requirements as not all had unique
id's. Still need to add a unique identifier to those requirements
            ('LME001'),
            ('LME002'),
            ('LME003'),
            ('LME010'),
            ('LME011'),
            ('LME012'),
            ('LME013'),
            ('LME014'),
            ('LME015'),
            ('LME016'),
            ('LME017'),
            ('LME020'),
            ('LME021'),
            ('LME022'),
            ('LME023'),
            ('LME024'),
            ('MWE001'),
            ('MWE002'),
            ('MWE003'),
            ('MWE004'),
            ('MWE005'),
            ('MWV001'),
            ('MWV002'),
            ('MWV003'),
            ('MWV004'),
            ('MWV005'),
            ('STORAGE001'),
            ('STORAGE002'),
            ('STORAGE003'),
            ('STORAGE004'),
            ('STORAGE005'),
            ('TIME001'),
            ('TIME002'),
            ('TIME003'),
```

```
('TIME004'),
('TIME005'),
('TIME006'),
('TIME007'),
('TIME008'),
('TIME009'),
('TIME010'),
('TIME011'),
('TIME012'),
('TIME013'),
('TIME014'),
('TIME015'),
('TIME016'),
('TIME017'),
('TIMESRV001'),
('TIMESRV002'),
('LOG001'),
('LOG002'),
('LOG003'),
('LOG004'),
('LOG005'),
('LOG006'),
('LOG007'),
('LOG008'),
('LOG009'),
('LOG010'),
('LOG011'),
('LOG012'),
('LOG013'),
('LOG014'),
('LOG015'),
('LOG016'),
('LOG017'),
('LOG018'),
('LOG019'),
('LOG020'),
('LOG021'),
('LOG022'),
('LOG023'),
('LOG024'),
('LOG025'),
('LOG026'),
('LOG027'),
('LOG028'),
('LOG029'),
('LOG030'),
('FWM001'),
('FWM002'),
('FWM003'),
('FWM004'),
('FWM005'),
('FWM006'),
('FWM007'),
('SIGN001'),
('SIGN002'),
('SIGN003'),
('SIGN004'),
('SIGN005'),
('SIGN006'),
('SIGN007'),
('SIGN008'),
('SIGN009'),
('SIGN010'),
('SIGN011'),
('VERIFY001'),
('VERIFY002'),
('VERIFY003'),
('VERIFY004'),
('VERIFY005'),
('VERIFY006'),
('ERA007'),
```

```
            ('ERA008'),
            ('ERA009'),
            ('ERA010'),
            ('ERA011'),
            ('ERA012'),
            ('ERA013'),
            ('ERA014'),
            ('ERA015'),
            ('ERA016'),
            ('ERA017'),
            ('ERA018'),
            ('ERA019'),
            ('ERA020'),
            ('ERA021'),
            ('ERA022'),
            ('ERA023'),
            ('ERA024'),
            ('UI001'),
            ('UI002'),
            ('AUTHZ001'),
            ('AUTHZ002'),
            ('AUTHN001'),
            ('AUTHN002'),
            ('AUTHN003'),
            ('AUTHN004')
    ) AS ext_req(vde_req)) AS external_requirements
LEFT OUTER JOIN
    (SELECT
         actionbody AS "External Requirement"
         ,ss_srs_link."$cd Requirement ID", ss_srs_link."$cd Summary"
         , srs_design_link."$designLabel Design ID", srs_design_link."$designLabel Status",
srs_design_link."$designLabel Summary"
    FROM
        (SELECT actionbody, CONCAT('$projectKey', '-', '$cd', '-', jiraissue.issuenum) AS "$cd Requirement ID",
jiraissue.id AS "$cd ID", jiraissue.summary "$cd Summary"
            FROM jiraissue, jiraaction
            WHERE
                project = (SELECT project_key.project_id FROM project_key WHERE project_key.project_key =
'$projectKey') AND -- Filter on project
                issuetype = (SELECT issuetype.id FROM issuetype WHERE issuetype.pname = 'New Feature') AND --
Filter on requirement types
                jiraissue.id IN (SELECT issueid FROM jiraaction WHERE actionbody SIMILAR TO '%[A-Z]{3,}[0-9]
{3}%') AND -- Filter on external requirement id pattern
                jiraissue.id IN (SELECT issue FROM label WHERE label in ('$cd')) AND -- Filter on requirement
type
                jiraaction.issueid = jiraissue.id AND
                jiraaction.actionbody SIMILAR TO '%[A-Z]{3,}[0-9]{3}%'
            ) ss_srs_link
    LEFT OUTER JOIN -- We want all rows from the above table. This way we get <null> result for any System
Specification Requirements addressing the specified external requirement that don't have a derived requirement
        (SELECT issuelink.source AS "$cd_$designLabel_src", issuelink.destination AS "$cd_$designLabel_dst"
FROM issuelink WHERE issuelink.linktype = (SELECT issuelinktype.id FROM issuelinktype WHERE linkname =
'Relates')) design_link
        ON ss_srs_link."$cd ID" = design_link."$cd_$designLabel_dst" AND design_link."$cd_$designLabel_src" in
( -- only want ids for issues of type SRS and
            (SELECT jiraissue.id
            FROM jiraissue
            WHERE
                project = (SELECT project_key.project_id FROM project_key WHERE project_key.project_key =
'$projectKey') AND -- Filter on project
                issuetype in (SELECT issuetype.id FROM issuetype WHERE issuetype.pname in ('Task', 'Sub-task'))
AND -- Filter on requirement types
                jiraissue.id IN (SELECT issue FROM label WHERE label in ('$designLabel')) -- Filter on
requirement type
            )
        )
    LEFT OUTER JOIN
        (SELECT CONCAT('$projectKey', '-', '$designLabel', '-', jiraissue.issuenum) AS "$designLabel Design
ID", (SELECT issuestatus.pname FROM issuestatus WHERE issuestatus.id = jiraissue.issuestatus) AS "$designLabel
Status", jiraissue.id AS "$designLabel ID", jiraissue.summary "$designLabel Summary"
            FROM jiraissue
```

```
            WHERE
                project = (SELECT project_key.project_id FROM project_key WHERE project_key.project_key =
'$projectKey') AND -- Filter on project
                issuetype in (SELECT issuetype.id FROM issuetype WHERE issuetype.pname in ('Task', 'Sub-task'))
AND -- Filter on requirement types
                jiraissue.id IN (SELECT issue FROM label WHERE label in ('$designLabel')) -- Filter on
requirement type
            ) srs_design_link ON design_link."$cd_$designLabel_src" = srs_design_link."$designLabel ID"
    ) AS internal_requirements ON internal_requirements."External Requirement" LIKE '%' ||
external_requirements.vde_req || '%'
```

WELMEC 7.2 SQL Query for mapping requirements to design

```
SELECT external_requirements.vde_req AS "External Requirements",
        internal_requirements."$cd Requirement ID", internal_requirements."$cd Summary",
internal_requirements."$designLabel Design ID", internal_requirements."$designLabel Status",
internal_requirements."$designLabel Summary" FROM
    (SELECT * FROM (
        VALUES -- the following list of VDE requirement id's are not all requirements as not all had unique
id's. Still need to add a unique identifier to those requirements
            ('WELMEC7.2:P1'),
            ('WELMEC7.2:P2'),
            ('WELMEC7.2:P3'),
            ('WELMEC7.2:P4'),
            ('WELMEC7.2:P5'),
            ('WELMEC7.2:P6'),
            ('WELMEC7.2:P7'),
            ('WELMEC7.2:L1'),
            ('WELMEC7.2:L2'),
            ('WELMEC7.2:L3'),
            ('WELMEC7.2:L4'),
            ('WELMEC7.2:L5'),
            ('WELMEC7.2:L6'),
            ('WELMEC7.2:L7'),
            ('WELMEC7.2:L8'),
            ('WELMEC7.2:T1'),
            ('WELMEC7.2:T2'),
            ('WELMEC7.2:T3'),
            ('WELMEC7.2:T4'),
            ('WELMEC7.2:T5'),
            ('WELMEC7.2:T6'),
            ('WELMEC7.2:T7'),
            ('WELMEC7.2:T8'),
            ('WELMEC7.2:S1'),
            ('WELMEC7.2:S2'),
            ('WELMEC7.2:S3'),
            ('WELMEC7.2:D1'),
            ('WELMEC7.2:D2'),
            ('WELMEC7.2:D3'),
            ('WELMEC7.2:D4')
    ) AS ext_req(vde_req)) AS external_requirements
LEFT OUTER JOIN
    (SELECT
         actionbody AS "External Requirement"
         ,ss_srs_link."$cd Requirement ID", ss_srs_link."$cd Summary"
         , srs_design_link."$designLabel Design ID", srs_design_link."$designLabel Status",
srs_design_link."$designLabel Summary"
    FROM
        (SELECT actionbody, CONCAT('$projectKey', '-', '$cd', '-', jiraissue.issuenum) AS "$cd Requirement ID",
jiraissue.id AS "$cd ID", jiraissue.summary "$cd Summary"
            FROM jiraissue, jiraaction
            WHERE
                project = (SELECT project_key.project_id FROM project_key WHERE project_key.project_key =
'$projectKey') AND -- Filter on project
                issuetype = (SELECT issuetype.id FROM issuetype WHERE issuetype.pname = 'New Feature') AND --
Filter on requirement types
                jiraissue.id IN (SELECT issueid FROM jiraaction WHERE actionbody SIMILAR TO '%[A-Z]{3,}[0-9]
{3}%') AND -- Filter on external requirement id pattern
                jiraissue.id IN (SELECT issue FROM label WHERE label in ('$cd')) AND -- Filter on requirement
```

```
type
                jiraaction.issueid = jiraissue.id AND
                jiraaction.actionbody SIMILAR TO '%[A-Z]{3,}[0-9]{3}%'
            ) ss_srs_link
    LEFT OUTER JOIN -- We want all rows from the above table. This way we get <null> result for any System
Specification Requirements addressing the specified external requirement that don't have a derived requirement
        (SELECT issuelink.source AS "$cd_$designLabel_src", issuelink.destination AS "$cd_$designLabel_dst"
FROM issuelink WHERE issuelink.linktype = (SELECT issuelinktype.id FROM issuelinktype WHERE linkname =
'Relates')) design_link
        ON ss_srs_link."$cd ID" = design_link."$cd_$designLabel_dst" AND design_link."$cd_$designLabel_src" in
( -- only want ids for issues of type SRS and
            (SELECT jiraissue.id
            FROM jiraissue
            WHERE
                project = (SELECT project_key.project_id FROM project_key WHERE project_key.project_key =
'$projectKey') AND -- Filter on project
                issuetype in (SELECT issuetype.id FROM issuetype WHERE issuetype.pname in ('Task', 'Sub-task'))
AND -- Filter on requirement types
                jiraissue.id IN (SELECT issue FROM label WHERE label in ('$designLabel')) -- Filter on
requirement type
            )
        )
    LEFT OUTER JOIN
        (SELECT CONCAT('$projectKey', '-', '$designLabel', '-', jiraissue.issuenum) AS "$designLabel Design
ID", (SELECT issuestatus.pname FROM issuestatus WHERE issuestatus.id = jiraissue.issuestatus) AS "$designLabel
Status", jiraissue.id AS "$designLabel ID", jiraissue.summary "$designLabel Summary"
            FROM jiraissue
            WHERE
                project = (SELECT project_key.project_id FROM project_key WHERE project_key.project_key =
'$projectKey') AND -- Filter on project
                issuetype in (SELECT issuetype.id FROM issuetype WHERE issuetype.pname in ('Task', 'Sub-task'))
AND -- Filter on requirement types
                jiraissue.id IN (SELECT issue FROM label WHERE label in ('$designLabel')) -- Filter on
requirement type
            ) srs_design_link ON design_link."$cd_$designLabel_src" = srs_design_link."$designLabel ID"
    ) AS internal_requirements ON internal_requirements."External Requirement" LIKE '%' ||
external_requirements.vde_req || '%'
```

System Overview

The need for the DC Meter is a consequence of the German authorities making it mandatory for all Electric Vehicle (EV) Charging Stations to contain a certified DC Meter suitable for billing purposes as specified in the standard document EN50470-1/-3. The values measured/calculated by the DC Meter are used by government authorities and end users to verify billing information from the Charge Point Operators (CPO) managing the EV Charging Stations.

The DC Meter is analogous to an AC Electricity Smart Meter but for EV Charging stations and instead of measuring the total AC usage to the EV Charging station, it measures the DC usage for charging the vehicle and specifically excludes power to run the EV Charging station, power lost in cabling and other losses.

Functionality includes (but not limited to):

1. accurately measure the DC usage when charging a vehicle
2. calculate the energy delivered to the user
3. keep track of charge session duration and time started
4. report the measured and calculated values to
    a. a display to the user and
    b. an independent server
5. be cryptographically secure in storage and transmission of information.

There are existing Tritium EV Charging stations installed at various site in Germany that will need to be retrofitted with the DC Meter Solution. As a consequence there are internal Tritium requirements that impact the DC Meter with regards to physical size and shape as it needs to fit in existing systems of different models and new systems yet to be built.

German installation sites include:

- Brohtal Ost
- Brohtal West
- Himmelkron
- Gruibingen Süd
- Ringdalskogen

- Hohenwarsleben
- Wunnenstein Ost
- Salzbergen
- Kjerlingland

- Wunnenstein West
- Mahlberg West
- Eichenzell
- Hohenlohe Süd

Different models of EV Charging Stations have different physical sizes and different sized Display Panels. The DC Meter also needs to fit inside these EV models and use their screens (as the DC Metering information is superimposed onto the existing Display Panel instead of having it's own screen).

System block diagram

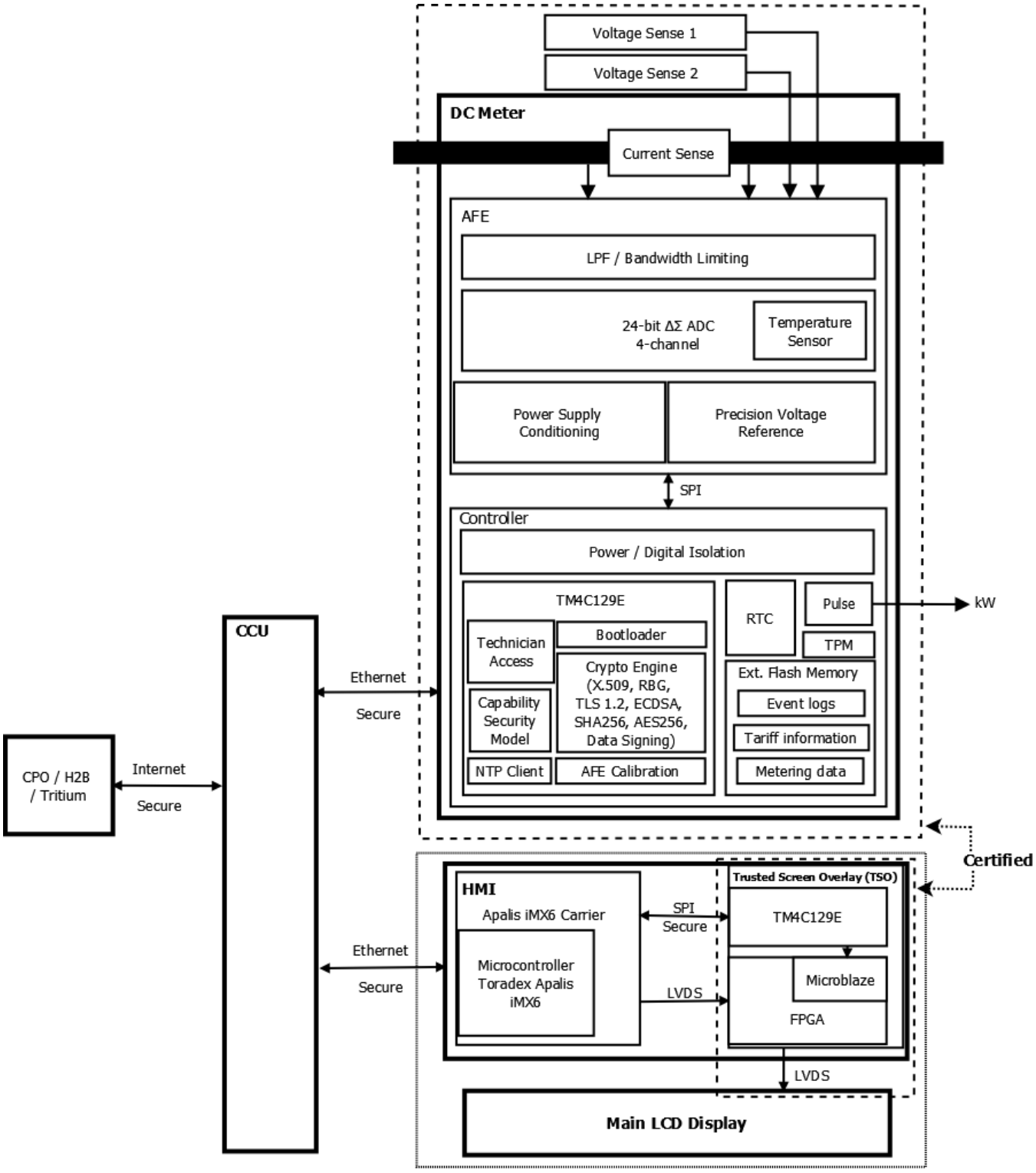


Figure – Block diagram of the DC Meter

Tariff Information

The DC Meter supports a tariff name to be displayed locally on the Trusted Screen Overlay and logged with the charge session information (as the OCMF optional "UT" field in the first record in the signed transaction data). No fiscal calculations are supported.

The DC Meter is configured in production with a default tariff name that will be used on power up.

The CPO can choose to update the active tariff throughout the meter's life through OCPP. Alternatively the Tariffs can be left with the default name.

Tariff name changes are signed with the CPO's private key. The signature is verified with the CPO public key which is pre-written into the meter's memory also during production. The CPO public key will conform to SECP256R1 (Key Curve), ASN.1 DER (Format), RFC-5480 (Standard).

Once tariffs are sent to the meter they will be used until new tariffs are sent or a power cycle. The tariff for a given charge session will be set once power delivery has started and changing the active tariff after this time will only apply to the next charge session.

Traceability Matrix

- 1 Table – Summary of Requirements
- 2 Table – Requirements from System Specification to Design Elements
- 3 Table – Requirements matrix from VDE to Design Elements
- 4 Table – Requirements matrix from WELMEC 7.2 to Design Elements

Table – Summary of Requirements

Source	Total Requirement Count in source	Done
Customer Contract	0 issues	0 issues
VDE Specification	157 Requirements	
System Specification	0 issues	0 issues
Software Specification	0 issues	0 issues
Hardware Specification	0 issues	0 issues

Table – Requirements from System Specification to Design Elements

Unknown macro: 'playsql-query'

Table – Requirements matrix from VDE to Design Elements

Unknown macro: 'playsql-query'

Table – Requirements matrix from WELMEC 7.2 to Design Elements

Unknown macro: 'playsql-query'

_Reusable Glossary

Sub pages of this contain reusable content (by using either the macro 'Include Page' or 'Include Excerpt') for displaying acronyms, abbreviations, terms, etc. For formal looking tables, use a 'link' to the page as the 'Term' column and the 'Include Excerpt' macro for the 'Description' column. If you are only after a very simple list not in a table, then use the 'Include Page' macro with settings "include excerpt" set to "simple" as per the list below.

- CPO — Charge Point Operator - An actor that is responsible for the operation and billing of charge stations. CPO uses CSMS software.
- CSMS — Charging Station Management System: Replaces previous references to CPO, when referring to software rather than the actor
- DC — The abbreviation for the term: Direct Current, which is an electric current that always flows in the same direction.
- EDSI — Energy Delivery Session Identifier. A practically-unique number generated by the DC Meter to identify individual charge sessions.
- EV — Abbreviation for the term "Electric Vehicle".
- EVSE — Electric Vehicle Supply Equipment. This is the equipment that connects with the vehicle to charge it. Colloquially referred to as "the charger".
- ISA — An abbreviation for the German company "Isabellahutte". ISA are also creating a DC Meter.
- SS — Acronym for: System Specification. A high level document containing requirements that will be derived by sub-systems.
- TDM — The project identifier for the DC Meter project as specified in JIRA.
- Tr_ISA — An identifier created to summarise that related information is due to a partnered activity between Tritium (Tr) and Isabelahutte (ISA). This is the alternative to Tritium only work as identified by Tr_Only.
- Tr_Only — An identifier created to summarise that related information work to be done solely by Tritium. This is the alternative option being partnered activity between Tritium (Tr) and Isabelahutte (ISA) as identified by Tr_ISA.

CPO

Charge Point Operator - An actor that is responsible for the operation and billing of charge stations. CPO uses CSMS software.

CSMS

Charging Station Management System: Replaces previous references to CPO, when referring to software rather than the actor

DC

The abbreviation for the term: Direct Current, which is an electric current that always flows in the same direction.

EDSI

Energy Delivery Session Identifier. A practically-unique number generated by the DC Meter to identify individual charge sessions.

EV

Abbreviation for the term "Electric Vehicle".

EVSE

Electric Vehicle Supply Equipment. This is the equipment that connects with the vehicle to charge it. Colloquially referred to as "the charger".

ISA

An abbreviation for the German company "Isabellahutte". ISA are also creating a DC Meter.

See also:

https://www.isabellenhuette.de/en/

SS

Acronym for: System Specification. A high level document containing requirements that will be derived by sub-systems.

TDM

The project identifier for the DC Meter project as specified in JIRA.

Tr_ISA

An identifier created to summarise that related information is due to a partnered activity between Tritium (Tr) and Isabelahutte (ISA). This is the alternative to Tritium only work as identified by Tr_Only.

Tr_Only

An identifier created to summarise that related information work to be done solely by Tritium. This is the alternative option being partnered activity between Tritium (Tr) and Isabelahutte (ISA) as identified by Tr_ISA.

_Reusable Images

This page is a central location for the DC Meter project to upload images. Once here they can be displayed on any page.
The benefit over attaching to a page, is that its easier to find images if they're in one place.

File	Modified
PDF File InterfaceContextDiagram.pdf An exported view of the Visio drawing of all external interfaces on the DC Meter	Jan 09, 2019 by Jeremy Kincaid
File DCMeterStates.vsdx Visio file for creating DC Meter states	Jan 09, 2019 by Jeremy Kincaid
PDF File DCMeterStates.pdf Exported diagram from Visio of DC Meter states.	Jan 09, 2019 by Jeremy Kincaid
File DetailedStateDiagram.dia Initial attempt to capture more detail on states of the DC Meter	Mar 04, 2019 by Jeremy Kincaid
PNG File DetailedStateDiagram.png PNG version of DetailedStateDiagram.dia so it can be displayed in a page	Mar 04, 2019 by Jeremy Kincaid
PNG File InterfaceContextDiagram.png	Jul 22, 2019 by Jeremy Kincaid
File InterfaceContextDiagram.vsdx	Jul 22, 2019 by Jeremy Kincaid
PNG File DC Meter Software Design.png Block diagram of the software subsystems with identification of tasks	Nov 05, 2019 by Jeremy Kincaid
PNG File TTI sequence.png Initial diagram depicting data flow from PC TTI app to DC Meter acting on the command	Nov 06, 2019 by Jeremy Kincaid
PNG File SAFE Initiative sequence.png Removal of extraction of previous logs to meter	Apr 30, 2020 by Jeremy Street-Thomas
PNG File SAFE Initiative.png Removal of extraction of previous logs to meter	Apr 30, 2020 by Jeremy Street-Thomas
PNG File Simplified_CSSM.png	Jul 16, 2020 by Jeremy Street-Thomas

Drag and drop to upload or browse for files

Download All

_Reusable References

- Introduction
- How-To
  - To add a new reference
  - To add a reference table in your desired page
  - To add a reference in your document to your table
- Current References
- Further Reading

Introduction

Sub pages contain details of referenced documents within various parts of the DC Meter Wiki pages. This provides one common place to list the details and re-reference as needed.

The steps below help you create a hyperlinked reference table in your document that reference the original so you'll always be updated with any changes.

How-To

To add a new reference

1. Create a sub page with the document name as the Title.
2. Add the "Excerpt" macro.
3. Exit the "Excerpt" macro and add the details of the document.

To add a reference table in your desired page

(N.B. reference tables in documents can be different from other documents, including the 'ID' of the reference):

1. Create a "References" table consisting of two columns labelled "ID" and "Reference".
2. In the "ID" column add the "Anchor" macro.
3. Edit the Anchor macro to add the id to the Anchor Name field. Anchor names for documents should be of the format "R" and a number that increments for each new reference in your sub pages (e.g. R1). The 'R' helps denote the difference between a document reference and a footnote, or table reference, or figure reference.
4. In the same column as the Anchor add the text that is identical to the anchor name. e.g. R1. This is because Anchor Names are not visible.
5. In the "Reference" column add the "Include Page" macro and specify the page with the title of the document you wish to reference.

To add a reference in your document to your table

1. Add the "Link" macro
2. In the 'Link' field type (assumes your not on the same page as the reference table) page:anchor_name.
3. In the 'Link Text' field type the id you entered into the reference table inside square brackets. e.g. [R1]

Current References

- AIS 20/31 — A proposal for: Functionality classes for random number generators.
- BSI TR-03116-3 — Technical Guideline BSI TR-03116 Cryptographic specifications for projects of the Federal government, Part 3: intelligent measuring systems. Referenced by the VDE AR E 2418 3 100 draft E document.
- IONITY Technical Specification High Power Charging — V2.2 26.06.2018
- ISO 18031 — Information technology -- Security techniques -- Random bit generation.
- NIST SP 800-90A — NIST Special Publication 800-90A Revision 1
- POL.003 Tritium C Coding Style & Guidelines — Document containing layout of C code, guides to naming convention
- VDE AR E 2418 3 100 draft E — Latest draft (as of 16 Oct 2018) of the system requirements from the German authorities (PTB) used for requirements on the DC Meter.
- WELMEC Guide 7.1 — Development of Software Requirements
- WELMEC Guide 7.2 — Technical guidance for the application of the Measuring Instruments Directive (MID) for software-equipped measuring instruments.

Further Reading

https://confluence.atlassian.com/conf610/anchors-952622202.html

https://confluence.atlassian.com/conf610/content-by-label-macro-952622566.html

https://confluence.atlassian.com/conf610/excerpt-macro-952622581.html

https://confluence.atlassian.com/conf610/include-page-macro-952622604.html

AIS 20/31

A proposal for: Functionality classes for random number generators.

Location: "Z:\Tritium\TRI110 - DC Meter\00 - Documents\Research\Software Security Guides\other documents NIST etc\AIS_31_Functionality_classes_for_random_number_generators_e.pdf"

BSI TR-03116-3

Technical Guideline BSI TR-03116 Cryptographic specifications for projects of the Federal government, Part 3: intelligent measuring systems. Referenced by the VDE AR E 2418 3 100 draft E document.

Location: "Z:\Tritium\TRI110 - DC Meter\00 - Documents\Research\Software Security Guides\Document referenced in German 6A\BSI TR_03116_3 Translated English.docx"

IONITY Technical Specification High Power Charging

V2.2 26.06.2018
Contractual requirements on delivering charging stations to IONITY including the DC Meter.

ISO 18031

Information technology -- Security techniques -- Random bit generation.

Location: TBD

NIST SP 800-90A

NIST Special Publication 800-90A Revision 1

Recommendation for Random Number Generation Using Deterministic Random Bit Generators

Location: "Z:\Tritium\TRI110 - DC Meter\00 - Documents\Research\Software Security Guides\other documents NIST etc\NIST.SP.800-90Ar1.pdf"

POL.003 Tritium C Coding Style & Guidelines

Document containing layout of C code, guides to naming conventionLocated: POL.003 Tritium C Coding Styles & Guidelines

VDE AR E 2418 3 100 draft E

Latest draft (as of 16 Oct 2018) of the system requirements from the German authorities (PTB) used for requirements on the DC Meter.

Located: "Z:\Tritium\TRI110 - DC Meter\00 - Documents\TRI110.012 VDE AR E 2418 3 100 draft\VDE AR E 2418 3 100 draft E.docx"

WELMEC Guide 7.1

Development of Software RequirementsLocated: "Z:\Tritium\TRI110 - DC Meter\00 - Documents\Research\WELMEC GUIDES\SOFTWARE GUIDES\DEVELOPMENT OF SOFTWRE REQUIREMENTS WELMAC 7.1.pdf"

WELMEC Guide 7.2

Technical guidance for the application of the Measuring Instruments Directive (MID) for software-equipped measuring instruments.

Located: "Z:\Tritium\TRI110 - DC Meter\00 - Documents\Research\WELMEC GUIDES\SOFTWARE GUIDES\Guide_7.2_2015__Software.pdf"

_DC Meter Development Pages

This contains the pages that are used during development that only have transient information which is expected to become out of date. i.e. Bugs/version faults/testing notes

2. Referenced Documents

The following table lists documents referenced within the DC Meter Firmware Design document.

ID	Reference
R1	**VDE AR E 2418 3 100 draft E** Latest draft (as of 16 Oct 2018) of the system requirements from the German authorities (PTB) used for requirements on the DC Meter.
R2	**BSI TR-03116-3** Technical Guideline BSI TR-03116 Cryptographic specifications for projects of the Federal government, Part 3: intelligent measuring systems. Referenced by the VDE AR E 2418 3 100 draft E document.
R3	**AIS 20/31** A proposal for: Functionality classes for random number generators.
R4	**ISO 18031** Information technology -- Security techniques -- Random bit generation.
R5	**NIST SP 800-90A** NIST Special Publication 800-90A Revision 1 Recommendation for Random Number Generation Using Deterministic Random Bit Generators

4. Requirements Traceability Matrix

General Requirements

VDE Requirement ID	Software Requirement Specification	Software/Firmware Design
LME001		

Key	Summary	Description
TDM-241	Display primitive error if meter knows it has failed unrecoverably	display "Err" if there is a fault with the meter. Tritium can log in to find details of the error or via technician access if available
TDM-243	session authentication id	display the unique id generated by the dc meter that identifies this charge session. This will be sent to Transparenz so that it can be mapped to the user.This unique id must also be shared with the charger so it can be used in the billing process.The session id will be created upon trigger of 'start-of-charge'. This could be current flow or an interrupt from the charger.
TDM-239	Charge time duration	Time in seconds for the charge duration. Must be with 1% of nominal value. Duration will be displayed in the format: [h]hh:mm:ss
TDM-242	comms between display and meter security status	display "SEC" if the encrypted channel is valid, "XXX" if there is an issue.
TDM-240	Meter firmware hash	Unique identifier of this release of software, which will be a 16 byte SHA2 hash of the binary. This hash can be stored at the very beginning of the binary and ignored when performing the hash calculation.
TDM-238	Meter Time Health Status	Display next to the meter time either as a symbol of confidence of accuracy of the meter clock: 1. successfully synchronised (within time period) 2. unsynchroised (hasn't sync'd since max sync duration (see config)) 3. invalid (time was sync'd but is out by 3%) 4. erred (time deviation >= 75 minutes)
TDM-231	Communications between DC Meter and remote display panel (IONITY Demo)	Ethernet to SPI bus to display module, with encryption, dedicated key, ack from display to check comms status, allows display to report errors to meter to forward to Tritium and audit log.To reduce work on FPGA, send a series of fixed-width strings always in the same order. FPGA knows how and where to display each string. Messages to the display are sent every second to keep the kWh view up to date.
TDM-237	Meter Time	1. Display local time. This means including DST and Timezone values. 2. The Time displayed SHALL update every second 3. Time SHALL be sent to the display as a string 4. Time SHALL be displayed in the format yyyy-mm-ddTHH:mm:ss z/+hh:mm
TDM-235	Energy usage for charge session	1. Displayed in kWh (Appendix B suggests could optionally use MWh but [ERA012] says must be kWh) 2. The format of the usage will be: `[0-9]{7}.[0-9]{2}` e.g. 0000000.00
TDM-167	Event Log	Provides persistent Logging functions for Meter.
TDM-236	Active Tariff	Tariffs can be used in multiple ways. Seasonal, time-of-day, or a combination. When not charging display the active tariff. When charging display the tariffs used so far during the charge period.

11 issues

LME002

Key	Summary	Description
TDM-241	Display primitive error if meter knows it has failed unrecoverably	display "Err" if there is a fault with the meter. Tritium can log in to find details of the error or via technician access if available
TDM-243	session authentication id	display the unique id generated by the dc meter that identifies this charge session. This will be sent to Transparenz so that it can be mapped to the user.This unique id must also be shared with the charger so it can be used in the billing process.The session id will be created upon trigger of 'start-of-charge'. This could be current flow or an interrupt from the charger.
TDM-239	Charge time duration	Time in seconds for the charge duration. Must be with 1% of nominal value. Duration will be displayed in the format: [h]hh:mm:ss
TDM-242	comms between display and meter security status	display "SEC" if the encrypted channel is valid, "XXX" if there is an issue.
TDM-240	Meter firmware hash	Unique identifier of this release of software, which will be a 16 byte SHA2 hash of the binary. This hash can be stored at the very beginning of the binary and ignored when performing the hash calculation.
TDM-238	Meter Time Health Status	Display next to the meter time either as a symbol of confidence of accuracy of the meter clock: 1. successfully synchronised (within time period) 2. unsynchroised (hasn't sync'd since max sync duration (see config)) 3. invalid (time was sync'd but is out by 3%) 4. erred (time deviation >= 75 minutes)
TDM-231	Communications between DC Meter and remote display panel (IONITY Demo)	Ethernet to SPI bus to display module, with encryption, dedicated key, ack from display to check comms status, allows display to report errors to meter to forward to Tritium and audit log.To reduce work on FPGA, send a series of fixed-width strings always in the same order. FPGA knows how and where to display each string. Messages to the display are sent every second to keep the kWh view up to date.
TDM-237	Meter Time	1. Display local time. This means including DST and Timezone values. 2. The Time displayed SHALL update every second 3. Time SHALL be sent to the display as a string 4. Time SHALL be displayed in the format yyyy-mm-ddTHH:mm:ss z/+hh:mm
TDM-235	Energy usage for charge session	1. Displayed in kWh (Appendix B suggests could optionally use MWh but [ERA012] says must be kWh) 2. The format of the usage will be: `[0-9]{7}.[0-9]{2}` e.g. 0000000.00
TDM-167	Event Log	Provides persistent Logging functions for Meter.
TDM-236	Active Tariff	Tariffs can be used in multiple ways. Seasonal, time-of-day, or a combination. When not charging display the active tariff. When charging display the tariffs used so far during the charge period.

11 issues

LME003

Key	Summary	Description
TDM-231	Communications between DC Meter and remote display panel (IONITY Demo)	Ethernet to SPI bus to display module, with encryption, dedicated key, ack from display to check comms status, allows display to report errors to meter to forward to Tritium and audit log.To reduce work on FPGA, send a series of fixed-width strings always in the same order. FPGA knows how and where to display each string. Messages to the display are sent every second to keep the kWh view up to date.
TDM-235	Energy usage for charge session	1. Displayed in kWh (Appendix B suggests could optionally use MWh but [ERA012] says must be kWh) 2. The format of the usage will be: `[0-9]{7}.[0-9]{2}` e.g. 0000000.00
TDM-236	Active Tariff	Tariffs can be used in multiple ways. Seasonal, time-of-day, or a combination. When not charging display the active tariff. When charging display the tariffs used so far during the charge period.

3 issues

Charging point measuring device with electricity metering function

VDE Requirement ID	Tritium Design Element ID

LME010

Key	Summary	Description
TDM-186	calibration table(s)	Used to convert readings from the DC Sensor to be accurate. At the moment this is: 1. offset 2. scale factor readings will be store as: raw value, offset, scaling factor, calculated result so original reads can be re-calculated if it was later determined the offset/scaling was incorrect.

1 issue

LME011

Key	Summary	Description
TDM-186	calibration table(s)	Used to convert readings from the DC Sensor to be accurate. At the moment this is: 1. offset 2. scale factor readings will be store as: raw value, offset, scaling factor, calculated result so original reads can be re-calculated if it was later determined the offset/scaling was incorrect.

1 issue

LME012

Key	Summary	Description

No issues found

LME013	Key Summary Description No issues found
LME014	Key Summary Description No issues found
LME015	Key Summary Description No issues found
LME016	Key Summary Description No issues found
LME017	Key Summary Description No issues found

Charging Point Measuring Device With Time Measurement Function	
VDE Requirement ID	**Tritium Design Element ID**
LME020	Key Summary Description TDM-196 time duration calculation Based on lessons learned with time synchronisation, calculating duration is a combination of a start time and a monotonic clock counting forwards. If an end time is required it is calculated from start time + duration.This time duration capability can also be used to implement timeouts within the system. 1 issue
LME021	Key Summary Description TDM-312 Determine the number of reads/interval to calculate usage how many samples per second do we need to meet accuracy requirements. 1 issue

LME022	Key Summary No issues found
LME023	Key Summary No issues found
LME024	Key Summary TDM-239 Charge time duration TDM-231 Communications between DC Meter and remote display panel (IONITY Demo) 2 issues

Measured Value Acquisition Module	
VDE Requirement ID	**Tritium Design Element ID**
MWE001	Key Summary Description TDM-195 Format of recorded usage measurements NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 1 issue
MWE002	Key Summary Description TDM-195 Format of recorded usage measurements NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 1 issue

MWE003	Key / Summary / Description TDM-195 / Format of recorded usage measurements / NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 1 issue
MWE004	Key / Summary / Description TDM-862 / Decimal Fixed Point Numbers / See https://wiki.tritium.com.au/display/RND/Decimal+Fixed+Point+Numbers TDM-195 / Format of recorded usage measurements / NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 2 issues
MWE005	Key / Summary / Description TDM-195 / Format of recorded usage measurements / NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 1 issue

Measured Value Processing Module	
VDE Requirement ID	**Tritium Design Element ID**
MWV001	Key / Summary / Description TDM-195 / Format of recorded usage measurements / NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 1 issue

MWV002

Key	Summary	Description
TDM-195	Format of recorded usage measurements	NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2

1 issue

MWV003

Key	Summary	Description
TDM-313	Determine smoothing algorithm	Post processing of data to remove anomalous reads (or spikes) but still maintain accuracy requirements. Errors in calculations shall be captured and logged/reported as part of telemetry.
TDM-154	Usage Session Measurements Recording /Reporting	stored measurements will be "as-is" without calculation but will contain enough information with each read so that a calculation may be performed at any time.use common 'circular buffer' mechanism for session related measurements
TDM-312	Determine the number of reads/interval to calculate usage	how many samples per second do we need to meet accuracy requirements.
TDM-160	Find Accurate Arithmetic Library	There are requirements regarding the accuracy of any calculation, which add pressure on the FPU because it inherently has error. So to reduce issues with certification a library that is more accurate needs to be found and used. The requirements regarding accuracy are: > *[MWV004] When calculating derived measurements, DO NOT add an error contribution of more than ± 1% of the least significant used digit of the calculated value.* > *[MWV005] A set of derived measurements may not degrade the resolution of the measurements obtained from the MWE component.*
TDM-195	Format of recorded usage measurements	NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2

5 issues

MWV005

Key	Summary	Description
TDM-154	Usage Session Measurements Recording/Reporting	stored measurements will be "as-is" without calculation but will contain enough information with each read so that a calculation may be performed at any time.use common 'circular buffer' mechanism for session related measurements
TDM-160	Find Accurate Arithmetic Library	There are requirements regarding the accuracy of any calculation, which add pressure on the FPU because it inherently has error. So to reduce issues with certification a library that is more accurate needs to be found and used. The requirements regarding accuracy are: *[MWV004] When calculating derived measurements, DO NOT add an error contribution of more than ± 1% of the least significant used digit of the calculated value.* *[MWV005] A set of derived measurements may not degrade the resolution of the measurements obtained from the MWE component.*
TDM-862	Decimal Fixed Point Numbers	See https://wiki.tritium.com.au/display/RND/Decimal+Fixed+Point+Numbers

3 issues

Measured Value Memory

VDE Requirement ID	Tritium Design Element ID

STORAGE001

Key	Summary	Description
TDM-153	UID generation for charge sessions	independent id. Maps to user somehow
TDM-195	Format of recorded usage measurements	NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2

2 issues

STORAGE002

Key	Summary	Description
TDM-153	UID generation for charge sessions	independent id. Maps to user somehow
TDM-195	Format of recorded usage measurements	NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2

2 issues

STORAGE003	<table><tr><th>Key</th><th>Summary</th><th>Description</th></tr><tr><td>TDM-195</td><td>Format of recorded usage measurements</td><td>NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2</td></tr></table> 1 issue
STORAGE004	<table><tr><th>Key</th><th>Summary</th><th>Description</th></tr><tr><td>TDM-195</td><td>Format of recorded usage measurements</td><td>NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2</td></tr></table> 1 issue
STORAGE005	<table><tr><th>Key</th><th>Summary</th><th>Description</th></tr><tr><td>TDM-197</td><td>packaging stored measurements into a SAFE Initiative payload</td><td>Using OCMF instead of SML due to time constraints and licensing issues with libraries.</td></tr></table> 1 issue

Time Management Method	
VDE Requirement ID	**Tritium Design Element ID**

TIME001

Key	Summary	Description
TDM-208	last time sync time	
TDM-184	Configuration of time	Time is automatically set as soon as the DC Meter has connection to a listed clock source. The configuration includes: • details of clock sources, • timezone, • time synchronisation frequency (specified in minutes, min 2 minutes [TIME012], max value = 7days (60*24*7) [TIME013]). • max time between sync before panicking When time is displayed it has to represent timezone and daylight savings, but when time is recorded, it SHALL be UTC.
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

3 issues

TIME002

Key	Summary	Description
TDM-238	Meter Time Health Status	Display next to the meter time either as a symbol of confidence of accuracy of the meter clock: 1. successfully synchronised (within time period) 2. unsynchroised (hasn't sync'd since max sync duration (see config)) 3. invalid (time was sync'd but is out by 3%) 4. erred (time deviation >= 75 minutes)
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

2 issues

TIME003

Key	Summary	Description
TDM-209	battery/supercap voltage (RTC and device)	
TDM-184	Configuration of time	Time is automatically set as soon as the DC Meter has connection to a listed clock source. The configuration includes: • details of clock sources, • timezone, • time synchronisation frequency (specified in minutes, min 2 minutes [TIME012], max value = 7days (60*24*7) [TIME013]). • max time between sync before panicking When time is displayed it has to represent timezone and daylight savings, but when time is recorded, it SHALL be UTC.
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

3 issues

TIME004

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

TIME005

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

TIME006

Key	Summary

No issues found

TIME007

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

TIME008

Key	Summary	Description
TDM-175	check RTC value	Tests to see if RTC value is approximately accurate. There will be the ability to store the current RTC value(s) when main power disappears. When power returns the stored (and time sync hasn't happened yet) time should be older than current time. Otherwise the RTC was reset completely.
TDM-174	check RTC battery /supercap voltage	TBD if using external RTC chip (with higher accuracy).1. communicate with RTC with read/write/interrupts2. react to battery bad flag on RTC
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

3 issues

TIME009

Key	Summary	Description
TDM-174	check RTC battery /supercap voltage	TBD if using external RTC chip (with higher accuracy).1. communicate with RTC with read/write/interrupts2. react to battery bad flag on RTC
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

2 issues

TIME010

Key	Summary	Description
TDM-238	Meter Time Health Status	Display next to the meter time either as a symbol of confidence of accuracy of the meter clock: 1. successfully synchronised (within time period) 2. unsynchroised (hasn't sync'd since max sync duration (see config)) 3. invalid (time was sync'd but is out by 3%) 4. erred (time deviation >= 75 minutes)
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

2 issues

TIME011

Key	Summary	Description

No issues found

TIME012

Key	Summary	Description
TDM-184	Configuration of time	Time is automatically set as soon as the DC Meter has connection to a listed clock source. The configuration includes: • details of clock sources, • timezone, • time synchronisation frequency (specified in minutes, min 2 minutes [TIME012], max value = 7days (60*24*7) [TIME013]). • max time between sync before panicking When time is displayed it has to represent timezone and daylight savings, but when time is recorded, it SHALL be UTC.

1 issue

TIME013

Key	Summary	Description
TDM-184	Configuration of time	Time is automatically set as soon as the DC Meter has connection to a listed clock source. The configuration includes: • details of clock sources, • timezone, • time synchronisation frequency (specified in minutes, min 2 minutes [TIME012], max value = 7days (60*24*7) [TIME013]). • max time between sync before panicking When time is displayed it has to represent timezone and daylight savings, but when time is recorded, it SHALL be UTC.

1 issue

TIME014

Key	Summary	Description
TDM-184	Configuration of time	Time is automatically set as soon as the DC Meter has connection to a listed clock source. The configuration includes: • details of clock sources, • timezone, • time synchronisation frequency (specified in minutes, min 2 minutes [TIME012], max value = 7days (60*24*7) [TIME013]). • max time between sync before panicking When time is displayed it has to represent timezone and daylight savings, but when time is recorded, it SHALL be UTC.

1 issue

TIME015

Key	Summary	Description
TDM-237	Meter Time	1. Display local time. This means including DST and Timezone values. 2. The Time displayed SHALL update every second 3. Time SHALL be sent to the display as a string 4. Time SHALL be displayed in the format yyyy-mm-ddTHH:mm:ss z/+hh:mm
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

2 issues

TIME016

Key	Summary	Description
TDM-184	Configuration of time	Time is automatically set as soon as the DC Meter has connection to a listed clock source. The configuration includes: • details of clock sources, • timezone, • time synchronisation frequency (specified in minutes, min 2 minutes [TIME012], max value = 7days (60*24*7) [TIME013]). • max time between sync before panicking When time is displayed it has to represent timezone and daylight savings, but when time is recorded, it SHALL be UTC.
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two-part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)
2 issues		

TIME017	

Key	Summary	Description
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

2 issues

Time Server	
VDE Requirement ID	**Tritium Design Element ID**

LOG001

Key	Summary	Description
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

2 issues

LOG002

Key	Summary	Description
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record

1 issue

LOG003

Key	Summary	Description
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record

1 issue

LOG004

Key	Summary	Description
TDM-155	Event Log Recording /Reporting	use common 'circular buffer' mechanism. The size of the event log must be large enough to store (8 + 2) years of events

1 issue

LOG005

Key	Summary	Description
TDM-230	Technician Access (VDE Demo)	A technician will require access to the DC meter. The DC meter has an Ethernet connection and is already required to have TLS 1.2 client functionality. • Implement TLS 1.2 server with required client side certificate authorisation. • Client will present to the server a certificate to prove it is approved to access the server. ~~Requires confirmation from James on direction of product. Need to know if we are expected to store cost on tariffs as a technician role could be to update tariff info. Technicians may perform a test locally on the meter via PC /Laptop login over local network to verify measurement accuracy with varying loads, so a live feed of usage (updating every 5 seconds???).~~

1 issue

LOG006

Key	Summary	Description
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record

1 issue

LOG007

Key	Summary	Description
TDM-155	Event Log Recording /Reporting	use common 'circular buffer' mechanism. The size of the event log must be large enough to store (8 + 2) years of events

1 issue

LOG008

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG009

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG010

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG011

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG012

Key	Summary	Description

No issues found

LOG013

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG014

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG015

Key	Summary	Description
TDM-183	DC Meter Run Time Self Test	Slow periodic tests performed every 6 hours. Continuous tests performed at variable rates depending on test (~30sec) Self test performed by supervisory state machine whilst not charging. Results of all tests need to be logged to the event log. Tests not presently contained in this task: • Check communications interfaces • MQTT connection • NTP connection • TSO connection • AFE comms • Flash comms Bootloader image integrity check (SHA256) is completed before jumping to it upon a FW update request
TDM-148	DC Meter Power On Self Test	Power up self-test run by supervisory state machine. Either runs or ensures result was valid of tests. Results of all tests need to be logged to the event log. • User image integrity check (SHA256) is completed by bootloader before jumping to it (Otherwise a FW update is attempted or BL holds operation of device)
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

3 issues

LOG016

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG017

Key	Summary	Description
TDM-223	statistics	1. max size (bytes) 2. max size (records) 3. used (bytes and records, as a %) 4. bad record count (fails hash test) 5. first record timestamp 6. last record timestamp 7. Thanks to WELMEC7.2:P7, may need to record/report a counter for number of times any configuration field was changed
TDM-226	The circular buffer shall provide notification events	notify listeners of events like: 1. size usage (75%, 80%, 90%, 100%) 2. reset/empty/delete/overwrite
TDM-155	Event Log Recording/Reporting	use common 'circular buffer' mechanism. The size of the event log must be large enough to store (8 + 2) years of events

3 issues

LOG018

Key	Summary	Description
TDM-234	share random user id for charge session with charger/CPO	
TDM-223	statistics	1. max size (bytes) 2. max size (records) 3. used (bytes and records, as a %) 4. bad record count (fails hash test) 5. first record timestamp 6. last record timestamp 7. Thanks to WELMEC7.2:P7, may need to record/report a counter for number of times any configuration field was changed
TDM-167	Event Log	Provides persistent Logging functions for Meter.

3 issues

LOG019

Key	Summary	Description
TDM-223	statistics	1. max size (bytes) 2. max size (records) 3. used (bytes and records, as a %) 4. bad record count (fails hash test) 5. first record timestamp 6. last record timestamp 7. Thanks to WELMEC7.2:P7, may need to record/report a counter for number of times any configuration field was changed

1 issue

LOG020

Key	Summary	Description
TDM-167	Event Log	Provides persistent Logging functions for Meter.

1 issue

LOG021

Key	Summary	Description
TDM-222	The circular buffer can be exported	export the whole buffer, using encryption, to an external interface. A separate flag will indicate if deletion of the contents is to happen after the export. Some kind of handshake is required to ensure what was exported is valid /readable before deleting. Exporting creates an entry in the event log. Exported data must contain a header which includes: 1. meter unique identifier 2. timestamp of the export (according to the meter) 3. name of the buffer (e.g. measurements, events, etc) Once data has been exported, it is marked as exported, which allows it to be automatically overwritten with newer records when space requires. Data can still be exported if marked for deletion.

1 issue

LOG022

Key	Summary	Description
TDM-222	The circular buffer can be exported	export the whole buffer, using encryption, to an external interface. A separate flag will indicate if deletion of the contents is to happen after the export. Some kind of handshake is required to ensure what was exported is valid /readable before deleting. Exporting creates an entry in the event log. Exported data must contain a header which includes: 1. meter unique identifier 2. timestamp of the export (according to the meter) 3. name of the buffer (e.g. measurements, events, etc) Once data has been exported, it is marked as exported, which allows it to be automatically overwritten with newer records when space requires. Data can still be exported if marked for deletion.

1 issue

LOG023

Key	Summary	Description
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record

1 issue

LOG024

Key	Summary	Description
TDM-225	record hashing the buffer internals (pointers, stats, etc)	To help detect if the buffer management becomes corrupt, we hash the internal variables used for management of records.
TDM-224	record hashing	Each record contains it's own hash to detect corruption. The hash includes all data fields.
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record

3 issues

LOG025

Key	Summary	Description

No issues found

LOG026

Key	Summary	Description
TDM-222	The circular buffer can be exported	export the whole buffer, using encryption, to an external interface. A separate flag will indicate if deletion of the contents is to happen after the export. Some kind of handshake is required to ensure what was exported is valid /readable before deleting. Exporting creates an entry in the event log. Exported data must contain a header which includes: 1. meter unique identifier 2. timestamp of the export (according to the meter) 3. name of the buffer (e.g. measurements, events, etc) Once data has been exported, it is marked as exported, which allows it to be automatically overwritten with newer records when space requires. Data can still be exported if marked for deletion.

1 issue

LOG027

Key	Summary	Description
TDM-230	Technician Access (VDE Demo)	A technician will require access to the DC meter. The DC meter has an Ethernet connection and is already required to have TLS 1.2 client functionality. • Implement TLS 1.2 server with required client side certificate authorisation. • Client will present to the server a certificate to prove it is approved to access the server. ~~Requires confirmation from James on direction of product. Need to know if we are expected to store cost on tariffs as a technician role could be to update tariff info. Technicians may perform a test locally on the meter via PC/Laptop login over local network to verify measurement accuracy with varying loads, so a live feed of usage (updating every 5 seconds???).~~
TDM-245	Technician Access messages to export/report parts of data	The private keys shall never be extracted or viewed. Public keys can be extracted or viewed if permissions permit. The state of the meter can be viewed. The CPO can export data. Technicians with the appropriate key can export measurements and event logs. Update live values to the Technician Access terminal
TDM-222	The circular buffer can be exported	export the whole buffer, using encryption, to an external interface. A separate flag will indicate if deletion of the contents is to happen after the export. Some kind of handshake is required to ensure what was exported is valid /readable before deleting. Exporting creates an entry in the event log. Exported data must contain a header which includes: 1. meter unique identifier 2. timestamp of the export (according to the meter) 3. name of the buffer (e.g. measurements, events, etc) Once data has been exported, it is marked as exported, which allows it to be automatically overwritten with newer records when space requires. Data can still be exported if marked for deletion.

3 issues

LOG029

Key	Summary	Description
TDM-155	Event Log Recording /Reporting	use common 'circular buffer' mechanism. The size of the event log must be large enough to store (8 + 2) years of events
TDM-222	The circular buffer can be exported	export the whole buffer, using encryption, to an external interface. A separate flag will indicate if deletion of the contents is to happen after the export. Some kind of handshake is required to ensure what was exported is valid /readable before deleting. Exporting creates an entry in the event log. Exported data must contain a header which includes: 1. meter unique identifier 2. timestamp of the export (according to the meter) 3. name of the buffer (e.g. measurements, events, etc) Once data has been exported, it is marked as exported, which allows it to be automatically overwritten with newer records when space requires. Data can still be exported if marked for deletion.

2 issues

LOG030	

Key	Summary	Description
TDM-222	The circular buffer can be exported	export the whole buffer, using encryption, to an external interface. A separate flag will indicate if deletion of the contents is to happen after the export. Some kind of handshake is required to ensure what was exported is valid /readable before deleting. Exporting creates an entry in the event log. Exported data must contain a header which includes: 1. meter unique identifier 2. timestamp of the export (according to the meter) 3. name of the buffer (e.g. measurements, events, etc) Once data has been exported, it is marked as exported, which allows it to be automatically overwritten with newer records when space requires. Data can still be exported if marked for deletion.

1 issue

Firmware Management Module	
VDE Requirement ID	**Tritium Design Element ID**
FWM001	

Key	Summary	Description
TDM-212	encrypted comms	~~Communicates to the Tritium server's to download updates~~
TDM-217	validate upgrade process and start metering software	no comms to charger means charger can't charge. Once metering software started, comms to charger recommences and a charge can occur.
TDM-216	validate upgrade binary	different parts of the meter can be upgraded (metering/event log can't be):1. firmware, 2. calibration tables, 3. configuration
TDM-215	validate binary provider	Only a trusted source can provide a binary upgrade. There needs to be a mechanism to authenticate the binary against the source of the binary, which is separate to the encryption of the binary.

4 issues

FWM002	

Key	Summary	Description
TDM-212	encrypted comms	~~Communicates to the Tritium server's to download updates~~
TDM-217	validate upgrade process and start metering software	no comms to charger means charger can't charge. Once metering software started, comms to charger recommences and a charge can occur.
TDM-216	validate upgrade binary	different parts of the meter can be upgraded (metering/event log can't be):1. firmware, 2. calibration tables, 3. configuration
TDM-215	validate binary provider	Only a trusted source can provide a binary upgrade. There needs to be a mechanism to authenticate the binary against the source of the binary, which is separate to the encryption of the binary.

4 issues

FWM003	

Key	Summary	Description

No issues found

FWM004

Key	Summary	Description
TDM-218	during upgrade, the display will show progress and upgrade version	requires code to communicate with the display (which should already be done). Create message to display regarding the upgrade process.
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. **REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS**

2 issues

FWM005

Key	Summary

No issues found

FWM006

Key	Summary	Description
TDM-217	validate upgrade process and start metering software	no comms to charger means charger can't charge. Once metering software started, comms to charger recommences and a charge can occur.

1 issue

FWM007

Key	Summary	Description
TDM-158	Upgrading (Bootloader)	Always runs on start up.Checks if there is a binary to install and performs necessary steps to install it safely. Always checks the validity of the software it is about to start.existing measurements/audit trail/etc must remain intact after an upgrade

1 issue

Signature Module	
VDE Requirement ID	**Tritium Design Element ID**

SIGN001

Key	Summary	Description
TDM-191	find/search for keys	to be used internally by the metering app. Private key values can't be displayed, exported, or printed.1. find/search for key by privilege/functionality2. find/search for key by user (a.k.a. public key)3. find/search for key by flag (active /invalidated/unused/private/public)4. find/search for key by id5. find/search for key based on key
TDM-185	Configuration of security key (s)	This task only addresses the basic get/set functionality. Further management needs are defined in task "Security Keys Management"

2 issues

SIGN002

Key	Summary	Description

No issues found

SIGN003

Key	Summary	Description
TDM-173	Prove random number generator	NIST have test code that can provide quality assurance on results from a random number generator. This is a combination of getting test code running and generating numbers from the hardware generator. This is an R&D task as the process for converting bits from the hardware RBG may require tuning to pass tests. 1. Download code 2. Get it to compile on Linux and run provided sample set to verify it works 3. Get it to run on target 4. Measure performance to see if we can run on target as part of run time validation (or a subset of)

1 issue

SIGN004

Key	Summary	Description

No issues found

SIGN005

Key	Summary	Description
TDM-189	unique identification	Ensure adding new keys doesn't result in a duplication of id.Removing/replacing keys deletes an id and allows it to be used again.Search algorithm to detect next available id
TDM-185	Configuration of security key(s)	This task only addresses the basic get/set functionality. Further management needs are defined in task "Security Keys Management"

2 issues

SIGN006

Key	Summary	Description
TDM-191	find/search for keys	to be used internally by the metering app. Private key values can't be displayed, exported, or printed.1. find/search for key by privilege/functionality2. find/search for key by user (a.k.a. public key)3. find/search for key by flag (active /invalidated/unused/private/public)4. find/search for key by id5. find/search for key based on key

1 issue

SIGN007

Key	Summary

No issues found

SIGN008	Key Summary No issues found
SIGN009	Key Summary No issues found
SIGN010	Key Summary Description TDM-192 invalidate key (s) by id Keeping a list of previously used pairs so they can't be reactivated by programming as if they are new TDM-191 find/search for keys to be used internally by the metering app. Private key values can't be displayed, exported, or printed.1. find/search for key by privilege/functionality2. find/search for key by user (a.k.a. public key)3. find/search for key by flag (active /invalidated/unused/private/public)4. find/search for key by id5. find/search for key based on key 2 issues
SIGN011	Key Summary Description No issues found

Verification Module	
VDE Requirement ID	**Tritium Design Element ID**
VERIFY001	Key Summary No issues found
VERIFY002	Key Summary No issues found

VERIFY003

Key	Summary	Description
TDM-172	add security library (mbed TLS) to the meter	ECDH, TLS, X.509, entropy management, etc. https://tls.mbed.org This library is capable of doing everything we want with regards to security signing, authentication, HMAC, encryption, random bit generator/entropy. However, we need to port it to the target. Some parts of the library need us to create code to fill in stubs and provide the missing functionality. We also need to performance test to see if we can still operate fast enough.

1 issue

VERIFY004

Key	Summary	Description
TDM-172	add security library (mbed TLS) to the meter	ECDH, TLS, X.509, entropy management, etc. https://tls.mbed.org This library is capable of doing everything we want with regards to security signing, authentication, HMAC, encryption, random bit generator/entropy. However, we need to port it to the target. Some parts of the library need us to create code to fill in stubs and provide the missing functionality. We also need to performance test to see if we can still operate fast enough.

1 issue

VERIFY005

Key	Summary	Description
TDM-172	add security library (mbed TLS) to the meter	ECDH, TLS, X.509, entropy management, etc. https://tls.mbed.org This library is capable of doing everything we want with regards to security signing, authentication, HMAC, encryption, random bit generator/entropy. However, we need to port it to the target. Some parts of the library need us to create code to fill in stubs and provide the missing functionality. We also need to performance test to see if we can still operate fast enough.

1 issue

VERIFY006

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. **REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS**

2 issues

Verification Compliant	
VDE Requirement ID	**Tritium Design Element ID**

ERA007

Key	Summary	Description
TDM-919	Display the cumulative (for life) kW register value	
TDM-152	Cumulative tariff registers summing all charge sessions	name/identifiercumulative storage, no resetting back to 0 unless counter overflow.Protection against duplicate tariffscreate/update/deletestart time (end time assumed by next tariff)
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

3 issues

ERA008

Key	Summary	Description
TDM-919	Display the cumulative (for life) kW register value	
TDM-152	Cumulative tariff registers summing all charge sessions	name/identifiercumulative storage, no resetting back to 0 unless counter overflow.Protection against duplicate tariffscreate/update/deletestart time (end time assumed by next tariff)
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

3 issues

ERA009

Key	Summary	Description
TDM-151	Tariff settings	Seasonal and/or time of day rate changes. 1. will need the fields: yyyy/mm/dd hh:mm, name, id, $/kWh, duration (minutes) 2. limit number of tariffs to 8 3. add ability to change tariffs through the terminal Further information on types of tariffs to cater for can be found in the following links (Note: these links are in German so translation is required): • https://wiki.scn.sap.com/wiki/pages/viewpage.action?pageId=446731342 • https://help.sap.com/viewer/3e88baa8ca4e4863a6203e00deb32df2/2.3.0.1/de-DE/3ba4e45461c2136de10000000a441470.html
TDM-194	maintain a cumulative measurement registers summing current charge session for each tariff	A single charging session may cover multiple tariffs. The user needs to be aware of this during the charge and end of charge.
TDM-236	Active Tariff	Tariffs can be used in multiple ways. Seasonal, time-of-day, or a combination. When not charging display the active tariff. When charging display the tariffs used so far during the charge period.
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

4 issues

ERA010

Key	Summary	Description
TDM-151	Tariff settings	Seasonal and/or time of day rate changes. 1. will need the fields: yyyy/mm/dd hh:mm, name, id, $/kWh, duration (minutes) 2. limit number of tariffs to 8 3. add ability to change tariffs through the terminal Further information on types of tariffs to cater for can be found in the following links (Note: these links are in German so translation is required): • https://wiki.scn.sap.com/wiki/pages/viewpage.action?pageId=446731342 • https://help.sap.com/viewer/3e88baa8ca4e4863a6203e00deb32df2/2.3.0.1/de-DE/3ba4e45461c2136de10000000a441470.html
TDM-194	maintain a cumulative measurement registers summing current charge session for each tariff	A single charging session may cover multiple tariffs. The user needs to be aware of this during the charge and end of charge.
TDM-152	Cumulative tariff registers summing all charge sessions	name/identifiercumulative storage, no resetting back to 0 unless counter overflow. Protection against duplicate tariffscreate/update/deletestart time (end time assumed by next tariff)
TDM-236	Active Tariff	Tariffs can be used in multiple ways. Seasonal, time-of-day, or a combination. When not charging display the active tariff. When charging display the tariffs used so far during the charge period.
TDM-924	Can you give us an example of a charge session with multiple tariffs?	The VDE spec doesn't contain detailed description on Tariffs. There are discussions around roaming tariffs, tariffs based on time, tariffs based on usage, but nothing concrete. See: • ~~TDM-151~~ • ~~TDM-152~~ • ~~TDM-194~~ • ~~TDM-236~~
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

6 issues

ERA011

Key	Summary	Description
TDM-235	Energy usage for charge session	1. Displayed in kWh (Appendix B suggests could optionally use MWh but [ERA012] says must be kWh) 2. The format of the usage will be: `[0-9]{7}.[0-9]{2}` e.g. 0000000.00

1 issue

ERA012

Key	Summary	Description
TDM-235	Energy usage for charge session	1. Displayed in kWh (Appendix B suggests could optionally use MWh but [ERA012] says must be kWh) 2. The format of the usage will be: `[0-9]{7}.[0-9]{2}` e.g. 0000000.00
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

2 issues

ERA013

Key	Summary	Description
TDM-239	Charge time duration	Time in seconds for the charge duration. Must be with 1% of nominal value. Duration will be displayed in the format: [h]hh:mm:ss
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

2 issues

ERA014

Key	Summary	Description
TDM-237	Meter Time	1. Display local time. This means including DST and Timezone values. 2. The Time displayed SHALL update every second 3. Time SHALL be sent to the display as a string 4. Time SHALL be displayed in the format yyyy-mm-ddTHH:mm:ss z/+hh:mm
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

2 issues

ERA015

Key	Summary	Description
TDM-237	Meter Time	1. Display local time. This means including DST and Timezone values. 2. The Time displayed SHALL update every second 3. Time SHALL be sent to the display as a string 4. Time SHALL be displayed in the format yyyy-mm-ddTHH:mm:ss z/+hh:mm
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

2 issues

ERA016

Key	Summary	Description
TDM-237	Meter Time	1. Display local time. This means including DST and Timezone values. 2. The Time displayed SHALL update every second 3. Time SHALL be sent to the display as a string 4. Time SHALL be displayed in the format yyyy-mm-ddTHH:mm:ss z/+hh:mm
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

2 issues

ERA017

Key	Summary	Description
TDM-241	Display primitive error if meter knows it has failed unrecoverably	display "Err" if there is a fault with the meter. Tritium can log in to find details of the error or via technician access if available
TDM-167	Event Log	Provides persistent Logging functions for Meter.
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

3 issues

ERA018

Key	Summary	Description
TDM-242	comms between display and meter security status	display "SEC" if the encrypted channel is valid, "XXX" if there is an issue.
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. **REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS**

2 issues

ERA019

Key	Summary	Description
TDM-242	comms between display and meter security status	display "SEC" if the encrypted channel is valid, "XXX" if there is an issue.
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. **REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS**

2 issues

ERA020

Key	Summary	Description
TDM-194	maintain a cumulative measurement registers summing current charge session for each tariff	A single charging session may cover multiple tariffs. The user needs to be aware of this during the charge and end of charge.
TDM-236	Active Tariff	Tariffs can be used in multiple ways. Seasonal, time-of-day, or a combination. When not charging display the active tariff. When charging display the tariffs used so far during the charge period.

2 issues

ERA021

Key	Summary	Description
TDM-243	session authentication id	display the unique id generated by the dc meter that identifies this charge session. This will be sent to Transparenz so that it can be mapped to the user.This unique id must also be shared with the charger so it can be used in the billing process.The session id will be created upon trigger of 'start-of-charge'. This could be current flow or an interrupt from the charger.

1 issue

ERA022

Key	Summary	Description
TDM-238	Meter Time Health Status	Display next to the meter time either as a symbol of confidence of accuracy of the meter clock: 1. successfully synchronised (within time period) 2. unsynchroised (hasn't sync'd since max sync duration (see config)) 3. invalid (time was sync'd but is out by 3%) 4. erred (time deviation >= 75 minutes)
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. **REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS**

2 issues

ERA023

Key	Summary	Description
TDM-243	session authentication id	display the unique id generated by the dc meter that identifies this charge session. This will be sent to Transparenz so that it can be mapped to the user.This unique id must also be shared with the charger so it can be used in the billing process.The session id will be created upon trigger of 'start-of-charge'. This could be current flow or an interrupt from the charger.
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. **REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS**

2 issues

ERA024	

Key	Summary	Description
TDM-179	check communications	
TDM-183	DC Meter Run Time Self Test	Slow periodic tests performed every 6 hours. Continuous tests performed at variable rates depending on test (~30sec) Self test performed by supervisory state machine whilst not charging. Results of all tests need to be logged to the event log. Tests not presently contained in this task: • Check communications interfaces • MQTT connection • NTP connection • TSO connection • AFE comms • Flash comms Bootloader image integrity check (SHA256) is completed before jumping to it upon a FW update request
TDM-190	different security keys for different functionality	A method to authenticate a user to a set of privileges/functionality. A separate key for has.to.be, Tritium, technician, etc.
TDM-231	Communications between DC Meter and remote display panel (IONITY Demo)	Ethernet to SPI bus to display module, with encryption, dedicated key, ack from display to check comms status, allows display to report errors to meter to forward to Tritium and audit log.To reduce work on FPGA, send a series of fixed-width strings always in the same order. FPGA knows how and where to display each string. Messages to the display are sent every second to keep the kWh view up to date.
TDM-185	Configuration of security key(s)	This task only addresses the basic get/set functionality. Further management needs are defined in task "Security Keys Management"
TDM-244	Client authorisation for access to the DC Meter	Anyone attempting access to the DC Meter as a "client" (i.e. the DC Meter is the server) to use the terminal interface of the DC Meter first is authenticated using X.509 certificates that were created by Tritium. An example is the technician must authorise themselves with the meter before seeing the menu and retrieving information. The CPO must authorise prior to updating tariffs.
TDM-973	Verification Compliancy	ERA024 - To what standard? Is TR-03107-1 an acceptable standard?

7 issues

User Interface

VDE Requirement ID	Tritium Design Element ID
UI001	

Key	Summary	Description
TDM-248	Terminal input validation of write commands for range testing and privileges for the user	

1 issue

UI002	

Key	Summary	Description

No issues found

Authorization Unit

VDE Requirement ID	Tritium Design Element ID
AUTHZ001	Key Summary No issues found
AUTHZ002	Key / Summary / Description TDM-195 — Format of recorded usage measurements — NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 TDM-199 — event id list 1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed) 2 issues

Authentication Unit	
VDE Requirement ID	**Tritium Design Element ID**

AUTHN001	Key Summary No issues found
AUTHN002	Key Summary No issues found
AUTHN003	Key Summary No issues found
AUTHN004	Key Summary No issues found

5. Inverse Traceability Matrix

Firmware and Software:

Key	Summary	T	Created	Updated	Due	Assignee	Reporter	P	Status	Resolution	Linked Issues

No issues found

DCM Box Testing

Following tests started from M2 hardware with firmware without calibration (30/09/2019).

Fault/Behaviour	Cause	Detailed Description	Closed
Removal of measured voltage does not stop charge session	ChargeSessionSM.cpp logic doesn't check this	Once a charge is started above current/voltage thresholds; voltage can be removed and charge session doesn't stop. Current being removed successfully stops session.	•
Enter Out of Service (OoS) mode	Underflow on power calculation	OoS cause reported as READ_ADC_FAIL. Goes OoS when no power is 0.000.	•
Energy displays ??? not value		Both debug Ws and TTI kWh display ??? instead of values. Occasionally a 'blip' will occur and what looks like power is shown for 1s.	•
TTI Energy displays Ws in a given loop not session energy	Temporary Ws shown	Needs to be total session energy	•
Doesn't support negative power. Rolls back total energy		Negative voltage doesn't latch in state machine. Energy counter rolls back when negative power is applied. Is this a problem? Does negative energy need independent counter	•
Negative current does not display correctly		AFE current shows negative but output current/debug current shows absolute.	•
Debug print of power shows VVVVV.VV when above 100kW	String print digit length	Does not have enough digits for power/Ws/voltage	•
Cable compensation voltage change can't be seen	Initialisation order	Voltage measured isn't changed by current when cable resistance is written to as expected.	•
Cable compensation seems to be based off opposite cable	Hardware abstraction opposite to labelled	Uses resistance value written to opposing cable	•
Uncertainty of sufficient energy transferred by IR LED	Non-standard test equipment	IR LED energy received is difficult to determine based on non-ideal receiver. Changed resistor to increase drive current.	•
AFE no longer reads/displays negative voltage /current	Typecast when calibrated to uint	Seems to have changed when calibration was introduced	•
Voltage calibration is the same for both voltage sense inputs		Calibration changing added to cable switching	•
Configuration isn't loaded once unit is reset		Not written and/or read in memory.	•
Temperature sensor reads false values at negative temps	Typecast to uint	Read ~6516 at negative temperatures	•
At -40'C one cable sense being shorted effects the measurement of the other (only shorting cable 2 with cable 1)	Leakage through pull down	Both sets of opto switches are turning on.	•
Bootloading resets the calibration information	Any JSON sent info writes to memory.	Changed so bootloading command does not	•

Non-issues/Accepted for first release

OoS doesn't recover		No OoS recovery mechanism yet	
Measurements displays ??? when out of range		E.g. If over 600A does not display the reading - should this be changed to maximum possible measurement rather than maximum expected? This is expected behaviour	
If no internet connection on startup cannot connect to TTI even on restore of internet		Unit powered up networked to router without external internet connectivity. Attempt to connect to TTI, fails as expected. Restore internet connectivity and attempt to connect to TTI, still fails.	
Charge start time/duration don't appear on TTI		Not yet implemented	
Whilst bootloading TTI must be closed or process crashes		Not yet fixed	

DC meter - overview of stories

All stories

Key	Summary	T	Updated	Assignee	Status	wp	Labels	Linked Issues
TDM-160	Find Accurate Arithmetic Library		Jul 01, 2019	Jeremy Kincaid	DONE	1841	accuracy investigation metering non-functional software	TDM-1921 , TDM-1083 , TDM-35 , TDM-1085
TDM-338	Draft of DC architecture spec		Mar 18, 2019	Jeremy Kincaid	DONE	1848 1850		TDM-452 , TDM-341 , TDM-340 , TDM-318 , TDM-339 , TDM-327
TDM-389	Create Functional, Environmental, Certification test plan		Oct 20, 2021	Jeremy Street-Thomas	DONE	1850 1902 1903 1904		TDM-984 , TDM-983 , TDM-452 , TDM-264 , TDM-390
TDM-387	Use existing SW tasks to publish SW architecture specifications		Mar 18, 2019	Jeremy Kincaid	DONE	1850 1902 1905		TDM-855 , TDM-459
TDM-296	Design Random Number Generator		Jul 01, 2019	Jeremy Kincaid	DONE	1850 1902 1903	Design software	TDM-1931 , TDM-385 , TDM-173 , TDM-35
TDM-388	Create specifications, test matrix & coverage status		Oct 20, 2021	Jeremy Street-Thomas	DONE	1905		TDM-452
TDM-453	Publish system requirements draft		Mar 18, 2019	Michael Walton	DONE	1905		TDM-458 , TDM-457 , TDM-455 , TDM-454 , TDM-452
TDM-896	Update planning		Mar 18, 2019	Jeremy Kincaid	DONE	1909		TDM-898 , TDM-897 , TDM-900 , TDM-899 , TDM-459
TDM-905	Certification Questions to VDE		Oct 20, 2021	Jeremy Street-Thomas	DONE	1915		TDM-975 , TDM-974 , TDM-973 , TDM-972 , TDM-971 , TDM-970 , TDM-452 , TDM-979 , TDM-978 , TDM-977 , TDM-976 , TDM-1025 , TDM-1306 , TDM-1307 , TDM-1948 , TDM-1949 , TDM-964 , TDM-963 , TDM-962 , TDM-961 , TDM-960 , TDM-927 , TDM-969 , TDM-924 , TDM-968 , TDM-923 , TDM-967 , TDM-966 , TDM-1318 , TDM-1319 , TDM-951 , TDM-950 , TDM-917 , TDM-916 , TDM-915 , TDM-959 , TDM-958 , TDM-957 , TDM-956 , TDM-954 , TDM-1001 , TDM-1320 , TDM-1321 , TDM-1004 , TDM-1368 , TDM-918 , TDM-982 , TDM-906 , TDM-949 , TDM-948 , TDM-1050 , TDM-947 , TDM-946 , TDM-945 , TDM-1334 , TDM-1016 , TDM-1019
TDM-548	DFMEA mechanical / industrial design scope		Oct 20, 2021	Bernie Walsh	DONE	1915	Main_Focus	TDM-932
TDM-530	DFMEA all electrical scope, including changes to charger bus bars		Oct 20, 2021	Jeremy Street-Thomas	DONE	1915		TDM-527
TDM-477	Publish electrical system specs draft		Oct 20, 2021	Unassigned	DONE	1915		TDM-466
TDM-467	Publish mechanical / industrial system specs draft		Apr 11, 2019	Bernie Walsh	DONE	1915		TDM-464

TDM-1314	Follow up on quoted standards		Oct 20, 2021	Jeremy Street-Thomas	DONE	1915		TDM-452
TDM-1051	[Cancelled] Release memory requirements		Nov 29, 2019	James Sneyd	DONE	1915		
TDM-1043	Release updated requirements		Oct 20, 2021	Jeremy Street-Thomas	DONE	1915		
TDM-1042	Release new PCB allowed dimensions to PCB designers		Nov 12, 2019	Oliver Gruber	DONE	1915		TDM-1041
TDM-1007	Review heat sink requirements for dispenser integration		Aug 12, 2019	Ben Guymer	DONE	1915		TDM-464 , TDM-1008
TDM-990	Update documentation FPGA		Apr 11, 2019	Alex Rudzki	DONE	1915		TDM-452 , TDM-913 , TDM-922
TDM-2198	General code reviews		Oct 20, 2021	James Sneyd	DONE			TDM-2199
TDM-2194	FPGA encryption		Mar 03, 2020	Alex Rudzki	DONE			TDM-2195 , TDM-2196 , TDM-2177
TDM-2193	Backlight display monitoring		Mar 03, 2020	Alex Rudzki	DONE			TDM-2201 , TDM-2200 , TDM-2177
TDM-2190	Automate Interceptor IP configuration		Dec 18, 2019	Alex Rudzki	DONE			TDM-2191 , TDM-2192 , TDM-2177
TDM-2178	Update pixel intercept module for PK		Dec 18, 2019	Alex Rudzki	DONE			TDM-2179 , TDM-2180 , TDM-2182 , TDM-2181 , TDM-2184 , TDM-2185 , TDM-2188 , TDM-2177
TDM-2058	FreeRTOS missing interrupts (ADS1220)		Nov 20, 2019	James Sneyd	DONE			
TDM-2055	Network-Attached Debug Printing		Nov 20, 2019	James Sneyd	DONE			TDM-936 , TDM-1915
TDM-1496	Revise accuracy calculations		Nov 20, 2019	Jeremy Street-Thomas	DONE			TDM-1497 , TDM-938
TDM-401	Draft NPI plan for DC meter project		Oct 20, 2021	Timo Bellgardt	DONE			
TDM-397	Create draft WBS and prepare		Mar 18, 2019	Alex Rudzki	DONE			TDM-452 , TDM-308
TDM-396	Create style guide for the Terminal		Oct 20, 2021	James Sneyd	DONE		software user-interface	TDM-1915
TDM-378	Networking /Comms Stack Integration: Explore Alternative RTOS		Nov 12, 2019	James Sneyd	DONE		Firmware	TDM-1028
TDM-372	Networking /Comms Stack Integration: lwIP Application		Nov 12, 2019	James Sneyd	DONE			TDM-931 , TDM-366 , TDM-377 , TDM-365 , TDM-375 , TDM-374 , TDM-1933 , TDM-373

TDM-366	Networking /Comms Stack Integration: mbedTLS Application		Nov 12, 2019	James Sneyd	DONE	Firmware	TDM-346 , TDM-1028 , TDM-230 , TDM-372 , TDM-371 , TDM-1914
TDM-365	Networking /Comms Stack Integration: FreeRTOS		Nov 12, 2019	James Sneyd	DONE	Firmware	TDM-369 , TDM-368 , TDM-367 , TDM-1028 , TDM-1933 , TDM-372 , TDM-370
TDM-364	Networking /Communicatio n Stack Integration Concept		Nov 12, 2019	James Sneyd	DONE		TDM-931 , TDM-1933
TDM-1466	Documentation M2 FW		Nov 29, 2019	James Sneyd	DONE		TDM-1469
TDM-1425	Document DC Meter software build process		Oct 20, 2021	James Sneyd	DONE	documentat ion process software	
TDM-1365	Design HW Controller		Nov 12, 2019	Richard Kosik	DONE		TDM-938
TDM-1364	Design HW AFE		Nov 20, 2019	Alex Rudzki	DONE		TDM-1464 , TDM-1366 , TDM-1465 , TDM-938 , TDM-1460 , TDM-1462
TDM-1363	Design HW FPGA		Nov 12, 2019	Vivien Wong	DONE		TDM-1367 , TDM-929 , TDM-938
TDM-1357	FPGA task planning		Nov 20, 2019	Alex Rudzki	DONE		TDM-1358 , TDM-452
TDM-1354	Verify FPGA operation		Oct 20, 2021	Alex Rudzki	DONE		TDM-1353 , TDM-1355 , TDM-1356
TDM-1337	Develop FPGA - Firmware		Oct 20, 2021	Jakub Tkaczyk	DONE		TDM-1342 , TDM-986 , TDM-1343 , TDM-61 , TDM-1344 , TDM-1356 , TDM-63 , TDM-62 , TDM-64 , TDM-1341 , TDM-1352
TDM-1336	Develop FPGA - Pixel Intercept module		Apr 17, 2020	Alex Rudzki	DONE		TDM-986 , TDM-2083 , TDM-2095 , TDM-2098 , TDM-1350 , TDM-1340 , TDM-1351 , TDM-2166 , TDM-2132 , TDM-2165 , TDM-2134 , TDM-1355 , TDM-60 , TDM-1840 , TDM-65 , TDM-1338 , TDM-1339
TDM-2075	FPGA interfacing		Nov 21, 2019	Alex Rudzki	DONE		TDM-986 , TDM-2076
TDM-7	Technical research		Mar 18, 2019	Alex Rudzki	DONE		TDM-325 , TDM-321 , TDM-376 , TDM-452 , TDM-361
TDM-354	lwIP Integration (No Operating System)		Jul 01, 2019	James Sneyd	DONE		TDM-359 , TDM-358 , TDM-357 , TDM-334 , TDM-356 , TDM-355 , TDM-1933 , TDM-35
TDM-329	Evaluate CAN bus message fragmentation (No Operating System)		Jul 01, 2019	James Sneyd	DONE		TDM-1908 , TDM-359 , TDM-333 , TDM-332 , TDM-331 , TDM-330 , TDM-35
TDM-310	Update/refine /test the ADC subsystem M1		Nov 25, 2019	James Sneyd	DONE		TDM-931 , TDM-1920 , TDM-24
TDM-187	configuration updates (or parts thereof)		Nov 12, 2019	Andy Meldrum	DONE	DC-Meter-Conf	TDM-1901 , TDM-936 , TDM-1915
TDM-183	DC Meter Run Time Self Test		Jul 01, 2020	Alex Rudzki	DONE	functional software	TDM-148 , TDM-2292 , TDM-1180 , TDM-2294 , TDM-2311 , TDM-2310 , TDM-2313 , TDM-1124 , TDM-2401 , TDM-2312 , TDM-2315 , TDM-2337 , TDM-2314 , TDM-2317 , TDM-76 , TDM-2316 , TDM-2308 , TDM-2319 , TDM-2318 , TDM-2309 , TDM-1904
TDM-169	SW Documentation X1		Mar 18, 2019	Jeremy Kincaid	DONE		TDM-35
TDM-168	Charger Software		Oct 20, 2021	Unassigned	DONE		TDM-1917
TDM-167	Event Log		Nov 12, 2019	Andy Meldrum	DONE		TDM-931 , TDM-1127 , TDM-1129 , TDM-1923 , TDM-1060 , TDM-1173

TDM-166	Test Environment X1		Mar 18, 2019	Jeremy Kincaid	DONE		TDM-35
TDM-165	Menu for technician		Oct 20, 2021	James Sneyd	DONE	SDD UI adaptation software status-reporting	
TDM-164	Displaying meter and charge information		Nov 26, 2019	James Sneyd	DONE		TDM-1908 , TDM-1004
TDM-163	Voltage Supervisor		Jul 12, 2021	James Sneyd	DONE		TDM-1907
TDM-162	User access /permissions scheme		Apr 17, 2020	Alex Rudzki	DONE		TDM-2266 , TDM-1925 , TDM-150 , TDM-190
TDM-161	Design Communications SW		Oct 20, 2021	James Sneyd	DONE		
TDM-159	Circular buffer mechanism X1		Jul 01, 2019	Jeremy Kincaid	DONE		TDM-35 , TDM-1937
TDM-158	Upgrading (Bootloader)		Oct 20, 2021	James Sneyd	DONE		TDM-1145 , TDM-2056
TDM-157	LED toggling (if no display)		Jul 01, 2019	James Sneyd	DONE	systems-engineering	TDM-980
TDM-156	Create a common set of status information for basic diagnostics		Oct 20, 2021	James Sneyd	DONE		TDM-1028 , TDM-1915
TDM-155	Event Log Recording /Reporting		Oct 20, 2021	James Sneyd	DONE	record-report	TDM-1113 , TDM-1126 , TDM-1137 , TDM-1116 , TDM-1924
TDM-154	Usage Session Measurements Recording /Reporting		Oct 20, 2021	James Sneyd	DONE	record-report	TDM-1083 , TDM-1085
TDM-153	UID generation for charge sessions		Oct 20, 2021	James Sneyd	DONE	record-report	TDM-1921 , TDM-1086 , TDM-1087
TDM-152	Cumulative tariff registers summing all charge sessions		Jul 01, 2019	Andy Meldrum	DONE	record-report	TDM-1166 , TDM-1921 , TDM-1945 , TDM-35 , TDM-1163 , TDM-1164
TDM-151	Tariff settings		Jul 01, 2019	Andy Meldrum	DONE	SDD tariff	TDM-1166 , TDM-561 , TDM-1945 , TDM-35 , TDM-1165
TDM-150	Security Keys Management		Oct 20, 2021	James Sneyd	DONE		TDM-162 , TDM-1929
TDM-149	Develop SW for setting and retrival of DC Meter Configuration		Oct 20, 2021	James Sneyd	DONE	SDD adaptation functional software	TDM-1901
TDM-148	DC Meter Power On Self Test		Oct 20, 2021	Alex Rudzki	DONE	software	TDM-1124 , TDM-2315 , TDM-2337 , TDM-76 , TDM-2316 , TDM-2319 , TDM-2292 , TDM-2318 , TDM-2294 , TDM-1903 , TDM-183 , TDM-2320
TDM-147	De-Risking activities		Mar 29, 2019	Jeremy Kincaid	DONE		TDM-35 , TDM-171 , TDM-170
TDM-92	X1 Design verification HW display interceptor		Apr 18, 2019	Alex Rudzki	DONE		TDM-998
TDM-81	Backend integration on controller		Oct 20, 2021	James Sneyd	DONE		TDM-1915

TDM-80	NTP packet forwarding for ethernet over CAN		Apr 26, 2019	James Sneyd	DONE		
TDM-72	Revisit AFE ovenisation control		Nov 08, 2019	Alex Rudzki	DONE		TDM-980
TDM-34	X1 Design verification HW controller		Apr 18, 2019	Alex Rudzki	DONE		TDM-998
TDM-33	X1 Design verification HW high current sensor		Apr 18, 2019	Alex Rudzki	DONE		TDM-998
TDM-32	Design voltage sense point enclosures		Nov 12, 2019	Bernie Walsh	DONE	Input_required	TDM-932
TDM-31	Design meter main enclosure		Mar 18, 2019	Bernie Walsh	DONE		TDM-489 , TDM-488 , TDM-487 , TDM-486 , TDM-29
TDM-30	Shunt resistor PCB mounting bracket		Apr 10, 2019	Bernie Walsh	DONE		TDM-29
TDM-27	Develop FPGA - CPU and Comms module		Nov 26, 2019	Jakub Tkaczyk	DONE		TDM-986 , TDM-353 , TDM-66 , TDM-2131 , TDM-59 , TDM-1350 , TDM-2133 , TDM-1344 , TDM-1355 , TDM-1345 , TDM-1346 , TDM-1347 , TDM-1348 , TDM-1349 , TDM-2109
TDM-26	Design FW for controller X1		Apr 10, 2019	Jeremy Kincaid	DONE		TDM-35
TDM-24	Design Controller		Apr 01, 2019	Jeremy Street-Thomas	DONE		TDM-315 , TDM-314 , TDM-303 , TDM-940 , TDM-311 , TDM-322 , TDM-310 , TDM-305
TDM-23	Design HW AFE		Mar 18, 2019	Alex Rudzki	DONE		TDM-50 , TDM-907 , TDM-44 , TDM-904 , TDM-49 , TDM-305
TDM-294	X1 Design HW FPGA		Mar 29, 2019	Vivien Wong	DONE		TDM-326 , TDM-348 , TDM-347 , TDM-985 , TDM-324 , TDM-323 , TDM-344 , TDM-342 , TDM-56 , TDM-284 , TDM-283 , TDM-272
TDM-272	Bring up current TRI105 HMI on Apalis module with FPGA overlay		Oct 20, 2021	Jakub Tkaczyk	DONE		TDM-2059 , TDM-286 , TDM-294
TDM-882	Working With Code Composer Studio		Mar 18, 2019	Andy Meldrum	DONE		TDM-459
TDM-862	Decimal Fixed Point Numbers		Jul 10, 2019	Andy Meldrum	DONE		TDM-1079 , TDM-1921 , TDM-35 , TDM-1085
TDM-854	Publish SW requirements		Mar 18, 2019	Jeremy Kincaid	DONE		TDM-560 , TDM-559 , TDM-459
TDM-607	Upload final BOM		Oct 20, 2021	Timo Bellgardt	DONE		TDM-562
TDM-606	Preliminary process capability study plan		Oct 20, 2021	Dion Schulz	DONE		TDM-562
TDM-605	Measurement system analysis plan		Oct 20, 2021	Dion Schulz	DONE		TDM-562
TDM-604	Process sheets		Oct 20, 2021	Timo Bellgardt	DONE		TDM-562
TDM-603	Pre-launch control plan		Oct 20, 2021	Dion Schulz	DONE		TDM-562
TDM-598	PFMEA		Oct 20, 2021	Timo Bellgardt	DONE		TDM-562
TDM-593	Characteristics matrix		Oct 20, 2021	Harry Watson	DONE		TDM-562

TDM-588	Floor plan layout		Oct 20, 2021	Timo Bellgardt	DONE		TDM-562
TDM-583	Process flow chart		Oct 20, 2021	Timo Bellgardt	DONE		TDM-562
TDM-578	Product / process qualiy system review		Oct 20, 2021	Dion Schulz	DONE		TDM-562
TDM-573	Packaging standards and specifications		Oct 20, 2021	Harry Watson	DONE		TDM-562
TDM-568	Develop DFM strategy		Oct 20, 2021	Timo Bellgardt	DONE		TDM-562
TDM-566	Develop manufacturing plan / design production line		Oct 20, 2021	Timo Bellgardt	DONE		TDM-567 , TDM-562
TDM-563	Develop Quality plan		Oct 20, 2021	Dion Schulz	DONE		TDM-565 , TDM-452
TDM-529	Design busbar changes		Mar 18, 2019	Bernie Walsh	DONE		TDM-535 , TDM-538 , TDM-537 , TDM-29 , TDM-536
TDM-528	Design harnesses		Oct 20, 2021	Cameron Swales	DONE		TDM-934
TDM-472	Publish FW / HW system specs draft		Nov 11, 2019	Jeremy Street-Thomas	DONE		TDM-476 , TDM-465 , TDM-475 , TDM-474 , TDM-473
TDM-450	Configurable language pack for setting and retrival of DC meter Configuration		Oct 20, 2021	James Sneyd	DONE	SDD UI adaptation	TDM-1901 , TDM-449
TDM-1322	Design process for handling of cryptographic keys		Oct 20, 2021	Jeremy Street-Thomas	DONE		TDM-562
TDM-1316	IONITY		Oct 20, 2021	Unassigned	DONE		TDM-1315 , TDM-1317
TDM-1052	[Cancelled] Review cooling options		Nov 29, 2019	Michael Walton	DONE		TDM-932
TDM-1049	Update specifications: need to monitor shunt temperature		Nov 12, 2019	Jeremy Street-Thomas	DONE		TDM-932
TDM-1048	Report ISD DC meter shunt cooling options to Bernie		Nov 12, 2019	Bernie Walsh	DONE		TDM-932
TDM-1035	Present cooling concept		Nov 12, 2019	Bernie Walsh	DONE		TDM-932
TDM-1031	Redesign meter enclosure		Nov 12, 2019	Bernie Walsh	DONE	Main_Focus	TDM-932
TDM-1024	Validate aliasing assumptions (for sigal sampling)		Oct 20, 2021	James Sneyd	DONE		TDM-1009
TDM-1023	Trial "Enterprise Architect" SW		Nov 08, 2019	Jeremy Kincaid	DONE		

TDM-1021	Do we need to be ready for dual cable charging pin voltage sensing?		May 07, 2019	Aidan Clarke	DONE	TDM-1020
TDM-1017	Component-level validation of TRI110-04v1 FPGA Interceptor PCB X1		Nov 20, 2019	Vivien Wong	DONE	TDM-1018 , TDM-998
TDM-1014	Fail to select appropriate RTOS		Nov 20, 2019	James Sneyd	DONE	TDM-1015 , TDM-1009
TDM-1011	FPIO ethernet comms is disturbed by electrical noise		Nov 14, 2019	Nicholas Keeling	DONE	TDM-1013 , TDM-1009
TDM-1010	CAN loading impact on meter / display / backend communcation		May 28, 2019	Dion Sumpton	DONE	TDM-1012 , TDM-1009
TDM-999	Component-level validation of TRI110-03v1 FPGA carrier PCB X1		Nov 20, 2019	Vivien Wong	DONE	TDM-1000 , TDM-998
TDM-989	Test Environment M1		Oct 20, 2021	James Sneyd	DONE	
TDM-988	SW Documentation M1		Nov 12, 2019	James Sneyd	DONE	TDM-931 , TDM-1467 , TDM-1468
TDM-981	Replace 5x gland between front and back by bulkhead connector		Oct 20, 2021	Cameron Swales	DONE	
TDM-953	Identify second source for shunt		Oct 20, 2021	Jeremy Street-Thomas	DONE	TDM-934
TDM-952	Redesign shunt mounting		Nov 12, 2019	Bernie Walsh	DONE	TDM-932
TDM-943	Design and Implement the ADC subsytem X1		Jul 01, 2019	Jeremy Kincaid	DONE	TDM-1920 , TDM-926 , TDM-35 , TDM-944 , TDM-965
TDM-942	Circular buffer mechanism M1		Oct 20, 2021	James Sneyd	DONE	
TDM-941	Design FW Controller M1		Oct 20, 2021	James Sneyd	DONE	
TDM-930	M1 Design HW FPGA		Nov 12, 2019	Vivien Wong	DONE	TDM-929
TDM-928	Design Remote Display		Nov 26, 2019	Vivien Wong	DONE	TDM-2059
TDM-920	Update documentation Meter		Mar 25, 2019	Alex Rudzki	DONE	TDM-452 , TDM-921

135 issues

List of key documents and references

Please use google to translate German material: https://translate.google.com/?client=firefox-b&um=1&ie=UTF-8&hl=en&client=tw-ob#view=home&op=translate&sl=de&tl=en

document / reference	comment	link
statement of work	written by Martin	Statement of Work - DC meter - 190219 - V1.2.docx
system requirements	written by Michael	TRI110.026v2 DC Meter system spec.docx
architecture specification	written by Deepak	
architecture specification	written by Jeremy	https://wiki.tritium.com.au/x/Owx9AQ
VDE draft standard	extracted from online tool and translated by google we have an original paper hardcopy in German	VDE AR E 2418 3 100 draft E.docx
WBS and requirements	in Jira	https://jira.tritium.com.au/projects/TDM?selectedItem=jp.ricksoft.plugins.wbsgantt-for-jira:wbsgantt-project
design folder	important subfolders: 001 engineering reference 003 block diagram 011 LCD screen integration 013 SAFE initative (transparency SW) 022 FPGA selection 024 Standards Compliance, esp. requirements matrix	Z:\Tritium\TRI110 - DC Meter\00 - Documents
description of "convenient solution"	This is an unofficially recommended solution (coming from PTB) Documents are in German and have been translated by google	E2E-Sicherheit early release rev 1 TRANSLATION.pdf.docx read me info4 rev 1 TRANSLATION.pdf.docx Z:\Tritium\TRI110 - DC Meter\00 - Documents\TRI110.025 Convenient Solution - info from Angelika and PTB 190123
Has2Be	Download beta version transparency SW (German) General info (German)	https://has-to-be.com/de/development-resources/transparenzsoftware/beta-download-transparenzsoftware/ https://has-to-be.com/de/development-resources/transparenzsoftware/
Testing labs	CSA (close to Munich) VDE (close to Frankfurt)	https://www.csagroup-bayern.org/ https://www.vde.com/tic-de
article about public key infrastructure	about Germany / in German	https://www.electrive.net/2019/01/12/bundesnetzagentur-erweitert-ladesaeulenkarte/
article about extension of deadline	about Germany / in German	https://www.electrive.net/2018/12/17/eichrecht-bei-ladeinfrastruktur-gemessener-gleichstrom/

M1 Prototype Hardware Issues

Design issues identified to be corrected:

#	Item	Issue	Remedy	Status	Completed
1	Controller PCBA	U9 (11345) has no opening on top paste.	Update footprint from library	Fixed in library. Part no longer on board	•
2	Controller PCBA	U9 (11345) is planned end of life	Replace with TC58CVG2S0HRAIG		•
3	Controller PCBA	Several single line items that don't seem critical	Reduce single line items where possible		•
4	Controller PCBA	Ringing on mosfet gates	Change DNF connecting mosfet driver ground from GND to GND_PWR (remove R82/R89 and fit R83/90)		•
5	Controller PCBA	Potential for mosfet gates to float	Add 10k pull down to mosfet gate		•
6	Controller PCBA	Fans/thermistor connector is SMD and won't be able to be picked up by ATE	Place test points on fans /thermistor lines		•
7	Controller PCBA	Fan feedback has minimum voltage of 0.3V. Inline resistor causes voltage divider.	Move pull up to connector side.		•
8	Controller PCBA	Fan feedback is being pulled up to above 3.3V when off. Fan power is trying to return through feedback.	Add optocoupler for speed using fan power		•
9	Controller PCBA	Requirement may have changed to need more memory	Add second flash part on bottom side (default DNF)		•
10	Controller PCBA	Will TPM be used?	Default is fitted.	Should be able to be removed later if not used.	•
11	Controller PCBA	External reference for ADC doesn't really provide value	Change to DNF, use 3.3V reference and change temp sensor bias resistors to 1%	Firmware must change to use 3.3V	•
12	Controller PCBA	Lower ovenisation power than expected. Track resistance is higher than expected and ovenisation voltage under load is lower.	Change transformer to 1:1.75		•
13	Controller PCBA	Cutouts for baffle have uncertain radius	Place known radius for cutout		•
14	Controller PCBA	New injected enclosure would interfere with corners of the board	Increase corner radius on board corners to avoid interference	Confirm to what size	•
15	Controller PCBA	Controller requires conformal coating for creepage	See if creepage (10mm) can be obtained to avoid this		•
16	AFE PCBA	Heat retention around AFE PCBA. Put foam inside the canister area, over the can, or over the whole board. Foam material? Silicon conformal coating?	Metal can by itself is sufficient for our needs	No action needed	•
17	AFE PCBA	Q1 (5528) footprint is quite wide	Change footprint		•
18	AFE PCBA	HV Detect circuitry can have voltage applied on other cable	Switch high voltage lines to allow two high accuracy voltage channels using ASSR-601JV-000E	Needs review	•

19	AFE PCBA	One signal across the barrier is currently unused	Now used for one wire interface		•
20	AFE PCBA	Could eliminate production needing to place formex over RTC cap	Increase cutout/copper-free distance around capacitor		•
21	AFE PCBA	Connection of temperature sensor to shunt needs to be reinforced isolation and well thermally coupled	Use one wire interface and place on AFE board.		•
22	AFE PCBA	Input power filters aren't used	Remove beads from board		•
23	Enclosure	Labelling panel isn't ideal for production solution	Design metal riveted plate on outside for required info	Confirm required information needed to be displayed	•
24	Enclosure	Inserts for board mounting need more material around them	Require more space around AFE mounting points	Need to support with minor AFE board change too.	•
25	Enclosure	Possibility for Fan wires to be pushed onto heatsink	Extend a cover over wires on the fan		•
26	FPGA Interceptor PCBA (v2)	Level shifter (TXB0106) should be replaced with an open-drain/push-pull level shifter (TXS0108) for TPM chip so that pull-up resistors can be used with it. The current level shifter (TXB0106) works for all signals except the open-drain pin PIRQ# on the TPM chip.	Unfit R63 and R65. Pin is driven by FPGA Fit R64 with 100ohm so that PIRQ# is held high. Possibly put in a buffer on PIRQ# between TPM and level shifter	Used remedy as of now.	•
27	FPGA Interceptor PCBA (v2)	EXT_CLK should be connected to "P-side" of clock capable diff pair input, (ref UG472 : page 49)	TPM_CLK_FPGA should be able to be moved to pin 'P4' as it probably doesnt need a dedicated clock pin	Presently implemented a workaround in FPGA fabric, but is sub-optimal and not recommended	•
28	FPGA Interceptor PCBA (v2)	FPGA part should be maintained as -2 speed grade (similar to v1)- CID 10867. On the V2, due to stock issue, -1 speed grade was used as a substitute CID 12658	Use CID 10867 -2 speed grade FPGA part for future builds.		•
29	FPGA Interceptor PCBA (v2)	Insufficient RAM for video buffer and microblaze CPU	Change to FPGA part with larger RAM with same package: XC7S50-2FTGB196I		•

M2 Prototype Hardware Issues

Design issues identified to be corrected:

#	Item	Issue	Remedy	Status	Completed
1	Controller PCBA	J3 has + and - swapped around from Tritium standard. -ve on top is standard	Should swap these pins around and also consider which location a 2 pin minifit would fit to allow harness re-use	Optional: Decided not needed change	•
2	Controller PCBA	Uncertainty if IR LED pulse energy is strong enough	Potentially decrease resistors (2 parallel x 390R used for testing works with less than ideal IR receiver)	R69 and R72 3164 on v3a	•
3	Controller PCBA	(Cosmetic) Note on UVLO has wrong comment on VREF setting	Change comment to VREF=0	To be done on next update	•
4	Controller PCBA	(Cosmetic) Note on fan speed low pass filter assumes only 1 pulse per revolution	Change comment to 10krpm = 333.33hz = 3ms (2 pulses per revolution)	To be done on next update	•
5	Controller PCBA	(Cosmetic) TP30 "TEMP_SNS" is missing overlay for TP name	Add name to overlay	To be done on next update	•
6	Controller PCBA	EMC Iso power noise	Fit Y caps; Change Y cap connection to GND_PWR and ISO_GND-prebeads (closer to transformer) Change Y caps to 2x CID:11147		•
7	Controller PCBA	EMC Iso power noise	Fit two 100pF caps on iso secondary near transformer (parallel C67/C68) (CID:143)		•
8	Controller PCBA	EMC Iso power noise	Change drive resistors to 15R (R71/R74) (CID: 2681)		•
9	Controller PCBA	EMC Iso power noise	Change snubber resistors to 4.7R (R73/R26) (CID: 101)		•
10	Controller PCBA	EMC noise	Fit 100nF caps to earth (C73/C75)		•
11	Controller PCBA	EMC ethernet noise	Fit 1nF to earth for ethernet termination (replace DNF R1) (CID: 7905)		•
12	Controller PCBA	EMC ethernet noise	Fit U4 ESD protection		•
13	Controller PCBA	EMC fixes may require addition space on board	Can remove the secondary DNF iso transformer circuitry		•
14	Controller PCBA	Compliance - CTI not specified	Specific that CTI >=175 in the manufacturing details		•
15	Controller PCBA	Compliance - creepage; for slot to extend creepage must be >1mm	Increase slot to 1.2mm		•
16	AFE PCBA	Q2-4 Fets have incorrect part number on schematic	Update comment	(documentation only, no physical change)	•
17	AFE PCBA	Both opto pairs turn on at once. Leakage through FET gate	Change R40: 100k pull down to 10Meg / Change to flip /flop;comparator?	R40 DNF (1259) on v3	•

18	AFE PCBA	Temperature sensor doesn't produce pulses.	Change pull up resistor to 10k and mosfet to npn transistor Possible future revision: change to comparator	R16 360 on v3 Q4 12540 on v3 This is reference design so no comparator needed.	•
19	AFE PCBA	Temperature sensor doesn't produce pulses.	Correct sensor footprint.	TODO on next board revision	•
20	AFE PCBA	Compliance - CTI not specified	Specific that CTI >=175 in the manufacturing details		•
21	DCM Assembly	EMC noise	Add earth cable to power harnesses		•
22	DCM Assembly	EMC noise	Change ethernet cable to shielded one - properly terminated		•
23	DCM Assembly	EMC noise	Add f19 (13184) to ethernet cable near DC meter end		•
24	DCM Assembly	EMC noise	Change RJ45 connector to one that connects the shield (17191)		•
25	DCM ISD HMI Assembly	EMC noise	Add f19 (13184) to ethernet cable near HMI end		•
26	DCM ISD HMI Assembly	EMC noise	Correctly terminate ethernet shield to connector		•
27	DCM ISD HMI Assembly	EMC noise	Spray paint HMI's plastic back cover internally with EMI shielding paint (CID: 5154)		•
28	DCM ISD HMI Assembly	EMC noise	Include f19 ferrites on 24V to HMI that already exist in total charger build	No design action needed, already exists	•
29	DCM ISD HMI Assembly	EMC immunity	Add f19 (17022) on 12V to DC meter		•
30	DCM ISD HMI Assembly	EMC immunity	Add f19 (17022) on 24V with a turn to HMI		•
31	DCM ISD HMI Assembly	EMC noise	Add earth braid from bottom right HMI bracket direct to ground point		•
32	DCM ISD HMI Assembly	EMC noise	Add jointing compound on HMI bracket feet		•
33	DCM PK HMI Assembly	EMC noise	Include 2xf19 ferrites on 24V rail that already exist in total charger build	No design action needed, already exists	•
34	DCM PK HMI Assembly	EMC noise	Place f19 ferrite (17022) on 24V and ethernet together close to HMI		•
35	DCM PK HMI Assembly	EMC noise	Connect braid directly to HMI earth stud		•
36	DCM PK HMI Assembly	EMC noise	Place f19 ferrite (17022) on ethernet from HMI with a turn		•

Mechanical placement requirements

The DC meter is comprised of two major mechanical components

1. The main metering module - This module is installed in the output power stage of the charger and measures the output current and voltage
2. Front display intercept module - This module is installed in the HMI to securely display information on the screen

The ambient temperature around the meter components must be within the range of **-40°C to +85°C**

Main metering module

The main metering module has a number of specific installation requirements to function correctly.

- All charger output current must flow through the Shunt (shunt resistor)
- The Shunt shall be installed in the negative output
- **The Shunt must not be connected close to any large heat source (Output fuse or diodes)**
    - Any thermal difference across the Shunt will cause a substantial error in measurement (This will likely be why the ISA meter had such long bus bars internally)
- **The Shunt will generate ~13W of heat when the charger output current is at 500A**
    - It is expected that the shunt will operate at a maximum temperature of 120°C
    - Adequate cooling of the shunt must be provided. The copper bus-bars will likely not be enough
    - Cooling solutions could be
        - Heat sinks to surrounding air, attached to the copper bus-bars
        - Water cooling block
    - Adequate air flow shall be provided around the DC meter
- At lower currents such as in the RT, the shunt will generate ~0.8W at 125A
- The module must measure charger output voltage. Only a positive measurement point is required
- The output voltage is measured by connecting a single wire to an external positive bolt terminal in the charger
- The positive voltage measurement sense point must be after any output diodes (Essentially needs to be as close to the cable connection point as possible)
- The module must have a tamper evident enclosure which encases all of its electronics (Not sure if the Shunt terminals need to be enclosed too)
- The Shunt bolt terminals and positive voltage measurement point will need to have some form of tamper evident seal

Refer to this datasheet for Shunt mechanical dimensions:

"Z:\Tritium\TRI110 - DC Meter\00 - Documents\Research\Technical\front end measurement\Shunt\Vishay\WSBS8536-40 29-May-2018.pdf"

Refer to this document for conceptual PCB layout. At present use the Shunt dimensions as a reference for size:

"Z:\Tritium\TRI110 - DC Meter\00 - Documents\TRI110.021 PK High Current Meter mechanical location\TRI110.021 DRAFT circuit board layout.pdf"

Refer to this document for the general thinking of the meter location in the PK back electronics enclosure:

"Z:\Tritium\TRI110 - DC Meter\00 - Documents\TRI110.021 PK High Current Meter mechanical location\Tritium DC meter location in PK.pdf"

Front display intercept module

- This must be installed between the HMI computer and LCD screen (For PK)
- The module is just a PCB. It will need some tamper evident seal

No solution present for RT. Possibly have a small LCD screen similar for ChargePoint HMI?

Refer to this PCB that is included in the Ionity HMI for mechanical dimensions:

"Z:\Tritium\TRI110 - DC Meter\05 - FPGA Carrier Dummy\Hardware v1\Project Outputs for TRI110-05-v01\TRI110-05-v01 - PCB MANUFACTURING DETAILS.PDF"

There is another PCB that will be on top of this one, it might add ~20mm of height.

Software Development Plan

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Scope

Identification

This document contains details about the software development process for the DC Meter project identified by the Tritium internal project code: TRI110. In the Tritium issue management system known as JIRA, this project has the code: TDM and can be found at https://jira.tritium.com.au/projects/TDM/summary. In Tritium's Wiki, documentation relating to this project can be found at https://wiki.tritium.com.au/display/RND/DC+Meter.

System Overview

The need for the DC Meter is a consequence of the German authorities making it mandatory for all Electric Vehicle (EV) Charging Stations to contain a certified DC Meter suitable for billing purposes as specified in the standard document EN50470-1/-3. The values measured/calculated by the DC Meter are used by government authorities and end users to verify billing information from the Charge Point Operators (CPO) managing the EV Charging Stations.

The DC Meter is analogous to an AC Electricity Smart Meter but for EV Charging stations and instead of measuring the total AC usage to the EV Charging station, it measures the DC usage for charging the vehicle and specifically excludes power to run the EV Charging station, power lost in cabling and other losses.

Functionality includes (but not limited to):

1. accurately measure the DC usage when charging a vehicle
2. calculate the energy delivered to the user
3. keep track of charge session duration and time started
4. report the measured and calculated values to
    a. a display to the user and
    b. an independent server
5. be cryptographically secure in storage and transmission of information.

There are existing Tritium EV Charging stations installed at various site in Germany that will need to be retrofitted with the DC Meter Solution. As a consequence there are internal Tritium requirements that impact the DC Meter with regards to physical size and shape as it needs to fit in existing systems of different models and new systems yet to be built.

German installation sites include:

- Brohtal Ost
- Brohtal West
- Himmelkron
- Gruibingen Süd
- Ringdalskogen

- Hohenwarsleben
- Wunnenstein Ost
- Salzbergen
- Kjerlingland

- Wunnenstein West
- Mahlberg West
- Eichenzell
- Hohenlohe Süd

Different models of EV Charging Stations have different physical sizes and different sized Display Panels. The DC Meter also needs to fit inside these EV models and use their screens (as the DC Metering information is superimposed onto the existing Display Panel instead of having it's own screen).

System block diagram

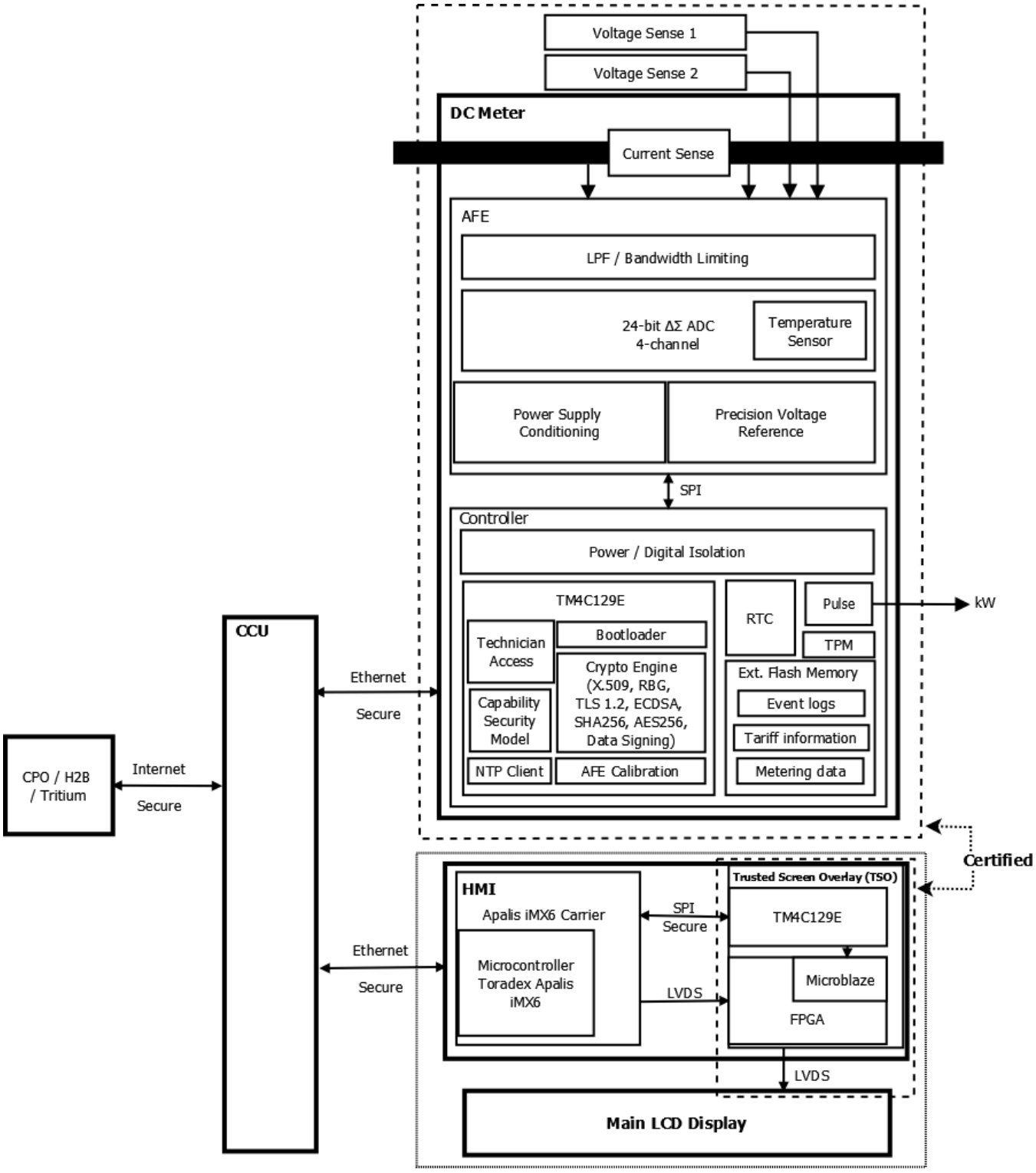


Figure – Block diagram of the DC Meter

Document Overview

As part of the certification requirements of the DC Meter, we need to show we followed a development plan, which outlines processes, tools, and deliverables as part of the development process. This document is commercially sensitive and shall be considered "Commercially in Confidence".

Relationship to Other Plans

Table - Relationship to Other Plans

Document	Subordinate/Equal/Superior
Project Management Plan	Subordinate
Test Plan	Superior
Certification Plan	Equal
Installation/Support Plan	Equal
DC Meter System Specification	Subordinate
DC Meter Firmware Design	Superior

Referenced Documents

Table - Referenced Documents

ID	Reference
R1	**VDE AR E 2418 3 100 draft E** Latest draft (as of 16 Oct 2018) of the system requirements from the German authorities (PTB) used for requirements on the DC Meter.
R2	**WELMEC Guide 7.2** Technical guidance for the application of the Measuring Instruments Directive (MID) for software-equipped measuring instruments.
R3	**WELMEC Guide 7.1** Development of Software Requirements
R4	**POL.003 Tritium C Coding Style & Guidelines** Document containing layout of C code, guides to naming convention

Overview of Required Work

Requirements as specified in the VDE[R1] are to be reviewed and used to create a software work breakdown structure. References to other documents will be reviewed for secondary requirements imposed on the system.

Documentation that needs to be submitted as part of the certification process needs to be identified and listed. Tritium have templates that will be used when providing these documents externally. All documents will need to be reviewed and approved before being released. All documents will initially be created in Confluence and later exported using Tritium Word Templates. Any released document will also be saved in Confluence with the project and optionally stored by Tritium's internal Quality and CI department.

Tests need to be performed on the software to verify the quality and reports are used as part of the deliverables for certification.

The DC Meter project is currently fully funded by Tritium and worked on by Tritium employees. The table below lists the staff working on it:

Table - DC Meter Team

Name	Position	Responsibilities

Aidan Clarke	Product Manager	
Nicholas Keeling	Power Electronics Module Lead	
Alex Rudzki	Hardware/Firmware Lead	AFE sensor board and TSO hardware
Jeremy Kincaid	Software Engineer	All firmware/software for the Controller board and remote LCD
Vivien Wong	Hardware Engineer	FPGA design
Andy Meldrum	Contract Software Developer	All firmware/software for the Controller board and remote LCD
James Sneyd	Firmware Engineer	All firmware for the Controller board and remote LCD
Jeremy Street-Thomas	Electronics Engineer	Hardware for the TM4C Controller board
John Huynh	Software Developer	Software for Trusted Technician Interface and Automated Testing
Hai Nuygen	Software/Electronics Developer	Automated Testing
Jakub Tkaczyk	Module Lead HMI	Trusted Screen Overlay TM4C, Microblaze, communications to DC Meter

Plans for Performing General Software Development Activities

Software Development Process

The software development process will follow the repetitive/iterative cycle as listed below:

1. Requirements Analysis
   Although this step is seen as the initial step, requirements are re-evaluated at regular intervals to catch anything that may have been missed the first time, or if the initial response to the requirement(s) were the best option, or if new information has been learned that leads to a different approach to meeting the requirement. This step should lead to a Work Breakdown Structure (WBS) identifying sub-tasks and level of effort with mapping to original requirements.
2. De-Risking Activities
   If there are requirements that are difficult to estimate the level of effort, or are considered high risk to the success of the project, then a dedicated activity is create to reduce the risk. This activity may be a proof of concept, or an investigation in to different technologies. The end result may be more requirements or a more solid design with estimations. These activities are optional.
3. Development Environment Setup
   Knowing what tools to use, and integrating them together to streamline the process. There may be an initial setup but as the development progresses new tools may be added and old tools replaced or tweaked.
4. Documentation
   The various documents relating to the project are not finalised until the project completes. There may be various versions released throughout but all documents should be considered "living" documents that change as new information comes to light. Documentation may be formal and released to 3rd parties, or may be informal such as JIRA tickets or Confluence/Wiki pages, or even loose documents in the project folder on the server. Formal documents are listed in later in this document, however it is up to the project members to add any other documents they see fit, especially if it help understand the system or explains the reasoning for direction of any part of the solution.
5. Implementation
   This is a combination of software implementation, testing, version control. Items are addressed based on the WBS.
6. Formal Testing
   Official results used in the certification process, peer reviews, also considered internal version of acceptance/functional testing. This step may not occur as often as the other steps as this requires more of the product to be completed before these are practicable.
7. Managing merging to the release branch
   The branch 'master' is the release branch. It should always reflect a working copy of the solution and updates to this branch should be carefully managed to ensure only wanted functionality is added, and the code is bug free.

Build/release cycles will be based on the Kanban methodology. The following releases are envisioned:

1. Prototype 1 - May 2019, using actual hardware but software not yet ready for certification although it should have near full functionality
2. Certification Candidate - Sept 2019, in testing to prove software is ready for certification.
3. Certification Release - Dec 2019, released to the PTB for certification

General Plans for Software Development

Software Development Tool-chain

Software/Firmware developers have the option of developing either in Windows or on a Linux VM running Ubuntu.

Table - Software Tools

Product	Version	Supplier	Notes

TI Code Composer Studio	Windows: 8.2.0.00007 Linux: 8.3.1.00004	http://processors.wiki.ti.com/index.php/Download_CCS#Code_Composer_Studio_Version_8_Downloads	IDE for authoring software/firmware, graphical debugging. This is the TI supported IDE.
TI ARM CGT	18.12.2.LTS	http://www.ti.com/tool/arm-cgt	The compiler created by TI to support the processor. The IDE uses this compiler in its configuration.
TM4C129E Development Board	A	http://www.ti.com/tool/EK-TM4C129EXL	The development kit used to test code on target until the DC Meter hardware is available.
TM4C129ENCPDT	Die Revision A2 Part Revision 3	http://www.ti.com/product/TM4C129ENCPDT?keyMatch=TM4C129ENCPDT&tisearch=Search-EN-everything&usecase=part-number	The core microprocessor on the DC Meter
Git	2.11.1 (Windows) 2.17.1 (Linux)	S:\GIT\Git-2.11.1-64-bit.exe https://git-scm.com/download/win - For Windows for latest built into Ubuntu (sudo apt-get update git)	Code versioning management tool.
TortoiseGit	2.4	S:\GIT\TortoiseGit-2.4.0.0-64bit.msi	The GUI frontend for Git to be used on Windows
GitKraken	6.0.1	http://gitkraken.com	The GUI frontend for Git to be used on Linux
Doxygen	1.8.13 (Linux)	http://www.doxygen.nl/download.html	Used for extracting source comments from code.
Ceedling	Ceedling:: 0.28.3 CMock:: 2.4.5 Unity:: 2.4.2 CException:: 1.3.1	http://www.throwtheswitch.org/ceedling	Used as the test harness. This is compiled into the project (as opposed to being a separate executable) so it is also listed below under reusable libraries. However, to run Ceedling you must install Ruby. Follow the instructions at the link provided. Ceedling includes Unity, CMock, and CException.
Putty		Tritium software drive	Used as the serial terminal to the development kit
PuttyGen		Tritium software drive	to create the key to access Git.
TRI70/00 - Git Setup	ssh://git@git.tritium.com.au/TRI70	Git respository setup scripts for Windows and Linux	Contains some setup scripts to more easily use Git and configures TortoiseGit

Using Code Composer Studio

Useful references: https://software-dl.ti.com/ccs/esd/documents/ccs_portable-projects.html

Code Composer Studio (CCS) is an IDE provided by TI for development of applications based on TI processors.

We are using a TM4C129ENCPDT device

CCS is provided in 32 and 64 bit versions. The 32 bit version is being deprecated. We are using the 64 bit version.

It provides several tool chains, a proprietary toolchain sourced from ARM and a GCC tool chain also sourced from ARM.

CCS is based on the Eclipse Project with TI customisation adding extra functionality for developing for TI microprocessors.

Java

CCS (and eclipse) are Java based, and require a Java Runtime Environment (JRE).CCS 64 bit requires the 64 bit version of Java.

Experience suggests that you are better off using a current version of Java 8. It seems to be Java 9 seems slower.

Project Integration

CCS supports a couple of flavor of projects

CCS Projects - using TI tool chain .. tightly integrated into the IDE .. Build, Debug all work as expected.

C/C++ Projects .. typically using gcc toolchain .. not to tightly integrated into the IDE .. in particular the debug process is not well integrated.

To get the most out of the IDE prefer CCS projects.

Workspace Configuration

Summary

Preferences\Show Advanced Settings
Preferences\ General\Always Run in background - uncheck
Preferences\General\Startup And Shutdown\RSE UI - uncheck
Preferences\General\Show Heap - check.. shown at bottom of screen
Preferences\ C/C++\Build\Console Uncheck Wrap Lines on the Console, Limit Console Output 20000, Delay Console Updates 100
Preferences\General\Workspace\Save Automatically before build
Preferences\General\Editors\Text Editors\Show Line Numbers
Preferences\C/C++/Code Style /Formatter - Import Tritium C Code Template Tritium_C.xml found in Repo/code_composer/templates - Set Tritium C as active profile
Preferences\C/C++/Code Style/Code Templates/Import Tritium Code Templates TritiumCodeTemplates-DoxygenComments.xml
Preferences\C/C++/Editor/Documentation Tool Comments - Doxygen
Preferences\General\Editors\Text Editors\Spelling\ Platform Dictionary - English (UK), User Defined Dictionary to your choosing

CCS (just like Eclipse) encourages the use of multiple work spaces. Work spaces are your friend. The less projects you have in a workspace the easier it will feel to navigate them.. and the less files CCS have to index.

Typically work spaces are not stored as part of the project repo .. they are maintained locally.

Treat work spaces as disposable items .. don't get attached to them. Most of the time this works out fine .. apart from debug configs..

Debug Configs are workspace specific .. but it is possible to import and export them. The real pain is that there a lot of settings in debug configs that have to be just right .. so the trick is to get one working.. then save the debug config somewhere in the project repo.

Remote System Explorer (RSE) - Disable It once per new workspace

Remote System Explorer is built into CCS. The idea is it lets you browse remote systems. The problem is that it tries to do this while you are compiling. It will hang the compile. There are better ways to remotely view files ..

Disable it by unchecking the Windows\Preferences\General\Startup And Shutdown\RSE UI checkbox.

This needs to be done once per Workspace. As far as I know there is no template to make this easier.

Upgrading the USB200 JTAG EMULATOR probe

For instructions refer to the site: http://dev.ti.com/tirex/explore/node?node=AADzJ8Y-La4f7Bi5Ga0TcA__FUz-xrs__LATEST

Compilation Issues

CCS seems to have a myriad of ways of having a compilation fail. Below are some tips to prevent it but another good resource is: http://software-dl.ti.com/ccs/esd/documents/sdto_ccs_build-errors.html#ccs-build-console

Console - uncheck wrap lines

This maybe a root cause of a lot of the compilation issues
A symptom of these issues is that the CCS UI becomes unresponsive. Investigation revealed that the compile/link cycle completed successfully .. but the the CCS UI was hung.
Further investigation revealed that Eclipse has long standing issues with the console being sluggish.. and it blocking when wrapping lines...

https://bugs.eclipse.org/bugs/show_bug.cgi?id=11601
https://bugs.eclipse.org/bugs/show_bug.cgi?id=519125

These issues are minimized by turning off console line wrap.

The settings for console line wrap are not shown by default .. In the preferences dialog the Show Advanced Settings Button must be pressed (The button then reads Hide Advanced Settings)

debug directory - Delete it if you have slow/hung compiles

For reasons unknown CCS will get to a point where a previously working build will either no longer work, or will compile really slowly. These issues can be often resolved by deleting the debug directory with the project folder.

This will typically happen for no obvious reason. Factors that seem to encourage this are things like shutting down CCS's host environment (in our case linux). Changing the source code underfoot also seems to encourage this. (i.e using git to change code versions.

A related issue is that if files are moved in the file system CCS will not delete compiled files in the build directory. Deleting the directory resolves this issue ..

Missing Paths - Check them if you have hung compiles

Compiles will hang if include paths are not found.. this seems to be associated with a warning that a path is not found.
Sometimes these warnings are false alarms .. sometimes not..

RSE - make sure it is disabled

As previously mentioned sometimes a build will stop when RSE tries to sync with a remote file system. Ensure it is disabled in the workspace

Troubleshooting CCS Issues

CCS Logs

CCS logs are your friends. Using a tail like tool is recommended .. one with a GUI is even better. glogg works well on Linux.
The logs are stored per CCS Workspace in the directory that CCS used to load the workspace

CCS Log
WorkspaceName/.metadata..log

Build Logs
WorkspaceName/.metadata/.plugins/org.eclipse.cdt.ui/xxx.log
There is a global build log and build logs for individual projects

CCS Progress Window

The CCS Progress Window shows the running tasks, and their progress. It will show if a task is blocking.
To show it Double Click at the bottom right hand corner of the CCS UI, to the left of the two vertical lines..



CCS Speed Optimizations

Parallel builds

All else being equal and working as expected this will have the biggest impact on build times.

In CCSProject\Properties\Build\Build Settings check enable parallel builds and choose use optimal jobs

(which should be the same as the number of cpus or virtual cpus in your PC most of the time unless you're running a VM then use:
vm_configured_thread_count - 2)

Don't bother setting it to a bigger number .. it has the effect of slowing the build down .. YMMV

Memory - you can never have enough

Out of the box CCS performs garbage collection during the build process ..

Increase Initial Heap Size for JVM Xms2048 (from 512)
Increase Max Heap Size -Xmx8192m (Modulo 1024) unless you only have 8192 system memory. -Xmx6144 is also good.

Doing this seems to make CCS much more stable

edit ccstudio.ini.. path ti/ccsv8/eclipse once per CCS

install-Xms2048m
-Xmx8192m

Software Development Methods

Writing Software

Code Composer Studio is the IDE for development as it caters to the Texas Instruments platform allowing easy debugging. It will also perform the code formatting required, and report the compiler's MISRA checking of code style, and it has it's own static code analysis reporting. All code will be generated manually with the exception of some testing framework which is able to be auto-generated in the beginning and later updated manually.

The DC Meter software folder structure will be as depicted in the figure below. Sub-modules will have their own src and test folders. This allows the modules to easily be extract and re-used in other projects keeping all related code and tests with it:

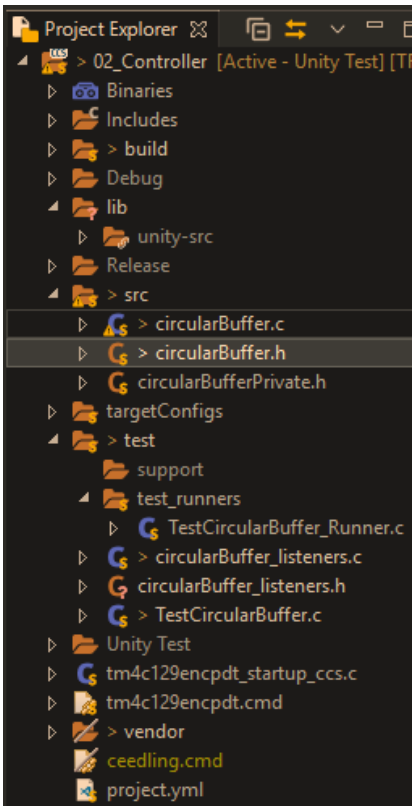


Figure - Project Folder Structure

Testing Software

Unit testing may be performed manually or automatically by the framework depending on the developer's preferences at the time. There will be a continuous integration server, which will perform a combination of nightly on target testing and off target testing.

Automated Testing

On Target Testing Vs PC/Server

Testing Metrics to Report

Code Version System

instructions for setting up Git can be found here: Git for Tritium Engineering Use. If you're using Linux then further instructions can be found in the "TRI70 /00 - Git Setup/install-prepare-commit-msg.ubuntu.sh" script.

Git will have some hooks added to allow auto testing of comments and code before permitting a commit, which is done as part of the Git preparation.

The Git repository for containing the actual DC Meter software/firmware is open to store anything. One folder will be the actual DC Meter software and everything else can be anything the project needs such as example projects, de-risking activities, documentation, etc.

Git has the command 'rebase' as a mechanism to pull from where you branched from and make all your changes happen after that rebase. For the DC Meter project, 'rebasing' is **NOT** encouraged. Git branch trees are good at showing when a change was started, how long it took, and how a merge may impact prior merges/functionality. There is also the potential for other problems when 'rebasing' especially around previous pushed commits and merging.

Prior to merging updated code back to the mainstream, a 'pull request' will be performed so a review of the update can be performed. The results of the pull request in git shall be emailed to the person responsible for performing the review and archived.

Frequently Used Git Commands

how-to: example for comparing a specific local file with the origin version of the same file. This is useful if you wanted to perform a diff prior to performing a pull. Note, difftool is used instead of diff to take advantage of 3rd party diff GUI's:

```
git difftool origin/master -- source/tri110-dcmeter-freertos-lwip2/src/cloud_task.h
```

how-to: example for comparing a specific local file of a previous version /revision with the origin, where the revision is represented as the short hash of the commit:

```
git difftool origin/master -- source/tri110-dcmeter-freertos-lwip2/src/cloud_task.h 2a03dd6
```

how-to: example list only the files that have changed for in the comparison:

```
git difftool --name-only origin/DCM-Demo -- source/tri110-dcmeter-freertos-lwip2/src/cloud_task.h 2a03dd6
```

how-to: example compare a file on one branch with the same file on another branch:

```
git difftool origin/DCM-Demo origin/TDM-366 -- source/tri110-dcmeter-freertos-lwip2/src/cloud_task.h
```

how-to: check which branches have been merged into master (the '-a' includes remote branches):

```
git branch -a --merged
```

how-to: remove a file from git that has wrongly commited and is now part of Git tracking:

```
git rm --cached <file_name>
```

how-to: merge a branch into your branch will full control over how the merge is performed:

```
git merge --no-commit --no-ff -s recursive -X patience 03728451c7099035c06c65f669b1af1656cf24e2
```

how-to: create a pull request via command line. This example requires you to have pushed your changes back to the server and the goal is to merge TDM-2104 to master.

```
git request-pull remotes/origin/master ssh://git@git.tritium.com.au/TRI110 remotes/origin/TDM-2104 >
PullRequest_TDM-2104.txt
Email the file PullRequest_TDM-2104.txt with the subject "[Pull Request] TDM-2104" to the person responsibe for
approving the pull request
```

how-to: create a pull request via TortoiseGit:

TortoiseGit has a button after the 'push' dialog, which creates a pull request.

Branching strategy

For the DC Meter project, the branching strategy used will be the feature branching strategy. In a nutshell, create a branch for the JIRA issue/task/feature you're going to work on. Do your changes. Merge back into master. For further information see: https://www.atlassian.com/git/tutorials/comparing-workflows/feature-branch-workflow

Guide to Commit Comments

The very first entry in a Git comment is the name of the branch you're working on. If you have setup Git properly, then the pre-commit hooks will do this for you. Some Git GUI applications will show you this value before you add your commit message, others will insert it after you've completed your message and you won't see it until the commit is done. Trust your GUI.

Documenting Code

Documenting code with Doxygen notation will allow auto-generation of code-documentation.

It will be up to the developer to maintain software design diagrams/documentation to keep in line with the implementation and vise versa.

Managing the release branch

Pre-requisites:

1. A JIRA ticket exists for the work
2. The JIRA ticket is assigned to a release

Merge Process (This process becomes more automated when you move to Bitbucket with its pipelines):

1. read pull request to verify
    a. only expected files were modified
    b. diff files to check modification
2. check some tests exist for the new code
3. merge code with the following command: git merge --no-commit --no-ff -s recursive -X patience <git hash of the branch to merge>
4. check differences between current and master prior to merge (maybe limit to specific files)
5. check differences between current and branch merged (maybe limit to specific files)
6. make sure it compiles in debug, release, UNITY_TEST (re-index, rebuild)
7. run in debug (no bootloader)
8. run in release with a bootloader
9. run unit tests after building for UNITY_TEST
10. checkout under Linux and build for CATCH2 and run tests (unless you get CATCH2 to run on target)
11. commit and push merge
12. ensure the related JIRA ticket is closed (consider if some comments need to be added for historical reasons)
13. update customer facing release notes (internal release notes are generated automatically)

Standards for Software Products

The VDE[R1] refers to WELMEC 7.2[R2] as a guide for development although not making it a requirement. Where possible, the software development process will comply to WELMEC 7.2[R2] as this also means we comply with MID requirements and should make certification easier.

The original requirements will not be captured in any other document than the original. Derived requirements will be captured in JIRA as tasks with a comment that links them to the parent requirement in the originating document. No formal standard is being followed for capturing requirements. In the absence of any formal guide, refer to https://confluence.atlassian.com/jirakb/using-jira-for-requirements-management-193300521.html combining JIRA and Confluence to manage requirements.

Coding style

Tritium have a formal document for this titled POL.003 -- Tritium C Coding Styles & Guidelines[R4]. Missing elements from Tritum's coding standard can be covered by:

1. https://wiki.sei.cmu.edu/confluence/display/c/SEI+CERT+C+Coding+Standard
2. https://wiki.sei.cmu.edu/confluence/pages/viewpage.action?pageId=88046682
3. https://isocpp.org/wiki/faq/coding-standards

The following deviations from POL.003 will be made:

Table - Tritium Coding Standard Deviations

POL.003 Section	Deviation	Rationale
2.2 - File description comment block	Style shall be changed to match Doxygen (http://www.doxygen.nl/manual/index.html) preferring to use the Qt style notation	Allow automatic extraction.
2.3 - Function description comment	Style shall be changed to match Doxygen (http://www.doxygen.nl/manual/index.html) preferring to use the Qt style notation	Allow automatic extraction.
2.5 – Code tags	Style shall be changed to match Doxygen Special Commands. Optionally use Markdown support to highlight existing tags	Supports automatic extraction.

Templates for Code Composer Studio can be found at `S:\Texas Instruments\CodeComposer Studio - Offline installer\Templates`.

Import existing Tritium Coding style by:

1. 'right-click' the project and select 'properties'
2. expand to: C/C++ General Formatter
3. click on "Configure Workplace Settings"
4. Click on "Import..."
5. Browse to the saved file "S:\Texas Instruments\CodeComposer Studio - Offline installer\Templates\Tritium_C.xml"
6. Click 'Apply'

To enable Doxygen support in CCS:

1. 'Right-Click' the project and select 'properties'
2. expand to: C/C++ General
3. 'Check' "Enable project specific settings"
4. Choose Documentation tool "Doxygen"
5. Click on 'OK'

To import code templates:

1. Open WindowPreferences
2. Expand to: C/C++ Code Style Code Templates
3. Click on "Import..."
4. Browse to: "S:\Texas Instruments\CodeComposer Studio - Offline installer\Templates\TritiumCodeTemplates.xml"
5. Click 'Open'
6. Click 'Apply'

The C and C++ language dialect used in writing software for the DC Meter are 'C11' and 'C++11'.

Incorporated Reusable Software Products

To reduce development time and help improve quality, some already existing and tested libraries will be used:

Table - Third Party Libraries Used in the DC Meter

Library Name	Version	License	Supplier	Description
FreeRTOS	8.2.3	MIT	https://www.freertos.org/	Real-time operating system kernel.
mbedTLS	2.13.0	Apache 2.0	https://tls.mbed.org	Elliptic curve implementations, random bit entropy generator, TLS support, signing /verification support.

lwIP	2.1.0	BSD	https://savannah.nongnu.org/projects/lwip/	Lightweight TCP/IP stack.
decNumber	0.1.1	Apache 2.0	https://github.com/amzn/ion-c	Decimal32 (base 10) arithmetic library to provide more accurate than float32/64 math operations.
Tivaware	2.1.4.178	Open	http://www.ti.com/tool/SW-TM4C	TivaWare library supporting the chosen processor. These provide access to the processor peripherals.
sts-2.1.2			https://csrc.nist.gov	Library to evaluate/test random number generation. This is used primarily as part of testing during development although some code may be used at run-time.
json-develop	3.7.0	MIT	https://github.com/nlohmann/json	Library to implement JSON, binary JSON, CBOR for configuration and messages between DC Meter and TTI
Unity/CMock/Ceedling		MIT	http://www.throwtheswitch.org	Testing framework for C
Catch2	2.6.1	Boost v1	https://github.com/catchorg/Catch2	Testing framework for C++
mqtt-c	3.1.1	MIT	http://docs.oasis-open.org/mqtt/mqtt/v3.1.1/os/mqtt-v3.1.1-os.html	Library for performing MQTT communications

Developed Reusable Software Products

Table - List of easily reusable modules for other products/projects

Software Module Name	Description
Bootloader	
crypto interface (SHA256, AES128/256, RBG, TLS 1.2, X.509, ECDSA)	
Ethernet over CAN	
BlockDevice, BlockFile, TritiumFiles	
Circular Buffer	
Software Downloader (aka updater)	
Remote Debug Printing	
Memory Allocators for STL and std::string	

Handling of Critical Requirements

The VDE[R1] addresses the requirement categories below either directly or by referring to other documents.

Safety Assurance

Safety is addressed through hardware requirements. The DC Meter is connected to high voltage, high current sources, but then fully enclosed to prevent physical access. The DC Meter software doesn't control anything safety related other than prevent a charge session starting if the log is full. There aren't any software requirements regarding safety assurance.

Security Assurance

Security assurance is addressed through either secure communications or physical restriction through tamper evident means.

Software Quality Assurance

The DC Meter software needs to be certified by the external body PTB. The following documents were used to determine what the certification requirements are.

Table - List of sources for the Product Requirements

Source	Actions	Enforceability (Guide/Strict)	Notes
"Z:\Tritium\TRI110 - DC Meter\00 - Documents\TRI110.005 PTB APPLICATION\QA Module D application and Research\PreliminaryList.pdf" – Preparation for a procedure to approve a quality system	ISO 9001:2000	Strict	For Harmonised Standards the example of OIML R49 is provided. However the VDE is to be our standard.

	Harmonised Standards, Normative Documents	Strict	Similarly for Directive 2004/22/EG (MID) the VDE is to be our standard as per the statement in: VDE, Appendix B.2 General: *"(mutatis mutandis, as no DC meters are covered)"*.
"Z:\Tritium\TRI110 - DC Meter\00 - Documents\TRI110.005 PTB APPLICATION\QA Module D application and Research\Criteria_adn_procedures_QS-Approval_March2018.pdf"	Self Conformity Assessment Document	Strict	This is the manufacturer's conformity assessment. Each requirement needs to be addressed with a description of how we conform to meet the requirement. This is partnered up with a Conformity Declaration.
	Conformity Declaration	Strict	
REA 6A	Product Requirements	Strict	
VDE refers to WELMAC 7.2 "Z:\Tritium\TRI110 - DC Meter\00 - Documents\Research\WELMEC GUIDES\SOFTWARE GUIDES\Guide_7.2_2015__Software.pdf"	Software Design Document	Strict	Must meet WELMEC 7.2 requirements: P1, P3, P4, P5, P7, L1, L2, L4 • notes on accuracy of measuring algorithms • description of the user interface • software identifiers and how they're constructed and different per version • system hardware overview • Data flow diagrams (comms and ui) • Protection against unintentional changes • Configuration parameters (set and secured) • Modes/states (relating to configuration changes) • Description of the stored data (usage, events, etc)
	Operating Manual	Strict	
	Source Code	Strict	

Addressing WELMEC 7.2

To assist in understanding what requirements apply to the DC Meter in WELMEC 7.2[R2], the DC Meter needs to be assigned a class. As the DC Meter is *" a built-for-purpose instrument with embedded software"* the WELMEC 7.2[R2] requirement type 'P' is applicable. The DC Meter also has functionality and hardware to support *"long term storage of measurement data"*, and *"transmission of measurement data"* and *"download of legally relevant software"* so the IT configuration (modular extensions) applicable include 'L', 'T', 'S', and 'D'. As this is an electricity meter the instrument specific extension of 'I.3' is applicable as described in Chapter 10.3 of WELMEC 7.2.

In accordance with P2/U2 we address detection of specific errors using recommendations from *WELMEC, Chapter 10 Extension I: Instrument Specific Software Requirements* as detailed below.

Important notes to consider when addressing WELMEC 7.2 concerns:

1. Some fault detections are possibly in hardware (e.g. MCU FaultISR, external flash Error Correction Codes)
2. Faults not detected by hardware are detected by software using either CRC or SHA256 hash calculations, or cryptographic signing.
3. Software detection is done regularly
4. The DC Meter is a dedicated instrument with an embedded system subject to legal control
5. The DC Meter depends on a network connection to the internet via a network switch providing firewall protection
6. The sensor is integerated
7. The user interface is always under legal control
8. There is long term, fixed (non-removable), local storage for audit logs
9. Uses the FreeRTOS operating system
10. The only other application residing on the system is the bootloader, which itself is part of the legally relevant software but separate to the metering software. Only one can run at a time, the only way to start the bootloader is to reboot, then it starts the metering application.
11. All software on the DC Meter is subject to legal control.

Using the risk table specified in the WELMEC 7.2[R2] guide in section I.3.6 the DC Meter has been identified in the risk class of **TBD**.

Table - WELMEC 7.2 Risk Class Definition

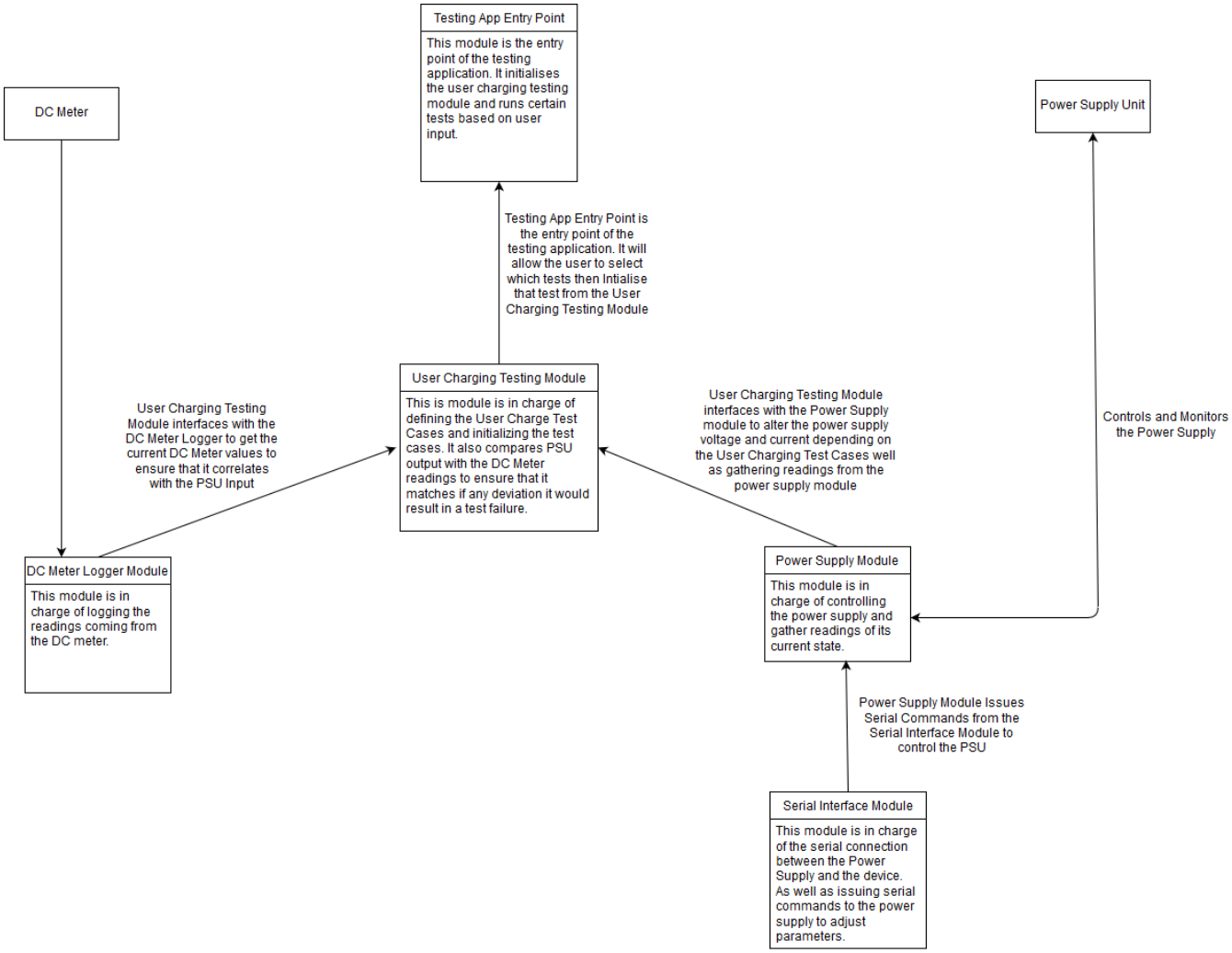
Risk Class	Software Protection	Software Examination	Software Conformity
A	Low	Low	Low
B	Middle	Middle	Low
C	Middle	Middle	Middle
D	High	Middle	Middle
E	High	High	Middle
F	High	High	High

Table - WELMEC 7.2 Risk Assessment

WELMEC 7.2 Requirement	Risk Assessment				Notes
	Software Protection	Software Examination	Software Conformity	Risk Class	
P1: Documentation					
P2: Software Identification					
P3: Influence via user interface					
P4: Influence via communication interface					
P5: Protection against accidental or unintentional changes					
P6: Protection against intentional changes					
P7: Parameter protection					
L1: Completeness of measurement data stored					
L2: Protection against accidental or unintentional changes					
L3: Integrity of data					
L4: Authenticity of measurement data stored					
L5: Confidentiality of keys					
L6: Retrieval, verification, and indication of stored data					
L7: Automatic storing					
L8: Storage capacity and continuity					
T1: Completeness of transmitted data					
T2: Protection against accidental or unintentional changes					
T3: Integrity of data					
T4: Authenticity of transmitted data					
T5: Confidentiality of keys					
T6: Handling of corrupted data					
T7: Transmission Delay					
T8: Availability of transmission services					
S1: Realisation of software separation					
S2: Mixed indication					
S3: Protective software interface					
D1: Download mechanism					
D2: Authentication of transmitted software					
D3: Integrity of downloaded software					
D4: Traceability of legally relevant software download					

Automated Test Server

Testing Application System Architecture Diagram



Controlling Diagnostic Messages

Overview

The C/C++ compiler provides diagnostic options to control compiler- and linker-generated diagnostic messages. The diagnostic options must be specified before the --run_linker option.

--diag_error=num	Categorises the diagnostic identified by num as an error. To determine the numeric identifier of a diagnostic message, use the --display_error_number option first in a separate compile. Then use --diag_error=num to recategorize the diagnostic as an error. You can only alter the severity of discretionary diagnostic messages.
--diag_remark=num	Categorises the diagnostic identified by num as a remark. To determine the numeric identifier of a diagnostic message, use the --display_error_number option first in a separate compile. Then use --diag_remark=num to recategorize the diagnostic as a remark. You can only alter the severity of discretionary diagnostic messages.
--diag_suppress=num	Suppresses the diagnostic identified by num. To determine the numeric identifier of a diagnostic message, use the --display_error_number option first in a separate compile. Then use --diag_suppress=num to suppress the diagnostic. You can only suppress discretionary diagnostic messages.
--diag_warning=num	Categorises the diagnostic identified by num as a warning. To determine the numeric identifier of a diagnostic message, use the --display_error_number option first in a separate compile. Then use --diag_warning=num to recategorize the diagnostic as a warning. You can only alter the severity of discretionary diagnostic messages.

Suppressed Warnings (3rd Party Libraries)

The DC Meter uses 3rd party libraries to deliver certain functionality. 3rd party libraries are evaluated to be fit-for-purpose and are not maintained by Tritium. Warnings generated from these 3rd party libraries are seen as beign and are suppressed.

MQTT Library

#112-D Statement is unreachable.

#190-D Enumerated type mixed with another type.

Doxygen

Introduction

Doxygen is being used as the tool to generate documentation for the Source Code for the DC Meter. Doxygen allows C & C++ code (among others) to be parsed and to generate

HTML (among others) files that document the source code.

Source Code can be annotated with Doxygen Commands to help in this process. See http://www.doxygen.nl/manual/commands.html for details.

Doxygen distribution

Doxygen is distributed as part of Ubuntu. The distributed version atm is 1.8.13

Doxygen may also distributed as part of tool chains that use it. This saves having to integrate it into the host OS.

Doxygen may be downloaded, compiled and installed to use the current version.

Building, Installing Doxygen on Ubuntu

The current version of Doxygen is 1.8.18

See http://www.doxygen.nl/download.html for instructions for downloading, building and installing the source from git

When installing on the UbuntuVM cmake, flex and bison may also needed to be installed.

Doxygen and Code Composer (CCS) / Eclipse

Out of the box both Eclipse and Doxygen allow the selection of which version of Doxygen to use amongst those that are installed on the host.

CCS/Eclipse integrate highlighting of Doxygen Commands in several different formats.

See http://www.doxygen.nl/manual/docblocks.html for details.

CCS/Eclipse will auto generate a formatting block for a function when the following text is typed (in bold italics) above a function declaration

//!<enter>

```
//! @param aaa
//! @param bbb
```

//! @return

/*!<enter>

/*!

*

 * @param aaa

 * @param bbb

 * @return

 */

/// <enter>

/// @param tariffDescriptor

/// @param theTime

/// @param elapsedTimeMinSec

/// @param meteredEnergyThisSession_kWh

/// @return

Note that the @param and @ return format are fixed. There is no easy means to customise them.

See https://stackoverflow.com/questions/4403192/change-doxygen-comment-style-in-eclipse for details

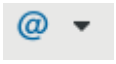
Doxygen and Eclox

Neither Eclipse or CCS are capable of running Doxygen out of the box to generate documentation.

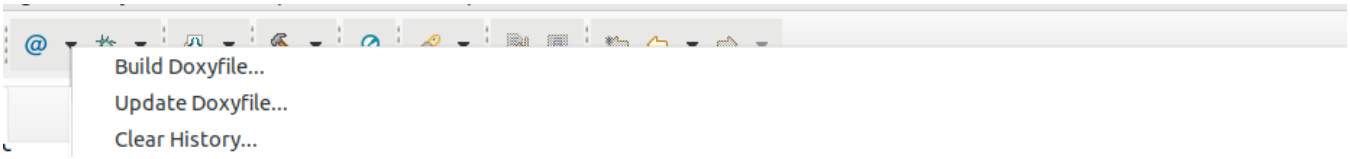
A third party add in, Eclox is used.

See https://anb0s.github.io/eclox/ for details

Eclox bundles a binary version of Doxygen (1.8.13??)

After installation the only hint that it has installed is the appearance of a Blue @ in the toolbar. 

Pressing the drop down arrow shows



Eclox allows the creation and modification of .doxyfile files.

A .doxyfile can be initially created by right mousing on a folder within CCS/Eclipse and choosing New\Other\Filter on dox then creating the file.

Once the file has been created double click on it to open the .doxyfile editor (courtesy of Eclox)

Project setting can be set on both the Basic and Advanced page.

ToDo .. come up with preferred settings ...

Doxygen and the DC Meter Code Base

The DC Meter is a mix of C and C++ Code. A recurring pattern is for a C++ class to have associated C code in the same header file.
The DC Meter uses .h for header files for both C + C++ code.

Doxygen is not a based on a compiler, it parses the code independently.

When a C++ compiler is compiling a file the macro__cplusplus is defined. When a C compiler is compiling a file the __cplusplus macro is not defined.

This is used in the header file to conditionally compile the C and C++ code by testing if __cplusplus is defined or not.

Doxygen does not do this, so consequently when a file that uses this technique is parsed by Doxygen the C code is parsed, and not the C++ Code.

There are a couple of workarounds for this.

One is to use Doxygen filters, which is a mechanism to pipe a file through a filter to preprocess it.

An alternative is to redefine a macro within Doxygen. This is the easier (for now) solution.

The Macro DOXYGEN is defined within Doxygen, and the following text in added to the top of header files that have a mix of C & C++ Code.

```
// When Doxygen is parsing ensure __cplusplus is defined

#ifdef DOXYGEN

#define __cplusplus

#endif //DOXYGEN
```

To define the DOXYGEN macro open the .doxyfile, go to the Advanced Dialog, Custom Tab. Search for Predefined Setting.

Predefined DOXYGEN=1

Doxygen vs Tritium C Coding Styles & Guidelines

In order to get the best out of Doxygen some of the Style Guidelines require a reinterpretation .. (Regard this as a process .. of which this is the first step ..)

2.1 Variable Names in comments : Doxygen explicitly uses Variable Names as part of a comment block. Once inserted there is no mechanism to update the comment if the variable name changes ..

```
//!
//! \param[in]          param1
//! \param[out]         param2
//! \param[in/out]      param3
//! \result             describe result
//! \brief              one line description of function
//! \details            details on separate line
//! \details            more details on separate line
//!
```

2.1 File Description Comment Block : If a \brief description, and then \details, are added under the copyright in an implementation file they are treated as documentation for the first thing that is declared in the file.

To associate with a specific class tag the text with \class

```
//! \class ClassName
//! Some Text
//! More Text
//! More Text
```

Extensions to the Tritium C Coding Standard

- Introduction
    - References
        - Web Links:
        - Books:
- Rules

Introduction

Here are some items not covered by the Tritum C coding standard that I think we should address for consistency sake and clarity. I encourage discussion of Tritium's coding standard with the goals of understanding it, improving it and us as developers, use it as a platform for:

1. better code-reuse,
2. more efficient with time on design, implementation, and testing,
3. better reliability

References

Web Links:

http://wiki.c2.com/?CodingStandardList
http://www.gnu.org/prep/standards/
https://isocpp.org/wiki/faq/coding-standards
https://users.ece.cmu.edu/~eno/coding/CppCodingStandard.html
https://wiki.sei.cmu.edu/confluence/display/c/SEI+CERT+C+Coding+Standard
https://wiki.sei.cmu.edu/confluence/pages/viewpage.action?pageId=88046682
https://www.approxion.com/capital-offenses-how-to-handle-abbreviations-in-camelcase/
https://stackoverflow.com/questions/15526107/acronyms-in-camelcase
http://cppunit.sourceforge.net/cppunit2/doc/coding_guidelines.html
http://wiki.c2.com/?CamelCase
https://isocpp.org/wiki/faq/mixing-c-and-cpp

Books:

"C++ Coding Standards: 101 Rules, Guidelines, and Best Practices". Herb Sutter and Andrei Alexandrescu, 8th August 2010.

Rules

Names containing acronyms when using camel case naming scheme often leads to an abomination in inconsistent names. For some it seems clear but adding some clarification to avoid confusion is a relatively simple task. Below I've identified what I think are the appropriate clarifications but I'm open to other suggestions. Unfortunately for me I have to change what was drummed into me at BAE Systems.

Basic Camel Case Capitalization Rules according to online references. This is provided as a starting point and does not take into account Tritium's coding standard.

1. The first letter is capitalized.
2. One or more letters in that word are also capitalised.
3. The word does not end on a capitalized letter: CamelCasE
4. No two capitalised letters shall follow directly each other: CamelCAse
5. No number in that word at any place: CamelCase1more
6. No dot(.), under_score or dash (-) within the word, only letters: Camel_Case
7. No ‘foreign’ letters in it like äöüß or accentuated like áéí. CämélCáße

Deviations for Tritium's coding standard:
point #1 needs to be changed to: "The first letter is capitalised only for Enumerations and Structures. Functions, variables (of any scope), function arguments, and file names, shall all begin with a lowercase letter."

Extensions to these rules to cater for ambiguities in dealing with abbreviations and acronyms in an attempt to better fit with Tritium's coding standard
8. Rules #4 and #6 shall prevail therefore acronyms and abbreviations shall be treated the same as words with the exception of units (see rule #9) and identifying the location of the function/global variable according to rule #9.1
8.1. If the acronym begins the name then it starts lower case and all acronym letters are lower case
8.2. If the acronym is mid name, it starts with an uppercase letter and remaining letters a lower case
8.3. If the acronym consists of only two letters, then capatilise both and try to place at the end unless rule #9 applies. (e.g. IO looks getter than Io but uV is proper)
9. Rule #6 can be broken under the following conditions:
9.1. to avoid name clashes in 'C', public functions and global variables (non-static) must be prefixed with a series of letters representing the file name they are in followed by an under_score. e.g. myFile_myGlobalVariable
9.2. to easily support units in names, the unit is postfixed to the name separated by an under_score with capitalisation conforming to SI guidelines (e.g. myfile_sampledData_mV).
9.3. the first letter following an under_score must be lower case unless it is an SI unit which is uppercase (e.g. V for Volts)
10. the first letter of the first word of the name is lowercase

Usage of the 'extern' keyword.
1. Whilst extern is automatically assumed in header files by the compiler so doesn't need to be explicitly stated except for clarification when the implementation is not in the 'c' file for that header.
2. In 'c' files, extern should be used to define functions if they're implemented outside that file and for modularity/portability/adaptability sake it is better to avoid including the header file. e.g. only the function API's are needed. This leave flexibility for the function to exist anywhere.

"#include" style.
two include styles exist:
#include <file>
#include "file"
They are different in that it tells the compiler where to start looking for the file. The first is for libraries and the second is for code you added. You can still reach a library header using double quotes as the library path is checked second. The former should not be used for code you added.

If your module allows for configuration, and that configuration can be done using #defines in a header file.
e.g. #include CONFIG_HEADER_FILE
elsewhere: #define CONFIG_HEADER_FILE "theConfigFile.h"
or in project settings -DCONFIG_HEADER_FILE="\"theConfigFile.h\""

Is it more important to enforce a strict coding layout across a company, or to be consistent within the file? Would it matter if there was a process to automatically format code on a commit hook using a tool like clang?

For a coding standard (not style guide) I recommend the Secure Coding Standard (https://wiki.sei.cmu.edu/confluence/display/c/SEI+CERT+C+Coding+Standard) over MISRA but from a company strategic direction we need to fill the gaps and address the concerns of the wider community regarding security as well as reliability.
For C++ I recommend the minimum of https://isocpp.org/wiki/faq/coding-standards and any of the 3 links to valid standards and recommend https://wiki.sei.cmu.edu/confluence/pages/viewpage.action?pageId=88046682.

Memory management needs to be consistent and it isn't always using the default malloc/calloc/free. So how do you neatly specify different memory management functions?

To avoid deeply nested if/else conditions and multiple return statements, wrap code in a do/while(0) and use break to exit.

Where to use the #include when mixing C and C++ code: as per https://isocpp.org/wiki/faq/mixing-c-and-cpp system headers are included outside the extern "C" and 'our' headers are inside. This also restricts the scope to just what is in the header that was created by 'us' instead of what is in the included headers. There may be a subtle side-effect otherwise.

Initialisation of structs at declaration should use use the ".member = value" notation to improve immunity during maintenance against change in the structure layout. new fields will be auto initialised to 0. Helps with self-documenting code. Prevents dependency on ordering of field initialisation. e.g)

```
struct myFile_sampleStruct
{
        uint32_t fieldA;
        uint8_t fieldB;
};

const struct myFile_sampleStruct sampleStruct =
{
        .fieldA = 1234;
        .fieldB = 56;
};
```

when an 'if' statement contains an immutable value as part of the condition, ensure the immutable is first in the condition. This helps prevent accidents where an '=' is missed from an equality check making it an assignemnt because the compiler will tell you it is an error, as opposed to the assignment taking place and if the result is == 0 or not.
e.g. "if(IMMUTABLE_VALUE == variableValue)" instead of "if(variableValue == IMMUTABLE_VALUE)"

Style Guide:

The following rules of thumb apply to any layout chosen. If in doubt ask these questions:
1) Code should be written once and read multiple times, so make it easy to read.
2) Be wary of fitting too much on one line. Typically people read in columns.
3) Avoid syntax that can lead to common problems that are difficult to find but easy to avoid
4) Code is typically easier to read when formatted in columns.

MQTT Broker Notes

Unsecure (port 1883)

.\mosquitto_sub.exe -t 'test/topic' -v

.\mosquitto_pub.exe -t 'test/topic' -m 'Host hello world!'

.\mosquitto.exe

Secure (port 8883)

.\mosquitto_sub.exe -t 'test/topic' -v --cafile "C:\Program Files\mosquitto\certs\ttiServerCA.crt" -p 8883 --insecure

.\mosquitto_pub.exe --cafile "C:\Program Files\mosquitto\certs\ttiServerCA.crt" -p 8883 -t 'test/topic' -m 'Host > hello world!' --insecure

.\mosquitto.exe -c mosquitto.conf

Getting the SAFE Certificate

.\mosquitto_sub.exe -t 'DCMS/CSMS/SAFE' -v --cafile .\dcmRoot-mqtt.crt -p 8883 --insecure

At end of charge, you get a data dump like this:


```
DCMS/CSMS/SAFE {"SessionID":"","EDSI":"0000000104","SAFE":"
PD94bWwgdmVyc2lvbj0iMS4wIiBlbmNvZGluZz0iVVRGLTgiPz4KPHZhbHVlcz4KCTx2YWx1ZT4KCQk8c2lnbmVkRGF0YSBmb3JtYXQ9Ik9DTUYi
IGVuY29kaW5nPSJwbGFpbiI+T0NNRnx7IkZWIjoiMS4wIiwiR0kiOiJUcml0aXVtIFB0eSBMdGQiLCJHUyI6IjEyMyIsIlBHIjoiVDAwMDAwMDAx
MDQiLCJNViI6IlRyaXRpdW0gUHR5IEx0ZCIsIk1TIjoiMTIzIiwiTUYiOiIxLjAwOSIsIklTIjoiZmFsc2UiLCJJTCI6IkhFQVJTQVkiLCJJRiI6
WyJSRklEX05PTkUiLCAiT0NQUF9OT05FIl0sIklUIjoiTE9DQUwiLCJJRCI6IiIsIlZTIjoiUlhaWU1YTVJQRVNBIiwiUkQiOlt7IlRNIjoiMjAy
MS0wOS0xNVQwNzo0ODowMCwwMDArMDIwMCBTIiwiVFgiOiJCIiwiUlYiOiIwLjAwMCIsIlJVIjoia1doIiwiU1QiOiJHIiwiUkkiOiIxLWI6MS44
LmUiLCJVVCI6IiAifSx7IlRNIjoiMjAyMS0wOS0xNVQwNzo0ODo0MywwMDArMDIwMCBTIiwiVFgiOiJDIiwiUlYiOiIzLjk1OSIsIlJVIjoia1do
IiwiU1QiOiJHIiwiUkkiOiIxLWI6MS44LmUiLCJVVCI6IiAifSx7IlRNIjoiMjAyMS0wOS0xNVQwNzo0ODo0MywwMDArMDIwMCBTIiwiVFgiOiJF
IiwiUlYiOiIzLjk1OSIsIlJVIjoia1doIiwiU1QiOiJHIiwiUkkiOiIxLWI6MS44LmUiLCJVVCI6IiAifV19fHsiU0QiOiIzMDQ0MDIyMDZGMTdG
RTQ0NzM1MDZEMUE5Qzk4MDgyOURGREZGNjNGNDdDRTM0RTQzOTFCMjM2NkYxQThFQkQ5NjRBMzgzRDUwMjIwMjUwNUYzQzdFNEFGNkQ0MThFRDE0
NkE4QURFQUQ1NTFCOTdFOUIwMTE1ODlERTY2RDNGRDkzM0NCRTc0MDYwMSJ9PC9zaWduZWREYXRhPgoJCTxwdWJsaWNLZXkgZW5jb2Rpbmc9InBs
YWluIj4zMDU5MzAxMzA2MDcyQTg2NDhDRTNEMDIwMTA2MDgyQTg2NDhDRTNEMDMwMTA3MDM0MjAwMDQ0N0FGRkFCNjU0QzMyQjhBODAyNjg5NDU2
MDUwRTRGRjM2RDVDRTIzOTYwRjdFMjBBMDgyNUQ3REI3RjY1QjFFQTA5MDg2QURCOTEwNDdCODkxNDIwQjJEMzAxRTc0NDFDMTQxQ0ZBRUVDRTJB
M0UwOUREQkZGNDlGODgzQUMzNzwvcHVibGljS2V5PgoJPC92YWx1ZT4KPC92YWx1ZXM+Cg==="}
```


Copy the "SAFE" data string to this page: https://codebeautify.org/base64-to-json-converter

Grab the XML output - this can be pasted in to Transparency Software.

MQTT - Mosquitto

Introduction

MQTT is a lightweight protocol used to publish and messages.

Clients are able to publish data to the broker on specified 'topic/s' or channel/s.

Multiple clients can connect to a single *broker,* acting as a server which then distribute the data to clients that have subscribed to specified 'topic/s'or channel/s.

The client source library was retrieved from Eclipse Paho MQTT, and installed with 'pip install paho-mqtt'

Broker was set up using Mosquitto, no security protocol was implemented, however it can be done using the mosquitto.conf file show below.

Publish

Client publish messages on specificied 'topic', no topic configuration is required. Similar to command prompt , topics have hierachy using / as separator. E. g topic= Measurement/inputVoltage/Watts

Prior to publishing, the payload must be encoded. the function 'process_msg' in client.py can be call. It detects the type of data to be publish and encode to JSON.

Publish method require a topic, and several optional parameters:

1. qos (quality of service). Default, qos = 0, the broker does not acknowledge the message have been received and the payload is not stored and re-transmitted by the client

   qos = 1. This requires an acknowledgement from broker and client stores the message until broker send a receipt (PUBACK)

   qos = 2. Message is only received once by the broker.

For the purpose of testings qos = 0 as data is sent at short interval and few messages are allowed to get lost occasionally.

2. retain. On publishing, retain = False (default). If client publish on an unsubscribed topic, the message is lost in the broker.

   if retain = True, the last payload sent by the client will be stored in the broker and will be distribute upon subscription.

e.g client.publish(topic='Test', payload='a string, dict, list, etc' , qos= 0, retain=False)

Subscribe

A client can subscribe to multiple topics to the broker. The required parameters is topic and qos

qos = 0. Default, the subscriber may not receive the payload, however it will not be received twice

qos = 1. The payload will be received at least once by the subscriber, but the same payload may be received again

qos = 2. The payload will be received exactly once. The slowest transfer as multiple 'handshakes' are required.

A publisher with qos=2 is not guaranteed that the payload will be received by Subcriber using qos = 0.

For the purpose of testings, qos =0 is used as large constant data transmission will be received constantly by the TTI and faster speed is preferred.

e.g client.subscribe(topic='Test' , qos = 0)

Callbacks:

ON_MESSAGE

When a message is received on a topic that the client subscribed to, the class method 'on_message' is automatically called. Thus on_message is used to process received payload.

```
def on_message(self,client,userdata,message):
    msg_in= message.payload.decode('utf-8')
    msg_in = json.loads(msg_in)
    if isinstance(msg_in,dict):
        self.payload_dict = msg_in
    if isinstance(msg_in,str):
        self.payload_str = msg_in
    if isinstance(msg_in,int):
        self.payload_num = int(msg_in)
    if isinstance(msg_in,list):
        self.payload_list = msg_in
```

The payload can be retrieved by calling the class attribute. E.g if the expected payload is of dict type, it can be called with *client.payload_dict.*

Since the expected payload sent by the Charger is expected to be of <dict>, the method on_message will provide the basic functionality of deserialising JSON payload and save the payload to the class attribute payload_dict.

ON_CONNECT

When the client connect to the broker, the broker send back a CONNACK message, which in response will call the method on_connect automatically. One of the parameter is rc. Which can be called for connection acknowledgement and debugging purposes.

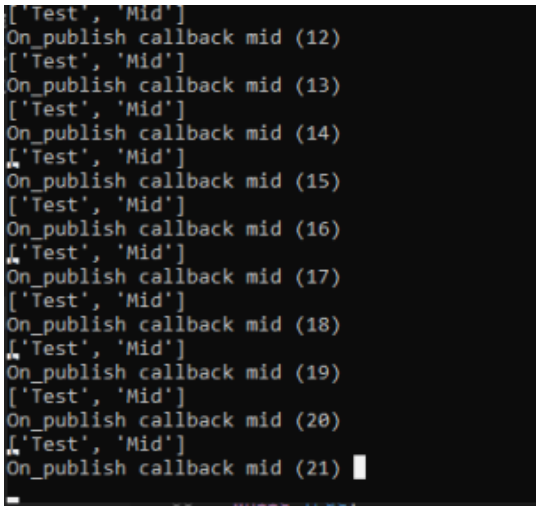
```
The value of rc indicates success or not:
    0: Connection successful
    1: Connection refused - incorrect protocol ver
    2: Connection refused - invalid client identif
    3: Connection refused - server unavailable
    4: Connection refused - bad username or passwo
    5: Connection refused - not authorised
    6-255: Currently unused.
```

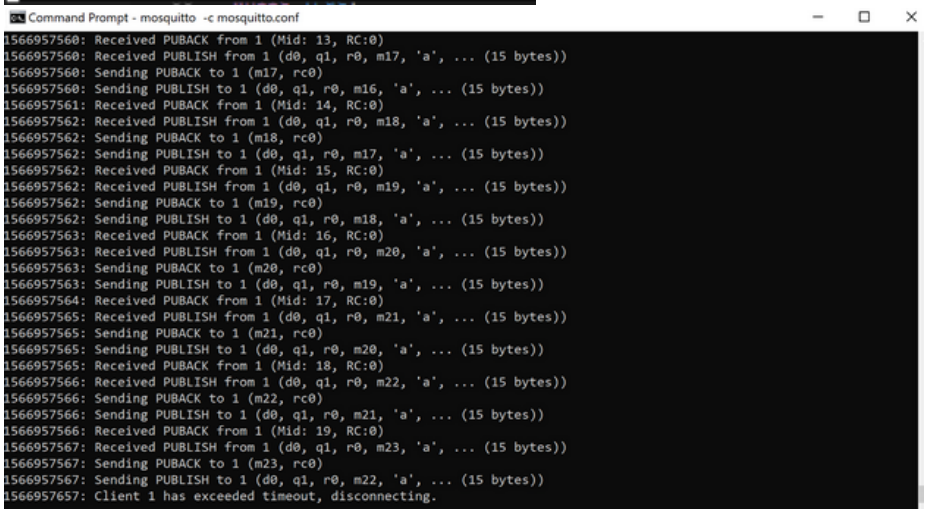
In the class method on_connect, the pass statement was used, however for debugging purposes, the rc value can be printed on the screen once the codes are uncommented.

ON_PUBLISH

Similar to on_connect, once the Client have published the payload to the broker, the on_publish callback will automatically be called. In the class method, the pass statement was used. However, in addition to rc, on_publish callback can also return 'mid' parameter

As shown above, rc == 0 signifies the message have been published successfully to the broker or else, an error have occurred. Mid value is an integer that corresponds to the published number as assigned by the client, it is essentially the message id.





Client_message_id Broker_log

By using the on_publish function, we can see that the payload transmission between broker and client is lagging. As the mid does not match between the two. The Quality of service (qos) for both client and broker

for the above logs was set at 1, which means that the payload must be received at least once by the subscriber at the cost of transmission speed. If the qos =0, the transmission speed will be faster, however the mid will always equal 0, as at qos =0, the broker and client will not set a unique payload_id.

During testing, the on_publish method simply have a pass statement, and qos is set to 0. However during debugging, qos should be set to 1 and the on_publish method should print the mid value.

ON_SUBSCRIBE

Similar to the above callbacks, on_subscribe is automatically called when the broker acknowledges. Pass statement was written here, however, the print function can be used to print the rc value.

Loop

Loop function must be called to initiate the client. If they are not called, incoming network data will not be processed and outgoing network data may not be sent in a timely fashion. There are three types of loop functionality

Loop(timeout=1.0, max_packets=1)

This is called regularly to process network events. While being called, it waits until the network socket is available for reading or writing, if appropriate, then handles incoming/outgoing data.

E.g:

client_1 = clients()

While True:

client_1.loop()

Loop_start() / Loop_stop(force=False)

These functions allow threaded interface to the network. Loop_start() must be called before connect(), then it wil runs a thread to call loop() automatically. This frees up the main thread for other work that may be blocking.

loop_stop() is called to stop the background thread.

E.g:

client_1 = clients()

client_1.connect('localhost')

client_1.loop_start()

While True:

client_1.publish('Topic',payload=None)

Loop_forever()

This is a blocking form of the network loop and will not return until the client calls disconnect. Additionally it will automatically handles reconnecting

loop_forever(timeout=1.0, max_packets=1, retry_first_connection=False)

It is recommended to simply uses loop() for testings unless threaded interface is required.

Usage:

The general usage is as follows:

1. Create client instance: client = *clients('client_id')*
2. Connect to a broker using one of the connect() functions: *client.connect('192.168.43.166' , port 1883, keepalive=60)*
3. Call one of the loop() functions to maintain network traffic flow with the broker, as shown above
4. Use subscribe to subscribe to a topic and receive messages published by the broker: *client.subscribe('Topic')*
5. Use publish() to publish messages to broker: *client.publish('Topic', payload='Test String')*
6. Use disconnect() to disconnect from broker: *client.disconnect()*

Callbacks will automatically be called upon receiving, publishing, connecting, subscribing to the broker.

MQTT - Broker set up

Installation

https://mosquitto.org/download/

After mosquitto package have been installed from the link above. Open command prompt and go to the mosquitto install directory and type *mosquitto*

You can use *mosquitto -h* for various command line switches

Mosquitto Configuration

The configuration file is included in the installation folder (mosquitto.conf). By default, MQTT Broker does not need the configuration file to run, as at default mosquitto.conf is completely commented out.

To start configuring the MQTT, edit the commented lines or create another blank config file.

Setting listener

bind_address set a specific IP address/hostname that can access the broker, if not set, the broker will be accessible to all network interfaces. In my config file, this was left empty.

port The recommended port for MQTT is 1883. For TLS the port is 8883 (This is currently not implemented). The default is 1883.

max_connections Set the maximum number of clients connection. Default is -1 which is unlimited

Logging

There are many type of messages log, which are displayed on the MQTT Broker console. Possible types are: debug, error, warning, notice, information, none, subscribe, unsubscribe, websocket and all. It is recommended that all of the log types are enable for debuging purposes.

e.g *log_type information*

Security

There are three ways Mosquitto broker verify the identity of MQTT client:

TLS

TLS can be enabled for the MQTT Broker, however it does not accept self signed certificate. Thus it was not enable for testing purposes between Charger - DcMeter and TTI.

Usernames and passwords

There is option to prompt the client to input username and password before a connection is allowed. However this is also not implemented.

Client ids

use_identity_as_clientid This replace the clientid that a client connected with with its username. This allows authentication to be tied to the clientid, which means that it is possible to prevent one client disconnecting

another by using the same clientid. For flexibility, this is set to true

```
#Define Port
port 1883

#Setting log displayed on console
log_type error
log_type notice
log_type information
log_type subscribe
log_timestamp true

#Set extra listener port
listener 8883

#Security
use_identity_as_clientid true
```

Mosquitto.conf

The Mosquitto broker does not need the config file to run. Thus it must be called explicitly to run the Broker with config file.

Ensure that the config file is in the same directory as mosquitto.exe and type the following: *mosquitto -c mosquitto.conf*

You should see the following screen if all is good:

For more information on MQTT Broker set up, https://mosquitto.org/man/mosquitto-conf-5.html

The GitHub for paho Client, https://github.com/eclipse/paho.mqtt.python

Notes on Measuring Software Performance

The documentation in this section is an informal recording of various information relating to measuring software performance as part of the development process. Any formally adopted processes will be captured in main 'Software Development Plan' page.

Table - References

Ref ID	Link/Reference	Description
PR1	https://www.freertos.org/FreeRTOS_Support_Forum_Archive/December_2011/freertos_Measuring_CPU_usage_4866946.html	Tips on using CPU usage metrics in FreeRTOS

Review of Enterprise Architect

- Introduction
- References
- Goals of the software
- Executive Summary
- Notes

Introduction

A review of Enterprise Architect (EA) from Sparx Systems was performed to see if it could help Tritium to become more efficient with time whilst improving quality and being able to supply documentation suitable for the certification requirements of the DC Meter. The information on this page contains information about how the EA software was evaluated and notes about observations of its usefulness.

References

ID	Name	Description
R1	Sparx Systems	Manufacturer of the Enterprise Architecture software found here: https://sparxsystems.com.au
R2	Tutorial Videos on EA	A useful link to videos that explain how to use EA: https://sparxsystems.com.au/resources/demos/index.html

Goals of the software

1. Easily synchronise design with implementation
   This involves generating code from design elements and the ability to round-trip code back into the design. In particular focusing on class diagrams, sequence diagrams, state diagrams.
2. Verification of state machines
   State machines are a common design element in software. Having the ability to verify state flow before implementation would greatly reduce time and improve quality
3. Ease of use
   If it takes weeks of configuration to get it right, and it feels like round-tripping is a fight every time, then maybe it isn't worth the effort.
4. Portability
   Not all software created at Tritium runs on windows and is compiled with MS Visual Studio. EA would need to do well compiling code using a cross-compiler, and work for the programming languages C and C++, assembler
5. Quality
   There must be a mechanism to ensure generated code conforms to a specified style guide or coding standard.
6. Integration with commonly used tools
   In particular JIRA, Eclipse (and hopefully Code Composer v8), MS Studio (free version), gcc compiler would be beneficial.

Executive Summary

Pros	Cons
	Doesn't run natively under Linux but will run under WINE

Notes

Version evaluated: v14, Professional

Ubuntu 18.x LTS VM- No Login Screen

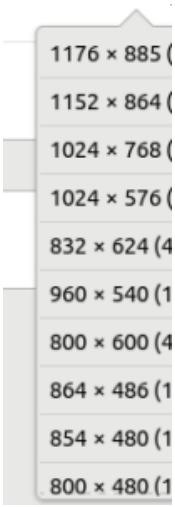
Introduction

After updating Ubuntu 18.x LTS running as a VMWare VM you are no longer able to log in to Ubuntu. Instead you are stuck on a purple screen.

Digging deeper it looks like the VMWare SVGA driver is no longer working as expected .. even though it was before the update.

It it possible to force a login by editing the grub command line and adding "nomodeset" to the linux command.
Adding the nomodeset parameter instructs the kernel to not load video drivers and use BIOS modes instead until X is loaded.
Doing this results in the login screen being shown.

Once logged in available screen sizes are limited to a maximum of 1176 x 885 pixels.



Investigation

Looking at the vmware.log shows the following entry

SVGA-GFB: Truncated maximum resolution for register modes to VRAM size: VRAM=4194304 bytes

For reasons that are not clear the SVGA driver is limiting itself to 4MB of RAM .. which is not enough to allocate screens bigger than 1176 x 885 pixels

I was able to replicate this behaviour on a clean install of Ubuntu 18 LTS .. updating resulted in exactly the same behaviour.
Whatever the root cause is is in the distro.

VMware have a flag that allows for minimum RAM allocation to be 8MB
svga.minVRAM8MB=TRUE – ensures 8Mb allocated to RAM rather than 4 .. does not go bigger than 8

Workround

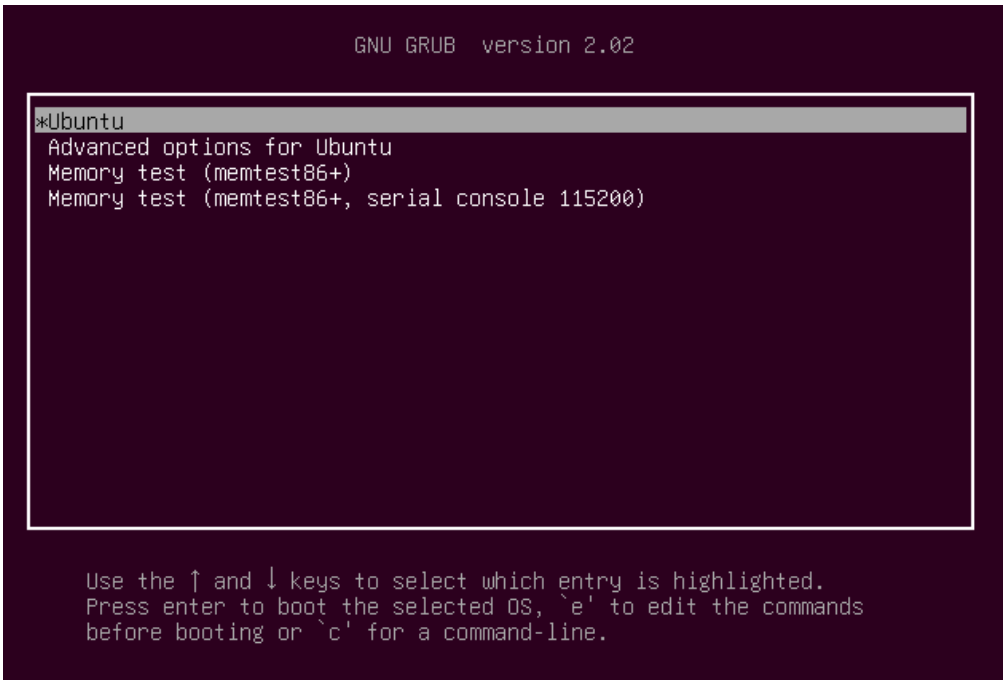
Edit the VM associated .vmx file. Do this with VMWare workstation NOT running.

Change svga.autodetect from TRUE to FALSE
svga.autodetect = "FALSE"
This will prevent the screens to be auto updated, and will force the use of config within the .vmx file

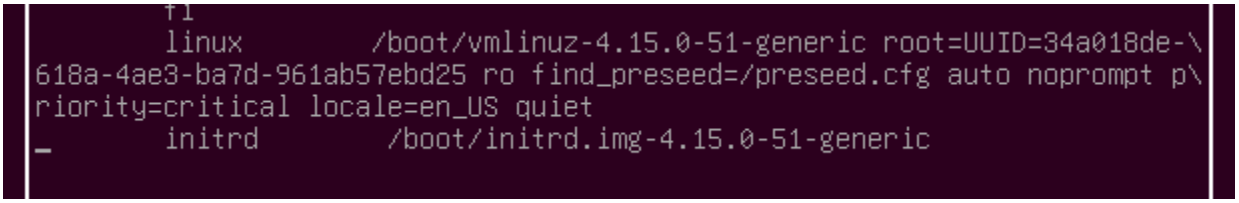
Add an entry for svga.minVRAM8MB, set to TRUE
svga.minVRAM8MB=TRUE
This will force the SVGA driver to allocate 8MB of RAM .. enough for 1920 x 1080

Add nomodeset command temporarily to grub command

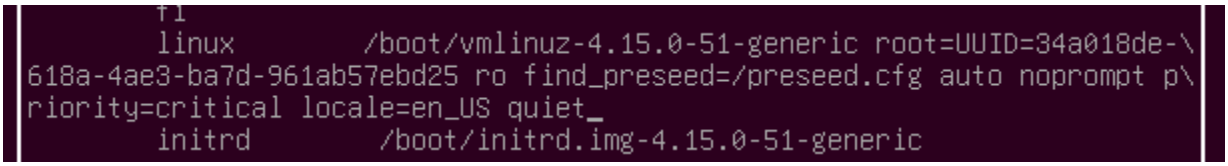
Boot VM
Click in the VM Window as it starts
Hold Down the Shift Key to get to GRUB Screen



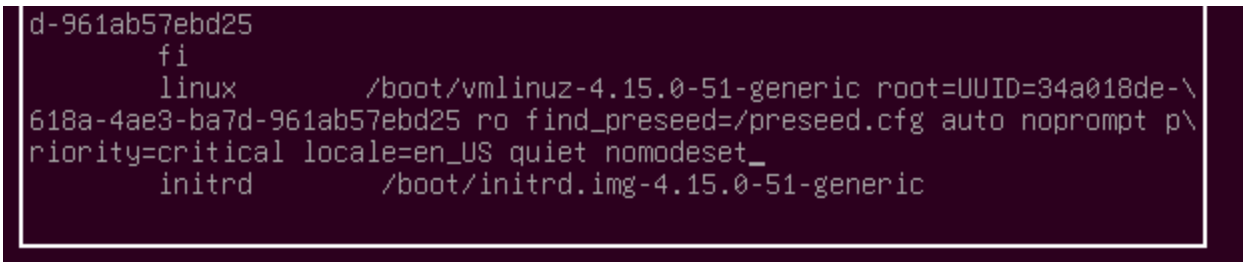
With *Ubuntu selected press e to edit grub options
Press down arrow until cursor is on line with initd on it ..



The press left arrow once .. cursor will move to end of previous line



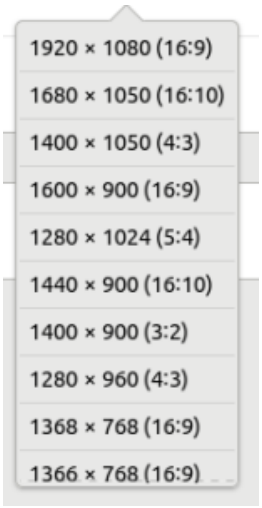
type " nomodeset"



Then press F10 key to continue boot process

This will bring up Login Screen .. Login as usual

Check screen resolutions .. expect to see resolution list like this ..



If you dont get resolutions bigger than 1176 x 885 then screen buffer is 4MB

To make nomodeset permanent

At a command prompt:

cd /etc/default
sudo nano grub

Modify
GRUB_CMDLINE_LINUX_DEFAULT="quiet"
to
GRUB_CMDLINE_LINUX_DEFAULT="quiet nomodeset"

```
GRUB_DEFAULT=0
GRUB_TIMEOUT_STYLE=hidden
GRUB_TIMEOUT=10
GRUB_DISTRIBUTOR=`lsb_release -i -s 2> /dev/null || echo Debian`
GRUB_CMDLINE_LINUX_DEFAULT="quiet"
GRUB_CMDLINE_LINUX="find_preseed=/preseed.cfg auto noprompt priority=critical l$
```

Save File
type sudo update-grub

```
dev@ubuntu:/etc/default$ sudo update-grub
Sourcing file `/etc/default/grub'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-4.15.0-51-generic
Found initrd image: /boot/initrd.img-4.15.0-51-generic
Found linux image: /boot/vmlinuz-4.15.0-50-generic
Found initrd image: /boot/initrd.img-4.15.0-50-generic
Found memtest86+ image: /boot/memtest86+.elf
Found memtest86+ image: /boot/memtest86+.bin
done
dev@ubuntu:/etc/default$
```

reboot VM .. should reboot to login screen

After making these changes the screen will maintain the resolution it has been set to .. it does not auto resize
This is expected to be a temporary workround till someone upstream notices this is broken !

X1 Prototype Hardware Issues

Design issues learnt that will be fixed on M1 prototype:

#	Item	Issue	Remedy	Status	Completed
1	Controller PCBA	eVref is connected to VDDA, Using external 2.5V reference means it won't boot.	Separate eVref and VDDA. Connect VDDA to 3v3 rail through ferrite bead.	Minor. To be done by the design engineer.	•
2	Controller PCBA	Creepage distance of 5.5mm around isolated section to AFE board does not meet pollution level 2, non potted requirements.	Decide on pollution degree of DC meter enclosure (to pot or not to pot)	Will require conformal coating. (Creepage: 3.2mm@PD1; 5.0 mm@PD2. Clearance: 3mm x2 for reinforced)	•
3	Controller PCBA	Choice of reinforced isolation or non-reinforced isolation components	Reinforced isolation will be used as reduce size and cost as appropriate components have been found	Decided.	•
4	Controller PCBA	Need to implement double isolation (or reinforced but not preferred) Consult IEC 61010. Rated to 1.2 kV requirements. Company practice should be that there is double isolation on any PCB that has both a low and high voltage connection. Each barrier needs to withstand 3kV dielectric (1 minute withstand) and 4 kV impulse.	Decide between option 1 and 2: 1 - Single isolation barrier on comms + single on AFE 2 - Double between MCU and AFE 3 - Reinforced isolation components	Solution 3 (duplicate of above)	•
5	Controller PCBA	Choice of DC/DC converter vs pulse transformer.	Pulse transformer will be used. Significant savings in cost and size.	Implemented	•
6	Controller PCBA	Choice of 12V or 5V DC/DC for AFE power	Replace the 12V/5V DC/DC with a 12V buck to 6V, install a 5V linear regulator on AFE, and drive ovenisation from 6V and AFE control circuitry off 5V.	Implemented	•
7	Controller PCBA	Power output of DC/DC converter is too low to drive ovenisation on AFE board to sufficient capacity.	Increase power to ovenisation when changing to pulse transformer.	2 pulse transformers in parallel gives up to 12W available.	•
8	Controller PCBA	D5 (diode to Super cap) doesn't meet expected pin out. Pin 2 appears to be NC.	Connect cathode to pin 3		•
9	Controller PCBA	Not enough component space as a single sided board for creepage isolation (safety) requirements	Make controller PCB double sided?	Single sided	•
10	Controller PCBA / TSO	Not considered industry best practice for a security device to not include a form of hardware security module. Cannot find level on any of the datasheets.	Add a Infineon or Microchip TPM 1.2 specification SPI /I2C IC with FIPS-140-2 level 2 minimum to both the DC meter controller and FPGA interceptor board	TPM: SLB9670 added	•
11	Controller PCBA	No silkscreen showing AFE isolation barrier	Add silkscreen to show isolation barrier	Determined it was not needed	•
12	Controller PCBA	Centre length board cutout no longer required for positioning in enclosure	Remove board cutout and replace with a mounting point		•
13	Controller PCBA	Designators missing from top overlay for J1-3	Add designators for J1-3 to top overlay	No action.	•
14	Controller PCBA	Sockets J1-3 are positioned too far inwards for proper connection to harnesses	Move sockets J1-3 to be flush with the edge of the PCB		•
15	Controller PCBA	Unnecessary IR LEDs for voltage and time	Remove IR LEDs for voltage and time. Possibility to remove IR LED completely because test pulse output is only needed for certification purposes. Still needs to be installed incase the DC meter is pulled out of a charger and verified on a bench.	Removed volt/time.	•
16	Controller PCBA	Missing test point for GND	Add test point for GND		•
17	Controller PCBA	Footprint for RJ45 housing (CID: 3371) is incorrect. Pin 7 is unnecessary.	Non issue. Left for compatibility to replacement parts.	Part changed anyway for consistency in company.	•
18	Controller PCBA	AFE SPI SCLK line needs 10k pull down	Add 10k pull down resistor		•
19	Controller PCBA	4Gbit external memory maybe insufficient for data purposes (can only be determined once regulation standards become clearer)	Investigate memory requirements	Sufficient given the information we currently have.	•

20	Controller PCBA	J6 and J7 thermistor connectors apparently are not easy to use	Check if these connectors work as expected. If not, replace with more reliable connectors.	Changed to 2 way pico. Fans are 3 way pico	•
21	Controller PCBA	Installed TM4C micro is currently rated 85'C	Purchase industrial version TM4C to ensure temperature range 105'C		•
22	Controller PCBA	No failsafe control if ovenisation is too hot	Micro can turn off isolated power if too hot.		•
23	Controller PCBA	100 MHz noise on negative connection points on both sides of RECOM DC/DC converter	Attach 2K at 100MHz beads on both positive and negative connection points on both input and output of DC/DC converter. Use Richard's RedExpert wondertool to decide on make and model of bead.	Removing Recom DC/DC converter, need to redesign filter.	•
24	Controller PCBA	Decide on new power supply options to replace CAN input power. Calculate power draw of new cooling systems? 12V or 24V input or wide range? Regulated or unregulated?	12V input. No power supply needed. 24V is not an option.	24V12V DCDC brick can always be added externally	•
25	Controller PCBA	Keep KK connectors for ambient and shunt thermisters on controller board?	Change connector type, keep measurement of both ambient and shunt, Richard to decide on which controller board placement side.	Ambient sensor will be onboard.	•
26	Controller PCBA	Debug feature: Software developers need to do timing verification of MCU Require 4xGPIO	Can use the existing debug header. SPI pins can be used as GPIO.		•
27	Controller PCBA	Remove the debug header?	No. Will be kept for GPIO. Configured whichever way firmware is configured.		•
28	Controller PCBA	Is a mounting post required on the long edge?	Yes - We will give location of mounting post to Bernie after placement of both controller and AFE boards.		•
29	Controller PCBA	Location of kW IR LED?	Depends on mechanical design by Bernie	Right edge. Can be vertical or right angle	•
30	Controller PCBA	Fans need to be (on/off) controlled by DC Meter	Add 2x1W Fan outputs.		•
31	AFE PCBA	ADC footprint is not the correct size (too small for actual size of pads). Thermal pad shorting to some surrounding pads.	Increase footprint size to be correct	22 Mar 2019 | A KR | Footprint corrected. Component/PCB needs to be updated	•
32	AFE PCBA	Testpoint for analogue ground near reference voltage location is not ideal. It was not re-located properly after changing ovenisation circuit for fabrication.	Move to better location		•
33	AFE PCBA	Ovenisation MOSFET footprint seems to be oversized for the part. It does work though.	Determine if footprint should be changed	Keeping as is	•
34	AFE PCBA	Socket J1 (maybe) positioned to far inwards for proper connection to harness	Move socket to be flush with PCB edge	22 Mar 2019 | A KR | Harness connects ok as is. Need to investigate tamper prevention requirements before any change made	•
35	AFE PCBA	Designator missing from top (*cough* bottom) overlay for J1	Add designator for J1 to top (*cough* bottom) overlay	22 Mar 2019 | A KR | NC - No designators are included on silk	•
36	AFE PCBA	T-section of the PCB near the mounting post could snap if uneven torque is applied when mounting shunt to busbars	Change AFE board shape on on the T-section past the solder points to the shunt to be narrower to make the AFE board physically more flexible to an unlevel shunt mounting	To be done by DRE after PCB size and shape requirements are finalised.	•
37	AFE PCBA	Clearance distance between C13 (controller board) and AFE components is less than minimal safety requirements	Increase board cutout for capacitor, or install Formex around capacitor	22 Mar 2019 | A KR | Already decided during PCB review to install Formex around capacitor	•
38	AFE PCBA	Which charge cable being used of a Dual PK needs to be determined.	Install a rough voltage sensor, PCB mount connector, optocoupler, trigger IC, and a 4 kV isolation barrier on the AFE.	Voltage sense impletented using comparator (Not isolated)	•
39	AFE PCBA	Temperature control of the ovenisation circuit could fail and cause a fire hazard.	Add an analog switch/relay to the AFE power supply.	Will be on controller board.	•

40	AFE PCBA	Kelvin sense points on future charge cables will not be able to be measured with current AFE design using a 4 channel ADC	Change 4 channel ADS1220 ADC for an 8 channel ADS1278	Scope creep - Decision made to stay with a 4 channel ADC and not include future proofing for Kelvin sense points. Power loss in charge cable will be calculated. DC meter will need to be configured to know which make & model of charge cable and its length is being used.	•
41	AFE PCBA	Analog vs Digital control signal for AFE	Error at 4 A at 25 degC is 2% with ovenisation. Can extrapolate data to generate calibration value. Can't use digital control signal because calibration will be more difficult.	Will remain analogue control. Have kept digital control hardware	•
42	AFE PCBA	Heat retention around AFE PCBA. Put foam inside the canister area, over the can, or over the whole board. Foam material?	Do additional below zero tests with foam over canister and over board. Repeat tests and check for major variances.	With 6W can get to 100'C from -35'C. Need to confirm foam to retain heat	•
43	AFE PCBA	At what power level is the ovenisation MOSFET switching on/off? We want to know whether there is a hard turn offs /ons at high power levels which could damage electronics. Get ADC upto operating temperature, then push it past this temperature and get a capture of power usage/switching signal.	Switching is reasonably slow but can be slowed with the RC on the gate.	Increased R1k. Can be tuned further if necessary	•
44	Both PCBAs	Overlay text for circuit title and company name uses Sans Serif font and looks like it's been written for a children's book and not an industrial application. Could create misunderstandings by technicians when reading (e.g. An I dashed with another I, I-I, could be interpreted as a H.)	`Use monospaced UPPERCASE Serif font for clear readability on all PCBs`	No action	•
45	Shunt	Temperature of shunt at 420 A reaches 85 °C after 25 minutes (and was still rising). 3D printed bottom part partially melted and warped (injection moulded plastic will have a different melting point but will still most likely melt in current design).	Need to investigate whether using a shunt is suitable in the first place because if it produces too much heat and consumes more power than allowed under B. 5.5.2 VDE AR E 2418 3 100 draft E. May need to look into using a fluxgate magnetometer type device. A Hall effect sensor is probably too inaccurate and will be very sensitive to the temperature of the metal strip inside the sensor (i.e. needs constant temperature). If a shunt is still to be used then a heatsink and/or fan will be required.	Decision made to stay with a shunt based current transducer for time being.	•
46	Enclosure	Ability to plug headers to 6 pin micro-molex and RJ45 sockets is restricted	Increase recess to allow better ability to unclip headers		•
47	Enclosure	AGND contact soldered to top of shunt protrudes into wall of top part enclosure	Cutout section around busbars will be expanded to increase airflow for heat dissipation		•
48	Enclosure	Weight of whole DC meter is supported by the shunt being sandwiched between the top and bottom enclosure parts. If enclosure cutout around the shunt is increased to allow additional ventilation air flow, the total weight of the DC meter will be supported only by the solder connection between the shunt contacts and AFE board. Solder contacts could crack or AFE board could snap if uneven torque applied when fastening the two studs in the shunt to busbars.	Investigate an external mounting frame option for the DC meter enclosure to off load the weight being put onto the shunt (could defeat the purpose of a universal shape enclosure for ISD/PK/RTD2 chargers if not done properly.)	Being designed by Bernie	•
49	Request	Provision of Diagnostic LEDs - right now there aren't any These are super handy during development .. also for fault diagnosis in the field If price/view ability is an issue in the field alternatively consider only populating development/early production boards.	Consider the provision of 4 LEDs Heartbeat (Green) - Flashes at 1s frequency when the Meter software is running as expected Out Of Service (Red) - Lit when the Meter is Out Of Service (for whatever reason) Remote Connection (Green) - Lit when connected to remote server (Transparency ??) Charging (Green) - Lit when the Meter considers the Charger is charging	Decision made to not be implemented. 4 debug IO available on spare header.	•
50	Flammability	Fire standards: UL 94 V2 for non live parts (e.g. plastic mounts for PCBS); UL 94 V0 for anything touching live parts.	Use V0 mounts around shunt. V2 everywhere else.		•
51	Ethernet Cables	Connecting shielded ethernet cables to an unearthed device is useless. Does the DC meter need an Earth tap? Connecting shielded would require changing the RJ45 jack.	Use shielded CAT5 cables and replace the RJ45 jack to a shielded jack. Connect shield pin to Bobsmith termination.	Made earth input possible, connected to shield.	•
52	CAN or Ethernet	CAN connectors consumes a lot of PCB surface area	Decision made to remove CAN and install a separate power connection.		•
53	FPGA Carrier PCBA	Incorrect capacitor was purchased for C106 (CID 8663). The footprint on PCB is 1206 which is correct but a 1210 part was purchased during the prototype run.	Nothing. 1210 part would still fit on the 1206 footprint.	Purchase correct part in next rev.	•
54	FPGA Interceptor PCBA	Silkscreen on J4 - P5/N5 is reversed.	N/A	Correct in next rev.	•

55	FPGA Interceptor PCBA	Replace R63 and R64 with a tact switches.	Currently they are DNP and resistor pads need to be shorted to invoke reset.	Correct in next rev.	•
56	FPGA Interceptor PCBA	Leave the 4 Vref pins as GPIOs- floating or tie to 1V. They are now tied to 2.5V VCCO which is incorrect.	Currently this is fixed in the FPGA design to generate internal 0.9V and pins are configured as GPIO.	Correct in next rev.	•
57	FPGA Interceptor PCBA	1.25V Voltage reference U2 chip is incorrectly connected in schematic.	Flip U2 chip around and soldered it to the right pads.	Correct in next rev.	•
58	FPGA Interceptor PCBA	Change R18 to 300 ohm CID 11475 for brighter LED illumination	Change over R18 from 2.2K to 330 ohm.	Correct in next rev.	•
59	FPGA Interceptor PCBA	Use larger pads for PMIC ADP5014. Library footprint needs to be updated.	N/A.	Correct in next rev.	•
60	FPGA Interceptor PCBA	Change R51 to 4.7k ohm resistor CID11468	Change over 10K resistor to 4.7K from stronger pull up.	Correct in next rev.	•
61	FPGA Interceptor PCBA	Add clock source 100MHz to MRCC pin of FPGA	• Possibly use the FPGA internal oscillator • Possibly use the EMCCLK • Possibly use LVDS differential clk • Green wire the ASEMB-100.000MHZ-LY-T to a FPGA clock capable pin	Include clock source ASEMB-100.000MHZ-LY-T in next rev.	•

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_DC Meter Development Pages

This contains the pages that are used during development that only have transient information which is expected to become out of date. i.e. Bugs/version faults/testing notes

2. Referenced Documents

The following table lists documents referenced within the DC Meter Firmware Design document.

ID	Reference
R1	**VDE AR E 2418 3 100 draft E** Latest draft (as of 16 Oct 2018) of the system requirements from the German authorities (PTB) used for requirements on the DC Meter.
R2	**BSI TR-03116-3** Technical Guideline BSI TR-03116 Cryptographic specifications for projects of the Federal government, Part 3: intelligent measuring systems. Referenced by the VDE AR E 2418 3 100 draft E document.
R3	**AIS 20/31** A proposal for: Functionality classes for random number generators.
R4	**ISO 18031** Information technology -- Security techniques -- Random bit generation.
R5	**NIST SP 800-90A** NIST Special Publication 800-90A Revision 1 Recommendation for Random Number Generation Using Deterministic Random Bit Generators

4. Requirements Traceability Matrix

General Requirements

VDE Requirement ID	Software Requirement Specification	Software/Firmware Design
LME001		

Key	Summary	Description
TDM-241	Display primitive error if meter knows it has failed unrecoverably	display "Err" if there is a fault with the meter. Tritium can log in to find details of the error or via technician access if available
TDM-243	session authentication id	display the unique id generated by the dc meter that identifies this charge session. This will be sent to Transparenz so that it can be mapped to the user.This unique id must also be shared with the charger so it can be used in the billing process.The session id will be created upon trigger of 'start-of-charge'. This could be current flow or an interrupt from the charger.
TDM-239	Charge time duration	Time in seconds for the charge duration. Must be with 1% of nominal value. Duration will be displayed in the format: [h]hh:mm:ss
TDM-242	comms between display and meter security status	display "SEC" if the encrypted channel is valid, "XXX" if there is an issue.
TDM-240	Meter firmware hash	Unique identifier of this release of software, which will be a 16 byte SHA2 hash of the binary. This hash can be stored at the very beginning of the binary and ignored when performing the hash calculation.
TDM-238	Meter Time Health Status	Display next to the meter time either as a symbol of confidence of accuracy of the meter clock: 1. successfully synchronised (within time period) 2. unsynchroised (hasn't sync'd since max sync duration (see config)) 3. invalid (time was sync'd but is out by 3%) 4. erred (time deviation >= 75 minutes)
TDM-231	Communications between DC Meter and remote display panel (IONITY Demo)	Ethernet to SPI bus to display module, with encryption, dedicated key, ack from display to check comms status, allows display to report errors to meter to forward to Tritium and audit log.To reduce work on FPGA, send a series of fixed-width strings always in the same order. FPGA knows how and where to display each string. Messages to the display are sent every second to keep the kWh view up to date.
TDM-237	Meter Time	1. Display local time. This means including DST and Timezone values. 2. The Time displayed SHALL update every second 3. Time SHALL be sent to the display as a string 4. Time SHALL be displayed in the format yyyy-mm-ddTHH:mm:ss z/+hh:mm
TDM-235	Energy usage for charge session	1. Displayed in kWh (Appendix B suggests could optionally use MWh but [ERA012] says must be kWh) 2. The format of the usage will be: `[0-9]{7}.[0-9]{2}` e.g. 0000000.00
TDM-167	Event Log	Provides persistent Logging functions for Meter.
TDM-236	Active Tariff	Tariffs can be used in multiple ways. Seasonal, time-of-day, or a combination. When not charging display the active tariff. When charging display the tariffs used so far during the charge period.

11 issues

LME002

Key	Summary	Description
TDM-241	Display primitive error if meter knows it has failed unrecoverably	display "Err" if there is a fault with the meter. Tritium can log in to find details of the error or via technician access if available
TDM-243	session authentication id	display the unique id generated by the dc meter that identifies this charge session. This will be sent to Transparenz so that it can be mapped to the user.This unique id must also be shared with the charger so it can be used in the billing process.The session id will be created upon trigger of 'start-of-charge'. This could be current flow or an interrupt from the charger.
TDM-239	Charge time duration	Time in seconds for the charge duration. Must be with 1% of nominal value. Duration will be displayed in the format: [h]hh:mm:ss
TDM-242	comms between display and meter security status	display "SEC" if the encrypted channel is valid, "XXX" if there is an issue.
TDM-240	Meter firmware hash	Unique identifier of this release of software, which will be a 16 byte SHA2 hash of the binary. This hash can be stored at the very beginning of the binary and ignored when performing the hash calculation.
TDM-238	Meter Time Health Status	Display next to the meter time either as a symbol of confidence of accuracy of the meter clock: 1. successfully synchronised (within time period) 2. unsynchroised (hasn't sync'd since max sync duration (see config)) 3. invalid (time was sync'd but is out by 3%) 4. erred (time deviation >= 75 minutes)
TDM-231	Communications between DC Meter and remote display panel (IONITY Demo)	Ethernet to SPI bus to display module, with encryption, dedicated key, ack from display to check comms status, allows display to report errors to meter to forward to Tritium and audit log.To reduce work on FPGA, send a series of fixed-width strings always in the same order. FPGA knows how and where to display each string. Messages to the display are sent every second to keep the kWh view up to date.
TDM-237	Meter Time	1. Display local time. This means including DST and Timezone values. 2. The Time displayed SHALL update every second 3. Time SHALL be sent to the display as a string 4. Time SHALL be displayed in the format yyyy-mm-ddTHH:mm:ss z/+hh:mm
TDM-235	Energy usage for charge session	1. Displayed in kWh (Appendix B suggests could optionally use MWh but [ERA012] says must be kWh) 2. The format of the usage will be: `[0-9]{7}.[0-9]{2}` e.g. 0000000.00
TDM-167	Event Log	Provides persistent Logging functions for Meter.
TDM-236	Active Tariff	Tariffs can be used in multiple ways. Seasonal, time-of-day, or a combination. When not charging display the active tariff. When charging display the tariffs used so far during the charge period.

11 issues

LME003

Key	Summary	Description
TDM-231	Communications between DC Meter and remote display panel (IONITY Demo)	Ethernet to SPI bus to display module, with encryption, dedicated key, ack from display to check comms status, allows display to report errors to meter to forward to Tritium and audit log.To reduce work on FPGA, send a series of fixed-width strings always in the same order. FPGA knows how and where to display each string. Messages to the display are sent every second to keep the kWh view up to date.
TDM-235	Energy usage for charge session	1. Displayed in kWh (Appendix B suggests could optionally use MWh but [ERA012] says must be kWh) 2. The format of the usage will be: `[0-9]{7}.[0-9]{2}` e.g. 0000000.00
TDM-236	Active Tariff	Tariffs can be used in multiple ways. Seasonal, time-of-day, or a combination. When not charging display the active tariff. When charging display the tariffs used so far during the charge period.

3 issues

Charging point measuring device with electricity metering function

VDE Requirement ID	Tritium Design Element ID

LME010

Key	Summary	Description
TDM-186	calibration table(s)	Used to convert readings from the DC Sensor to be accurate. At the moment this is: 1. offset 2. scale factor readings will be store as: raw value, offset, scaling factor, calculated result so original reads can be re-calculated if it was later determined the offset/scaling was incorrect.

1 issue

LME011

Key	Summary	Description
TDM-186	calibration table(s)	Used to convert readings from the DC Sensor to be accurate. At the moment this is: 1. offset 2. scale factor readings will be store as: raw value, offset, scaling factor, calculated result so original reads can be re-calculated if it was later determined the offset/scaling was incorrect.

1 issue

LME012

Key	Summary	Description

No issues found

LME013	Key Summary Description No issues found
LME014	Key Summary Description No issues found
LME015	Key Summary Description No issues found
LME016	Key Summary Description No issues found
LME017	Key Summary Description No issues found

Charging Point Measuring Device With Time Measurement Function	
VDE Requirement ID	**Tritium Design Element ID**
LME020	Key Summary Description TDM-196 time duration calculation Based on lessons learned with time synchronisation, calculating duration is a combination of a start time and a monotonic clock counting forwards. If an end time is required it is calculated from start time + duration.This time duration capability can also be used to implement timeouts within the system. 1 issue
LME021	Key Summary Description TDM-312 Determine the number of reads/interval to calculate usage how many samples per second do we need to meet accuracy requirements. 1 issue

LME022	Key Summary No issues found
LME023	Key Summary No issues found
LME024	Key Summary TDM-239 Charge time duration TDM-231 Communications between DC Meter and remote display panel (IONITY Demo) 2 issues

Measured Value Acquisition Module	
VDE Requirement ID	**Tritium Design Element ID**
MWE001	Key Summary Description TDM-195 Format of recorded usage measurements NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 1 issue
MWE002	Key Summary Description TDM-195 Format of recorded usage measurements NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 1 issue

MWE003	Key / Summary / Description TDM-195 / Format of recorded usage measurements / NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 1 issue
MWE004	Key / Summary / Description TDM-862 / Decimal Fixed Point Numbers / See https://wiki.tritium.com.au/display/RND/Decimal+Fixed+Point+Numbers TDM-195 / Format of recorded usage measurements / NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 2 issues
MWE005	Key / Summary / Description TDM-195 / Format of recorded usage measurements / NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 1 issue

Measured Value Processing Module	
VDE Requirement ID	**Tritium Design Element ID**
MWV001	Key / Summary / Description TDM-195 / Format of recorded usage measurements / NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 1 issue

MWV002

Key	Summary	Description
TDM-195	Format of recorded usage measurements	NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2

1 issue

MWV003

Key	Summary	Description
TDM-313	Determine smoothing algorithm	Post processing of data to remove anomalous reads (or spikes) but still maintain accuracy requirements. Errors in calculations shall be captured and logged/reported as part of telemetry.
TDM-154	Usage Session Measurements Recording /Reporting	stored measurements will be "as-is" without calculation but will contain enough information with each read so that a calculation may be performed at any time.use common 'circular buffer' mechanism for session related measurements
TDM-312	Determine the number of reads/interval to calculate usage	how many samples per second do we need to meet accuracy requirements.
TDM-160	Find Accurate Arithmetic Library	There are requirements regarding the accuracy of any calculation, which add pressure on the FPU because it inherently has error. So to reduce issues with certification a library that is more accurate needs to be found and used. The requirements regarding accuracy are: > *[MWV004] When calculating derived measurements, DO NOT add an error contribution of more than ± 1% of the least significant used digit of the calculated value.* > *[MWV005] A set of derived measurements may not degrade the resolution of the measurements obtained from the MWE component.*
TDM-195	Format of recorded usage measurements	NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2

5 issues

MWV005

Key	Summary	Description
TDM-154	Usage Session Measurements Recording/Reporting	stored measurements will be "as-is" without calculation but will contain enough information with each read so that a calculation may be performed at any time.use common 'circular buffer' mechanism for session related measurements
TDM-160	Find Accurate Arithmetic Library	There are requirements regarding the accuracy of any calculation, which add pressure on the FPU because it inherently has error. So to reduce issues with certification a library that is more accurate needs to be found and used. The requirements regarding accuracy are: *[MWV004] When calculating derived measurements, DO NOT add an error contribution of more than ± 1% of the least significant used digit of the calculated value.* *[MWV005] A set of derived measurements may not degrade the resolution of the measurements obtained from the MWE component.*
TDM-862	Decimal Fixed Point Numbers	See https://wiki.tritium.com.au/display/RND/Decimal+Fixed+Point+Numbers

3 issues

Measured Value Memory

VDE Requirement ID	Tritium Design Element ID

STORAGE001

Key	Summary	Description
TDM-153	UID generation for charge sessions	independent id. Maps to user somehow
TDM-195	Format of recorded usage measurements	NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2

2 issues

STORAGE002

Key	Summary	Description
TDM-153	UID generation for charge sessions	independent id. Maps to user somehow
TDM-195	Format of recorded usage measurements	NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2

2 issues

STORAGE003	<table><tr><th>Key</th><th>Summary</th><th>Description</th></tr><tr><td>TDM-195</td><td>Format of recorded usage measurements</td><td>NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2</td></tr></table> 1 issue
STORAGE004	<table><tr><th>Key</th><th>Summary</th><th>Description</th></tr><tr><td>TDM-195</td><td>Format of recorded usage measurements</td><td>NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2</td></tr></table> 1 issue
STORAGE005	<table><tr><th>Key</th><th>Summary</th><th>Description</th></tr><tr><td>TDM-197</td><td>packaging stored measurements into a SAFE Initiative payload</td><td>Using OCMF instead of SML due to time constraints and licensing issues with libraries.</td></tr></table> 1 issue

Time Management Method	
VDE Requirement ID	**Tritium Design Element ID**

TIME001

Key	Summary	Description
TDM-208	last time sync time	
TDM-184	Configuration of time	Time is automatically set as soon as the DC Meter has connection to a listed clock source. The configuration includes: • details of clock sources, • timezone, • time synchronisation frequency (specified in minutes, min 2 minutes [TIME012], max value = 7days (60*24*7) [TIME013]). • max time between sync before panicking When time is displayed it has to represent timezone and daylight savings, but when time is recorded, it SHALL be UTC.
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

3 issues

TIME002

Key	Summary	Description
TDM-238	Meter Time Health Status	Display next to the meter time either as a symbol of confidence of accuracy of the meter clock: 1. successfully synchronised (within time period) 2. unsynchroised (hasn't sync'd since max sync duration (see config)) 3. invalid (time was sync'd but is out by 3%) 4. erred (time deviation >= 75 minutes)
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

2 issues

TIME003

Key	Summary	Description
TDM-209	battery/supercap voltage (RTC and device)	
TDM-184	Configuration of time	Time is automatically set as soon as the DC Meter has connection to a listed clock source. The configuration includes: • details of clock sources, • timezone, • time synchronisation frequency (specified in minutes, min 2 minutes [TIME012], max value = 7days (60*24*7) [TIME013]). • max time between sync before panicking When time is displayed it has to represent timezone and daylight savings, but when time is recorded, it SHALL be UTC.
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

3 issues

TIME004

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

TIME005

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

TIME006

Key	Summary

No issues found

TIME007

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

TIME008

Key	Summary	Description
TDM-175	check RTC value	Tests to see if RTC value is approximately accurate. There will be the ability to store the current RTC value(s) when main power disappears. When power returns the stored (and time sync hasn't happened yet) time should be older than current time. Otherwise the RTC was reset completely.
TDM-174	check RTC battery /supercap voltage	TBD if using external RTC chip (with higher accuracy).1. communicate with RTC with read/write/interrupts2. react to battery bad flag on RTC
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

3 issues

TIME009

Key	Summary	Description
TDM-174	check RTC battery /supercap voltage	TBD if using external RTC chip (with higher accuracy).1. communicate with RTC with read/write/interrupts2. react to battery bad flag on RTC
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

2 issues

TIME010

Key	Summary	Description
TDM-238	Meter Time Health Status	Display next to the meter time either as a symbol of confidence of accuracy of the meter clock: 1. successfully synchronised (within time period) 2. unsynchroised (hasn't sync'd since max sync duration (see config)) 3. invalid (time was sync'd but is out by 3%) 4. erred (time deviation >= 75 minutes)
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

2 issues

TIME011

Key	Summary	Description

No issues found

TIME012

Key	Summary	Description
TDM-184	Configuration of time	Time is automatically set as soon as the DC Meter has connection to a listed clock source. The configuration includes: • details of clock sources, • timezone, • time synchronisation frequency (specified in minutes, min 2 minutes [TIME012], max value = 7days (60*24*7) [TIME013]). • max time between sync before panicking When time is displayed it has to represent timezone and daylight savings, but when time is recorded, it SHALL be UTC.

1 issue

TIME013

Key	Summary	Description
TDM-184	Configuration of time	Time is automatically set as soon as the DC Meter has connection to a listed clock source. The configuration includes: • details of clock sources, • timezone, • time synchronisation frequency (specified in minutes, min 2 minutes [TIME012], max value = 7days (60*24*7) [TIME013]). • max time between sync before panicking When time is displayed it has to represent timezone and daylight savings, but when time is recorded, it SHALL be UTC.

1 issue

TIME014

Key	Summary	Description
TDM-184	Configuration of time	Time is automatically set as soon as the DC Meter has connection to a listed clock source. The configuration includes: • details of clock sources, • timezone, • time synchronisation frequency (specified in minutes, min 2 minutes [TIME012], max value = 7days (60*24*7) [TIME013]). • max time between sync before panicking When time is displayed it has to represent timezone and daylight savings, but when time is recorded, it SHALL be UTC.

1 issue

TIME015

Key	Summary	Description
TDM-237	Meter Time	1. Display local time. This means including DST and Timezone values. 2. The Time displayed SHALL update every second 3. Time SHALL be sent to the display as a string 4. Time SHALL be displayed in the format yyyy-mm-ddTHH:mm:ss z/+hh:mm
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

2 issues

TIME016

Key	Summary	Description
TDM-184	Configuration of time	Time is automatically set as soon as the DC Meter has connection to a listed clock source. The configuration includes: • details of clock sources, • timezone, • time synchronisation frequency (specified in minutes, min 2 minutes [TIME012], max value = 7days (60*24*7) [TIME013]). • max time between sync before panicking When time is displayed it has to represent timezone and daylight savings, but when time is recorded, it SHALL be UTC.
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two-part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)
2 issues		

TIME017	

Key	Summary	Description
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

2 issues

Time Server	
VDE Requirement ID	**Tritium Design Element ID**

LOG001

Key	Summary	Description
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

2 issues

LOG002

Key	Summary	Description
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record

1 issue

LOG003

Key	Summary	Description
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record

1 issue

LOG004

Key	Summary	Description
TDM-155	Event Log Recording /Reporting	use common 'circular buffer' mechanism. The size of the event log must be large enough to store (8 + 2) years of events

1 issue

LOG005

Key	Summary	Description
TDM-230	Technician Access (VDE Demo)	A technician will require access to the DC meter. The DC meter has an Ethernet connection and is already required to have TLS 1.2 client functionality. • Implement TLS 1.2 server with required client side certificate authorisation. • Client will present to the server a certificate to prove it is approved to access the server. ~~Requires confirmation from James on direction of product. Need to know if we are expected to store cost on tariffs as a technician role could be to update tariff info. Technicians may perform a test locally on the meter via PC /Laptop login over local network to verify measurement accuracy with varying loads, so a live feed of usage (updating every 5 seconds???).~~

1 issue

LOG006

Key	Summary	Description
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record

1 issue

LOG007

Key	Summary	Description
TDM-155	Event Log Recording /Reporting	use common 'circular buffer' mechanism. The size of the event log must be large enough to store (8 + 2) years of events

1 issue

LOG008

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG009

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG010

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG011

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG012

Key	Summary	Description

No issues found

LOG013

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG014

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG015

Key	Summary	Description
TDM-183	DC Meter Run Time Self Test	Slow periodic tests performed every 6 hours. Continuous tests performed at variable rates depending on test (~30sec) Self test performed by supervisory state machine whilst not charging. Results of all tests need to be logged to the event log. Tests not presently contained in this task: • Check communications interfaces • MQTT connection • NTP connection • TSO connection • AFE comms • Flash comms Bootloader image integrity check (SHA256) is completed before jumping to it upon a FW update request
TDM-148	DC Meter Power On Self Test	Power up self-test run by supervisory state machine. Either runs or ensures result was valid of tests. Results of all tests need to be logged to the event log. • User image integrity check (SHA256) is completed by bootloader before jumping to it (Otherwise a FW update is attempted or BL holds operation of device)
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

3 issues

LOG016

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG017

Key	Summary	Description
TDM-223	statistics	1. max size (bytes) 2. max size (records) 3. used (bytes and records, as a %) 4. bad record count (fails hash test) 5. first record timestamp 6. last record timestamp 7. Thanks to WELMEC7.2:P7, may need to record/report a counter for number of times any configuration field was changed
TDM-226	The circular buffer shall provide notification events	notify listeners of events like: 1. size usage (75%, 80%, 90%, 100%) 2. reset/empty/delete/overwrite
TDM-155	Event Log Recording/Reporting	use common 'circular buffer' mechanism. The size of the event log must be large enough to store (8 + 2) years of events

3 issues

LOG018

Key	Summary	Description
TDM-234	share random user id for charge session with charger/CPO	
TDM-223	statistics	1. max size (bytes) 2. max size (records) 3. used (bytes and records, as a %) 4. bad record count (fails hash test) 5. first record timestamp 6. last record timestamp 7. Thanks to WELMEC7.2:P7, may need to record/report a counter for number of times any configuration field was changed
TDM-167	Event Log	Provides persistent Logging functions for Meter.

3 issues

LOG019

Key	Summary	Description
TDM-223	statistics	1. max size (bytes) 2. max size (records) 3. used (bytes and records, as a %) 4. bad record count (fails hash test) 5. first record timestamp 6. last record timestamp 7. Thanks to WELMEC7.2:P7, may need to record/report a counter for number of times any configuration field was changed

1 issue

LOG020

Key	Summary	Description
TDM-167	Event Log	Provides persistent Logging functions for Meter.

1 issue

LOG021

Key	Summary	Description
TDM-222	The circular buffer can be exported	export the whole buffer, using encryption, to an external interface. A separate flag will indicate if deletion of the contents is to happen after the export. Some kind of handshake is required to ensure what was exported is valid /readable before deleting. Exporting creates an entry in the event log. Exported data must contain a header which includes: 1. meter unique identifier 2. timestamp of the export (according to the meter) 3. name of the buffer (e.g. measurements, events, etc) Once data has been exported, it is marked as exported, which allows it to be automatically overwritten with newer records when space requires. Data can still be exported if marked for deletion.

1 issue

LOG022

Key	Summary	Description
TDM-222	The circular buffer can be exported	export the whole buffer, using encryption, to an external interface. A separate flag will indicate if deletion of the contents is to happen after the export. Some kind of handshake is required to ensure what was exported is valid /readable before deleting. Exporting creates an entry in the event log. Exported data must contain a header which includes: 1. meter unique identifier 2. timestamp of the export (according to the meter) 3. name of the buffer (e.g. measurements, events, etc) Once data has been exported, it is marked as exported, which allows it to be automatically overwritten with newer records when space requires. Data can still be exported if marked for deletion.

1 issue

LOG023

Key	Summary	Description
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record

1 issue

LOG024

Key	Summary	Description
TDM-225	record hashing the buffer internals (pointers, stats, etc)	To help detect if the buffer management becomes corrupt, we hash the internal variables used for management of records.
TDM-224	record hashing	Each record contains it's own hash to detect corruption. The hash includes all data fields.
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record

3 issues

LOG025

Key	Summary	Description

No issues found

LOG026

Key	Summary	Description
TDM-222	The circular buffer can be exported	export the whole buffer, using encryption, to an external interface. A separate flag will indicate if deletion of the contents is to happen after the export. Some kind of handshake is required to ensure what was exported is valid /readable before deleting. Exporting creates an entry in the event log. Exported data must contain a header which includes: 1. meter unique identifier 2. timestamp of the export (according to the meter) 3. name of the buffer (e.g. measurements, events, etc) Once data has been exported, it is marked as exported, which allows it to be automatically overwritten with newer records when space requires. Data can still be exported if marked for deletion.

1 issue

LOG027

Key	Summary	Description
TDM-230	Technician Access (VDE Demo)	A technician will require access to the DC meter. The DC meter has an Ethernet connection and is already required to have TLS 1.2 client functionality. • Implement TLS 1.2 server with required client side certificate authorisation. • Client will present to the server a certificate to prove it is approved to access the server. ~~Requires confirmation from James on direction of product. Need to know if we are expected to store cost on tariffs as a technician role could be to update tariff info. Technicians may perform a test locally on the meter via PC/Laptop login over local network to verify measurement accuracy with varying loads, so a live feed of usage (updating every 5 seconds???).~~
TDM-245	Technician Access messages to export/report parts of data	The private keys shall never be extracted or viewed. Public keys can be extracted or viewed if permissions permit. The state of the meter can be viewed. The CPO can export data. Technicians with the appropriate key can export measurements and event logs. Update live values to the Technician Access terminal
TDM-222	The circular buffer can be exported	export the whole buffer, using encryption, to an external interface. A separate flag will indicate if deletion of the contents is to happen after the export. Some kind of handshake is required to ensure what was exported is valid /readable before deleting. Exporting creates an entry in the event log. Exported data must contain a header which includes: 1. meter unique identifier 2. timestamp of the export (according to the meter) 3. name of the buffer (e.g. measurements, events, etc) Once data has been exported, it is marked as exported, which allows it to be automatically overwritten with newer records when space requires. Data can still be exported if marked for deletion.

3 issues

LOG029

Key	Summary	Description
TDM-155	Event Log Recording /Reporting	use common 'circular buffer' mechanism. The size of the event log must be large enough to store (8 + 2) years of events
TDM-222	The circular buffer can be exported	export the whole buffer, using encryption, to an external interface. A separate flag will indicate if deletion of the contents is to happen after the export. Some kind of handshake is required to ensure what was exported is valid /readable before deleting. Exporting creates an entry in the event log. Exported data must contain a header which includes: 1. meter unique identifier 2. timestamp of the export (according to the meter) 3. name of the buffer (e.g. measurements, events, etc) Once data has been exported, it is marked as exported, which allows it to be automatically overwritten with newer records when space requires. Data can still be exported if marked for deletion.

2 issues

LOG030	

Key	Summary	Description
TDM-222	The circular buffer can be exported	export the whole buffer, using encryption, to an external interface. A separate flag will indicate if deletion of the contents is to happen after the export. Some kind of handshake is required to ensure what was exported is valid /readable before deleting. Exporting creates an entry in the event log. Exported data must contain a header which includes: 1. meter unique identifier 2. timestamp of the export (according to the meter) 3. name of the buffer (e.g. measurements, events, etc) Once data has been exported, it is marked as exported, which allows it to be automatically overwritten with newer records when space requires. Data can still be exported if marked for deletion.

1 issue

Firmware Management Module	
VDE Requirement ID	**Tritium Design Element ID**
FWM001	

Key	Summary	Description
TDM-212	encrypted comms	~~Communicates to the Tritium server's to download updates~~
TDM-217	validate upgrade process and start metering software	no comms to charger means charger can't charge. Once metering software started, comms to charger recommences and a charge can occur.
TDM-216	validate upgrade binary	different parts of the meter can be upgraded (metering/event log can't be):1. firmware, 2. calibration tables, 3. configuration
TDM-215	validate binary provider	Only a trusted source can provide a binary upgrade. There needs to be a mechanism to authenticate the binary against the source of the binary, which is separate to the encryption of the binary.

4 issues

FWM002	

Key	Summary	Description
TDM-212	encrypted comms	~~Communicates to the Tritium server's to download updates~~
TDM-217	validate upgrade process and start metering software	no comms to charger means charger can't charge. Once metering software started, comms to charger recommences and a charge can occur.
TDM-216	validate upgrade binary	different parts of the meter can be upgraded (metering/event log can't be):1. firmware, 2. calibration tables, 3. configuration
TDM-215	validate binary provider	Only a trusted source can provide a binary upgrade. There needs to be a mechanism to authenticate the binary against the source of the binary, which is separate to the encryption of the binary.

4 issues

FWM003	

Key	Summary	Description

No issues found

FWM004

Key	Summary	Description
TDM-218	during upgrade, the display will show progress and upgrade version	requires code to communicate with the display (which should already be done). Create message to display regarding the upgrade process.
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. **REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS**

2 issues

FWM005

Key	Summary

No issues found

FWM006

Key	Summary	Description
TDM-217	validate upgrade process and start metering software	no comms to charger means charger can't charge. Once metering software started, comms to charger recommences and a charge can occur.

1 issue

FWM007

Key	Summary	Description
TDM-158	Upgrading (Bootloader)	Always runs on start up.Checks if there is a binary to install and performs necessary steps to install it safely. Always checks the validity of the software it is about to start.existing measurements/audit trail/etc must remain intact after an upgrade

1 issue

Signature Module	
VDE Requirement ID	**Tritium Design Element ID**

SIGN001

Key	Summary	Description
TDM-191	find/search for keys	to be used internally by the metering app. Private key values can't be displayed, exported, or printed.1. find/search for key by privilege/functionality2. find/search for key by user (a.k.a. public key)3. find/search for key by flag (active /invalidated/unused/private/public)4. find/search for key by id5. find/search for key based on key
TDM-185	Configuration of security key (s)	This task only addresses the basic get/set functionality. Further management needs are defined in task "Security Keys Management"

2 issues

SIGN002

Key	Summary	Description

No issues found

SIGN003

Key	Summary	Description
TDM-173	Prove random number generator	NIST have test code that can provide quality assurance on results from a random number generator. This is a combination of getting test code running and generating numbers from the hardware generator. This is an R&D task as the process for converting bits from the hardware RBG may require tuning to pass tests. 1. Download code 2. Get it to compile on Linux and run provided sample set to verify it works 3. Get it to run on target 4. Measure performance to see if we can run on target as part of run time validation (or a subset of)

1 issue

SIGN004

Key	Summary	Description

No issues found

SIGN005

Key	Summary	Description
TDM-189	unique identification	Ensure adding new keys doesn't result in a duplication of id.Removing/replacing keys deletes an id and allows it to be used again.Search algorithm to detect next available id
TDM-185	Configuration of security key(s)	This task only addresses the basic get/set functionality. Further management needs are defined in task "Security Keys Management"

2 issues

SIGN006

Key	Summary	Description
TDM-191	find/search for keys	to be used internally by the metering app. Private key values can't be displayed, exported, or printed.1. find/search for key by privilege/functionality2. find/search for key by user (a.k.a. public key)3. find/search for key by flag (active /invalidated/unused/private/public)4. find/search for key by id5. find/search for key based on key

1 issue

SIGN007

Key	Summary

No issues found

SIGN008	Key Summary No issues found
SIGN009	Key Summary No issues found
SIGN010	Key Summary Description TDM-192 invalidate key (s) by id Keeping a list of previously used pairs so they can't be reactivated by programming as if they are new TDM-191 find/search for keys to be used internally by the metering app. Private key values can't be displayed, exported, or printed.1. find/search for key by privilege/functionality2. find/search for key by user (a.k.a. public key)3. find/search for key by flag (active /invalidated/unused/private/public)4. find/search for key by id5. find/search for key based on key 2 issues
SIGN011	Key Summary Description No issues found

Verification Module	
VDE Requirement ID	**Tritium Design Element ID**
VERIFY001	Key Summary No issues found
VERIFY002	Key Summary No issues found

VERIFY003

Key	Summary	Description
TDM-172	add security library (mbed TLS) to the meter	ECDH, TLS, X.509, entropy management, etc. https://tls.mbed.org This library is capable of doing everything we want with regards to security signing, authentication, HMAC, encryption, random bit generator/entropy. However, we need to port it to the target. Some parts of the library need us to create code to fill in stubs and provide the missing functionality. We also need to performance test to see if we can still operate fast enough.

1 issue

VERIFY004

Key	Summary	Description
TDM-172	add security library (mbed TLS) to the meter	ECDH, TLS, X.509, entropy management, etc. https://tls.mbed.org This library is capable of doing everything we want with regards to security signing, authentication, HMAC, encryption, random bit generator/entropy. However, we need to port it to the target. Some parts of the library need us to create code to fill in stubs and provide the missing functionality. We also need to performance test to see if we can still operate fast enough.

1 issue

VERIFY005

Key	Summary	Description
TDM-172	add security library (mbed TLS) to the meter	ECDH, TLS, X.509, entropy management, etc. https://tls.mbed.org This library is capable of doing everything we want with regards to security signing, authentication, HMAC, encryption, random bit generator/entropy. However, we need to port it to the target. Some parts of the library need us to create code to fill in stubs and provide the missing functionality. We also need to performance test to see if we can still operate fast enough.

1 issue

VERIFY006

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. **REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS**

2 issues

Verification Compliant	
VDE Requirement ID	**Tritium Design Element ID**

ERA007

Key	Summary	Description
TDM-919	Display the cumulative (for life) kW register value	
TDM-152	Cumulative tariff registers summing all charge sessions	name/identifiercumulative storage, no resetting back to 0 unless counter overflow.Protection against duplicate tariffscreate/update/deletestart time (end time assumed by next tariff)
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

3 issues

ERA008

Key	Summary	Description
TDM-919	Display the cumulative (for life) kW register value	
TDM-152	Cumulative tariff registers summing all charge sessions	name/identifiercumulative storage, no resetting back to 0 unless counter overflow.Protection against duplicate tariffscreate/update/deletestart time (end time assumed by next tariff)
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

3 issues

ERA009

Key	Summary	Description
TDM-151	Tariff settings	Seasonal and/or time of day rate changes. 1. will need the fields: yyyy/mm/dd hh:mm, name, id, $/kWh, duration (minutes) 2. limit number of tariffs to 8 3. add ability to change tariffs through the terminal Further information on types of tariffs to cater for can be found in the following links (Note: these links are in German so translation is required): • https://wiki.scn.sap.com/wiki/pages/viewpage.action?pageId=446731342 • https://help.sap.com/viewer/3e88baa8ca4e4863a6203e00deb32df2/2.3.0.1/de-DE/3ba4e45461c2136de10000000a441470.html
TDM-194	maintain a cumulative measurement registers summing current charge session for each tariff	A single charging session may cover multiple tariffs. The user needs to be aware of this during the charge and end of charge.
TDM-236	Active Tariff	Tariffs can be used in multiple ways. Seasonal, time-of-day, or a combination. When not charging display the active tariff. When charging display the tariffs used so far during the charge period.
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

4 issues

ERA010

Key	Summary	Description
TDM-151	Tariff settings	Seasonal and/or time of day rate changes. 1. will need the fields: yyyy/mm/dd hh:mm, name, id, $/kWh, duration (minutes) 2. limit number of tariffs to 8 3. add ability to change tariffs through the terminal Further information on types of tariffs to cater for can be found in the following links (Note: these links are in German so translation is required): • https://wiki.scn.sap.com/wiki/pages/viewpage.action?pageId=446731342 • https://help.sap.com/viewer/3e88baa8ca4e4863a6203e00deb32df2/2.3.0.1/de-DE/3ba4e45461c2136de10000000a441470.html
TDM-194	maintain a cumulative measurement registers summing current charge session for each tariff	A single charging session may cover multiple tariffs. The user needs to be aware of this during the charge and end of charge.
TDM-152	Cumulative tariff registers summing all charge sessions	name/identifiercumulative storage, no resetting back to 0 unless counter overflow. Protection against duplicate tariffscreate/update/deletestart time (end time assumed by next tariff)
TDM-236	Active Tariff	Tariffs can be used in multiple ways. Seasonal, time-of-day, or a combination. When not charging display the active tariff. When charging display the tariffs used so far during the charge period.
TDM-924	Can you give us an example of a charge session with multiple tariffs?	The VDE spec doesn't contain detailed description on Tariffs. There are discussions around roaming tariffs, tariffs based on time, tariffs based on usage, but nothing concrete. See: • ~~TDM-151~~ • ~~TDM-152~~ • ~~TDM-194~~ • ~~TDM-236~~
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

6 issues

ERA011

Key	Summary	Description
TDM-235	Energy usage for charge session	1. Displayed in kWh (Appendix B suggests could optionally use MWh but [ERA012] says must be kWh) 2. The format of the usage will be: `[0-9]{7}.[0-9]{2}` e.g. 0000000.00

1 issue

ERA012

Key	Summary	Description
TDM-235	Energy usage for charge session	1. Displayed in kWh (Appendix B suggests could optionally use MWh but [ERA012] says must be kWh) 2. The format of the usage will be: `[0-9]{7}.[0-9]{2}` e.g. 0000000.00
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

2 issues

ERA013

Key	Summary	Description
TDM-239	Charge time duration	Time in seconds for the charge duration. Must be with 1% of nominal value. Duration will be displayed in the format: [h]hh:mm:ss
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

2 issues

ERA014

Key	Summary	Description
TDM-237	Meter Time	1. Display local time. This means including DST and Timezone values. 2. The Time displayed SHALL update every second 3. Time SHALL be sent to the display as a string 4. Time SHALL be displayed in the format yyyy-mm-ddTHH:mm:ss z/+hh:mm
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

2 issues

ERA015

Key	Summary	Description
TDM-237	Meter Time	1. Display local time. This means including DST and Timezone values. 2. The Time displayed SHALL update every second 3. Time SHALL be sent to the display as a string 4. Time SHALL be displayed in the format yyyy-mm-ddTHH:mm:ss z/+hh:mm
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

2 issues

ERA016

Key	Summary	Description
TDM-237	Meter Time	1. Display local time. This means including DST and Timezone values. 2. The Time displayed SHALL update every second 3. Time SHALL be sent to the display as a string 4. Time SHALL be displayed in the format yyyy-mm-ddTHH:mm:ss z/+hh:mm
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

2 issues

ERA017

Key	Summary	Description
TDM-241	Display primitive error if meter knows it has failed unrecoverably	display "Err" if there is a fault with the meter. Tritium can log in to find details of the error or via technician access if available
TDM-167	Event Log	Provides persistent Logging functions for Meter.
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

3 issues

ERA018

Key	Summary	Description
TDM-242	comms between display and meter security status	display "SEC" if the encrypted channel is valid, "XXX" if there is an issue.
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. **REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS**

2 issues

ERA019

Key	Summary	Description
TDM-242	comms between display and meter security status	display "SEC" if the encrypted channel is valid, "XXX" if there is an issue.
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. **REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS**

2 issues

ERA020

Key	Summary	Description
TDM-194	maintain a cumulative measurement registers summing current charge session for each tariff	A single charging session may cover multiple tariffs. The user needs to be aware of this during the charge and end of charge.
TDM-236	Active Tariff	Tariffs can be used in multiple ways. Seasonal, time-of-day, or a combination. When not charging display the active tariff. When charging display the tariffs used so far during the charge period.

2 issues

ERA021

Key	Summary	Description
TDM-243	session authentication id	display the unique id generated by the dc meter that identifies this charge session. This will be sent to Transparenz so that it can be mapped to the user.This unique id must also be shared with the charger so it can be used in the billing process.The session id will be created upon trigger of 'start-of-charge'. This could be current flow or an interrupt from the charger.

1 issue

ERA022

Key	Summary	Description
TDM-238	Meter Time Health Status	Display next to the meter time either as a symbol of confidence of accuracy of the meter clock: 1. successfully synchronised (within time period) 2. unsynchroised (hasn't sync'd since max sync duration (see config)) 3. invalid (time was sync'd but is out by 3%) 4. erred (time deviation >= 75 minutes)
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. **REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS**

2 issues

ERA023

Key	Summary	Description
TDM-243	session authentication id	display the unique id generated by the dc meter that identifies this charge session. This will be sent to Transparenz so that it can be mapped to the user.This unique id must also be shared with the charger so it can be used in the billing process.The session id will be created upon trigger of 'start-of-charge'. This could be current flow or an interrupt from the charger.
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. **REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS**

2 issues

ERA024

Key	Summary	Description
TDM-179	check communications	
TDM-183	DC Meter Run Time Self Test	Slow periodic tests performed every 6 hours. Continuous tests performed at variable rates depending on test (~30sec) Self test performed by supervisory state machine whilst not charging. Results of all tests need to be logged to the event log. Tests not presently contained in this task: • Check communications interfaces • MQTT connection • NTP connection • TSO connection • AFE comms • Flash comms Bootloader image integrity check (SHA256) is completed before jumping to it upon a FW update request
TDM-190	different security keys for different functionality	A method to authenticate a user to a set of privileges/functionality. A separate key for has.to.be, Tritium, technician, etc.
TDM-231	Communications between DC Meter and remote display panel (IONITY Demo)	Ethernet to SPI bus to display module, with encryption, dedicated key, ack from display to check comms status, allows display to report errors to meter to forward to Tritium and audit log.To reduce work on FPGA, send a series of fixed-width strings always in the same order. FPGA knows how and where to display each string. Messages to the display are sent every second to keep the kWh view up to date.
TDM-185	Configuration of security key(s)	This task only addresses the basic get/set functionality. Further management needs are defined in task "Security Keys Management"
TDM-244	Client authorisation for access to the DC Meter	Anyone attempting access to the DC Meter as a "client" (i.e. the DC Meter is the server) to use the terminal interface of the DC Meter first is authenticated using X.509 certificates that were created by Tritium. An example is the technician must authorise themselves with the meter before seeing the menu and retrieving information. The CPO must authorise prior to updating tariffs.
TDM-973	Verification Compliancy	ERA024 - To what standard? Is TR-03107-1 an acceptable standard?

7 issues

User Interface

VDE Requirement ID	Tritium Design Element ID
UI001	Key: TDM-248 — Summary: Terminal input validation of write commands for range testing and privileges for the user — Description: 1 issue
UI002	Key — Summary — Description No issues found

Authorization Unit

VDE Requirement ID	Tritium Design Element ID
AUTHZ001	Key Summary No issues found
AUTHZ002	Key / Summary / Description TDM-195 — Format of recorded usage measurements — NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 TDM-199 — event id list 1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed) 2 issues

Authentication Unit	
VDE Requirement ID	**Tritium Design Element ID**

AUTHN001	Key Summary No issues found
AUTHN002	Key Summary No issues found
AUTHN003	Key Summary No issues found
AUTHN004	Key Summary No issues found

5. Inverse Traceability Matrix

Firmware and Software:

Key	Summary	T	Created	Updated	Due	Assignee	Reporter	P	Status	Resolution	Linked Issues

No issues found

DCM Box Testing

Following tests started from M2 hardware with firmware without calibration (30/09/2019).

Fault/Behaviour	Cause	Detailed Description	Closed
Removal of measured voltage does not stop charge session	ChargeSessionSM.cpp logic doesn't check this	Once a charge is started above current/voltage thresholds; voltage can be removed and charge session doesn't stop. Current being removed successfully stops session.	•
Enter Out of Service (OoS) mode	Underflow on power calculation	OoS cause reported as READ_ADC_FAIL. Goes OoS when no power is 0.000.	•
Energy displays ??? not value		Both debug Ws and TTI kWh display ??? instead of values. Occasionally a 'blip' will occur and what looks like power is shown for 1s.	•
TTI Energy displays Ws in a given loop not session energy	Temporary Ws shown	Needs to be total session energy	•
Doesn't support negative power. Rolls back total energy		Negative voltage doesn't latch in state machine. Energy counter rolls back when negative power is applied. Is this a problem? Does negative energy need independent counter	•
Negative current does not display correctly		AFE current shows negative but output current/debug current shows absolute.	•
Debug print of power shows VVVVV.VV when above 100kW	String print digit length	Does not have enough digits for power/Ws/voltage	•
Cable compensation voltage change can't be seen	Initialisation order	Voltage measured isn't changed by current when cable resistance is written to as expected.	•
Cable compensation seems to be based off opposite cable	Hardware abstraction opposite to labelled	Uses resistance value written to opposing cable	•
Uncertainty of sufficient energy transferred by IR LED	Non-standard test equipment	IR LED energy received is difficult to determine based on non-ideal receiver. Changed resistor to increase drive current.	•
AFE no longer reads/displays negative voltage /current	Typecast when calibrated to uint	Seems to have changed when calibration was introduced	•
Voltage calibration is the same for both voltage sense inputs		Calibration changing added to cable switching	•
Configuration isn't loaded once unit is reset		Not written and/or read in memory.	•
Temperature sensor reads false values at negative temps	Typecast to uint	Read ~6516 at negative temperatures	•
At -40'C one cable sense being shorted effects the measurement of the other (only shorting cable 2 with cable 1)	Leakage through pull down	Both sets of opto switches are turning on.	•
Bootloading resets the calibration information	Any JSON sent info writes to memory.	Changed so bootloading command does not	•

Non-issues/Accepted for first release

OoS doesn't recover		No OoS recovery mechanism yet	
Measurements displays ??? when out of range		E.g. If over 600A does not display the reading - should this be changed to maximum possible measurement rather than maximum expected? This is expected behaviour	
If no internet connection on startup cannot connect to TTI even on restore of internet		Unit powered up networked to router without external internet connectivity. Attempt to connect to TTI, fails as expected. Restore internet connectivity and attempt to connect to TTI, still fails.	
Charge start time/duration don't appear on TTI		Not yet implemented	
Whilst bootloading TTI must be closed or process crashes		Not yet fixed	

DC meter - overview of stories

All stories

Key	Summary	T	Updated	Assignee	Status	wp	Labels	Linked Issues
TDM-160	Find Accurate Arithmetic Library		Jul 01, 2019	Jeremy Kincaid	DONE	1841	accuracy investigation metering non-functional software	TDM-1921 , TDM-1083 , TDM-35 , TDM-1085
TDM-338	Draft of DC architecture spec		Mar 18, 2019	Jeremy Kincaid	DONE	1848 1850		TDM-452 , TDM-341 , TDM-340 , TDM-318 , TDM-339 , TDM-327
TDM-389	Create Functional, Environmental, Certification test plan		Oct 20, 2021	Jeremy Street-Thomas	DONE	1850 1902 1903 1904		TDM-984 , TDM-983 , TDM-452 , TDM-264 , TDM-390
TDM-387	Use existing SW tasks to publish SW architecture specifications		Mar 18, 2019	Jeremy Kincaid	DONE	1850 1902 1905		TDM-855 , TDM-459
TDM-296	Design Random Number Generator		Jul 01, 2019	Jeremy Kincaid	DONE	1850 1902 1903	Design software	TDM-1931 , TDM-385 , TDM-173 , TDM-35
TDM-388	Create specifications, test matrix & coverage status		Oct 20, 2021	Jeremy Street-Thomas	DONE	1905		TDM-452
TDM-453	Publish system requirements draft		Mar 18, 2019	Michael Walton	DONE	1905		TDM-458 , TDM-457 , TDM-455 , TDM-454 , TDM-452
TDM-896	Update planning		Mar 18, 2019	Jeremy Kincaid	DONE	1909		TDM-898 , TDM-897 , TDM-900 , TDM-899 , TDM-459
TDM-905	Certification Questions to VDE		Oct 20, 2021	Jeremy Street-Thomas	DONE	1915		TDM-975 , TDM-974 , TDM-973 , TDM-972 , TDM-971 , TDM-970 , TDM-452 , TDM-979 , TDM-978 , TDM-977 , TDM-976 , TDM-1025 , TDM-1306 , TDM-1307 , TDM-1948 , TDM-1949 , TDM-964 , TDM-963 , TDM-962 , TDM-961 , TDM-960 , TDM-927 , TDM-969 , TDM-924 , TDM-968 , TDM-923 , TDM-967 , TDM-966 , TDM-1318 , TDM-1319 , TDM-951 , TDM-950 , TDM-917 , TDM-916 , TDM-915 , TDM-959 , TDM-958 , TDM-957 , TDM-956 , TDM-954 , TDM-1001 , TDM-1320 , TDM-1321 , TDM-1004 , TDM-1368 , TDM-918 , TDM-982 , TDM-906 , TDM-949 , TDM-948 , TDM-1050 , TDM-947 , TDM-946 , TDM-945 , TDM-1334 , TDM-1016 , TDM-1019
TDM-548	DFMEA mechanical / industrial design scope		Oct 20, 2021	Bernie Walsh	DONE	1915	Main_Focus	TDM-932
TDM-530	DFMEA all electrical scope, including changes to charger bus bars		Oct 20, 2021	Jeremy Street-Thomas	DONE	1915		TDM-527
TDM-477	Publish electrical system specs draft		Oct 20, 2021	Unassigned	DONE	1915		TDM-466
TDM-467	Publish mechanical / industrial system specs draft		Apr 11, 2019	Bernie Walsh	DONE	1915		TDM-464

TDM-1314	Follow up on quoted standards		Oct 20, 2021	Jeremy Street-Thomas	DONE	1915		TDM-452
TDM-1051	[Cancelled] Release memory requirements		Nov 29, 2019	James Sneyd	DONE	1915		
TDM-1043	Release updated requirements		Oct 20, 2021	Jeremy Street-Thomas	DONE	1915		
TDM-1042	Release new PCB allowed dimensions to PCB designers		Nov 12, 2019	Oliver Gruber	DONE	1915		TDM-1041
TDM-1007	Review heat sink requirements for dispenser integration		Aug 12, 2019	Ben Guymer	DONE	1915		TDM-464 , TDM-1008
TDM-990	Update documentation FPGA		Apr 11, 2019	Alex Rudzki	DONE	1915		TDM-452 , TDM-913 , TDM-922
TDM-2198	General code reviews		Oct 20, 2021	James Sneyd	DONE			TDM-2199
TDM-2194	FPGA encryption		Mar 03, 2020	Alex Rudzki	DONE			TDM-2195 , TDM-2196 , TDM-2177
TDM-2193	Backlight display monitoring		Mar 03, 2020	Alex Rudzki	DONE			TDM-2201 , TDM-2200 , TDM-2177
TDM-2190	Automate Interceptor IP configuration		Dec 18, 2019	Alex Rudzki	DONE			TDM-2191 , TDM-2192 , TDM-2177
TDM-2178	Update pixel intercept module for PK		Dec 18, 2019	Alex Rudzki	DONE			TDM-2179 , TDM-2180 , TDM-2182 , TDM-2181 , TDM-2184 , TDM-2185 , TDM-2188 , TDM-2177
TDM-2058	FreeRTOS missing interrupts (ADS1220)		Nov 20, 2019	James Sneyd	DONE			
TDM-2055	Network-Attached Debug Printing		Nov 20, 2019	James Sneyd	DONE			TDM-936 , TDM-1915
TDM-1496	Revise accuracy calculations		Nov 20, 2019	Jeremy Street-Thomas	DONE			TDM-1497 , TDM-938
TDM-401	Draft NPI plan for DC meter project		Oct 20, 2021	Timo Bellgardt	DONE			
TDM-397	Create draft WBS and prepare		Mar 18, 2019	Alex Rudzki	DONE			TDM-452 , TDM-308
TDM-396	Create style guide for the Terminal		Oct 20, 2021	James Sneyd	DONE		software user-interface	TDM-1915
TDM-378	Networking /Comms Stack Integration: Explore Alternative RTOS		Nov 12, 2019	James Sneyd	DONE		Firmware	TDM-1028
TDM-372	Networking /Comms Stack Integration: lwIP Application		Nov 12, 2019	James Sneyd	DONE			TDM-931 , TDM-366 , TDM-377 , TDM-365 , TDM-375 , TDM-374 , TDM-1933 , TDM-373

TDM-366	Networking /Comms Stack Integration: mbedTLS Application		Nov 12, 2019	James Sneyd	DONE	Firmware	TDM-346 , TDM-1028 , TDM-230 , TDM-372 , TDM-371 , TDM-1914
TDM-365	Networking /Comms Stack Integration: FreeRTOS		Nov 12, 2019	James Sneyd	DONE	Firmware	TDM-369 , TDM-368 , TDM-367 , TDM-1028 , TDM-1933 , TDM-372 , TDM-370
TDM-364	Networking /Communicatio n Stack Integration Concept		Nov 12, 2019	James Sneyd	DONE		TDM-931 , TDM-1933
TDM-1466	Documentation M2 FW		Nov 29, 2019	James Sneyd	DONE		TDM-1469
TDM-1425	Document DC Meter software build process		Oct 20, 2021	James Sneyd	DONE	documentat ion process software	
TDM-1365	Design HW Controller		Nov 12, 2019	Richard Kosik	DONE		TDM-938
TDM-1364	Design HW AFE		Nov 20, 2019	Alex Rudzki	DONE		TDM-1464 , TDM-1366 , TDM-1465 , TDM-938 , TDM-1460 , TDM-1462
TDM-1363	Design HW FPGA		Nov 12, 2019	Vivien Wong	DONE		TDM-1367 , TDM-929 , TDM-938
TDM-1357	FPGA task planning		Nov 20, 2019	Alex Rudzki	DONE		TDM-1358 , TDM-452
TDM-1354	Verify FPGA operation		Oct 20, 2021	Alex Rudzki	DONE		TDM-1353 , TDM-1355 , TDM-1356
TDM-1337	Develop FPGA - Firmware		Oct 20, 2021	Jakub Tkaczyk	DONE		TDM-1342 , TDM-986 , TDM-1343 , TDM-61 , TDM-1344 , TDM-1356 , TDM-63 , TDM-62 , TDM-64 , TDM-1341 , TDM-1352
TDM-1336	Develop FPGA - Pixel Intercept module		Apr 17, 2020	Alex Rudzki	DONE		TDM-986 , TDM-2083 , TDM-2095 , TDM-2098 , TDM-1350 , TDM-1340 , TDM-1351 , TDM-2166 , TDM-2132 , TDM-2165 , TDM-2134 , TDM-1355 , TDM-60 , TDM-1840 , TDM-65 , TDM-1338 , TDM-1339
TDM-2075	FPGA interfacing		Nov 21, 2019	Alex Rudzki	DONE		TDM-986 , TDM-2076
TDM-7	Technical research		Mar 18, 2019	Alex Rudzki	DONE		TDM-325 , TDM-321 , TDM-376 , TDM-452 , TDM-361
TDM-354	lwIP Integration (No Operating System)		Jul 01, 2019	James Sneyd	DONE		TDM-359 , TDM-358 , TDM-357 , TDM-334 , TDM-356 , TDM-355 , TDM-1933 , TDM-35
TDM-329	Evaluate CAN bus message fragmentation (No Operating System)		Jul 01, 2019	James Sneyd	DONE		TDM-1908 , TDM-359 , TDM-333 , TDM-332 , TDM-331 , TDM-330 , TDM-35
TDM-310	Update/refine /test the ADC subsystem M1		Nov 25, 2019	James Sneyd	DONE		TDM-931 , TDM-1920 , TDM-24
TDM-187	configuration updates (or parts thereof)		Nov 12, 2019	Andy Meldrum	DONE	DC-Meter-Conf	TDM-1901 , TDM-936 , TDM-1915
TDM-183	DC Meter Run Time Self Test		Jul 01, 2020	Alex Rudzki	DONE	functional software	TDM-148 , TDM-2292 , TDM-1180 , TDM-2294 , TDM-2311 , TDM-2310 , TDM-2313 , TDM-1124 , TDM-2401 , TDM-2312 , TDM-2315 , TDM-2337 , TDM-2314 , TDM-2317 , TDM-76 , TDM-2316 , TDM-2308 , TDM-2319 , TDM-2318 , TDM-2309 , TDM-1904
TDM-169	SW Documentation X1		Mar 18, 2019	Jeremy Kincaid	DONE		TDM-35
TDM-168	Charger Software		Oct 20, 2021	Unassigned	DONE		TDM-1917
TDM-167	Event Log		Nov 12, 2019	Andy Meldrum	DONE		TDM-931 , TDM-1127 , TDM-1129 , TDM-1923 , TDM-1060 , TDM-1173

TDM-166	Test Environment X1		Mar 18, 2019	Jeremy Kincaid	DONE		TDM-35
TDM-165	Menu for technician		Oct 20, 2021	James Sneyd	DONE	SDD UI adaptation software status-reporting	
TDM-164	Displaying meter and charge information		Nov 26, 2019	James Sneyd	DONE		TDM-1908 , TDM-1004
TDM-163	Voltage Supervisor		Jul 12, 2021	James Sneyd	DONE		TDM-1907
TDM-162	User access /permissions scheme		Apr 17, 2020	Alex Rudzki	DONE		TDM-2266 , TDM-1925 , TDM-150 , TDM-190
TDM-161	Design Communications SW		Oct 20, 2021	James Sneyd	DONE		
TDM-159	Circular buffer mechanism X1		Jul 01, 2019	Jeremy Kincaid	DONE		TDM-35 , TDM-1937
TDM-158	Upgrading (Bootloader)		Oct 20, 2021	James Sneyd	DONE		TDM-1145 , TDM-2056
TDM-157	LED toggling (if no display)		Jul 01, 2019	James Sneyd	DONE	systems-engineering	TDM-980
TDM-156	Create a common set of status information for basic diagnostics		Oct 20, 2021	James Sneyd	DONE		TDM-1028 , TDM-1915
TDM-155	Event Log Recording /Reporting		Oct 20, 2021	James Sneyd	DONE	record-report	TDM-1113 , TDM-1126 , TDM-1137 , TDM-1116 , TDM-1924
TDM-154	Usage Session Measurements Recording /Reporting		Oct 20, 2021	James Sneyd	DONE	record-report	TDM-1083 , TDM-1085
TDM-153	UID generation for charge sessions		Oct 20, 2021	James Sneyd	DONE	record-report	TDM-1921 , TDM-1086 , TDM-1087
TDM-152	Cumulative tariff registers summing all charge sessions		Jul 01, 2019	Andy Meldrum	DONE	record-report	TDM-1166 , TDM-1921 , TDM-1945 , TDM-35 , TDM-1163 , TDM-1164
TDM-151	Tariff settings		Jul 01, 2019	Andy Meldrum	DONE	SDD tariff	TDM-1166 , TDM-561 , TDM-1945 , TDM-35 , TDM-1165
TDM-150	Security Keys Management		Oct 20, 2021	James Sneyd	DONE		TDM-162 , TDM-1929
TDM-149	Develop SW for setting and retrival of DC Meter Configuration		Oct 20, 2021	James Sneyd	DONE	SDD adaptation functional software	TDM-1901
TDM-148	DC Meter Power On Self Test		Oct 20, 2021	Alex Rudzki	DONE	software	TDM-1124 , TDM-2315 , TDM-2337 , TDM-76 , TDM-2316 , TDM-2319 , TDM-2292 , TDM-2318 , TDM-2294 , TDM-1903 , TDM-183 , TDM-2320
TDM-147	De-Risking activities		Mar 29, 2019	Jeremy Kincaid	DONE		TDM-35 , TDM-171 , TDM-170
TDM-92	X1 Design verification HW display interceptor		Apr 18, 2019	Alex Rudzki	DONE		TDM-998
TDM-81	Backend integration on controller		Oct 20, 2021	James Sneyd	DONE		TDM-1915

TDM-80	NTP packet forwarding for ethernet over CAN		Apr 26, 2019	James Sneyd	DONE		
TDM-72	Revisit AFE ovenisation control		Nov 08, 2019	Alex Rudzki	DONE		TDM-980
TDM-34	X1 Design verification HW controller		Apr 18, 2019	Alex Rudzki	DONE		TDM-998
TDM-33	X1 Design verification HW high current sensor		Apr 18, 2019	Alex Rudzki	DONE		TDM-998
TDM-32	Design voltage sense point enclosures		Nov 12, 2019	Bernie Walsh	DONE	Input_required	TDM-932
TDM-31	Design meter main enclosure		Mar 18, 2019	Bernie Walsh	DONE		TDM-489 , TDM-488 , TDM-487 , TDM-486 , TDM-29
TDM-30	Shunt resistor PCB mounting bracket		Apr 10, 2019	Bernie Walsh	DONE		TDM-29
TDM-27	Develop FPGA - CPU and Comms module		Nov 26, 2019	Jakub Tkaczyk	DONE		TDM-986 , TDM-353 , TDM-66 , TDM-2131 , TDM-59 , TDM-1350 , TDM-2133 , TDM-1344 , TDM-1355 , TDM-1345 , TDM-1346 , TDM-1347 , TDM-1348 , TDM-1349 , TDM-2109
TDM-26	Design FW for controller X1		Apr 10, 2019	Jeremy Kincaid	DONE		TDM-35
TDM-24	Design Controller		Apr 01, 2019	Jeremy Street-Thomas	DONE		TDM-315 , TDM-314 , TDM-303 , TDM-940 , TDM-311 , TDM-322 , TDM-310 , TDM-305
TDM-23	Design HW AFE		Mar 18, 2019	Alex Rudzki	DONE		TDM-50 , TDM-907 , TDM-44 , TDM-904 , TDM-49 , TDM-305
TDM-294	X1 Design HW FPGA		Mar 29, 2019	Vivien Wong	DONE		TDM-326 , TDM-348 , TDM-347 , TDM-985 , TDM-324 , TDM-323 , TDM-344 , TDM-342 , TDM-56 , TDM-284 , TDM-283 , TDM-272
TDM-272	Bring up current TRI105 HMI on Apalis module with FPGA overlay		Oct 20, 2021	Jakub Tkaczyk	DONE		TDM-2059 , TDM-286 , TDM-294
TDM-882	Working With Code Composer Studio		Mar 18, 2019	Andy Meldrum	DONE		TDM-459
TDM-862	Decimal Fixed Point Numbers		Jul 10, 2019	Andy Meldrum	DONE		TDM-1079 , TDM-1921 , TDM-35 , TDM-1085
TDM-854	Publish SW requirements		Mar 18, 2019	Jeremy Kincaid	DONE		TDM-560 , TDM-559 , TDM-459
TDM-607	Upload final BOM		Oct 20, 2021	Timo Bellgardt	DONE		TDM-562
TDM-606	Preliminary process capability study plan		Oct 20, 2021	Dion Schulz	DONE		TDM-562
TDM-605	Measurement system analysis plan		Oct 20, 2021	Dion Schulz	DONE		TDM-562
TDM-604	Process sheets		Oct 20, 2021	Timo Bellgardt	DONE		TDM-562
TDM-603	Pre-launch control plan		Oct 20, 2021	Dion Schulz	DONE		TDM-562
TDM-598	PFMEA		Oct 20, 2021	Timo Bellgardt	DONE		TDM-562
TDM-593	Characteristics matrix		Oct 20, 2021	Harry Watson	DONE		TDM-562

TDM-588	Floor plan layout		Oct 20, 2021	Timo Bellgardt	DONE		TDM-562
TDM-583	Process flow chart		Oct 20, 2021	Timo Bellgardt	DONE		TDM-562
TDM-578	Product / process qualiy system review		Oct 20, 2021	Dion Schulz	DONE		TDM-562
TDM-573	Packaging standards and specifications		Oct 20, 2021	Harry Watson	DONE		TDM-562
TDM-568	Develop DFM strategy		Oct 20, 2021	Timo Bellgardt	DONE		TDM-562
TDM-566	Develop manufacturing plan / design production line		Oct 20, 2021	Timo Bellgardt	DONE		TDM-567 , TDM-562
TDM-563	Develop Quality plan		Oct 20, 2021	Dion Schulz	DONE		TDM-565 , TDM-452
TDM-529	Design busbar changes		Mar 18, 2019	Bernie Walsh	DONE		TDM-535 , TDM-538 , TDM-537 , TDM-29 , TDM-536
TDM-528	Design harnesses		Oct 20, 2021	Cameron Swales	DONE		TDM-934
TDM-472	Publish FW / HW system specs draft		Nov 11, 2019	Jeremy Street-Thomas	DONE		TDM-476 , TDM-465 , TDM-475 , TDM-474 , TDM-473
TDM-450	Configurable language pack for setting and retrival of DC meter Configuration		Oct 20, 2021	James Sneyd	DONE	SDD UI adaptation	TDM-1901 , TDM-449
TDM-1322	Design process for handling of cryptographic keys		Oct 20, 2021	Jeremy Street-Thomas	DONE		TDM-562
TDM-1316	IONITY		Oct 20, 2021	Unassigned	DONE		TDM-1315 , TDM-1317
TDM-1052	[Cancelled] Review cooling options		Nov 29, 2019	Michael Walton	DONE		TDM-932
TDM-1049	Update specifications: need to monitor shunt temperature		Nov 12, 2019	Jeremy Street-Thomas	DONE		TDM-932
TDM-1048	Report ISD DC meter shunt cooling options to Bernie		Nov 12, 2019	Bernie Walsh	DONE		TDM-932
TDM-1035	Present cooling concept		Nov 12, 2019	Bernie Walsh	DONE		TDM-932
TDM-1031	Redesign meter enclosure		Nov 12, 2019	Bernie Walsh	DONE	Main_Focus	TDM-932
TDM-1024	Validate aliasing assumptions (for sigal sampling)		Oct 20, 2021	James Sneyd	DONE		TDM-1009
TDM-1023	Trial "Enterprise Architect" SW		Nov 08, 2019	Jeremy Kincaid	DONE		

TDM-1021	Do we need to be ready for dual cable charging pin voltage sensing?		May 07, 2019	Aidan Clarke	DONE	TDM-1020
TDM-1017	Component-level validation of TRI110-04v1 FPGA Interceptor PCB X1		Nov 20, 2019	Vivien Wong	DONE	TDM-1018 , TDM-998
TDM-1014	Fail to select appropriate RTOS		Nov 20, 2019	James Sneyd	DONE	TDM-1015 , TDM-1009
TDM-1011	FPIO ethernet comms is disturbed by electrical noise		Nov 14, 2019	Nicholas Keeling	DONE	TDM-1013 , TDM-1009
TDM-1010	CAN loading impact on meter / display / backend communcation		May 28, 2019	Dion Sumpton	DONE	TDM-1012 , TDM-1009
TDM-999	Component-level validation of TRI110-03v1 FPGA carrier PCB X1		Nov 20, 2019	Vivien Wong	DONE	TDM-1000 , TDM-998
TDM-989	Test Environment M1		Oct 20, 2021	James Sneyd	DONE	
TDM-988	SW Documentation M1		Nov 12, 2019	James Sneyd	DONE	TDM-931 , TDM-1467 , TDM-1468
TDM-981	Replace 5x gland between front and back by bulkhead connector		Oct 20, 2021	Cameron Swales	DONE	
TDM-953	Identify second source for shunt		Oct 20, 2021	Jeremy Street-Thomas	DONE	TDM-934
TDM-952	Redesign shunt mounting		Nov 12, 2019	Bernie Walsh	DONE	TDM-932
TDM-943	Design and Implement the ADC subsytem X1		Jul 01, 2019	Jeremy Kincaid	DONE	TDM-1920 , TDM-926 , TDM-35 , TDM-944 , TDM-965
TDM-942	Circular buffer mechanism M1		Oct 20, 2021	James Sneyd	DONE	
TDM-941	Design FW Controller M1		Oct 20, 2021	James Sneyd	DONE	
TDM-930	M1 Design HW FPGA		Nov 12, 2019	Vivien Wong	DONE	TDM-929
TDM-928	Design Remote Display		Nov 26, 2019	Vivien Wong	DONE	TDM-2059
TDM-920	Update documentation Meter		Mar 25, 2019	Alex Rudzki	DONE	TDM-452 , TDM-921

135 issues

List of key documents and references

Please use google to translate German material: https://translate.google.com/?client=firefox-b&um=1&ie=UTF-8&hl=en&client=tw-ob#view=home&op=translate&sl=de&tl=en

document / reference	comment	link
statement of work	written by Martin	Statement of Work - DC meter - 190219 - V1.2.docx
system requirements	written by Michael	TRI110.026v2 DC Meter system spec.docx
architecture specification	written by Deepak	
architecture specification	written by Jeremy	https://wiki.tritium.com.au/x/Owx9AQ
VDE draft standard	extracted from online tool and translated by google we have an original paper hardcopy in German	VDE AR E 2418 3 100 draft E.docx
WBS and requirements	in Jira	https://jira.tritium.com.au/projects/TDM?selectedItem=jp.ricksoft.plugins.wbsgantt-for-jira:wbsgantt-project
design folder	important subfolders: 001 engineering reference 003 block diagram 011 LCD screen integration 013 SAFE initative (transparency SW) 022 FPGA selection 024 Standards Compliance, esp. requirements matrix	Z:\Tritium\TRI110 - DC Meter\00 - Documents
description of "convenient solution"	This is an unofficially recommended solution (coming from PTB) Documents are in German and have been translated by google	E2E-Sicherheit early release rev 1 TRANSLATION.pdf.docx read me info4 rev 1 TRANSLATION.pdf.docx Z:\Tritium\TRI110 - DC Meter\00 - Documents\TRI110.025 Convenient Solution - info from Angelika and PTB 190123
Has2Be	Download beta version transparency SW (German) General info (German)	https://has-to-be.com/de/development-resources/transparenzsoftware/beta-download-transparenzsoftware/ https://has-to-be.com/de/development-resources/transparenzsoftware/
Testing labs	CSA (close to Munich) VDE (close to Frankfurt)	https://www.csagroup-bayern.org/ https://www.vde.com/tic-de
article about public key infrastructure	about Germany / in German	https://www.electrive.net/2019/01/12/bundesnetzagentur-erweitert-ladesaeulenkarte/
article about extension of deadline	about Germany / in German	https://www.electrive.net/2018/12/17/eichrecht-bei-ladeinfrastruktur-gemessener-gleichstrom/

M1 Prototype Hardware Issues

Design issues identified to be corrected:

#	Item	Issue	Remedy	Status	Completed
1	Controller PCBA	U9 (11345) has no opening on top paste.	Update footprint from library	Fixed in library. Part no longer on board	•
2	Controller PCBA	U9 (11345) is planned end of life	Replace with TC58CVG2S0HRAIG		•
3	Controller PCBA	Several single line items that don't seem critical	Reduce single line items where possible		•
4	Controller PCBA	Ringing on mosfet gates	Change DNF connecting mosfet driver ground from GND to GND_PWR (remove R82/R89 and fit R83/90)		•
5	Controller PCBA	Potential for mosfet gates to float	Add 10k pull down to mosfet gate		•
6	Controller PCBA	Fans/thermistor connector is SMD and won't be able to be picked up by ATE	Place test points on fans /thermistor lines		•
7	Controller PCBA	Fan feedback has minimum voltage of 0.3V. Inline resistor causes voltage divider.	Move pull up to connector side.		•
8	Controller PCBA	Fan feedback is being pulled up to above 3.3V when off. Fan power is trying to return through feedback.	Add optocoupler for speed using fan power		•
9	Controller PCBA	Requirement may have changed to need more memory	Add second flash part on bottom side (default DNF)		•
10	Controller PCBA	Will TPM be used?	Default is fitted.	Should be able to be removed later if not used.	•
11	Controller PCBA	External reference for ADC doesn't really provide value	Change to DNF, use 3.3V reference and change temp sensor bias resistors to 1%	Firmware must change to use 3.3V	•
12	Controller PCBA	Lower ovenisation power than expected. Track resistance is higher than expected and ovenisation voltage under load is lower.	Change transformer to 1:1.75		•
13	Controller PCBA	Cutouts for baffle have uncertain radius	Place known radius for cutout		•
14	Controller PCBA	New injected enclosure would interfere with corners of the board	Increase corner radius on board corners to avoid interference	Confirm to what size	•
15	Controller PCBA	Controller requires conformal coating for creepage	See if creepage (10mm) can be obtained to avoid this		•
16	AFE PCBA	Heat retention around AFE PCBA. Put foam inside the canister area, over the can, or over the whole board. Foam material? Silicon conformal coating?	Metal can by itself is sufficient for our needs	No action needed	•
17	AFE PCBA	Q1 (5528) footprint is quite wide	Change footprint		•
18	AFE PCBA	HV Detect circuitry can have voltage applied on other cable	Switch high voltage lines to allow two high accuracy voltage channels using ASSR-601JV-000E	Needs review	•

19	AFE PCBA	One signal across the barrier is currently unused	Now used for one wire interface		•
20	AFE PCBA	Could eliminate production needing to place formex over RTC cap	Increase cutout/copper-free distance around capacitor		•
21	AFE PCBA	Connection of temperature sensor to shunt needs to be reinforced isolation and well thermally coupled	Use one wire interface and place on AFE board.		•
22	AFE PCBA	Input power filters aren't used	Remove beads from board		•
23	Enclosure	Labelling panel isn't ideal for production solution	Design metal riveted plate on outside for required info	Confirm required information needed to be displayed	•
24	Enclosure	Inserts for board mounting need more material around them	Require more space around AFE mounting points	Need to support with minor AFE board change too.	•
25	Enclosure	Possibility for Fan wires to be pushed onto heatsink	Extend a cover over wires on the fan		•
26	FPGA Interceptor PCBA (v2)	Level shifter (TXB0106) should be replaced with an open-drain/push-pull level shifter (TXS0108) for TPM chip so that pull-up resistors can be used with it. The current level shifter (TXB0106) works for all signals except the open-drain pin PIRQ# on the TPM chip.	Unfit R63 and R65. Pin is driven by FPGA Fit R64 with 100ohm so that PIRQ# is held high. Possibly put in a buffer on PIRQ# between TPM and level shifter	Used remedy as of now.	•
27	FPGA Interceptor PCBA (v2)	EXT_CLK should be connected to "P-side" of clock capable diff pair input, (ref UG472 : page 49)	TPM_CLK_FPGA should be able to be moved to pin 'P4' as it probably doesnt need a dedicated clock pin	Presently implemented a workaround in FPGA fabric, but is sub-optimal and not recommended	•
28	FPGA Interceptor PCBA (v2)	FPGA part should be maintained as -2 speed grade (similar to v1)- CID 10867. On the V2, due to stock issue, -1 speed grade was used as a substitute CID 12658	Use CID 10867 -2 speed grade FPGA part for future builds.		•
29	FPGA Interceptor PCBA (v2)	Insufficient RAM for video buffer and microblaze CPU	Change to FPGA part with larger RAM with same package: XC7S50-2FTGB196I		•

M2 Prototype Hardware Issues

Design issues identified to be corrected:

#	Item	Issue	Remedy	Status	Completed
1	Controller PCBA	J3 has + and - swapped around from Tritium standard. -ve on top is standard	Should swap these pins around and also consider which location a 2 pin minifit would fit to allow harness re-use	Optional: Decided not needed change	•
2	Controller PCBA	Uncertainty if IR LED pulse energy is strong enough	Potentially decrease resistors (2 parallel x 390R used for testing works with less than ideal IR receiver)	R69 and R72 3164 on v3a	•
3	Controller PCBA	(Cosmetic) Note on UVLO has wrong comment on VREF setting	Change comment to VREF=0	To be done on next update	•
4	Controller PCBA	(Cosmetic) Note on fan speed low pass filter assumes only 1 pulse per revolution	Change comment to 10krpm = 333.33hz = 3ms (2 pulses per revolution)	To be done on next update	•
5	Controller PCBA	(Cosmetic) TP30 "TEMP_SNS" is missing overlay for TP name	Add name to overlay	To be done on next update	•
6	Controller PCBA	EMC Iso power noise	Fit Y caps; Change Y cap connection to GND_PWR and ISO_GND-prebeads (closer to transformer) Change Y caps to 2x CID:11147		•
7	Controller PCBA	EMC Iso power noise	Fit two 100pF caps on iso secondary near transformer (parallel C67/C68) (CID:143)		•
8	Controller PCBA	EMC Iso power noise	Change drive resistors to 15R (R71/R74) (CID: 2681)		•
9	Controller PCBA	EMC Iso power noise	Change snubber resistors to 4.7R (R73/R26) (CID: 101)		•
10	Controller PCBA	EMC noise	Fit 100nF caps to earth (C73/C75)		•
11	Controller PCBA	EMC ethernet noise	Fit 1nF to earth for ethernet termination (replace DNF R1) (CID: 7905)		•
12	Controller PCBA	EMC ethernet noise	Fit U4 ESD protection		•
13	Controller PCBA	EMC fixes may require addition space on board	Can remove the secondary DNF iso transformer circuitry		•
14	Controller PCBA	Compliance - CTI not specified	Specific that CTI >=175 in the manufacturing details		•
15	Controller PCBA	Compliance - creepage; for slot to extend creepage must be >1mm	Increase slot to 1.2mm		•
16	AFE PCBA	Q2-4 Fets have incorrect part number on schematic	Update comment	(documentation only, no physical change)	•
17	AFE PCBA	Both opto pairs turn on at once. Leakage through FET gate	Change R40: 100k pull down to 10Meg / Change to flip /flop;comparator?	R40 DNF (1259) on v3	•

18	AFE PCBA	Temperature sensor doesn't produce pulses.	Change pull up resistor to 10k and mosfet to npn transistor Possible future revision: change to comparator	R16 360 on v3 Q4 12540 on v3 This is reference design so no comparator needed.	•
19	AFE PCBA	Temperature sensor doesn't produce pulses.	Correct sensor footprint.	TODO on next board revision	•
20	AFE PCBA	Compliance - CTI not specified	Specific that CTI >=175 in the manufacturing details		•
21	DCM Assembly	EMC noise	Add earth cable to power harnesses		•
22	DCM Assembly	EMC noise	Change ethernet cable to shielded one - properly terminated		•
23	DCM Assembly	EMC noise	Add f19 (13184) to ethernet cable near DC meter end		•
24	DCM Assembly	EMC noise	Change RJ45 connector to one that connects the shield (17191)		•
25	DCM ISD HMI Assembly	EMC noise	Add f19 (13184) to ethernet cable near HMI end		•
26	DCM ISD HMI Assembly	EMC noise	Correctly terminate ethernet shield to connector		•
27	DCM ISD HMI Assembly	EMC noise	Spray paint HMI's plastic back cover internally with EMI shielding paint (CID: 5154)		•
28	DCM ISD HMI Assembly	EMC noise	Include f19 ferrites on 24V to HMI that already exist in total charger build	No design action needed, already exists	•
29	DCM ISD HMI Assembly	EMC immunity	Add f19 (17022) on 12V to DC meter		•
30	DCM ISD HMI Assembly	EMC immunity	Add f19 (17022) on 24V with a turn to HMI		•
31	DCM ISD HMI Assembly	EMC noise	Add earth braid from bottom right HMI bracket direct to ground point		•
32	DCM ISD HMI Assembly	EMC noise	Add jointing compound on HMI bracket feet		•
33	DCM PK HMI Assembly	EMC noise	Include 2xf19 ferrites on 24V rail that already exist in total charger build	No design action needed, already exists	•
34	DCM PK HMI Assembly	EMC noise	Place f19 ferrite (17022) on 24V and ethernet together close to HMI		•
35	DCM PK HMI Assembly	EMC noise	Connect braid directly to HMI earth stud		•
36	DCM PK HMI Assembly	EMC noise	Place f19 ferrite (17022) on ethernet from HMI with a turn		•

Mechanical placement requirements

The DC meter is comprised of two major mechanical components

1. The main metering module - This module is installed in the output power stage of the charger and measures the output current and voltage
2. Front display intercept module - This module is installed in the HMI to securely display information on the screen

The ambient temperature around the meter components must be within the range of **-40°C to +85°C**

Main metering module

The main metering module has a number of specific installation requirements to function correctly.

- All charger output current must flow through the Shunt (shunt resistor)
- The Shunt shall be installed in the negative output
- **The Shunt must not be connected close to any large heat source (Output fuse or diodes)**
    - Any thermal difference across the Shunt will cause a substantial error in measurement (This will likely be why the ISA meter had such long bus bars internally)
- **The Shunt will generate ~13W of heat when the charger output current is at 500A**
    - It is expected that the shunt will operate at a maximum temperature of 120°C
    - Adequate cooling of the shunt must be provided. The copper bus-bars will likely not be enough
    - Cooling solutions could be
        - Heat sinks to surrounding air, attached to the copper bus-bars
        - Water cooling block
    - Adequate air flow shall be provided around the DC meter
- At lower currents such as in the RT, the shunt will generate ~0.8W at 125A
- The module must measure charger output voltage. Only a positive measurement point is required
- The output voltage is measured by connecting a single wire to an external positive bolt terminal in the charger
- The positive voltage measurement sense point must be after any output diodes (Essentially needs to be as close to the cable connection point as possible)
- The module must have a tamper evident enclosure which encases all of its electronics (Not sure if the Shunt terminals need to be enclosed too)
- The Shunt bolt terminals and positive voltage measurement point will need to have some form of tamper evident seal

Refer to this datasheet for Shunt mechanical dimensions:

"Z:\Tritium\TRI110 - DC Meter\00 - Documents\Research\Technical\front end measurement\Shunt\Vishay\WSBS8536-40 29-May-2018.pdf"

Refer to this document for conceptual PCB layout. At present use the Shunt dimensions as a reference for size:

"Z:\Tritium\TRI110 - DC Meter\00 - Documents\TRI110.021 PK High Current Meter mechanical location\TRI110.021 DRAFT circuit board layout.pdf"

Refer to this document for the general thinking of the meter location in the PK back electronics enclosure:

"Z:\Tritium\TRI110 - DC Meter\00 - Documents\TRI110.021 PK High Current Meter mechanical location\Tritium DC meter location in PK.pdf"

Front display intercept module

- This must be installed between the HMI computer and LCD screen (For PK)
- The module is just a PCB. It will need some tamper evident seal

No solution present for RT. Possibly have a small LCD screen similar for ChargePoint HMI?

Refer to this PCB that is included in the Ionity HMI for mechanical dimensions:

"Z:\Tritium\TRI110 - DC Meter\05 - FPGA Carrier Dummy\Hardware v1\Project Outputs for TRI110-05-v01\TRI110-05-v01 - PCB MANUFACTURING DETAILS.PDF"

There is another PCB that will be on top of this one, it might add ~20mm of height.

Software Development Plan

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Scope

Identification

This document contains details about the software development process for the DC Meter project identified by the Tritium internal project code: TRI110. In the Tritium issue management system known as JIRA, this project has the code: TDM and can be found at https://jira.tritium.com.au/projects/TDM/summary. In Tritium's Wiki, documentation relating to this project can be found at https://wiki.tritium.com.au/display/RND/DC+Meter.

System Overview

The need for the DC Meter is a consequence of the German authorities making it mandatory for all Electric Vehicle (EV) Charging Stations to contain a certified DC Meter suitable for billing purposes as specified in the standard document EN50470-1/-3. The values measured/calculated by the DC Meter are used by government authorities and end users to verify billing information from the Charge Point Operators (CPO) managing the EV Charging Stations.

The DC Meter is analogous to an AC Electricity Smart Meter but for EV Charging stations and instead of measuring the total AC usage to the EV Charging station, it measures the DC usage for charging the vehicle and specifically excludes power to run the EV Charging station, power lost in cabling and other losses.

Functionality includes (but not limited to):

1. accurately measure the DC usage when charging a vehicle
2. calculate the energy delivered to the user
3. keep track of charge session duration and time started
4. report the measured and calculated values to
    a. a display to the user and
    b. an independent server
5. be cryptographically secure in storage and transmission of information.

There are existing Tritium EV Charging stations installed at various site in Germany that will need to be retrofitted with the DC Meter Solution. As a consequence there are internal Tritium requirements that impact the DC Meter with regards to physical size and shape as it needs to fit in existing systems of different models and new systems yet to be built.

German installation sites include:

- Brohtal Ost
- Brohtal West
- Himmelkron
- Gruibingen Süd
- Ringdalskogen

- Hohenwarsleben
- Wunnenstein Ost
- Salzbergen
- Kjerlingland

- Wunnenstein West
- Mahlberg West
- Eichenzell
- Hohenlohe Süd

Different models of EV Charging Stations have different physical sizes and different sized Display Panels. The DC Meter also needs to fit inside these EV models and use their screens (as the DC Metering information is superimposed onto the existing Display Panel instead of having it's own screen).

System block diagram

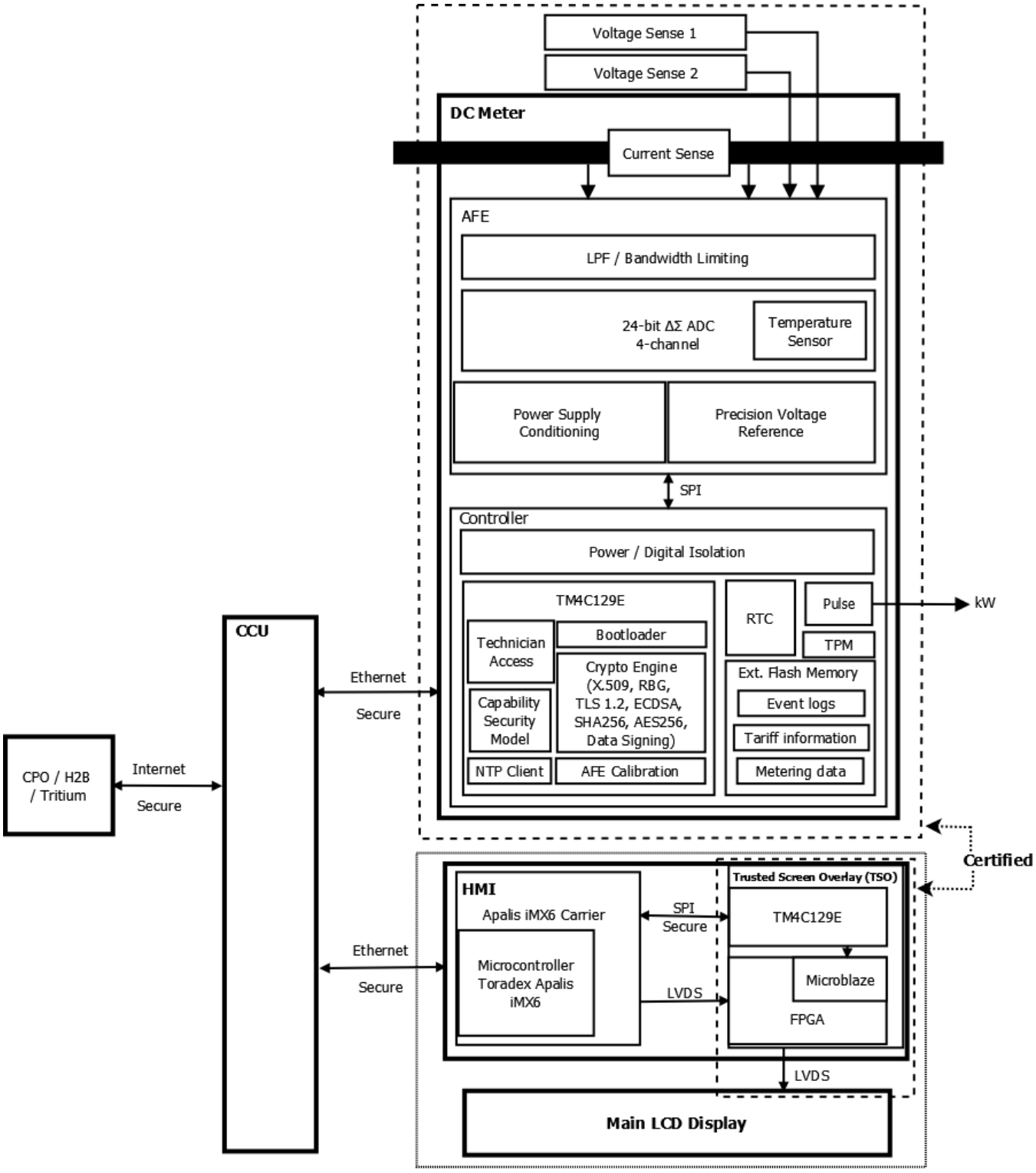


Figure – Block diagram of the DC Meter

Document Overview

As part of the certification requirements of the DC Meter, we need to show we followed a development plan, which outlines processes, tools, and deliverables as part of the development process. This document is commercially sensitive and shall be considered "Commercially in Confidence".

Relationship to Other Plans

Table - Relationship to Other Plans

Document	Subordinate/Equal/Superior
Project Management Plan	Subordinate
Test Plan	Superior
Certification Plan	Equal
Installation/Support Plan	Equal
DC Meter System Specification	Subordinate
DC Meter Firmware Design	Superior

Referenced Documents

Table - Referenced Documents

ID	Reference
R1	**VDE AR E 2418 3 100 draft E** Latest draft (as of 16 Oct 2018) of the system requirements from the German authorities (PTB) used for requirements on the DC Meter.
R2	**WELMEC Guide 7.2** Technical guidance for the application of the Measuring Instruments Directive (MID) for software-equipped measuring instruments.
R3	**WELMEC Guide 7.1** Development of Software Requirements
R4	**POL.003 Tritium C Coding Style & Guidelines** Document containing layout of C code, guides to naming convention

Overview of Required Work

Requirements as specified in the VDE[R1] are to be reviewed and used to create a software work breakdown structure. References to other documents will be reviewed for secondary requirements imposed on the system.

Documentation that needs to be submitted as part of the certification process needs to be identified and listed. Tritium have templates that will be used when providing these documents externally. All documents will need to be reviewed and approved before being released. All documents will initially be created in Confluence and later exported using Tritium Word Templates. Any released document will also be saved in Confluence with the project and optionally stored by Tritium's internal Quality and CI department.

Tests need to be performed on the software to verify the quality and reports are used as part of the deliverables for certification.

The DC Meter project is currently fully funded by Tritium and worked on by Tritium employees. The table below lists the staff working on it:

Table - DC Meter Team

Name	Position	Responsibilities

Aidan Clarke	Product Manager	
Nicholas Keeling	Power Electronics Module Lead	
Alex Rudzki	Hardware/Firmware Lead	AFE sensor board and TSO hardware
Jeremy Kincaid	Software Engineer	All firmware/software for the Controller board and remote LCD
Vivien Wong	Hardware Engineer	FPGA design
Andy Meldrum	Contract Software Developer	All firmware/software for the Controller board and remote LCD
James Sneyd	Firmware Engineer	All firmware for the Controller board and remote LCD
Jeremy Street-Thomas	Electronics Engineer	Hardware for the TM4C Controller board
John Huynh	Software Developer	Software for Trusted Technician Interface and Automated Testing
Hai Nuygen	Software/Electronics Developer	Automated Testing
Jakub Tkaczyk	Module Lead HMI	Trusted Screen Overlay TM4C, Microblaze, communications to DC Meter

Plans for Performing General Software Development Activities

Software Development Process

The software development process will follow the repetitive/iterative cycle as listed below:

1. Requirements Analysis
   Although this step is seen as the initial step, requirements are re-evaluated at regular intervals to catch anything that may have been missed the first time, or if the initial response to the requirement(s) were the best option, or if new information has been learned that leads to a different approach to meeting the requirement. This step should lead to a Work Breakdown Structure (WBS) identifying sub-tasks and level of effort with mapping to original requirements.
2. De-Risking Activities
   If there are requirements that are difficult to estimate the level of effort, or are considered high risk to the success of the project, then a dedicated activity is create to reduce the risk. This activity may be a proof of concept, or an investigation in to different technologies. The end result may be more requirements or a more solid design with estimations. These activities are optional.
3. Development Environment Setup
   Knowing what tools to use, and integrating them together to streamline the process. There may be an initial setup but as the development progresses new tools may be added and old tools replaced or tweaked.
4. Documentation
   The various documents relating to the project are not finalised until the project completes. There may be various versions released throughout but all documents should be considered "living" documents that change as new information comes to light. Documentation may be formal and released to 3rd parties, or may be informal such as JIRA tickets or Confluence/Wiki pages, or even loose documents in the project folder on the server. Formal documents are listed in later in this document, however it is up to the project members to add any other documents they see fit, especially if it help understand the system or explains the reasoning for direction of any part of the solution.
5. Implementation
   This is a combination of software implementation, testing, version control. Items are addressed based on the WBS.
6. Formal Testing
   Official results used in the certification process, peer reviews, also considered internal version of acceptance/functional testing. This step may not occur as often as the other steps as this requires more of the product to be completed before these are practicable.
7. Managing merging to the release branch
   The branch 'master' is the release branch. It should always reflect a working copy of the solution and updates to this branch should be carefully managed to ensure only wanted functionality is added, and the code is bug free.

Build/release cycles will be based on the Kanban methodology. The following releases are envisioned:

1. Prototype 1 - May 2019, using actual hardware but software not yet ready for certification although it should have near full functionality
2. Certification Candidate - Sept 2019, in testing to prove software is ready for certification.
3. Certification Release - Dec 2019, released to the PTB for certification

General Plans for Software Development

Software Development Tool-chain

Software/Firmware developers have the option of developing either in Windows or on a Linux VM running Ubuntu.

Table - Software Tools

Product	Version	Supplier	Notes

TI Code Composer Studio	Windows: 8.2.0.00007 Linux: 8.3.1.00004	http://processors.wiki.ti.com/index.php/Download_CCS#Code_Composer_Studio_Version_8_Downloads	IDE for authoring software/firmware, graphical debugging. This is the TI supported IDE.
TI ARM CGT	18.12.2.LTS	http://www.ti.com/tool/arm-cgt	The compiler created by TI to support the processor. The IDE uses this compiler in its configuration.
TM4C129E Development Board	A	http://www.ti.com/tool/EK-TM4C129EXL	The development kit used to test code on target until the DC Meter hardware is available.
TM4C129ENCPDT	Die Revision A2 Part Revision 3	http://www.ti.com/product/TM4C129ENCPDT?keyMatch=TM4C129ENCPDT&tisearch=Search-EN-everything&usecase=part-number	The core microprocessor on the DC Meter
Git	2.11.1 (Windows) 2.17.1 (Linux)	S:\GIT\Git-2.11.1-64-bit.exe https://git-scm.com/download/win - For Windows for latest built into Ubuntu (sudo apt-get update git)	Code versioning management tool.
TortoiseGit	2.4	S:\GIT\TortoiseGit-2.4.0.0-64bit.msi	The GUI frontend for Git to be used on Windows
GitKraken	6.0.1	http://gitkraken.com	The GUI frontend for Git to be used on Linux
Doxygen	1.8.13 (Linux)	http://www.doxygen.nl/download.html	Used for extracting source comments from code.
Ceedling	Ceedling:: 0.28.3 CMock:: 2.4.5 Unity:: 2.4.2 CException:: 1.3.1	http://www.throwtheswitch.org/ceedling	Used as the test harness. This is compiled into the project (as opposed to being a separate executable) so it is also listed below under reusable libraries. However, to run Ceedling you must install Ruby. Follow the instructions at the link provided. Ceedling includes Unity, CMock, and CException.
Putty		Tritium software drive	Used as the serial terminal to the development kit
PuttyGen		Tritium software drive	to create the key to access Git.
TRI70/00 - Git Setup	ssh://git@git.tritium.com.au/TRI70	Git respository setup scripts for Windows and Linux	Contains some setup scripts to more easily use Git and configures TortoiseGit

Using Code Composer Studio

Useful references: https://software-dl.ti.com/ccs/esd/documents/ccs_portable-projects.html

Code Composer Studio (CCS) is an IDE provided by TI for development of applications based on TI processors.

We are using a TM4C129ENCPDT device

CCS is provided in 32 and 64 bit versions. The 32 bit version is being deprecated. We are using the 64 bit version.

It provides several tool chains, a proprietary toolchain sourced from ARM and a GCC tool chain also sourced from ARM.

CCS is based on the Eclipse Project with TI customisation adding extra functionality for developing for TI microprocessors.

Java

CCS (and eclipse) are Java based, and require a Java Runtime Environment (JRE).CCS 64 bit requires the 64 bit version of Java.

Experience suggests that you are better off using a current version of Java 8. It seems to be Java 9 seems slower.

Project Integration

CCS supports a couple of flavor of projects

CCS Projects - using TI tool chain .. tightly integrated into the IDE .. Build, Debug all work as expected.

C/C++ Projects .. typically using gcc toolchain .. not to tightly integrated into the IDE .. in particular the debug process is not well integrated.

To get the most out of the IDE prefer CCS projects.

Workspace Configuration

Summary

Preferences\Show Advanced Settings
Preferences\ General\Always Run in background - uncheck
Preferences\General\Startup And Shutdown\RSE UI - uncheck
Preferences\General\Show Heap - check.. shown at bottom of screen
Preferences\ C/C++\Build\Console Uncheck Wrap Lines on the Console, Limit Console Output 20000, Delay Console Updates 100
Preferences\General\Workspace\Save Automatically before build
Preferences\General\Editors\Text Editors\Show Line Numbers
Preferences\C/C++/Code Style /Formatter - Import Tritium C Code Template Tritium_C.xml found in Repo/code_composer/templates - Set Tritium C as active profile
Preferences\C/C++/Code Style/Code Templates/Import Tritium Code Templates TritiumCodeTemplates-DoxygenComments.xml
Preferences\C/C++/Editor/Documentation Tool Comments - Doxygen
Preferences\General\Editors\Text Editors\Spelling\ Platform Dictionary - English (UK), User Defined Dictionary to your choosing

CCS (just like Eclipse) encourages the use of multiple work spaces. Work spaces are your friend. The less projects you have in a workspace the easier it will feel to navigate them.. and the less files CCS have to index.

Typically work spaces are not stored as part of the project repo .. they are maintained locally.

Treat work spaces as disposable items .. don't get attached to them. Most of the time this works out fine .. apart from debug configs..

Debug Configs are workspace specific .. but it is possible to import and export them. The real pain is that there a lot of settings in debug configs that have to be just right .. so the trick is to get one working.. then save the debug config somewhere in the project repo.

Remote System Explorer (RSE) - Disable It once per new workspace

Remote System Explorer is built into CCS. The idea is it lets you browse remote systems. The problem is that it tries to do this while you are compiling. It will hang the compile. There are better ways to remotely view files ..

Disable it by unchecking the Windows\Preferences\General\Startup And Shutdown\RSE UI checkbox.

This needs to be done once per Workspace. As far as I know there is no template to make this easier.

Upgrading the USB200 JTAG EMULATOR probe

For instructions refer to the site: http://dev.ti.com/tirex/explore/node?node=AADzJ8Y-La4f7Bi5Ga0TcA__FUz-xrs__LATEST

Compilation Issues

CCS seems to have a myriad of ways of having a compilation fail. Below are some tips to prevent it but another good resource is: http://software-dl.ti.com/ccs/esd/documents/sdto_ccs_build-errors.html#ccs-build-console

Console - uncheck wrap lines

This maybe a root cause of a lot of the compilation issues
A symptom of these issues is that the CCS UI becomes unresponsive. Investigation revealed that the compile/link cycle completed successfully .. but the the CCS UI was hung.
Further investigation revealed that Eclipse has long standing issues with the console being sluggish.. and it blocking when wrapping lines...

https://bugs.eclipse.org/bugs/show_bug.cgi?id=11601
https://bugs.eclipse.org/bugs/show_bug.cgi?id=519125

These issues are minimized by turning off console line wrap.

The settings for console line wrap are not shown by default .. In the preferences dialog the Show Advanced Settings Button must be pressed (The button then reads Hide Advanced Settings)

debug directory - Delete it if you have slow/hung compiles

For reasons unknown CCS will get to a point where a previously working build will either no longer work, or will compile really slowly. These issues can be often resolved by deleting the debug directory with the project folder.

This will typically happen for no obvious reason. Factors that seem to encourage this are things like shutting down CCS's host environment (in our case linux). Changing the source code underfoot also seems to encourage this. (i.e using git to change code versions.

A related issue is that if files are moved in the file system CCS will not delete compiled files in the build directory. Deleting the directory resolves this issue ..

Missing Paths - Check them if you have hung compiles

Compiles will hang if include paths are not found.. this seems to be associated with a warning that a path is not found.
Sometimes these warnings are false alarms .. sometimes not..

RSE - make sure it is disabled

As previously mentioned sometimes a build will stop when RSE tries to sync with a remote file system. Ensure it is disabled in the workspace

Troubleshooting CCS Issues

CCS Logs

CCS logs are your friends. Using a tail like tool is recommended .. one with a GUI is even better. glogg works well on Linux.
The logs are stored per CCS Workspace in the directory that CCS used to load the workspace

CCS Log
WorkspaceName/.metadata..log

Build Logs
WorkspaceName/.metadata/.plugins/org.eclipse.cdt.ui/xxx.log
There is a global build log and build logs for individual projects

CCS Progress Window

The CCS Progress Window shows the running tasks, and their progress. It will show if a task is blocking.
To show it Double Click at the bottom right hand corner of the CCS UI, to the left of the two vertical lines..



CCS Speed Optimizations

Parallel builds

All else being equal and working as expected this will have the biggest impact on build times.

In CCSProject\Properties\Build\Build Settings check enable parallel builds and choose use optimal jobs

(which should be the same as the number of cpus or virtual cpus in your PC most of the time unless you're running a VM then use:
vm_configured_thread_count - 2)

Don't bother setting it to a bigger number .. it has the effect of slowing the build down .. YMMV

Memory - you can never have enough

Out of the box CCS performs garbage collection during the build process ..

Increase Initial Heap Size for JVM Xms2048 (from 512)
Increase Max Heap Size -Xmx8192m (Modulo 1024) unless you only have 8192 system memory. -Xmx6144 is also good.

Doing this seems to make CCS much more stable

edit ccstudio.ini.. path ti/ccsv8/eclipse once per CCS

install-Xms2048m
-Xmx8192m

Software Development Methods

Writing Software

Code Composer Studio is the IDE for development as it caters to the Texas Instruments platform allowing easy debugging. It will also perform the code formatting required, and report the compiler's MISRA checking of code style, and it has it's own static code analysis reporting. All code will be generated manually with the exception of some testing framework which is able to be auto-generated in the beginning and later updated manually.

The DC Meter software folder structure will be as depicted in the figure below. Sub-modules will have their own src and test folders. This allows the modules to easily be extract and re-used in other projects keeping all related code and tests with it:

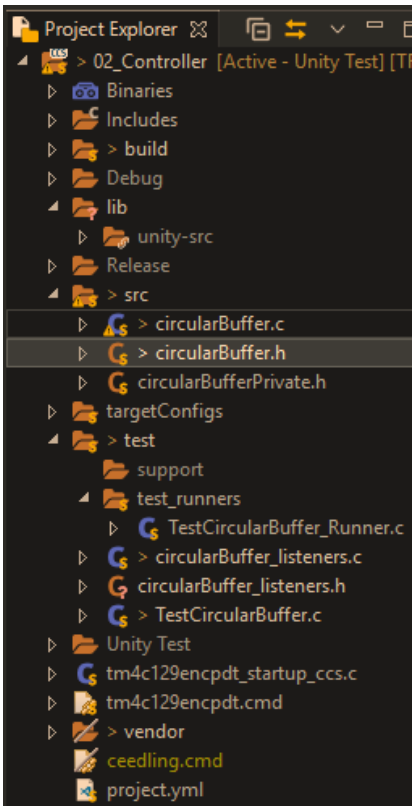


Figure - Project Folder Structure

Testing Software

Unit testing may be performed manually or automatically by the framework depending on the developer's preferences at the time. There will be a continuous integration server, which will perform a combination of nightly on target testing and off target testing.

Automated Testing

On Target Testing Vs PC/Server

Testing Metrics to Report

Code Version System

instructions for setting up Git can be found here: Git for Tritium Engineering Use. If you're using Linux then further instructions can be found in the "TRI70 /00 - Git Setup/install-prepare-commit-msg.ubuntu.sh" script.

Git will have some hooks added to allow auto testing of comments and code before permitting a commit, which is done as part of the Git preparation.

The Git repository for containing the actual DC Meter software/firmware is open to store anything. One folder will be the actual DC Meter software and everything else can be anything the project needs such as example projects, de-risking activities, documentation, etc.

Git has the command 'rebase' as a mechanism to pull from where you branched from and make all your changes happen after that rebase. For the DC Meter project, 'rebasing' is **NOT** encouraged. Git branch trees are good at showing when a change was started, how long it took, and how a merge may impact prior merges/functionality. There is also the potential for other problems when 'rebasing' especially around previous pushed commits and merging.

Prior to merging updated code back to the mainstream, a 'pull request' will be performed so a review of the update can be performed. The results of the pull request in git shall be emailed to the person responsible for performing the review and archived.

Frequently Used Git Commands

how-to: example for comparing a specific local file with the origin version of the same file. This is useful if you wanted to perform a diff prior to performing a pull. Note, difftool is used instead of diff to take advantage of 3rd party diff GUI's:

```
git difftool origin/master -- source/tri110-dcmeter-freertos-lwip2/src/cloud_task.h
```

how-to: example for comparing a specific local file of a previous version /revision with the origin, where the revision is represented as the short hash of the commit:

```
git difftool origin/master -- source/tri110-dcmeter-freertos-lwip2/src/cloud_task.h 2a03dd6
```

how-to: example list only the files that have changed for in the comparison:

```
git difftool --name-only origin/DCM-Demo -- source/tri110-dcmeter-freertos-lwip2/src/cloud_task.h 2a03dd6
```

how-to: example compare a file on one branch with the same file on another branch:

```
git difftool origin/DCM-Demo origin/TDM-366 -- source/tri110-dcmeter-freertos-lwip2/src/cloud_task.h
```

how-to: check which branches have been merged into master (the '-a' includes remote branches):

```
git branch -a --merged
```

how-to: remove a file from git that has wrongly commited and is now part of Git tracking:

```
git rm --cached <file_name>
```

how-to: merge a branch into your branch will full control over how the merge is performed:

```
git merge --no-commit --no-ff -s recursive -X patience 03728451c7099035c06c65f669b1af1656cf24e2
```

how-to: create a pull request via command line. This example requires you to have pushed your changes back to the server and the goal is to merge TDM-2104 to master.

```
git request-pull remotes/origin/master ssh://git@git.tritium.com.au/TRI110 remotes/origin/TDM-2104 >
PullRequest_TDM-2104.txt
Email the file PullRequest_TDM-2104.txt with the subject "[Pull Request] TDM-2104" to the person responsibe for
approving the pull request
```

how-to: create a pull request via TortoiseGit:

TortoiseGit has a button after the 'push' dialog, which creates a pull request.

Branching strategy

For the DC Meter project, the branching strategy used will be the feature branching strategy. In a nutshell, create a branch for the JIRA issue/task/feature you're going to work on. Do your changes. Merge back into master. For further information see: https://www.atlassian.com/git/tutorials/comparing-workflows/feature-branch-workflow

Guide to Commit Comments

The very first entry in a Git comment is the name of the branch you're working on. If you have setup Git properly, then the pre-commit hooks will do this for you. Some Git GUI applications will show you this value before you add your commit message, others will insert it after you've completed your message and you won't see it until the commit is done. Trust your GUI.

Documenting Code

Documenting code with Doxygen notation will allow auto-generation of code-documentation.

It will be up to the developer to maintain software design diagrams/documentation to keep in line with the implementation and vise versa.

Managing the release branch

Pre-requisites:

1. A JIRA ticket exists for the work
2. The JIRA ticket is assigned to a release

Merge Process (This process becomes more automated when you move to Bitbucket with its pipelines):

1. read pull request to verify
    a. only expected files were modified
    b. diff files to check modification
2. check some tests exist for the new code
3. merge code with the following command: git merge --no-commit --no-ff -s recursive -X patience <git hash of the branch to merge>
4. check differences between current and master prior to merge (maybe limit to specific files)
5. check differences between current and branch merged (maybe limit to specific files)
6. make sure it compiles in debug, release, UNITY_TEST (re-index, rebuild)
7. run in debug (no bootloader)
8. run in release with a bootloader
9. run unit tests after building for UNITY_TEST
10. checkout under Linux and build for CATCH2 and run tests (unless you get CATCH2 to run on target)
11. commit and push merge
12. ensure the related JIRA ticket is closed (consider if some comments need to be added for historical reasons)
13. update customer facing release notes (internal release notes are generated automatically)

Standards for Software Products

The VDE[R1] refers to WELMEC 7.2[R2] as a guide for development although not making it a requirement. Where possible, the software development process will comply to WELMEC 7.2[R2] as this also means we comply with MID requirements and should make certification easier.

The original requirements will not be captured in any other document than the original. Derived requirements will be captured in JIRA as tasks with a comment that links them to the parent requirement in the originating document. No formal standard is being followed for capturing requirements. In the absence of any formal guide, refer to https://confluence.atlassian.com/jirakb/using-jira-for-requirements-management-193300521.html combining JIRA and Confluence to manage requirements.

Coding style

Tritium have a formal document for this titled POL.003 -- Tritium C Coding Styles & Guidelines[R4]. Missing elements from Tritum's coding standard can be covered by:

1. https://wiki.sei.cmu.edu/confluence/display/c/SEI+CERT+C+Coding+Standard
2. https://wiki.sei.cmu.edu/confluence/pages/viewpage.action?pageId=88046682
3. https://isocpp.org/wiki/faq/coding-standards

The following deviations from POL.003 will be made:

Table - Tritium Coding Standard Deviations

POL.003 Section	Deviation	Rationale
2.2 - File description comment block	Style shall be changed to match Doxygen (http://www.doxygen.nl/manual/index.html) preferring to use the Qt style notation	Allow automatic extraction.
2.3 - Function description comment	Style shall be changed to match Doxygen (http://www.doxygen.nl/manual/index.html) preferring to use the Qt style notation	Allow automatic extraction.
2.5 – Code tags	Style shall be changed to match Doxygen Special Commands. Optionally use Markdown support to highlight existing tags	Supports automatic extraction.

Templates for Code Composer Studio can be found at `S:\Texas Instruments\CodeComposer Studio - Offline installer\Templates`.

Import existing Tritium Coding style by:

1. 'right-click' the project and select 'properties'
2. expand to: C/C++ General Formatter
3. click on "Configure Workplace Settings"
4. Click on "Import..."
5. Browse to the saved file "S:\Texas Instruments\CodeComposer Studio - Offline installer\Templates\Tritium_C.xml"
6. Click 'Apply'

To enable Doxygen support in CCS:

1. 'Right-Click' the project and select 'properties'
2. expand to: C/C++ General
3. 'Check' "Enable project specific settings"
4. Choose Documentation tool "Doxygen"
5. Click on 'OK'

To import code templates:

1. Open WindowPreferences
2. Expand to: C/C++ Code Style Code Templates
3. Click on "Import..."
4. Browse to: "S:\Texas Instruments\CodeComposer Studio - Offline installer\Templates\TritiumCodeTemplates.xml"
5. Click 'Open'
6. Click 'Apply'

The C and C++ language dialect used in writing software for the DC Meter are 'C11' and 'C++11'.

Incorporated Reusable Software Products

To reduce development time and help improve quality, some already existing and tested libraries will be used:

Table - Third Party Libraries Used in the DC Meter

Library Name	Version	License	Supplier	Description
FreeRTOS	8.2.3	MIT	https://www.freertos.org/	Real-time operating system kernel.
mbedTLS	2.13.0	Apache 2.0	https://tls.mbed.org	Elliptic curve implementations, random bit entropy generator, TLS support, signing /verification support.

lwIP	2.1.0	BSD	https://savannah.nongnu.org/projects/lwip/	Lightweight TCP/IP stack.
decNumber	0.1.1	Apache 2.0	https://github.com/amzn/ion-c	Decimal32 (base 10) arithmetic library to provide more accurate than float32/64 math operations.
Tivaware	2.1.4.178	Open	http://www.ti.com/tool/SW-TM4C	TivaWare library supporting the chosen processor. These provide access to the processor peripherals.
sts-2.1.2			https://csrc.nist.gov	Library to evaluate/test random number generation. This is used primarily as part of testing during development although some code may be used at run-time.
json-develop	3.7.0	MIT	https://github.com/nlohmann/json	Library to implement JSON, binary JSON, CBOR for configuration and messages between DC Meter and TTI
Unity/CMock/Ceedling		MIT	http://www.throwtheswitch.org	Testing framework for C
Catch2	2.6.1	Boost v1	https://github.com/catchorg/Catch2	Testing framework for C++
mqtt-c	3.1.1	MIT	http://docs.oasis-open.org/mqtt/mqtt/v3.1.1/os/mqtt-v3.1.1-os.html	Library for performing MQTT communications

Developed Reusable Software Products

Table - List of easily reusable modules for other products/projects

Software Module Name	Description
Bootloader	
crypto interface (SHA256, AES128/256, RBG, TLS 1.2, X.509, ECDSA)	
Ethernet over CAN	
BlockDevice, BlockFile, TritiumFiles	
Circular Buffer	
Software Downloader (aka updater)	
Remote Debug Printing	
Memory Allocators for STL and std::string	

Handling of Critical Requirements

The VDE[R1] addresses the requirement categories below either directly or by referring to other documents.

Safety Assurance

Safety is addressed through hardware requirements. The DC Meter is connected to high voltage, high current sources, but then fully enclosed to prevent physical access. The DC Meter software doesn't control anything safety related other than prevent a charge session starting if the log is full. There aren't any software requirements regarding safety assurance.

Security Assurance

Security assurance is addressed through either secure communications or physical restriction through tamper evident means.

Software Quality Assurance

The DC Meter software needs to be certified by the external body PTB. The following documents were used to determine what the certification requirements are.

Table - List of sources for the Product Requirements

Source	Actions	Enforceability (Guide/Strict)	Notes
"Z:\Tritium\TRI110 - DC Meter\00 - Documents\TRI110.005 PTB APPLICATION\QA Module D application and Research\PreliminaryList.pdf" – Preparation for a procedure to approve a quality system	ISO 9001:2000	Strict	For Harmonised Standards the example of OIML R49 is provided. However the VDE is to be our standard.

	Harmonised Standards, Normative Documents	Strict	Similarly for Directive 2004/22/EG (MID) the VDE is to be our standard as per the statement in: VDE, Appendix B.2 General: *"(mutatis mutandis, as no DC meters are covered)"*.
"Z:\Tritium\TRI110 - DC Meter\00 - Documents\TRI110.005 PTB APPLICATION\QA Module D application and Research\Criteria_adn_procedures_QS-Approval_March2018.pdf"	Self Conformity Assessment Document	Strict	This is the manufacturer's conformity assessment. Each requirement needs to be addressed with a description of how we conform to meet the requirement. This is partnered up with a Conformity Declaration.
	Conformity Declaration	Strict	
REA 6A	Product Requirements	Strict	
VDE refers to WELMAC 7.2 "Z:\Tritium\TRI110 - DC Meter\00 - Documents\Research\WELMEC GUIDES\SOFTWARE GUIDES\Guide_7.2_2015__Software.pdf"	Software Design Document	Strict	Must meet WELMEC 7.2 requirements: P1, P3, P4, P5, P7, L1, L2, L4 • notes on accuracy of measuring algorithms • description of the user interface • software identifiers and how they're constructed and different per version • system hardware overview • Data flow diagrams (comms and ui) • Protection against unintentional changes • Configuration parameters (set and secured) • Modes/states (relating to configuration changes) • Description of the stored data (usage, events, etc)
	Operating Manual	Strict	
	Source Code	Strict	

Addressing WELMEC 7.2

To assist in understanding what requirements apply to the DC Meter in WELMEC 7.2[R2], the DC Meter needs to be assigned a class. As the DC Meter is *" a built-for-purpose instrument with embedded software"* the WELMEC 7.2[R2] requirement type 'P' is applicable. The DC Meter also has functionality and hardware to support *"long term storage of measurement data"*, and *"transmission of measurement data"* and *"download of legally relevant software"* so the IT configuration (modular extensions) applicable include 'L', 'T', 'S', and 'D'. As this is an electricity meter the instrument specific extension of 'I.3' is applicable as described in Chapter 10.3 of WELMEC 7.2.

In accordance with P2/U2 we address detection of specific errors using recommendations from *WELMEC, Chapter 10 Extension I: Instrument Specific Software Requirements* as detailed below.

Important notes to consider when addressing WELMEC 7.2 concerns:

1. Some fault detections are possibly in hardware (e.g. MCU FaultISR, external flash Error Correction Codes)
2. Faults not detected by hardware are detected by software using either CRC or SHA256 hash calculations, or cryptographic signing.
3. Software detection is done regularly
4. The DC Meter is a dedicated instrument with an embedded system subject to legal control
5. The DC Meter depends on a network connection to the internet via a network switch providing firewall protection
6. The sensor is integerated
7. The user interface is always under legal control
8. There is long term, fixed (non-removable), local storage for audit logs
9. Uses the FreeRTOS operating system
10. The only other application residing on the system is the bootloader, which itself is part of the legally relevant software but separate to the metering software. Only one can run at a time, the only way to start the bootloader is to reboot, then it starts the metering application.
11. All software on the DC Meter is subject to legal control.

Using the risk table specified in the WELMEC 7.2[R2] guide in section I.3.6 the DC Meter has been identified in the risk class of **TBD**.

Table - WELMEC 7.2 Risk Class Definition

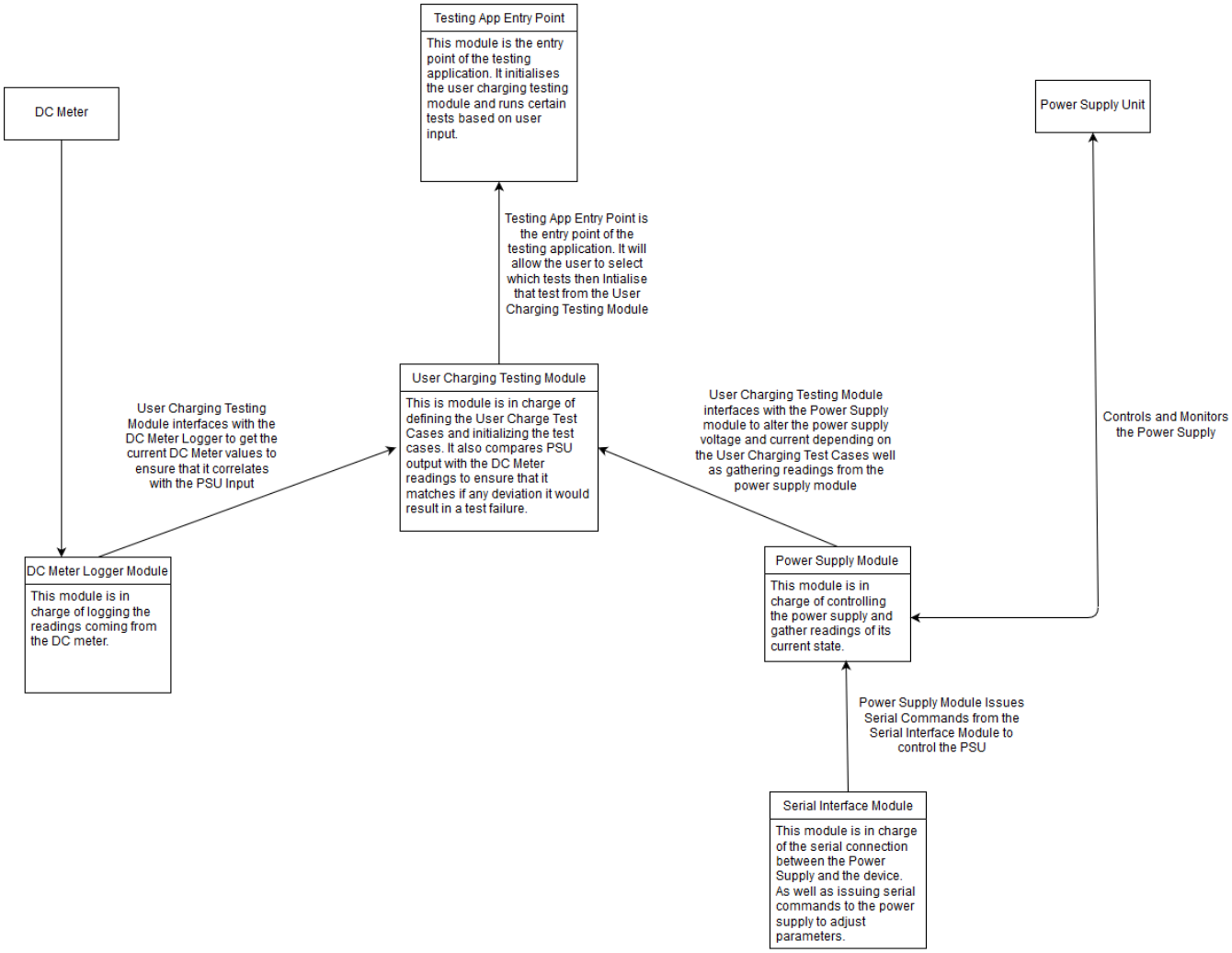
Risk Class	Software Protection	Software Examination	Software Conformity
A	Low	Low	Low
B	Middle	Middle	Low
C	Middle	Middle	Middle
D	High	Middle	Middle
E	High	High	Middle
F	High	High	High

Table - WELMEC 7.2 Risk Assessment

WELMEC 7.2 Requirement	Risk Assessment				Notes
	Software Protection	Software Examination	Software Conformity	Risk Class	
P1: Documentation					
P2: Software Identification					
P3: Influence via user interface					
P4: Influence via communication interface					
P5: Protection against accidental or unintentional changes					
P6: Protection against intentional changes					
P7: Parameter protection					
L1: Completeness of measurement data stored					
L2: Protection against accidental or unintentional changes					
L3: Integrity of data					
L4: Authenticity of measurement data stored					
L5: Confidentiality of keys					
L6: Retrieval, verification, and indication of stored data					
L7: Automatic storing					
L8: Storage capacity and continuity					
T1: Completeness of transmitted data					
T2: Protection against accidental or unintentional changes					
T3: Integrity of data					
T4: Authenticity of transmitted data					
T5: Confidentiality of keys					
T6: Handling of corrupted data					
T7: Transmission Delay					
T8: Availability of transmission services					
S1: Realisation of software separation					
S2: Mixed indication					
S3: Protective software interface					
D1: Download mechanism					
D2: Authentication of transmitted software					
D3: Integrity of downloaded software					
D4: Traceability of legally relevant software download					

Automated Test Server

Testing Application System Architecture Diagram



Controlling Diagnostic Messages

Overview

The C/C++ compiler provides diagnostic options to control compiler- and linker-generated diagnostic messages. The diagnostic options must be specified before the --run_linker option.

--diag_error=num	Categorises the diagnostic identified by num as an error. To determine the numeric identifier of a diagnostic message, use the --display_error_number option first in a separate compile. Then use --diag_error=num to recategorize the diagnostic as an error. You can only alter the severity of discretionary diagnostic messages.
--diag_remark=num	Categorises the diagnostic identified by num as a remark. To determine the numeric identifier of a diagnostic message, use the --display_error_number option first in a separate compile. Then use --diag_remark=num to recategorize the diagnostic as a remark. You can only alter the severity of discretionary diagnostic messages.
--diag_suppress=num	Suppresses the diagnostic identified by num. To determine the numeric identifier of a diagnostic message, use the --display_error_number option first in a separate compile. Then use --diag_suppress=num to suppress the diagnostic. You can only suppress discretionary diagnostic messages.
--diag_warning=num	Categorises the diagnostic identified by num as a warning. To determine the numeric identifier of a diagnostic message, use the --display_error_number option first in a separate compile. Then use --diag_warning=num to recategorize the diagnostic as a warning. You can only alter the severity of discretionary diagnostic messages.

Suppressed Warnings (3rd Party Libraries)

The DC Meter uses 3rd party libraries to deliver certain functionality. 3rd party libraries are evaluated to be fit-for-purpose and are not maintained by Tritium. Warnings generated from these 3rd party libraries are seen as beign and are suppressed.

MQTT Library

#112-D Statement is unreachable.

#190-D Enumerated type mixed with another type.

Doxygen

Introduction

Doxygen is being used as the tool to generate documentation for the Source Code for the DC Meter. Doxygen allows C & C++ code (among others) to be parsed and to generate

HTML (among others) files that document the source code.

Source Code can be annotated with Doxygen Commands to help in this process. See http://www.doxygen.nl/manual/commands.html for details.

Doxygen distribution

Doxygen is distributed as part of Ubuntu. The distributed version atm is 1.8.13

Doxygen may also distributed as part of tool chains that use it. This saves having to integrate it into the host OS.

Doxygen may be downloaded, compiled and installed to use the current version.

Building, Installing Doxygen on Ubuntu

The current version of Doxygen is 1.8.18

See http://www.doxygen.nl/download.html for instructions for downloading, building and installing the source from git

When installing on the UbuntuVM cmake, flex and bison may also needed to be installed.

Doxygen and Code Composer (CCS) / Eclipse

Out of the box both Eclipse and Doxygen allow the selection of which version of Doxygen to use amongst those that are installed on the host.

CCS/Eclipse integrate highlighting of Doxygen Commands in several different formats.

See http://www.doxygen.nl/manual/docblocks.html for details.

CCS/Eclipse will auto generate a formatting block for a function when the following text is typed (in bold italics) above a function declaration

//!<enter>

```
//! @param aaa
//! @param bbb
```

//! @return

/*!<enter>

/*!

*

 * @param aaa

 * @param bbb

 * @return

 */

/// <enter>

/// @param tariffDescriptor

/// @param theTime

/// @param elapsedTimeMinSec

/// @param meteredEnergyThisSession_kWh

/// @return

Note that the @param and @ return format are fixed. There is no easy means to customise them.

See https://stackoverflow.com/questions/4403192/change-doxygen-comment-style-in-eclipse for details

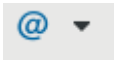
Doxygen and Eclox

Neither Eclipse or CCS are capable of running Doxygen out of the box to generate documentation.

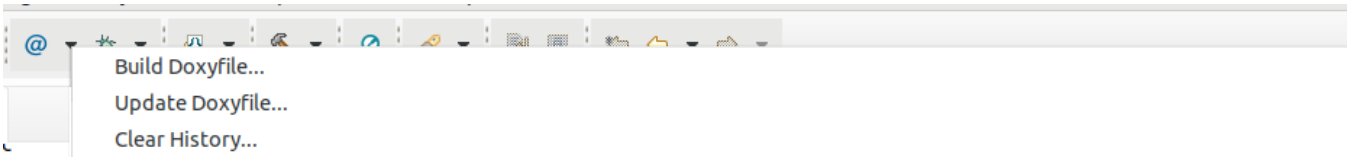
A third party add in, Eclox is used.

See https://anb0s.github.io/eclox/ for details

Eclox bundles a binary version of Doxygen (1.8.13??)

After installation the only hint that it has installed is the appearance of a Blue @ in the toolbar. 

Pressing the drop down arrow shows



Eclox allows the creation and modification of .doxyfile files.

A .doxyfile can be initially created by right mousing on a folder within CCS/Eclipse and choosing New\Other\Filter on dox then creating the file.

Once the file has been created double click on it to open the .doxyfile editor (courtesy of Eclox)

Project setting can be set on both the Basic and Advanced page.

ToDo .. come up with preferred settings ...

Doxygen and the DC Meter Code Base

The DC Meter is a mix of C and C++ Code. A recurring pattern is for a C++ class to have associated C code in the same header file.
The DC Meter uses .h for header files for both C + C++ code.

Doxygen is not a based on a compiler, it parses the code independently.

When a C++ compiler is compiling a file the macro__cplusplus is defined. When a C compiler is compiling a file the __cplusplus macro is not defined.

This is used in the header file to conditionally compile the C and C++ code by testing if __cplusplus is defined or not.

Doxygen does not do this, so consequently when a file that uses this technique is parsed by Doxygen the C code is parsed, and not the C++ Code.

There are a couple of workarounds for this.

One is to use Doxygen filters, which is a mechanism to pipe a file through a filter to preprocess it.

An alternative is to redefine a macro within Doxygen. This is the easier (for now) solution.

The Macro DOXYGEN is defined within Doxygen, and the following text in added to the top of header files that have a mix of C & C++ Code.

```
// When Doxygen is parsing ensure __cplusplus is defined

#ifdef DOXYGEN

#define __cplusplus

#endif //DOXYGEN
```

To define the DOXYGEN macro open the .doxyfile, go to the Advanced Dialog, Custom Tab. Search for Predefined Setting.

Predefined DOXYGEN=1

Doxygen vs Tritium C Coding Styles & Guidelines

In order to get the best out of Doxygen some of the Style Guidelines require a reinterpretation .. (Regard this as a process .. of which this is the first step ..)

2.1 Variable Names in comments : Doxygen explicitly uses Variable Names as part of a comment block. Once inserted there is no mechanism to update the comment if the variable name changes ..

```
//!
//! \param[in]          param1
//! \param[out]         param2
//! \param[in/out]      param3
//! \result             describe result
//! \brief              one line description of function
//! \details            details on separate line
//! \details            more details on separate line
//!
```

2.1 File Description Comment Block : If a \brief description, and then \details, are added under the copyright in an implementation file they are treated as documentation for the first thing that is declared in the file.

To associate with a specific class tag the text with \class

```
//! \class ClassName
//! Some Text
//! More Text
//! More Text
```

Extensions to the Tritium C Coding Standard

- Introduction
    - References
        - Web Links:
        - Books:
- Rules

Introduction

Here are some items not covered by the Tritum C coding standard that I think we should address for consistency sake and clarity. I encourage discussion of Tritium's coding standard with the goals of understanding it, improving it and us as developers, use it as a platform for:

1. better code-reuse,
2. more efficient with time on design, implementation, and testing,
3. better reliability

References

Web Links:

http://wiki.c2.com/?CodingStandardList
http://www.gnu.org/prep/standards/
https://isocpp.org/wiki/faq/coding-standards
https://users.ece.cmu.edu/~eno/coding/CppCodingStandard.html
https://wiki.sei.cmu.edu/confluence/display/c/SEI+CERT+C+Coding+Standard
https://wiki.sei.cmu.edu/confluence/pages/viewpage.action?pageId=88046682
https://www.approxion.com/capital-offenses-how-to-handle-abbreviations-in-camelcase/
https://stackoverflow.com/questions/15526107/acronyms-in-camelcase
http://cppunit.sourceforge.net/cppunit2/doc/coding_guidelines.html
http://wiki.c2.com/?CamelCase
https://isocpp.org/wiki/faq/mixing-c-and-cpp

Books:

"C++ Coding Standards: 101 Rules, Guidelines, and Best Practices". Herb Sutter and Andrei Alexandrescu, 8th August 2010.

Rules

Names containing acronyms when using camel case naming scheme often leads to an abomination in inconsistent names. For some it seems clear but adding some clarification to avoid confusion is a relatively simple task. Below I've identified what I think are the appropriate clarifications but I'm open to other suggestions. Unfortunately for me I have to change what was drummed into me at BAE Systems.

Basic Camel Case Capitalization Rules according to online references. This is provided as a starting point and does not take into account Tritium's coding standard.

1. The first letter is capitalized.
2. One or more letters in that word are also capitalised.
3. The word does not end on a capitalized letter: CamelCasE
4. No two capitalised letters shall follow directly each other: CamelCAse
5. No number in that word at any place: CamelCase1more
6. No dot(.), under_score or dash (-) within the word, only letters: Camel_Case
7. No ‘foreign’ letters in it like äöüß or accentuated like áéí. CämélCáße

Deviations for Tritium's coding standard:
point #1 needs to be changed to: "The first letter is capitalised only for Enumerations and Structures. Functions, variables (of any scope), function arguments, and file names, shall all begin with a lowercase letter."

Extensions to these rules to cater for ambiguities in dealing with abbreviations and acronyms in an attempt to better fit with Tritium's coding standard
8. Rules #4 and #6 shall prevail therefore acronyms and abbreviations shall be treated the same as words with the exception of units (see rule #9) and identifying the location of the function/global variable according to rule #9.1
8.1. If the acronym begins the name then it starts lower case and all acronym letters are lower case
8.2. If the acronym is mid name, it starts with an uppercase letter and remaining letters a lower case
8.3. If the acronym consists of only two letters, then capatilise both and try to place at the end unless rule #9 applies. (e.g. IO looks getter than Io but uV is proper)
9. Rule #6 can be broken under the following conditions:
9.1. to avoid name clashes in 'C', public functions and global variables (non-static) must be prefixed with a series of letters representing the file name they are in followed by an under_score. e.g. myFile_myGlobalVariable
9.2. to easily support units in names, the unit is postfixed to the name separated by an under_score with capitalisation conforming to SI guidelines (e.g. myfile_sampledData_mV).
9.3. the first letter following an under_score must be lower case unless it is an SI unit which is uppercase (e.g. V for Volts)
10. the first letter of the first word of the name is lowercase

Usage of the 'extern' keyword.
1. Whilst extern is automatically assumed in header files by the compiler so doesn't need to be explicitly stated except for clarification when the implementation is not in the 'c' file for that header.
2. In 'c' files, extern should be used to define functions if they're implemented outside that file and for modularity/portability/adaptability sake it is better to avoid including the header file. e.g. only the function API's are needed. This leave flexibility for the function to exist anywhere.

"#include" style.
two include styles exist:
#include <file>
#include "file"
They are different in that it tells the compiler where to start looking for the file. The first is for libraries and the second is for code you added. You can still reach a library header using double quotes as the library path is checked second. The former should not be used for code you added.

If your module allows for configuration, and that configuration can be done using #defines in a header file.
e.g. #include CONFIG_HEADER_FILE
elsewhere: #define CONFIG_HEADER_FILE "theConfigFile.h"
or in project settings -DCONFIG_HEADER_FILE="\"theConfigFile.h\""

Is it more important to enforce a strict coding layout across a company, or to be consistent within the file? Would it matter if there was a process to automatically format code on a commit hook using a tool like clang?

For a coding standard (not style guide) I recommend the Secure Coding Standard (https://wiki.sei.cmu.edu/confluence/display/c/SEI+CERT+C+Coding+Standard) over MISRA but from a company strategic direction we need to fill the gaps and address the concerns of the wider community regarding security as well as reliability.
For C++ I recommend the minimum of https://isocpp.org/wiki/faq/coding-standards and any of the 3 links to valid standards and recommend https://wiki.sei.cmu.edu/confluence/pages/viewpage.action?pageId=88046682.

Memory management needs to be consistent and it isn't always using the default malloc/calloc/free. So how do you neatly specify different memory management functions?

To avoid deeply nested if/else conditions and multiple return statements, wrap code in a do/while(0) and use break to exit.

Where to use the #include when mixing C and C++ code: as per https://isocpp.org/wiki/faq/mixing-c-and-cpp system headers are included outside the extern "C" and 'our' headers are inside. This also restricts the scope to just what is in the header that was created by 'us' instead of what is in the included headers. There may be a subtle side-effect otherwise.

Initialisation of structs at declaration should use use the ".member = value" notation to improve immunity during maintenance against change in the structure layout. new fields will be auto initialised to 0. Helps with self-documenting code. Prevents dependency on ordering of field initialisation. e.g)

```
struct myFile_sampleStruct
{
        uint32_t fieldA;
        uint8_t fieldB;
};

const struct myFile_sampleStruct sampleStruct =
{
        .fieldA = 1234;
        .fieldB = 56;
};
```

when an 'if' statement contains an immutable value as part of the condition, ensure the immutable is first in the condition. This helps prevent accidents where an '=' is missed from an equality check making it an assignemnt because the compiler will tell you it is an error, as opposed to the assignment taking place and if the result is == 0 or not.
e.g. "if(IMMUTABLE_VALUE == variableValue)" instead of "if(variableValue == IMMUTABLE_VALUE)"

Style Guide:

The following rules of thumb apply to any layout chosen. If in doubt ask these questions:
1) Code should be written once and read multiple times, so make it easy to read.
2) Be wary of fitting too much on one line. Typically people read in columns.
3) Avoid syntax that can lead to common problems that are difficult to find but easy to avoid
4) Code is typically easier to read when formatted in columns.

MQTT Broker Notes

Unsecure (port 1883)

.\mosquitto_sub.exe -t 'test/topic' -v

.\mosquitto_pub.exe -t 'test/topic' -m 'Host hello world!'

.\mosquitto.exe

Secure (port 8883)

.\mosquitto_sub.exe -t 'test/topic' -v --cafile "C:\Program Files\mosquitto\certs\ttiServerCA.crt" -p 8883 --insecure

.\mosquitto_pub.exe --cafile "C:\Program Files\mosquitto\certs\ttiServerCA.crt" -p 8883 -t 'test/topic' -m 'Host > hello world!' --insecure

.\mosquitto.exe -c mosquitto.conf

Getting the SAFE Certificate

.\mosquitto_sub.exe -t 'DCMS/CSMS/SAFE' -v --cafile .\dcmRoot-mqtt.crt -p 8883 --insecure

At end of charge, you get a data dump like this:


```
DCMS/CSMS/SAFE {"SessionID":"","EDSI":"0000000104","SAFE":"
PD94bWwgdmVyc2lvbj0iMS4wIiBlbmNvZGluZz0iVVRGLTgiPz4KPHZhbHVlcz4KCTx2YWx1ZT4KCQk8c2lnbmVkRGF0YSBmb3JtYXQ9Ik9DTUYi
IGVuY29kaW5nPSJwbGFpbiI+T0NNRnx7IkZWIjoiMS4wIiwiR0kiOiJUcml0aXVtIFB0eSBMdGQiLCJHUyI6IjEyMyIsIlBHIjoiVDAwMDAwMDAx
MDQiLCJNViI6IlRyaXRpdW0gUHR5IEx0ZCIsIk1TIjoiMTIzIiwiTUYiOiIxLjAwOSIsIklTIjoiZmFsc2UiLCJJTCI6IkhFQVJTQVkiLCJJRiI6
WyJSRklEX05PTkUiLCAiT0NQUF9OT05FIl0sIklUIjoiTE9DQUwiLCJJRCI6IiIsIlZTIjoiUlhaWU1YTVJQRVNBIiwiUkQiOlt7IlRNIjoiMjAy
MS0wOS0xNVQwNzo0ODowMCwwMDArMDIwMCBTIiwiVFgiOiJCIiwiUlYiOiIwLjAwMCIsIlJVIjoia1doIiwiU1QiOiJHIiwiUkkiOiIxLWI6MS44
LmUiLCJVVCI6IiAifSx7IlRNIjoiMjAyMS0wOS0xNVQwNzo0ODo0MywwMDArMDIwMCBTIiwiVFgiOiJDIiwiUlYiOiIzLjk1OSIsIlJVIjoia1do
IiwiU1QiOiJHIiwiUkkiOiIxLWI6MS44LmUiLCJVVCI6IiAifSx7IlRNIjoiMjAyMS0wOS0xNVQwNzo0ODo0MywwMDArMDIwMCBTIiwiVFgiOiJF
IiwiUlYiOiIzLjk1OSIsIlJVIjoia1doIiwiU1QiOiJHIiwiUkkiOiIxLWI6MS44LmUiLCJVVCI6IiAifV19fHsiU0QiOiIzMDQ0MDIyMDZGMTdG
RTQ0NzM1MDZEMUE5Qzk4MDgyOURGREZGNjNGNDdDRTM0RTQzOTFCMjM2NkYxQThFQkQ5NjRBMzgzRDUwMjIwMjUwNUYzQzdFNEFGNkQ0MThFRDE0
NkE4QURFQUQ1NTFCOTdFOUIwMTE1ODlERTY2RDNGRDkzM0NCRTc0MDYwMSJ9PC9zaWduZWREYXRhPgoJCTxwdWJsaWNLZXkgZW5jb2Rpbmc9InBs
YWluIj4zMDU5MzAxMzA2MDcyQTg2NDhDRTNEMDIwMTA2MDgyQTg2NDhDRTNEMDMwMTA3MDM0MjAwMDQ0N0FGRkFCNjU0QzMyQjhBODAyNjg5NDU2
MDUwRTRGRjM2RDVDRTIzOTYwRjdFMjBBMDgyNUQ3REI3RjY1QjFFQTA5MDg2QURCOTEwNDdCODkxNDIwQjJEMzAxRTc0NDFDMTQxQ0ZBRUVDRTJB
M0UwOUREQkZGNDlGODgzQUMzNzwvcHVibGljS2V5PgoJPC92YWx1ZT4KPC92YWx1ZXM+Cg==="}
```


Copy the "SAFE" data string to this page: https://codebeautify.org/base64-to-json-converter

Grab the XML output - this can be pasted in to Transparency Software.

MQTT - Mosquitto

Introduction

MQTT is a lightweight protocol used to publish and messages.

Clients are able to publish data to the broker on specified 'topic/s' or channel/s.

Multiple clients can connect to a single *broker,* acting as a server which then distribute the data to clients that have subscribed to specified 'topic/s'or channel/s.

The client source library was retrieved from Eclipse Paho MQTT, and installed with 'pip install paho-mqtt'

Broker was set up using Mosquitto, no security protocol was implemented, however it can be done using the mosquitto.conf file show below.

Publish

Client publish messages on specificied 'topic', no topic configuration is required. Similar to command prompt , topics have hierachy using / as separator. E. g topic= Measurement/inputVoltage/Watts

Prior to publishing, the payload must be encoded. the function 'process_msg' in client.py can be call. It detects the type of data to be publish and encode to JSON.

Publish method require a topic, and several optional parameters:

1. qos (quality of service). Default, qos = 0, the broker does not acknowledge the message have been received and the payload is not stored and re-transmitted by the client

   qos = 1. This requires an acknowledgement from broker and client stores the message until broker send a receipt (PUBACK)

   qos = 2. Message is only received once by the broker.

For the purpose of testings qos = 0 as data is sent at short interval and few messages are allowed to get lost occasionally.

2. retain. On publishing, retain = False (default). If client publish on an unsubscribed topic, the message is lost in the broker.

   if retain = True, the last payload sent by the client will be stored in the broker and will be distribute upon subscription.

e.g client.publish(topic='Test', payload='a string, dict, list, etc' , qos= 0, retain=False)

Subscribe

A client can subscribe to multiple topics to the broker. The required parameters is topic and qos

qos = 0. Default, the subscriber may not receive the payload, however it will not be received twice

qos = 1. The payload will be received at least once by the subscriber, but the same payload may be received again

qos = 2. The payload will be received exactly once. The slowest transfer as multiple 'handshakes' are required.

A publisher with qos=2 is not guaranteed that the payload will be received by Subcriber using qos = 0.

For the purpose of testings, qos =0 is used as large constant data transmission will be received constantly by the TTI and faster speed is preferred.

e.g client.subscribe(topic='Test' , qos = 0)

Callbacks:

ON_MESSAGE

When a message is received on a topic that the client subscribed to, the class method 'on_message' is automatically called. Thus on_message is used to process received payload.

```
def on_message(self,client,userdata,message):
    msg_in= message.payload.decode('utf-8')
    msg_in = json.loads(msg_in)
    if isinstance(msg_in,dict):
        self.payload_dict = msg_in
    if isinstance(msg_in,str):
        self.payload_str = msg_in
    if isinstance(msg_in,int):
        self.payload_num = int(msg_in)
    if isinstance(msg_in,list):
        self.payload_list = msg_in
```

The payload can be retrieved by calling the class attribute. E.g if the expected payload is of dict type, it can be called with *client.payload_dict.*

Since the expected payload sent by the Charger is expected to be of <dict>, the method on_message will provide the basic functionality of deserialising JSON payload and save the payload to the class attribute payload_dict.

ON_CONNECT

When the client connect to the broker, the broker send back a CONNACK message, which in response will call the method on_connect automatically. One of the parameter is rc. Which can be called for connection acknowledgement and debugging purposes.

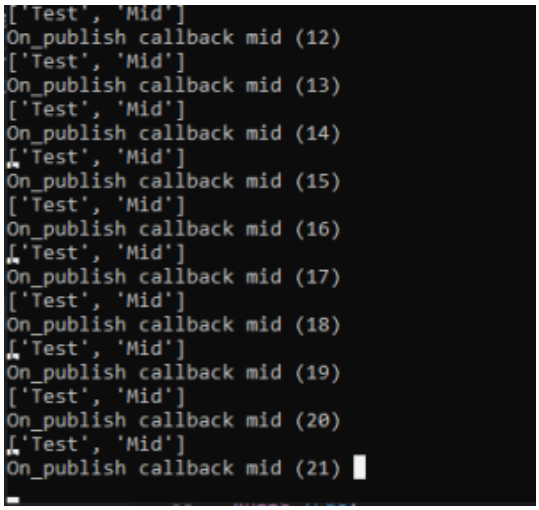
```
The value of rc indicates success or not:
    0: Connection successful
    1: Connection refused - incorrect protocol ver
    2: Connection refused - invalid client identif
    3: Connection refused - server unavailable
    4: Connection refused - bad username or passwo
    5: Connection refused - not authorised
    6-255: Currently unused.
```

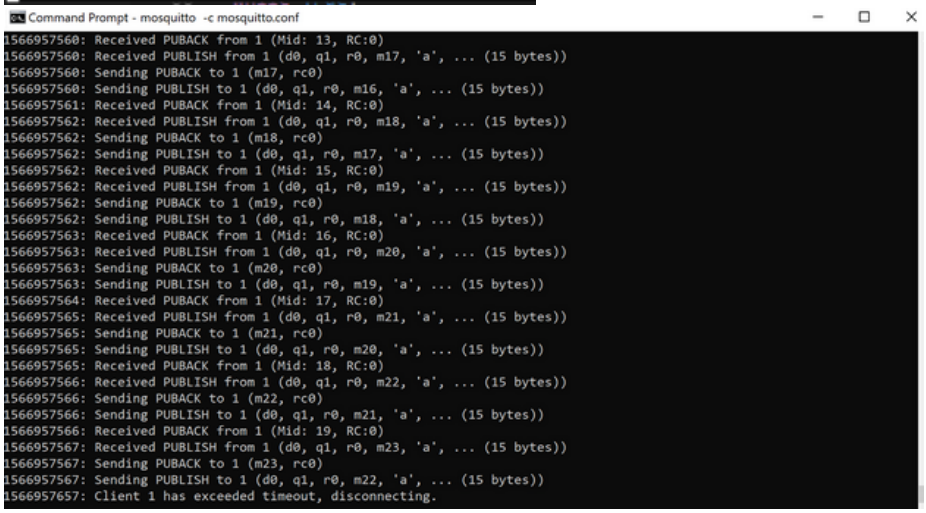
In the class method on_connect, the pass statement was used, however for debugging purposes, the rc value can be printed on the screen once the codes are uncommented.

ON_PUBLISH

Similar to on_connect, once the Client have published the payload to the broker, the on_publish callback will automatically be called. In the class method, the pass statement was used. However, in addition to rc, on_publish callback can also return 'mid' parameter

As shown above, rc == 0 signifies the message have been published successfully to the broker or else, an error have occurred. Mid value is an integer that corresponds to the published number as assigned by the client, it is essentially the message id.





Client_message_id Broker_log

By using the on_publish function, we can see that the payload transmission between broker and client is lagging. As the mid does not match between the two. The Quality of service (qos) for both client and broker

for the above logs was set at 1, which means that the payload must be received at least once by the subscriber at the cost of transmission speed. If the qos =0, the transmission speed will be faster, however the mid will always equal 0, as at qos =0, the broker and client will not set a unique payload_id.

During testing, the on_publish method simply have a pass statement, and qos is set to 0. However during debugging, qos should be set to 1 and the on_publish method should print the mid value.

ON_SUBSCRIBE

Similar to the above callbacks, on_subscribe is automatically called when the broker acknowledges. Pass statement was written here, however, the print function can be used to print the rc value.

Loop

Loop function must be called to initiate the client. If they are not called, incoming network data will not be processed and outgoing network data may not be sent in a timely fashion. There are three types of loop functionality

Loop(timeout=1.0, max_packets=1)

This is called regularly to process network events. While being called, it waits until the network socket is available for reading or writing, if appropriate, then handles incoming/outgoing data.

E.g:

client_1 = clients()

While True:

client_1.loop()

Loop_start() / Loop_stop(force=False)

These functions allow threaded interface to the network. Loop_start() must be called before connect(), then it wil runs a thread to call loop() automatically. This frees up the main thread for other work that may be blocking.

loop_stop() is called to stop the background thread.

E.g:

client_1 = clients()

client_1.connect('localhost')

client_1.loop_start()

While True:

client_1.publish('Topic',payload=None)

Loop_forever()

This is a blocking form of the network loop and will not return until the client calls disconnect. Additionally it will automatically handles reconnecting

loop_forever(timeout=1.0, max_packets=1, retry_first_connection=False)

It is recommended to simply uses loop() for testings unless threaded interface is required.

Usage:

The general usage is as follows:

1. Create client instance: client = *clients('client_id')*
2. Connect to a broker using one of the connect() functions: *client.connect('192.168.43.166' , port 1883, keepalive=60)*
3. Call one of the loop() functions to maintain network traffic flow with the broker, as shown above
4. Use subscribe to subscribe to a topic and receive messages published by the broker: *client.subscribe('Topic')*
5. Use publish() to publish messages to broker: *client.publish('Topic', payload='Test String')*
6. Use disconnect() to disconnect from broker: *client.disconnect()*

Callbacks will automatically be called upon receiving, publishing, connecting, subscribing to the broker.

MQTT - Broker set up

Installation

https://mosquitto.org/download/

After mosquitto package have been installed from the link above. Open command prompt and go to the mosquitto install directory and type *mosquitto*

You can use *mosquitto -h* for various command line switches

Mosquitto Configuration

The configuration file is included in the installation folder (mosquitto.conf). By default, MQTT Broker does not need the configuration file to run, as at default mosquitto.conf is completely commented out.

To start configuring the MQTT, edit the commented lines or create another blank config file.

Setting listener

bind_address set a specific IP address/hostname that can access the broker, if not set, the broker will be accessible to all network interfaces. In my config file, this was left empty.

port The recommended port for MQTT is 1883. For TLS the port is 8883 (This is currently not implemented). The default is 1883.

max_connections Set the maximum number of clients connection. Default is -1 which is unlimited

Logging

There are many type of messages log, which are displayed on the MQTT Broker console. Possible types are: debug, error, warning, notice, information, none, subscribe, unsubscribe, websocket and all. It is recommended that all of the log types are enable for debuging purposes.

e.g *log_type information*

Security

There are three ways Mosquitto broker verify the identity of MQTT client:

TLS

TLS can be enabled for the MQTT Broker, however it does not accept self signed certificate. Thus it was not enable for testing purposes between Charger - DcMeter and TTI.

Usernames and passwords

There is option to prompt the client to input username and password before a connection is allowed. However this is also not implemented.

Client ids

use_identity_as_clientid This replace the clientid that a client connected with with its username. This allows authentication to be tied to the clientid, which means that it is possible to prevent one client disconnecting

another by using the same clientid. For flexibility, this is set to true

```
#Define Port
port 1883

#Setting log displayed on console
log_type error
log_type notice
log_type information
log_type subscribe
log_timestamp true

#Set extra listener port
listener 8883

#Security
use_identity_as_clientid true
```

Mosquitto.conf

The Mosquitto broker does not need the config file to run. Thus it must be called explicitly to run the Broker with config file.

Ensure that the config file is in the same directory as mosquitto.exe and type the following: *mosquitto -c mosquitto.conf*

You should see the following screen if all is good:

For more information on MQTT Broker set up, https://mosquitto.org/man/mosquitto-conf-5.html

The GitHub for paho Client, https://github.com/eclipse/paho.mqtt.python

Notes on Measuring Software Performance

The documentation in this section is an informal recording of various information relating to measuring software performance as part of the development process. Any formally adopted processes will be captured in main 'Software Development Plan' page.

Table - References

Ref ID	Link/Reference	Description
PR1	https://www.freertos.org/FreeRTOS_Support_Forum_Archive/December_2011/freertos_Measuring_CPU_usage_4866946.html	Tips on using CPU usage metrics in FreeRTOS

Review of Enterprise Architect

- Introduction
- References
- Goals of the software
- Executive Summary
- Notes

Introduction

A review of Enterprise Architect (EA) from Sparx Systems was performed to see if it could help Tritium to become more efficient with time whilst improving quality and being able to supply documentation suitable for the certification requirements of the DC Meter. The information on this page contains information about how the EA software was evaluated and notes about observations of its usefulness.

References

ID	Name	Description
R1	Sparx Systems	Manufacturer of the Enterprise Architecture software found here: https://sparxsystems.com.au
R2	Tutorial Videos on EA	A useful link to videos that explain how to use EA: https://sparxsystems.com.au/resources/demos/index.html

Goals of the software

1. Easily synchronise design with implementation
   This involves generating code from design elements and the ability to round-trip code back into the design. In particular focusing on class diagrams, sequence diagrams, state diagrams.
2. Verification of state machines
   State machines are a common design element in software. Having the ability to verify state flow before implementation would greatly reduce time and improve quality
3. Ease of use
   If it takes weeks of configuration to get it right, and it feels like round-tripping is a fight every time, then maybe it isn't worth the effort.
4. Portability
   Not all software created at Tritium runs on windows and is compiled with MS Visual Studio. EA would need to do well compiling code using a cross-compiler, and work for the programming languages C and C++, assembler
5. Quality
   There must be a mechanism to ensure generated code conforms to a specified style guide or coding standard.
6. Integration with commonly used tools
   In particular JIRA, Eclipse (and hopefully Code Composer v8), MS Studio (free version), gcc compiler would be beneficial.

Executive Summary

Pros	Cons
	Doesn't run natively under Linux but will run under WINE

Notes

Version evaluated: v14, Professional

Ubuntu 18.x LTS VM- No Login Screen

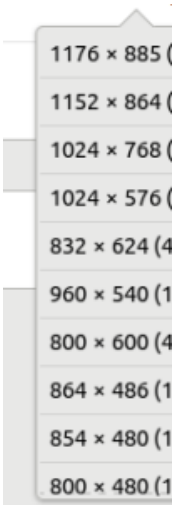
Introduction

After updating Ubuntu 18.x LTS running as a VMWare VM you are no longer able to log in to Ubuntu. Instead you are stuck on a purple screen.

Digging deeper it looks like the VMWare SVGA driver is no longer working as expected .. even though it was before the update.

It it possible to force a login by editing the grub command line and adding "nomodeset" to the linux command.
Adding the nomodeset parameter instructs the kernel to not load video drivers and use BIOS modes instead until X is loaded.
Doing this results in the login screen being shown.

Once logged in available screen sizes are limited to a maximum of 1176 x 885 pixels.



Investigation

Looking at the vmware.log shows the following entry

SVGA-GFB: Truncated maximum resolution for register modes to VRAM size: VRAM=4194304 bytes

For reasons that are not clear the SVGA driver is limiting itself to 4MB of RAM .. which is not enough to allocate screens bigger than 1176 x 885 pixels

I was able to replicate this behaviour on a clean install of Ubuntu 18 LTS .. updating resulted in exactly the same behaviour.
Whatever the root cause is is in the distro.

VMware have a flag that allows for minimum RAM allocation to be 8MB
svga.minVRAM8MB=TRUE – ensures 8Mb allocated to RAM rather than 4 .. does not go bigger than 8

Workround

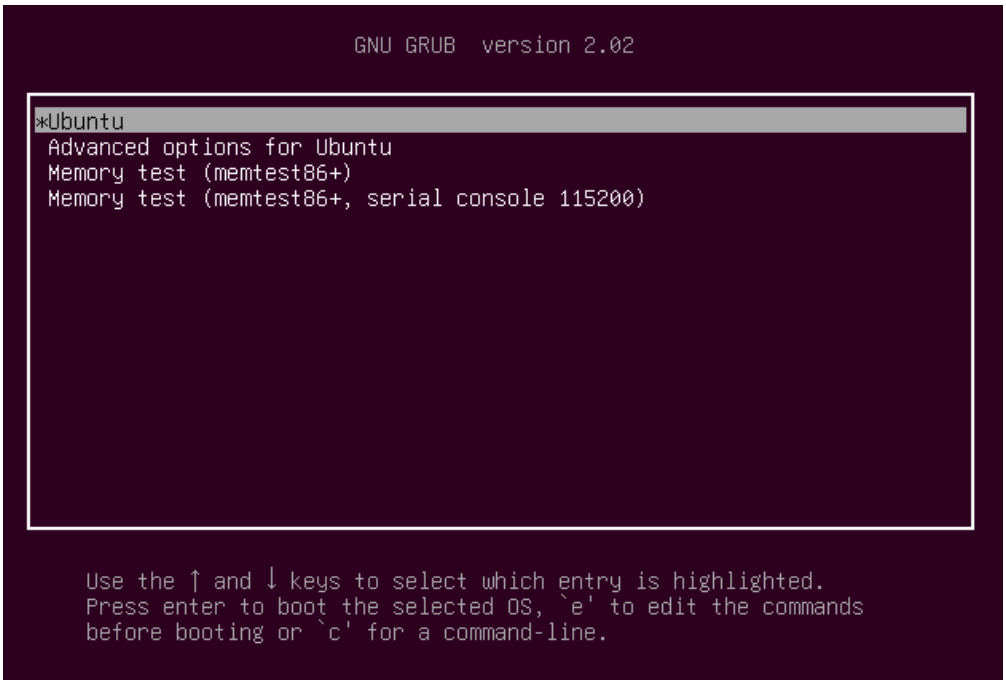
Edit the VM associated .vmx file. Do this with VMWare workstation NOT running.

Change svga.autodetect from TRUE to FALSE
svga.autodetect = "FALSE"
This will prevent the screens to be auto updated, and will force the use of config within the .vmx file

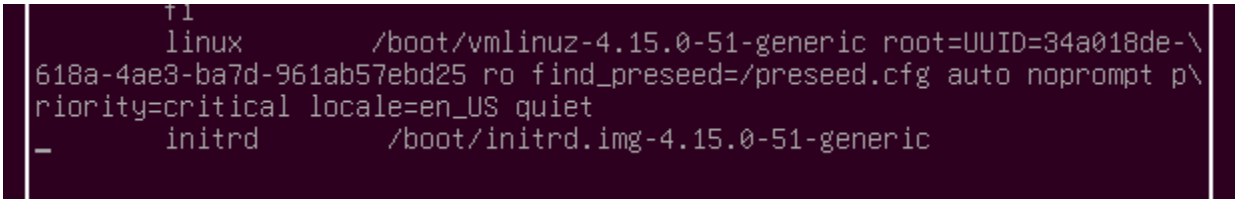
Add an entry for svga.minVRAM8MB, set to TRUE
svga.minVRAM8MB=TRUE
This will force the SVGA driver to allocate 8MB of RAM .. enough for 1920 x 1080

Add nomodeset command temporarily to grub command

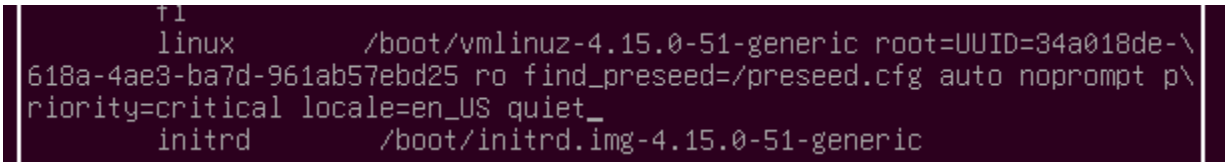
Boot VM
Click in the VM Window as it starts
Hold Down the Shift Key to get to GRUB Screen



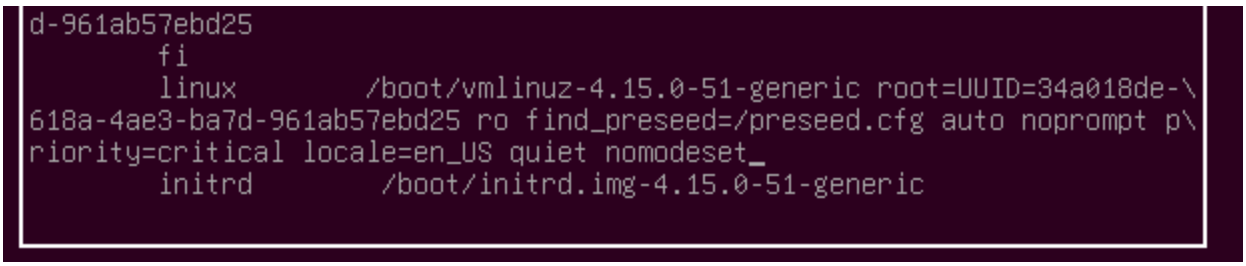
With *Ubuntu selected press e to edit grub options
Press down arrow until cursor is on line with initd on it ..



The press left arrow once .. cursor will move to end of previous line



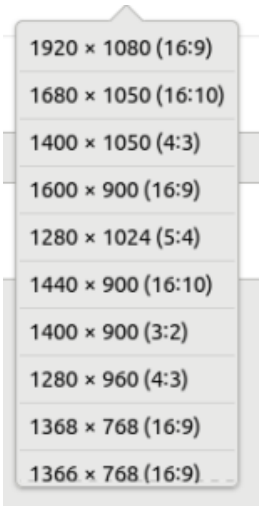
type " nomodeset"



Then press F10 key to continue boot process

This will bring up Login Screen .. Login as usual

Check screen resolutions .. expect to see resolution list like this ..



If you dont get resolutions bigger than 1176 x 885 then screen buffer is 4MB

To make nomodeset permanent

At a command prompt:

cd /etc/default
sudo nano grub

Modify
GRUB_CMDLINE_LINUX_DEFAULT="quiet"
to
GRUB_CMDLINE_LINUX_DEFAULT="quiet nomodeset"

```
GRUB_DEFAULT=0
GRUB_TIMEOUT_STYLE=hidden
GRUB_TIMEOUT=10
GRUB_DISTRIBUTOR=`lsb_release -i -s 2> /dev/null || echo Debian`
GRUB_CMDLINE_LINUX_DEFAULT="quiet"
GRUB_CMDLINE_LINUX="find_preseed=/preseed.cfg auto noprompt priority=critical l$
```

Save File
type sudo update-grub

```
dev@ubuntu:/etc/default$ sudo update-grub
Sourcing file `/etc/default/grub'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-4.15.0-51-generic
Found initrd image: /boot/initrd.img-4.15.0-51-generic
Found linux image: /boot/vmlinuz-4.15.0-50-generic
Found initrd image: /boot/initrd.img-4.15.0-50-generic
Found memtest86+ image: /boot/memtest86+.elf
Found memtest86+ image: /boot/memtest86+.bin
done
dev@ubuntu:/etc/default$
```

reboot VM .. should reboot to login screen

After making these changes the screen will maintain the resolution it has been set to .. it does not auto resize
This is expected to be a temporary workround till someone upstream notices this is broken !

X1 Prototype Hardware Issues

Design issues learnt that will be fixed on M1 prototype:

#	Item	Issue	Remedy	Status	Completed
1	Controller PCBA	eVref is connected to VDDA, Using external 2.5V reference means it won't boot.	Separate eVref and VDDA. Connect VDDA to 3v3 rail through ferrite bead.	Minor. To be done by the design engineer.	•
2	Controller PCBA	Creepage distance of 5.5mm around isolated section to AFE board does not meet pollution level 2, non potted requirements.	Decide on pollution degree of DC meter enclosure (to pot or not to pot)	Will require conformal coating. (Creepage: 3.2mm@PD1; 5.0 mm@PD2. Clearance: 3mm x2 for reinforced)	•
3	Controller PCBA	Choice of reinforced isolation or non-reinforced isolation components	Reinforced isolation will be used as reduce size and cost as appropriate components have been found	Decided.	•
4	Controller PCBA	Need to implement double isolation (or reinforced but not preferred) Consult IEC 61010. Rated to 1.2 kV requirements. Company practice should be that there is double isolation on any PCB that has both a low and high voltage connection. Each barrier needs to withstand 3kV dielectric (1 minute withstand) and 4 kV impulse.	Decide between option 1 and 2: 1 - Single isolation barrier on comms + single on AFE 2 - Double between MCU and AFE 3 - Reinforced isolation components	Solution 3 (duplicate of above)	•
5	Controller PCBA	Choice of DC/DC converter vs pulse transformer.	Pulse transformer will be used. Significant savings in cost and size.	Implemented	•
6	Controller PCBA	Choice of 12V or 5V DC/DC for AFE power	Replace the 12V/5V DC/DC with a 12V buck to 6V, install a 5V linear regulator on AFE, and drive ovenisation from 6V and AFE control circuitry off 5V.	Implemented	•
7	Controller PCBA	Power output of DC/DC converter is too low to drive ovenisation on AFE board to sufficient capacity.	Increase power to ovenisation when changing to pulse transformer.	2 pulse transformers in parallel gives up to 12W available.	•
8	Controller PCBA	D5 (diode to Super cap) doesn't meet expected pin out. Pin 2 appears to be NC.	Connect cathode to pin 3		•
9	Controller PCBA	Not enough component space as a single sided board for creepage isolation (safety) requirements	Make controller PCB double sided?	Single sided	•
10	Controller PCBA / TSO	Not considered industry best practice for a security device to not include a form of hardware security module. Cannot find level on any of the datasheets.	Add a Infineon or Microchip TPM 1.2 specification SPI /I2C IC with FIPS-140-2 level 2 minimum to both the DC meter controller and FPGA interceptor board	TPM: SLB9670 added	•
11	Controller PCBA	No silkscreen showing AFE isolation barrier	Add silkscreen to show isolation barrier	Determined it was not needed	•
12	Controller PCBA	Centre length board cutout no longer required for positioning in enclosure	Remove board cutout and replace with a mounting point		•
13	Controller PCBA	Designators missing from top overlay for J1-3	Add designators for J1-3 to top overlay	No action.	•
14	Controller PCBA	Sockets J1-3 are positioned too far inwards for proper connection to harnesses	Move sockets J1-3 to be flush with the edge of the PCB		•
15	Controller PCBA	Unnecessary IR LEDs for voltage and time	Remove IR LEDs for voltage and time. Possibility to remove IR LED completely because test pulse output is only needed for certification purposes. Still needs to be installed incase the DC meter is pulled out of a charger and verified on a bench.	Removed volt/time.	•
16	Controller PCBA	Missing test point for GND	Add test point for GND		•
17	Controller PCBA	Footprint for RJ45 housing (CID: 3371) is incorrect. Pin 7 is unnecessary.	Non issue. Left for compatibility to replacement parts.	Part changed anyway for consistency in company.	•
18	Controller PCBA	AFE SPI SCLK line needs 10k pull down	Add 10k pull down resistor		•
19	Controller PCBA	4Gbit external memory maybe insufficient for data purposes (can only be determined once regulation standards become clearer)	Investigate memory requirements	Sufficient given the information we currently have.	•

20	Controller PCBA	J6 and J7 thermistor connectors apparently are not easy to use	Check if these connectors work as expected. If not, replace with more reliable connectors.	Changed to 2 way pico. Fans are 3 way pico	•
21	Controller PCBA	Installed TM4C micro is currently rated 85'C	Purchase industrial version TM4C to ensure temperature range 105'C		•
22	Controller PCBA	No failsafe control if ovenisation is too hot	Micro can turn off isolated power if too hot.		•
23	Controller PCBA	100 MHz noise on negative connection points on both sides of RECOM DC/DC converter	Attach 2K at 100MHz beads on both positive and negative connection points on both input and output of DC/DC converter. Use Richard's RedExpert wondertool to decide on make and model of bead.	Removing Recom DC/DC converter, need to redesign filter.	•
24	Controller PCBA	Decide on new power supply options to replace CAN input power. Calculate power draw of new cooling systems? 12V or 24V input or wide range? Regulated or unregulated?	12V input. No power supply needed. 24V is not an option.	24V12V DCDC brick can always be added externally	•
25	Controller PCBA	Keep KK connectors for ambient and shunt thermisters on controller board?	Change connector type, keep measurement of both ambient and shunt, Richard to decide on which controller board placement side.	Ambient sensor will be onboard.	•
26	Controller PCBA	Debug feature: Software developers need to do timing verification of MCU Require 4xGPIO	Can use the existing debug header. SPI pins can be used as GPIO.		•
27	Controller PCBA	Remove the debug header?	No. Will be kept for GPIO. Configured whichever way firmware is configured.		•
28	Controller PCBA	Is a mounting post required on the long edge?	Yes - We will give location of mounting post to Bernie after placement of both controller and AFE boards.		•
29	Controller PCBA	Location of kW IR LED?	Depends on mechanical design by Bernie	Right edge. Can be vertical or right angle	•
30	Controller PCBA	Fans need to be (on/off) controlled by DC Meter	Add 2x1W Fan outputs.		•
31	AFE PCBA	ADC footprint is not the correct size (too small for actual size of pads). Thermal pad shorting to some surrounding pads.	Increase footprint size to be correct	22 Mar 2019 | A KR | Footprint corrected. Component/PCB needs to be updated	•
32	AFE PCBA	Testpoint for analogue ground near reference voltage location is not ideal. It was not re-located properly after changing ovenisation circuit for fabrication.	Move to better location		•
33	AFE PCBA	Ovenisation MOSFET footprint seems to be oversized for the part. It does work though.	Determine if footprint should be changed	Keeping as is	•
34	AFE PCBA	Socket J1 (maybe) positioned to far inwards for proper connection to harness	Move socket to be flush with PCB edge	22 Mar 2019 | A KR | Harness connects ok as is. Need to investigate tamper prevention requirements before any change made	•
35	AFE PCBA	Designator missing from top (*cough* bottom) overlay for J1	Add designator for J1 to top (*cough* bottom) overlay	22 Mar 2019 | A KR | NC - No designators are included on silk	•
36	AFE PCBA	T-section of the PCB near the mounting post could snap if uneven torque is applied when mounting shunt to busbars	Change AFE board shape on on the T-section past the solder points to the shunt to be narrower to make the AFE board physically more flexible to an unlevel shunt mounting	To be done by DRE after PCB size and shape requirements are finalised.	•
37	AFE PCBA	Clearance distance between C13 (controller board) and AFE components is less than minimal safety requirements	Increase board cutout for capacitor, or install Formex around capacitor	22 Mar 2019 | A KR | Already decided during PCB review to install Formex around capacitor	•
38	AFE PCBA	Which charge cable being used of a Dual PK needs to be determined.	Install a rough voltage sensor, PCB mount connector, optocoupler, trigger IC, and a 4 kV isolation barrier on the AFE.	Voltage sense impletented using comparator (Not isolated)	•
39	AFE PCBA	Temperature control of the ovenisation circuit could fail and cause a fire hazard.	Add an analog switch/relay to the AFE power supply.	Will be on controller board.	•

40	AFE PCBA	Kelvin sense points on future charge cables will not be able to be measured with current AFE design using a 4 channel ADC	Change 4 channel ADS1220 ADC for an 8 channel ADS1278	Scope creep - Decision made to stay with a 4 channel ADC and not include future proofing for Kelvin sense points. Power loss in charge cable will be calculated. DC meter will need to be configured to know which make & model of charge cable and its length is being used.	•
41	AFE PCBA	Analog vs Digital control signal for AFE	Error at 4 A at 25 degC is 2% with ovenisation. Can extrapolate data to generate calibration value. Can't use digital control signal because calibration will be more difficult.	Will remain analogue control. Have kept digital control hardware	•
42	AFE PCBA	Heat retention around AFE PCBA. Put foam inside the canister area, over the can, or over the whole board. Foam material?	Do additional below zero tests with foam over canister and over board. Repeat tests and check for major variances.	With 6W can get to 100'C from -35'C. Need to confirm foam to retain heat	•
43	AFE PCBA	At what power level is the ovenisation MOSFET switching on/off? We want to know whether there is a hard turn offs /ons at high power levels which could damage electronics. Get ADC upto operating temperature, then push it past this temperature and get a capture of power usage/switching signal.	Switching is reasonably slow but can be slowed with the RC on the gate.	Increased R1k. Can be tuned further if necessary	•
44	Both PCBAs	Overlay text for circuit title and company name uses Sans Serif font and looks like it's been written for a children's book and not an industrial application. Could create misunderstandings by technicians when reading (e.g. An I dashed with another I, I-I, could be interpreted as a H.)	`Use monospaced UPPERCASE Serif font for clear readability on all PCBs`	No action	•
45	Shunt	Temperature of shunt at 420 A reaches 85 °C after 25 minutes (and was still rising). 3D printed bottom part partially melted and warped (injection moulded plastic will have a different melting point but will still most likely melt in current design).	Need to investigate whether using a shunt is suitable in the first place because if it produces too much heat and consumes more power than allowed under B. 5.5.2 VDE AR E 2418 3 100 draft E. May need to look into using a fluxgate magnetometer type device. A Hall effect sensor is probably too inaccurate and will be very sensitive to the temperature of the metal strip inside the sensor (i.e. needs constant temperature). If a shunt is still to be used then a heatsink and/or fan will be required.	Decision made to stay with a shunt based current transducer for time being.	•
46	Enclosure	Ability to plug headers to 6 pin micro-molex and RJ45 sockets is restricted	Increase recess to allow better ability to unclip headers		•
47	Enclosure	AGND contact soldered to top of shunt protrudes into wall of top part enclosure	Cutout section around busbars will be expanded to increase airflow for heat dissipation		•
48	Enclosure	Weight of whole DC meter is supported by the shunt being sandwiched between the top and bottom enclosure parts. If enclosure cutout around the shunt is increased to allow additional ventilation air flow, the total weight of the DC meter will be supported only by the solder connection between the shunt contacts and AFE board. Solder contacts could crack or AFE board could snap if uneven torque applied when fastening the two studs in the shunt to busbars.	Investigate an external mounting frame option for the DC meter enclosure to off load the weight being put onto the shunt (could defeat the purpose of a universal shape enclosure for ISD/PK/RTD2 chargers if not done properly.)	Being designed by Bernie	•
49	Request	Provision of Diagnostic LEDs - right now there aren't any These are super handy during development .. also for fault diagnosis in the field If price/view ability is an issue in the field alternatively consider only populating development/early production boards.	Consider the provision of 4 LEDs Heartbeat (Green) - Flashes at 1s frequency when the Meter software is running as expected Out Of Service (Red) - Lit when the Meter is Out Of Service (for whatever reason) Remote Connection (Green) - Lit when connected to remote server (Transparency ??) Charging (Green) - Lit when the Meter considers the Charger is charging	Decision made to not be implemented. 4 debug IO available on spare header.	•
50	Flammability	Fire standards: UL 94 V2 for non live parts (e.g. plastic mounts for PCBS); UL 94 V0 for anything touching live parts.	Use V0 mounts around shunt. V2 everywhere else.		•
51	Ethernet Cables	Connecting shielded ethernet cables to an unearthed device is useless. Does the DC meter need an Earth tap? Connecting shielded would require changing the RJ45 jack.	Use shielded CAT5 cables and replace the RJ45 jack to a shielded jack. Connect shield pin to Bobsmith termination.	Made earth input possible, connected to shield.	•
52	CAN or Ethernet	CAN connectors consumes a lot of PCB surface area	Decision made to remove CAN and install a separate power connection.		•
53	FPGA Carrier PCBA	Incorrect capacitor was purchased for C106 (CID 8663). The footprint on PCB is 1206 which is correct but a 1210 part was purchased during the prototype run.	Nothing. 1210 part would still fit on the 1206 footprint.	Purchase correct part in next rev.	•
54	FPGA Interceptor PCBA	Silkscreen on J4 - P5/N5 is reversed.	N/A	Correct in next rev.	•

55	FPGA Interceptor PCBA	Replace R63 and R64 with a tact switches.	Currently they are DNP and resistor pads need to be shorted to invoke reset.	Correct in next rev.	•
56	FPGA Interceptor PCBA	Leave the 4 Vref pins as GPIOs- floating or tie to 1V. They are now tied to 2.5V VCCO which is incorrect.	Currently this is fixed in the FPGA design to generate internal 0.9V and pins are configured as GPIO.	Correct in next rev.	•
57	FPGA Interceptor PCBA	1.25V Voltage reference U2 chip is incorrectly connected in schematic.	Flip U2 chip around and soldered it to the right pads.	Correct in next rev.	•
58	FPGA Interceptor PCBA	Change R18 to 300 ohm CID 11475 for brighter LED illumination	Change over R18 from 2.2K to 330 ohm.	Correct in next rev.	•
59	FPGA Interceptor PCBA	Use larger pads for PMIC ADP5014. Library footprint needs to be updated.	N/A.	Correct in next rev.	•
60	FPGA Interceptor PCBA	Change R51 to 4.7k ohm resistor CID11468	Change over 10K resistor to 4.7K from stronger pull up.	Correct in next rev.	•
61	FPGA Interceptor PCBA	Add clock source 100MHz to MRCC pin of FPGA	• Possibly use the FPGA internal oscillator • Possibly use the EMCCLK • Possibly use LVDS differential clk • Green wire the ASEMB-100.000MHZ-LY-T to a FPGA clock capable pin	Include clock source ASEMB-100.000MHZ-LY-T in next rev.	•